Holomorphic Feynman diagrams in four dimensions and the quantum BPS ring

Botao Li[1], Kangning Liu and Chi-Ming Chang

[1]Peking University

Abstract

We compute, at arbitrary loop order, the kernels of the holomorphic Feynman graphs that carry the loop corrections to supersymmetry transformations and to regularized products of local operators in four-dimensional $\mathcal{N} = 1$ supersymmetric gauge theories. Building on the holomorphic graph integrals of Wang and the operatope of Budzik, Gaiotto, Kulp, Wu and Yu, we read the operatope current equations as those of an electrical network, with Schwinger times as resistances and holomorphic momenta as injected currents. Series, parallel and star–triangle reductions, used as changes of variables on Schwinger space, reduce each graph step by step to one loop; iterating them gives the fan, spider and cubic product-forest kernels at arbitrary loop order, and extends the reduction scheme of earlier work while giving it a physical interpretation. These kernels determine the loop-corrected, anomalous supersymmetry transformation of composite operators and the renormalized quantum product that defines the BPS ring. As an explicit application, we give every perturbative correction to the supersymmetry transformation in $\mathcal{N} = 4$ Yang–Mills theory, including its action on operators with arbitrary holomorphic derivatives. Using the marked higher operations of Gaiotto, Kulp and Wu, we also construct and compute the quantum product of operators and prove that, on $\mathbf{Q}$-cohomology, it is associative, graded commutative and unital, supplying the multiplication of the quantum BPS ring. As a consequence of this ring structure, every multi-graviton operator lifts to a cocycle of the full quantum supercharge, with an explicit and finite all-order completion. We also report a computer census of fortuitous classes and of their lifting: new fortuitous classes and one-loop lifts in $\mathcal{N} = 4$ Yang–Mills theory with several gauge groups, where we have found no two-loop lift so far, two-loop lifts in the Leigh–Strassler and β-deformations of $\mathcal{N} = 4$ Yang–Mills theory, two-, three- and four-loop lifts in abelian gauge theories, and two- and three-loop lifts in $\mathcal{N} = 2$ $SU(2)$ and $SU(3)$ gauge theories and their singlet deformations.

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1 Introduction

In a unitary superconformal theory, the cohomology of a nilpotent supercharge Q is isomorphic to the corresponding BPS sector. Each class has a unique harmonic representative, annihilated by Q and its radial adjoint $Q^\dagger = S$. For $Q = Q^4_-$ in $\mathcal{N} = 4$ Yang–Mills theory, this is the 1/16-BPS sector [1, Section 2]. The scaling dimensions of these states are fixed by their angular momenta

and R-charges through the BPS bound. Quantum corrections to the supercharge, the anomalous supersymmetry transformation, determine which classical classes persist in the quantum BPS spectrum and constrain the microscopic states of dual AdS black holes [2].

In the physical theory, a complete treatment requires a careful definition of the renormalized products of local operators and of their counterterms, which is technically demanding. The holomorphic twist of the theory is a much more efficient framework. The holomorphic twist organizes supersymmetry cohomology in the Batalin–Vilkovisky (BV) formalism [3, 4]. Homotopy transfer takes the physical BV complex to the complex of holomorphic local operators, where the action and the BV differential[1] simplify considerably. The graph integrals of the twisted theory are ultraviolet finite [5, Theorem 5 and Corollary 4]. The anomalous supersymmetry transformation of composite operators is then a quantum correction to the BV differential, organized into the higher brackets of the twisted theory [4, 6]. Its perturbative cohomology determines the perturbative quantum BPS states, and computing it requires the higher brackets at every loop order.[2] Quantum corrections to operator relations already appear in the Konishi anomaly and the four-dimensional chiral ring [7, 8]; the complete one-loop correction Q_1 was computed in [4].

We further develop the holomorphic graph-integral and higher-operation constructions of [4–6, 9] to evaluate these corrections at higher loops. The graph integrals that evaluate the higher brackets are integrals over a region of Schwinger space called the operatope [9].[3] Intuitively, these are cut Feynman diagrams, a familiar structure in the computation of anomalies [4, 5].

We observe that the current equations and the loop constraints of the operatope [9, Equations (2.26), (2.31), (2.33)] are Ohm's and Kirchhoff's laws of an electrical network: the Schwinger time of each edge becomes a resistance, the holomorphic momenta of the output letters become injected currents, and Wang's weighted Laplacian [5, Definition 12, Remark 6] becomes the conductance matrix.[4] Building on this analogy and on the locality of Ohm's law, we use the circuit reductions iteratively, transferring the sources at each step, as changes of variables on Schwinger space. They cancel the spanning-tree denominators of the earlier formulations and reduce a higher-loop graph step by step to lower loops, and finally to one loop or to a tree.[5] Our circuit iteration interprets and extends the cube-integral recursion of Gui, Wang and Williams [13, Theorem 4.13]: it gives the recursion an explicit physical meaning as series and parallel reductions of resistors, and adds the star–triangle transformations.

The operatope is defined in arbitrary dimension d, for all graphs that satisfy the Laman condition together with the corresponding bound on every subgraph. Two families of graphs are of particular interest in the holomorphic twist of four-dimensional $\mathcal{N} = 1$ gauge theories, the fans and the spiders: they give all perturbative corrections to the supersymmetry transformation of composite operators, at arbitrary loop order, in every such theory with a Lagrangian description whose superpotential is treated perturbatively, as a polynomial or a formal power series. They are defined in Sections 5 and 7. In a theory whose superpotential is at most cubic, with an arbitrary Kähler potential, such as $\mathcal{N} = 4$ Yang–Mills theory, every loop correction to the supersymmetry transformation is a fan. We derive the fan and spider kernels in closed form at

[1] Here the BV differential is the sum of the Koszul–Tate differential, which imposes the equations of motion, the gauge BRST transformation, and the nilpotent supercharge under study.

[2] The exact spectrum may also receive nonperturbative corrections: instanton effects can lift perturbative cohomology classes [3, Section 1] or, equivalently, correct the supercharges [4, Section 2.2.1].

[3] Wang defined the heat-kernel regularized graph integrals on $\mathbb{C}^d$ and established their ultraviolet finiteness, the holomorphic locality of their boundary integrals and the Kirchhoff structure of their Schwinger integrands [5, Sections 2–4]; the operatope supplied the combined Schwinger-space propagator, the edge-current variables and the Laman constraints [9, Section 2]; the higher operations are those of Gaiotto, Kulp and Wu [6], and the holomorphic supercharge construction is that of [3, 4].

[4] The reading of Schwinger parameters as resistances of a network is classical for Feynman integrals [10, 11].

[5] A single star–triangle move is a classical change of Schwinger variables [12, Lemma 2.4.23]; here, series, parallel, and star–triangle moves are iterated.

every loop order. For these graphs, the successive series and parallel reductions, read in reverse, construct the positive edge resistances from dimensionless ratios, the fractions in which a current divides between parallel resistors, and the network reduction gives the kernels in these simplex coordinates.[6]

The anomalous supersymmetry transformation[7] violates the Leibniz rule: at one loop it acts on two letters at once, like a second-order differential operator, and at n loops on $n+1$ letters. The holomorphic twist also provides a perturbative quantum product, built from the higher brackets with a pair of auxiliary fields [6] and analogous to normal ordering in an interacting conformal field theory. When the quantum master equation holds, that is, when the gauge anomaly cancels, the perturbative BV differential obeys the Leibniz rule for this product. We prove that the analogy is exact: the quantum product is cohomologous to the integral over an arbitrary three-sphere against the Bochner–Martinelli form, the propagator. It is associative and graded commutative on $\mathbf{Q}$-cohomology, but neither on the full BV complex. The quantum product is not needed to compute the cohomology as a vector space; it supplies the ring structure. For example, the test of quantum lifting [3, Section 1] then needs to be carried out only on the generators of the BPS ring. The explicit form of the product also shows that the multi-gravitons of the 1/16-BPS sector of $SU(N)$ $\mathcal{N}=4$ Yang–Mills theory remain quantum cocycles at arbitrary loop order. The quantum product is evaluated by the same kind of graph integrals; it involves more types of kernels than the supersymmetry transformation. From three loops on, the quantum product also contains interaction forests that are neither fans nor spiders; for renormalizable theories, series, parallel, and zero-source star–triangle moves keep their currents polynomial.[8]

As an explicit application, we give the all-order supercharge of $\mathcal{N}=4$ Yang–Mills theory in (8.3). We also improve the enumeration algorithm for fortuitous classes and find new classes in $\mathcal{N}=4$ Yang–Mills theory with gauge groups $SU(N)$, $SO(2N+1)$, $Sp(N)$ and G_2. The results and a description of the algorithm are collected in the repository [14]. With the explicit $Q_2, Q_3, \ldots$ derived here, we test the lifting of these classes beyond one loop. We find no two-loop lift in $\mathcal{N}=4$ Yang–Mills theory. We do find many one-loop lifts in the $SO(2N+1)$ theories; through the levels computed, they restore the agreement of the $SO(2N+1)$ and $Sp(N)$ spectra required by S-duality. We also find two-loop lifts in the Leigh–Strassler deformation [15] of $SU(3)$ $\mathcal{N}=4$ Yang–Mills theory, with superpotential

$$W = \operatorname{tr}(\gamma^1\gamma^2\gamma^3) - q\operatorname{tr}(\gamma^1\gamma^3\gamma^2) + \frac{h}{3}\left[\operatorname{tr}((\gamma^1)^3) + \operatorname{tr}((\gamma^2)^3) + \operatorname{tr}((\gamma^3)^3)\right]$$

two-, three- and four-loop lifts in abelian theories of SQED type, and two- and three-loop lifts in $\mathcal{N}=2$ $SU(2)$ and $SU(3)$ gauge theories and their singlet deformations.

Section 2 summarizes our main results, including the all-loop supercharge Q_n in $\mathcal{N}=4$ Yang–Mills theory, the quantum BPS ring and multi-graviton protection, the census of fortuitous classes and higher-loop lifting tests (in which we also find the first fortuitous class of $SU(4)$, at $E+J_L=14$), and the electrical-circuit dictionary; the proofs are in the later sections. Section 3 reviews the holomorphic twist of the physical theory, the higher brackets, and their reduction

[6] In these coordinates, the currents are polynomial and the spanning-tree denominator of the Kirchhoff solution cancels, so every coefficient in the kernel's displacement expansion is a polynomial integral over explicit products of intervals and simplices, hence a finite factorial sum.

[7] We use the terms anomalous supersymmetry transformation, loop correction to the supercharge and quantum BV differential interchangeably; strictly they are different notions, but here they refer to the same operator.

[8] In the original Schwinger times the same integrands are rational, with powers of the spanning-tree polynomial in the denominator [5, Corollaries 1–2 and proof of Theorem 6], while the operatope coordinates simplify the integrand at the price of an intricate integration region [9, Sections 2.3 and 2.6]. Fans and spiders are Laman graphs of type I′; Gui, Wang and Williams wrote their weights recursively from the L_∞ relations as integrals over unit cubes with polynomial integrands at each order in the displacements [13, Theorem 4.13], leaving the relation of the cube parameters to Schwinger parameters open [13, Remark 4.14], and evaluated the fans explicitly through three loops, the last with computer algebra [13, Sections 4.4–4.6]. We supply this relation and its physical interpretation.

to a current integral, the starting point of our computations. Section 4 proves constraints on the contributing graphs in four dimensions and classifies the graphs of Q_n. Section 5 computes the fan kernel at arbitrary loop order via local series–parallel reductions. Section 6 treats the quantum product and its Leibniz rule, with explicit products through two loops. Section 7 evaluates the remaining kernels, those of spiders and cubic product forests, using star–triangle and series–parallel operators. Section 8 writes the n-loop supercharge Q_n of $\mathcal{N}=4$ Yang–Mills theory as a differential operator of order $n+1$. Section 9 proves that the multi-gravitons remain closed under the full supercharge and sets up the successive homotopy transfer on which the practical computation of the cohomology ring rests. Section 10 concludes with open directions, and Appendix A collects the $\mathcal{N}=4$ superfield conventions and the charge sectors and gradings used in the lifting tests.

2 Summary of results

We summarize our main results in four parts: the closed integral and Taylor-series formulas for the all-loop supercharge Q_n in $\mathcal{N}=4$ Yang–Mills theory (Section 2.1), the algebraic theorems governing quantum lifting, multi-graviton protection, and the quantum BPS ring as a contraction of the classical ring (Section 2.2), the enumeration of fortuitous classes and higher-loop lifting census across gauge theories (Section 2.3), and the electrical-circuit dictionary underlying local iterative reduction of holomorphic graph kernels (Section 2.4). Graph-classification theorems and general fan, spider, and product-forest kernel derivations are presented directly in Sections 4–8.

2.1 Complete expression of Q_n in $\mathcal{N}=4$ Yang–Mills theory

In four-dimensional $\mathcal{N}=4$ Yang–Mills theory, the twisted fields assemble into a single polynomial superfield $C(x,\theta) \in \mathfrak{g} \otimes \Omega^{0,\bullet}(\mathbb{C}^2)[\theta^1,\theta^2,\theta^3]$ (Appendix A.1), and only the n-loop fan kernel contributes to Q_n for every $n \geq 1$. Our first main result evaluates Q_n in closed form at arbitrary loop order, including its action on operators carrying arbitrary holomorphic ∂_+ derivatives.

Theorem 2.1 (All-loop supercharge of $\mathcal{N}=4$ Yang–Mills theory)**.** *With the superfield action (A.3), superfield C, structure constants C_{ABC}, odd coordinates $\theta=(\theta^1,\theta^2,\theta^3)$, and parameter domain*

$$0<r_j<1 \quad (1\leq j\leq 2n-2), \qquad r_{2n-1}>0, \quad r_{2n}>0, \quad r_{2n-1}+r_{2n}<1,$$

as in (5.13), the n-loop supercharge for $n\geq 1$ acting on a local operator at x is

$$\begin{aligned} Q_n = \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n} \int_{(5.13)} dr_1\cdots dr_{2n} \left[\prod_{j=1}^{n-1} r_{2j-1}^{n-j} r_{2j}(1-r_{2j})^{j-1}\right] r_{2n}^{n-1} \int d^3\theta \int \prod_{j=1}^{n+1} d^4 z_j \\ \times C_{A_1 D R_1}\left(\prod_{j=2}^{n} C_{A_j R_{j-1} R_j}\right) C_{A_{n+1} R_n E} \\ \times (\partial_{+\dot a_1}\cdots\partial_{+\dot a_n} C^D)(x+x_1(r,z),\theta)(\partial_+{}^{\dot a_1}\cdots\partial_+{}^{\dot a_n} C^E)(x+x_{n+1}(r,z),\theta) \\ \times \frac{\delta}{\delta C^{A_1}(x+z_1,\theta)}\cdots\frac{\delta}{\delta C^{A_{n+1}}(x+z_{n+1},\theta)}, \end{aligned} \tag{2.1}$$

where the displayed derivatives act on the two output superfields before evaluation at the translated

positions

$$
\begin{aligned}
x_1^{+\dot a}(r,z) &= \sum_{\substack{j=1\\ \text{path vertices}}}^{n} (1-r_{2j-1})\left(\prod_{k=1}^{j-1} r_{2k-1}\right)\left[\left(\prod_{k=1}^{j-1}(1-r_{2k})\right) z_1^{+\dot a} + \sum_{\substack{i=2\\ \text{path vertices}}}^{j} r_{2i-2}\left(\prod_{k=i}^{j-1}(1-r_{2k})\right) z_i^{+\dot a}\right] \\
&\quad + \left(\prod_{k=1}^{n} r_{2k-1}\right) z_{n+1}^{+\dot a}, \\
x_{n+1}^{+\dot a}(r,z) &= r_{2n}\left[\left(\prod_{k=1}^{n-1}(1-r_{2k})\right) z_1^{+\dot a} + \sum_{\substack{i=2\\ \text{path vertices}}}^{n} r_{2i-2}\left(\prod_{k=i}^{n-1}(1-r_{2k})\right) z_i^{+\dot a}\right] + (1-r_{2n}) z_{n+1}^{+\dot a}.
\end{aligned}
\tag{2.2}
$$

Equations (2.1)–(2.2) are proved in Section 8 ((8.3) and (5.24)). Each coordinate r_j is a current-division ratio in the iterative series–parallel reduction of the n-loop fan (Section 2.4), and Q_n replaces $n+1$ input letters by two output letters.[9]

Because the integrand of (2.1) is polynomial in $r_1,\dots,r_{2n}$ at every order in the displacements $z_1,\dots,z_{n+1}$, integrating term by term via (5.26) yields an explicit rational Taylor expansion of Q_n. Writing $\partial_{1,+\dot a}$ and $\partial_{n+1,+\dot a}$ for $\partial_{+\dot a}$ acting on the first output factor $(\partial_{+\dot a_1}\cdots\partial_{+\dot a_n} C^D)(x,\theta)$ and the second output factor $(\partial_+{}^{\dot a_1}\cdots\partial_+{}^{\dot a_n} C^E)(x,\theta)$, respectively, we obtain

$$
\begin{aligned}
Q_n = \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n} \sum_{m=0}^{\infty} \sum_{\substack{r_0,s_0,r_{ik},s_i\ge 0\\ r_0+s_0+\sum_{i=1}^n s_i+\sum_{1\le i\le k\le n} r_{ik}=m}} & c^{(n)}_{r_0,s_0,\{r_{ik}\},\{s_i\}} \int d^3\theta \int \prod_{j=1}^{n+1} d^4 z_j \\
&\times C_{A_1 D R_1}\left(\prod_{j=2}^{n} C_{A_j R_{j-1} R_j}\right) C_{A_{n+1} R_n E} \\
&\times \frac{\left(-\frac12 z_1^{+\dot b}\partial_{1,+\dot b}\right)^{r_0}\left(-\frac12 z_{n+1}^{+\dot c}\partial_{n+1,+\dot c}\right)^{s_0}}{r_0!\, s_0!} \prod_{1\le i\le k\le n} \frac{\left[-\frac12 (z_{i+1}-z_i)^{+\dot d}\partial_{1,+\dot d}\right]^{r_{ik}}}{r_{ik}!} \\
&\times \prod_{i=1}^{n} \frac{\left[-\frac12 (z_i - z_{i+1})^{+\dot e}\partial_{n+1,+\dot e}\right]^{s_i}}{s_i!} (\partial_{+\dot a_1}\cdots\partial_{+\dot a_n} C^D)(x,\theta)(\partial_+{}^{\dot a_1}\cdots\partial_+{}^{\dot a_n} C^E)(x,\theta) \\
&\times \frac{\delta}{\delta C^{A_1}(x+z_1,\theta)}\cdots\frac{\delta}{\delta C^{A_{n+1}}(x+z_{n+1},\theta)},
\end{aligned}
\tag{2.3}
$$

with exact rational factorial coefficient

$$
\begin{aligned}
c^{(n)}_{r_0,s_0,\{r_{ik}\},\{s_i\}} = \frac{\left(\sum_{i=1}^n r_{in}\right)!\,\left(n-1+\sum_{i=1}^n s_i\right)!}{\left(n+1+\sum_{i=1}^n r_{in}+\sum_{i=1}^n s_i\right)!} \prod_{j=1}^{n-1} \frac{1}{n-j+1+\sum_{k=j}^n \sum_{i=1}^k r_{ik}} \\
\times \prod_{j=1}^{n-1} \frac{\left(1+\sum_{i=1}^j r_{ij}\right)!\,\left(j-1+\sum_{i=1}^j\left[s_i+\sum_{k=j+1}^n r_{ik}\right]\right)!}{\left(j+1+\sum_{i=1}^j\left[s_i+\sum_{k=j}^n r_{ik}\right]\right)!}.
\end{aligned}
\tag{2.4}
$$

On any polynomial local operator, expanding the z_j polynomials converts each displacement monomial directly into a jet derivative via

$$
\int d^4 z_j \, \frac{\left(-\frac12 z_j^{++}\right)^{p_j}\left(-\frac12 z_j^{+-}\right)^{q_j}}{p_j!\, q_j!} \frac{\delta}{\delta C^{A_j}(x+z_j,\theta)} = \frac{\delta}{\delta(\partial_{++}^{p_j}\partial_{+-}^{q_j} C^{A_j})(x,\theta)},
$$

so only finitely many terms m contribute to any given operator. In particular, on operators without input holomorphic derivatives ($m=0$, where $c^{(n)}_{0,0,\{0\},\{0\}} = 1/[(n!)^3(n+1)]$), (2.3)

[9] At one loop, Q_1 agrees with the operator of Budzik and Kulp [4, Equations (3.64)–(3.70)] up to normalization; its component (8.6) reproduces the Konishi anomaly.

specializes to (8.4):

$$Q_n|_{\text{input jets}=0} = \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n (n!)^3 (n+1)} \int d^3\theta\, C_{A_1 D R_1} \left(\prod_{j=2}^{n} C_{A_j R_{j-1} R_j} \right) C_{A_{n+1} R_n E} \times (\partial_{+\dot{a}_1} \cdots \partial_{+\dot{a}_n} C^D)(\partial_+{}^{\dot{a}_1} \cdots \partial_+{}^{\dot{a}_n} C^E) \prod_{j=1}^{n+1} \frac{\delta}{\delta C^{A_j}(\theta)}. \tag{2.5}$$

For the companion marked higher operations $\{\mathcal{I}^{\otimes n}, A, B\}^{\gamma,0}$ defining the n-loop quantum product $A \star B$ in (2.8), the explicit one- and two-loop operators are given in Section 6.4, while the machine-readable n-loop product-forest kernels, exact rational zero-displacement values, and Taylor-expansion tables (explicitly enumerated and evaluated through seven loops) are archived on GitHub at `All-loop-holomorphic-kernels/product_kernels` [14].

2.2 Quantum lifting, multi-graviton protection, and the quantum BPS ring

We next summarize the general structural results governing how classical Q_0-cohomology classes lift to the quantum supercharge $\mathbf{Q} = \sum_{n\geq 0} \hbar^n Q_n$ and how the quantum BPS ring inherits its multiplication from the classical BPS ring by homotopy contraction.

Lemma 2.2 (Derivative grading)**.** *For an interaction without holomorphic derivatives, Q_n raises each holomorphic derivative total—the numbers of ∂_{++} and ∂_{+-}—by n. With homotopy contractions homogeneous in these totals, the successive obstruction differential d_r of Section 9.3 raises each by r; a nonzero class can therefore be d_r-exact only if both totals of its leading representative are at least r. In particular, a class whose leading representative has no holomorphic derivatives is d_r-exact for no $r \geq 1$.*

The $\mathcal{N} = 4$ grading is Corollary A.2; the same derivative count applies to the massive SQED and mass-deformed examples in Section 2.3.

Theorem 2.3 (Finite loop order)**.** *For the interaction (3.38) with $\bar{W}$ of any degree, and every monomial $\mathcal{O}$,*

$$Q_n \mathcal{O} = 0 \qquad \text{for } n > \#(\mathsf{b}, \beta \text{ in } \mathcal{O}). \tag{2.6}$$

This is Corollary 4.12, which ensures that $\mathbf{Q}$ truncates at finite loop order on every local operator of fixed classical charges. Moreover, the quadratic higher-bracket identities imply $\mathbf{Q}^2 = 0$ whenever the interaction satisfies the quantum master equation (6.4) (Lemma 6.1). For a general off-shell $\mathcal{N} = 1$ gauge theory with gauge-invariant $\bar{W}$, the interaction transforms as (Theorem 6.2)

$$Q_0 \mathcal{I} = 0, \qquad Q_1 \mathcal{I} = \frac{ig^3}{128\pi^2} \operatorname{Tr}_R(t_A\{t_B, t_C\})\, \mathsf{c}^A (\partial_{+\dot{a}} \mathsf{c}^B)(\partial_+{}^{\dot{a}} \mathsf{c}^C), \qquad Q_{n\geq 2} \mathcal{I} = 0, \tag{2.7}$$

so standard chiral gauge-anomaly cancellation $\operatorname{Tr}_R(t_A\{t_B, t_C\}) = 0$ implies $Q_n \mathcal{I} = 0$ for all $n \geq 0$, establishing $\mathbf{Q}^2 = 0$ at every perturbative order.

Because Q_n ($n \geq 1$) acts simultaneously on $n+1$ input letters, pointwise multiplication of local operators does not satisfy the Leibniz rule under $\mathbf{Q}$. Following Gaiotto, Kulp and Wu [6, Sections 4.3 and 4.5], we deform pointwise multiplication into the marked quantum product

$$A \star B = \sum_{\substack{n\geq 0 \\ \text{loop orders}}} \frac{\hbar^n}{n!} \{\mathcal{I}^{\otimes n}, A, B\}^{\gamma,0}, \tag{2.8}$$

where $\gamma = \{(A, B)\}$ marks the edge joining the two insertions, contracted last with $z_{AB} = 0$ (6.26); for polynomial inputs, the order-$\hbar^n$ term vanishes once n exceeds their combined number of b, β letters (6.3).

Theorem 2.4 (Quantum BPS ring as a contraction of the classical ring)**.** *For the holomorphic graph operations of Sections 3.7 and 6, suppose that the interaction satisfies the quantum master equation (6.4).*

(i) $\mathbf{Q}(A \star B) = (\mathbf{Q}A) \star B + (-1)^{|A|} A \star (\mathbf{Q}B)$*, with* $|A|$ *the fermion number of* A*.*

(ii) *The quantum product induces an associative, graded-commutative, unital multiplication on* $H(\mathbf{Q})$*.*[10]

(iii) *Under successive homotopy transfer (Section 9.3), the spectral-sequence pages* $E_1 = H(Q_0)$ *and* $E_{r+1} = H(E_r, d_r)$ *are obtained by iterative homotopy contractions of the classical BPS ring* E_1*: on every page* E_r*,* $A \star B$ *induces the ordinary product inherited from the* $\hbar^0$ *classical representatives, and every obstruction differential* d_r *is a derivation of that ring structure.*

(iv) *For homogeneous quantum cocycles* A, B *and every* $L > 0$*,* $A \star B$ *is cohomologous, in the effective-observable complex at scale* L*, to the quantum normal-ordering product of [3, Section 2.3], defined by the contour integral (6.41) of the scale-*L *two-insertion expansion.*

These statements combine Theorems 6.3, 6.4 and 6.6.[11] Crucially, part (iii) means that the perturbative quantum BPS ring is a *homotopy contraction* (and, as an algebra, a subquotient) of the classical BPS ring $E_1 = H(Q_0)$: each page $E_{r+1} = H(E_r, d_r)$ is the cohomology of the graded-commutative ring E_r with respect to the derivation d_r, so the surviving classes inherit classical multiplication directly on their leading terms while classes lifted by $d_r \neq 0$ or rendered d_r-exact drop out.

This ring structure has immediate consequences for protected operators in $\mathcal{N} = 4$ Yang–Mills theory (proved in Section 9):

- *All-order protection of multi-gravitons.* Every single-graviton generator G—a component of $\mathrm{Tr}(dC)^p$ [1, Section 4]—satisfies $Q_n G = 0$ for all $n \geq 0$ (9.5). By Theorem 2.4(i), any classical multi-graviton product $G_1 \cdots G_m$ lifts to an exact $\mathbf{Q}$-cocycle via iterated $\star$-multiplication (with any bracketing),

$$\mathcal{O} = (\cdots((G_1 \star G_2) \star G_3) \cdots \star G_m), \qquad \mathbf{Q}\mathcal{O} = 0, \qquad \mathcal{O}|_{\hbar=0} = G_1 \cdots G_m, \tag{2.9}$$

  and its perturbative completion terminates at finite order: with $\deg_\theta = 0, 1, 2, 3$ on $\mathsf{c}, \gamma, \beta, \mathsf{b}$ (A.7), the coefficient of $\hbar^n$ vanishes once $3n > \deg_\theta(G_1 \cdots G_m)$ (Theorem 9.1).

- *Dressed classes and generator reduction.* If $\mathcal{O}[\mathcal{O}_0]$ is a closed quantum completion of $\mathcal{O}_0$, then $G \star \mathcal{O}[\mathcal{O}_0]$ is a closed completion of the graviton-dressed operator $G\mathcal{O}_0$ [14, supplement, Section S8.3]. More generally, because every d_r is a derivation on E_r, outgoing lifting tests reduce to the non-graviton (fortuitous) algebra generators of each page (9.43).

- *Fortuitous bound on lifted ranks.* Every completed multi-graviton class has $d_r = 0$ on every page (9.47).[12] Consequently, in any finite-dimensional charge sector of $\mathcal{N} = 4$ Yang–Mills theory in cohomological degree k, the total rank lifted through page R is bounded by the fortuitous dimension (9.51):

$$\sum_{\substack{r=1 \\ \text{pages}}}^{R} \operatorname{rank} d_r^k \leq \dim E_1^k - \dim \operatorname{span}\left\{ [\mathcal{O}_0]_1 \in E_1^k : \mathcal{O}_0 \text{ is a multi-graviton} \right\}. \tag{2.10}$$

[10] On cochains, $\star$ is associative and graded-commutative up to explicit $\mathbf{Q}$-homotopies constructed in Section 6.

[11] Part (i) follows from the quadratic higher-bracket identities [9, Section 3.2], [5, Theorem 7], [6, Equations (4.19), (4.42)] with one marked edge; the Leibniz rule and ring structure for quantum normal ordering were anticipated in [3, Section 2.2 and footnote 28], and the two-mark associator graph is drawn in [6, Figure 7]. The all-order associativity homotopy, strict unit, contraction-page ring inheritance, and normal-ordering equivalence are proved in Section 6.

[12] A closed completion can only be killed if its leading term becomes d_r-exact from a fortuitous source with nonzero holomorphic derivatives: if $\mathbf{Q}X = \hbar^r \mathcal{O}$, then $[\mathcal{O}]$ is $\hbar^r$-torsion over $\mathbb{C}[[\hbar]]$ and trivial after inverting $\hbar$ (9.52).

2.3 Fortuitous enumeration and quantum lifting results

Armed with the explicit all-loop operators Q_n and the successive homotopy-transfer algorithm (Section 9.3), we carry out a systematic computer census of classical Q_0-cohomology classes and their higher-loop obstruction maps d_1, d_2, d_3, d_4 across $\mathcal{N} = 4$, $\mathcal{N} = 2$, and $\mathcal{N} = 1$ gauge theories. Full algorithmic details, complete sector-by-sector tables, and machine-verifiable rational certificates are archived in the companion repository and supplementary material [14].

Before presenting the theory-by-theory tables, we highlight our two overarching findings:

1. *New fortuitous classes and absence of higher-loop lifts in $\mathcal{N} = 4$ Yang–Mills theory.* In $\mathcal{N} = 4$ Yang–Mills theory, we discover new fortuitous classes in $SU(2)$ (P39, X40), $SU(3)$ (the charge-conjugation-odd multiplet X_7 and four primary sectors at ten and eleven letters), $SU(4)$ (the first fortuitous class at finite rank, outside the scalar/BMN sector at $n_+ = n_- = 2$),[13] and $SO(5) = Sp(2)$, $Sp(3)$, $SO(9)$, and $SO(11)$, together with a monotone scalar class (2.15) in G_2 that vanishes upon restriction to the Cartan subalgebra. At one loop (d_1), we find new rank-one lifts in $SO(9)$ and $SO(11)$ alongside the known lifts of $\mathcal{O}_1$ in $SU(2)$ [20] and O_f in $SO(7)$ [2, 27]; through all levels computed, these d_1 pairings restore exact charge-by-charge agreement between S-dual $SO(2N+1)$ and $Sp(N)$ spectra. Crucially, across *all* $\mathcal{N} = 4$ gauge groups and sectors tested, we find *no lift beyond one loop* ($d_{r\geq 2} = 0$).

2. *Genuine two-, three-, and four-loop lifts in $\mathcal{N} = 1$ and $\mathcal{N} = 2$ gauge theories.* Once supersymmetry is reduced below $\mathcal{N} = 4$, higher-loop differentials d_r ($r \geq 2$) become nontrivial and lift classical Q_0-cohomology classes that survive one loop:

   - two-loop lifts ($d_2 \neq 0$) in the $SU(3)$ Leigh–Strassler deformation (starting from P_{14}, certified over $\mathbb{Q}$) and in the $SU(3)$ β-deformation;
   - two-, three-, and four-loop lifts ($d_2 \neq 0$, $d_3 \neq 0$, $d_4 \neq 0$, all certified over $\mathbb{Q}$ on their lowest representatives) in abelian $U(1)$ and $U(1)^2$ theories coupled to neutral singlets;
   - non-abelian two-loop lifts ($d_2 \neq 0$) in pure $\mathcal{N} = 2$ $SU(2)$ and $SU(3)$ Yang–Mills theories, and both two-loop ($d_2 \neq 0$) and three-loop ($d_3 \neq 0$, at weights 13 and 14, certified over $\mathbb{Q}$) lifts in $\mathcal{N} = 2$ $SU(2)$ Yang–Mills coupled to a flipping singlet.

The census builds on [1, 19, 21–23, 27–29], reproduces the published $SU(2)$ counts [30], the one-loop lifts of [2, 20], and the mass-deformed classes of [31], and uses commuting-scheme and invariant-theory results [8, 32–51] to analyze non-Cartan classes. The Leigh–Strassler family follows [15, 52]; our one- and two-loop operators were cross-checked against [4, 20], [13, Section 4.5], and [25], with exact integer and rational linear algebra performed via FLINT [53].

Conventions, sector labels, and rational certification. All theories below share the twisted action (3.37),

$$S_{\text{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \Big[\mathsf{b}_A \left(\bar\partial \mathsf{c}^A - \frac{g}{2} C^A{}_{BC} \mathsf{c}^B \mathsf{c}^C \right) + \left(\bar\partial \gamma_i - i g \mathsf{c}^A (t_A)^j{}_i \gamma_j \right) \beta^i + \bar W(\gamma, \mathsf{w}) \Big]_{(0,2)}, \tag{2.11}$$

specialized by choosing the gauge algebra $\mathfrak{g}$, matter representation, and antichiral potential $\bar W(\gamma, \mathsf{w})$. In $\mathcal{N} = 4$ theories, conserved charges decompose the complex into finite-dimensional sectors $(\lambda)_Y = (\theta_1, \theta_2, \theta_3; n_+, n_-)_Y$, recording the three θ^I weights, the numbers (n_+, n_-) of

[13] Earlier $SU(4)$ searches [22, 24, 25] cover the BMN sector without derivatives or gaugini; a fortuitous $SU(4)$ class built from two $SU(2)$ cores at much higher charges is constructed in [26].

$(\partial_{+\dot{+}}, \partial_{+\dot{-}})$ derivatives, and the letter count Y (Appendix A, (A.6)–(A.10)). In each sector we write $h = \dim E_1$ for the classical Q_0-cohomology dimension, f for the *fortuitous dimension*—the dimension of E_1 modulo the image of stable large-N (monotone/multi-graviton) classes [19, 28], computed as in [14, supplement, Section S1.5]—and ρ for the rank of restriction to the Cartan subalgebra, so $h - \rho$ counts classes in the Cartan kernel. While most monotone classes are visible to Cartan restriction and most fortuitous classes vanish on the Cartan subalgebra, neither implication holds universally: X40 ($SU(2)$) and X_7 ($SU(3)$) are fortuitous yet Cartan-visible ($h = \rho$), whereas the G_2 Gram determinant (2.15) is monotone yet lies in the Cartan kernel ($\rho = 0$).

To establish whether a class $[\mathcal{O}_0]_1 \in E_1$ survives to $H(\mathbf{Q})$, two sequences of tests are required on successive pages E_r: *outgoing tests*, which check whether $d_r[\mathcal{O}_0]_r = 0$ (equivalently, whether the order-r residual $R_r = \sum_{k=1}^{r} Q_k \mathcal{O}_{r-k}$ lies in $\operatorname{im} Q_0$ for a suitable chain $\mathcal{O}_0, \ldots, \mathcal{O}_{r-1}$), and *incoming tests*, which check whether $[\mathcal{O}_0]_r \notin \operatorname{im} d_r$ from lower-letter sectors. Every entry carries an explicit verification tag [14, supplement, Section S1.7]:

- $[\mathbb{Q}]$ denotes an *unconditional proof over* $\mathbb{Q}$—established either by an explicit closed rational completion / exact algebraic proof, by a machine-verified rational lifting certificate $(\mathcal{O}_0, \ldots, \mathcal{O}_{r-1}; \phi_1, \ldots, \phi_r)$ over $\mathbb{Q}$, or by exact emptiness / derivative-grading exclusion (Lemma 2.2);
- $[p, p']$ denotes independent modular rank computations modulo two large primes ($p = 2^{31} - 1$ and a second prime p');
- $[p]$ (and unmarked census ranks) denotes exact linear algebra over $\mathbb{F}_p$ with $p = 2^{31} - 1$, giving a certified lower bound on rational rank that is exact unless p divides a pivot minor.

2.3.1 $\mathcal{N} = 4$ Yang–Mills: $SU(N)$

With three adjoint chiral multiplets assembled into the superfield C of (A.2), the twisted action is (A.3) ([14, results report, §1.1]):

$$S_{\text{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \left[\int d^3\theta \left(\frac{1}{2} \delta_{AB} C^A \bar{\partial} C^B + \frac{i}{3!} C_{ABC} C^A C^B C^C \right) \right]_{(0,2)}, \qquad \mathfrak{g} = \mathfrak{su}(N). \tag{2.12}$$

By Theorem 2.4(iii), d_r is a derivation on E_r, so Table 2.1 lists only *fortuitous generators*—classes that are neither $\mathfrak{psu}(1,2|3)$ descendants of lower classes nor products of lower fortuitous classes with gravitons ([14, supplement, Sections S2.2 and S2.3]). No lift beyond one loop occurs in any tested $SU(N)$ sector.

Table 2.1: Fortuitous generators of $SU(N)$ $\mathcal{N} = 4$ Yang–Mills theory and their quantum lifting status.

G	**Class / sector**	**Classical result**	**Lifting result**
$SU(2)$	$\mathcal{O}_0 : (4,4,4;0,0)_7$	Known threshold class [20, 21]	Closed completion through $\hbar^4$; incoming excluded ($n_\pm = 0$). All-order survival $[\mathbb{Q}]$
$SU(2)$	$\mathcal{O}_1 : (6,6,6;0,0)_9$	Known class [19]	$d_1 \neq 0$ [20], lifted at one loop $[\mathbb{Q}]$
$SU(2)$	P39 : $(5,5,5;3,0)_9$	1 new class beyond 12 products and descendants	$d_1 = 0$ $[\mathbb{Q}]$; $R_2, R_3, R_4 \in \operatorname{im} Q_0$; d_5 target empty; incoming excluded ($n_- = 0$): survival $[p, p']$

G	**Class / sector**	**Classical result**	**Lifting result**
$SU(2)$	X40 : $(7,5,5;1,1)_{10}$	$h=\rho=8$, graviton dim. 7 ($f=1$, Cartan-visible)	All admissible outgoing obstructions vanish; incoming sources empty ($[p,p']$ for $r\le 2$, $[p]$ for $r\ge 3$): survival $[p,p']$
$SU(3)$	X_7 : $(2,2,2;2,2)_7$	$h=\rho=26$, graviton dim. 25 ($f=1$ [$\mathbb{Q}$], C-odd)	All admissible incoming and outgoing maps vanish $[p]$: all-order survival $[p]$
$SU(3)$	X_7 top: $(0,0,0;5,5)_7$	Fortuitous relative Lie-algebra class [$\mathbb{Q}$]	Outgoing targets empty [$\mathbb{Q}$]; no incoming map $[p]$: survival in X_7 multiplet $[p]$
$SU(3)$	F_1 $(n=3)$: $(7,4,4;0,0)_{10}$	Known tower [22, 23], $f=1$ [$\mathbb{Q}$]	Closed completion through $\hbar^4$, incoming excluded ($n_\pm=0$): all-order survival $[p]$
$SU(3)$	$(5,4,4;1,1)_{10}$	New primary sector; $h=2$, $\rho=0$	$d_1=d_2=d_3=0$ on both classes; $d_4=0$ conditional; no incoming map $[p]$
$SU(3)$	$(7,4,4;1,0)_{11}$	$h=2$, $\rho=0$; 1 class beyond F_1 descendant	$d_1=d_3=0$; $d_2=d_4=0$ conditional; incoming excluded ($n_-=0$) [$\mathbb{Q}$]; tested $[p]$
$SU(3)$	$(6,4,4;1,1)_{11}$	New primary sector; $h=3$, $\rho=0$	$d_3=0$; $d_4=0$ conditional; d_1,d_2 open; no incoming map $[p]$
$SU(3)$	$(7,5,4;1,0)_{11}$	New primary sector; $h=4$, $\rho=0$	$d_4=0$ conditional; d_1,d_2,d_3 open; incoming excluded ($n_-=0$) [$\mathbb{Q}$]
$SU(3)$	B_1 $(n=3)$: $(8,5,4;0,0)_{11}$	Known tower [22, 23]	Incoming excluded ($n_\pm=0$) [$\mathbb{Q}$]; d_1 open
$SU(4)$	$(4,2,2;2,2)_9$	First $SU(4)$ fortuitous class: $f=1$ [$\mathbb{Q}$], C-odd	$R_1,R_2\in\operatorname{im}Q_0$; $r\ge 3$ outgoing targets empty [$\mathbb{Q}$]; incoming ranks zero $[p]$

Here "conditional" indicates that $d_r=0$ follows if the Cartan restriction of R_r vanishes, because in the corresponding target sector $h=\rho$ implies $\ker(\text{Cartan})\cap\ker Q_0=\operatorname{im}Q_0$ [14, results report, §1.6]. Complete census frontiers through $SU(15)$ (where no fortuitous classes appear up to the scanned bounds) and explicit representatives of $\mathcal{O}_0$ and its graviton hair are given in [14, supplement, Sections S2.1, S8.2 and S8.3].

2.3.2 $\mathcal{N}=4$ Yang–Mills: $SO(2N+1)$ and $Sp(N)$

Montonen–Olive S-duality relates $SO(2N+1)$ and $Sp(N)$ $\mathcal{N}=4$ Yang–Mills theories, whose Lie algebras differ for $N\ge 3$:

$$S_{\text{tw}}=\int_{\mathbb{C}^2}\Omega\wedge\left[\int d^3\theta\left(\frac{1}{2}\delta_{AB}C^A\bar{\partial}C^B+\frac{i}{3!}C_{ABC}C^AC^BC^C\right)\right]_{(0,2)},\qquad \mathfrak{g}=\mathfrak{so}_{2N+1}\text{ or }\mathfrak{sp}_{2N}. \tag{2.13}$$

Classically (E_1), $SO(2N+1)$ possesses extra fortuitous classes without $Sp(N)$ counterparts; as shown in Table 2.2 ([14, supplement, Sections S3.1–S3.3]), rank-one one-loop differentials d_1 lift these excess pairs and restore charge-by-charge S-duality agreement on E_2 through $E+J_L=11$, with no d_2 candidate pairs and vanishing d_3 on all candidates.

Table 2.2: First fortuitous sectors of $SO(2N+1)$ and $Sp(N)$ $\mathcal{N}=4$ Yang–Mills theory and their lifting.

G	**Source sector**	**Classical result**	**Lifting result**
$SO(5)=Sp(2)$	$(3,2,2;1,1)_7$	New fortuitous class: $f=1$ [$\mathbb{Q}$], $\rho=0$	All admissible incoming and outgoing maps vanish [p]
$SO(7)$	$O_f:(3,3,3;0,0)_8$	Known fortuitous source [27], $f=1$ [$\mathbb{Q}$]	$\operatorname{rank} d_1=1$ into $(2,2,2;1,1)_8$ [2] [$\mathbb{Q}$]
$Sp(3)$	$(5,2,2;1,1)_9$	New fortuitous class: $h=f=1$ [$\mathbb{Q}$]	All admissible incoming and outgoing maps vanish [p]
$Sp(3)$	$(5,0,0;3,3)_9$	New class: $h=1$ [$\mathbb{Q}$], $\rho=0$	Outgoing maps excluded [$\mathbb{Q}$]; incoming d_2 unresolved
$SO(9)$	$(5,3,3;0,0)_{10}$	$h=6$, $f=1$: one-class excess over $Sp(4)$	$\operatorname{rank} d_1=1$ into $(4,2,2;1,1)_{10}$ [p]
$SO(11)$	$(7,3,3;0,0)_{12}$	$h=23$, $f=1$: excess over $Sp(5)$	$\operatorname{rank} d_1=1$ into $(6,2,2;1,1)_{12}$ [p]

2.3.3 $\mathcal{N}=4$ Yang–Mills: G_2

For the exceptional gauge algebra $\mathfrak{g}=\mathfrak{g}_2$,

$$S_{\mathrm{tw}}=\int_{\mathbb{C}^2}\Omega\wedge\left[\int d^3\theta\left(\frac{1}{2}\delta_{AB}C^A\bar{\partial}C^B+\frac{i}{3!}C_{ABC}C^AC^BC^C\right)\right]_{(0,2)},\qquad \mathfrak{g}=\mathfrak{g}_2, \tag{2.14}$$

we find a striking classical and quantum phenomenon ([14, supplement, Sections S5.2–S5.3]): the six-letter scalar Gram determinant in the 7-dimensional representation,

$$\left[\det_{I,J}\operatorname{tr}_7(\gamma^I\gamma^J)\right]=\left[\frac{1}{6}\varepsilon_{IJK}\varepsilon_{LMN}\operatorname{tr}_7(\gamma^I\gamma^L)\operatorname{tr}_7(\gamma^J\gamma^M)\operatorname{tr}_7(\gamma^K\gamma^N)\right]\neq 0, \tag{2.15}$$

defines a nonzero monotone multi-graviton class that vanishes identically on the rank-2 Cartan subalgebra ($\rho=0$), yet survives to all loop orders over $\mathbb{Q}$.

Table 2.3: Results for G_2 $\mathcal{N}=4$ Yang–Mills theory.

Sector	**Classical result**	**Lifting result**
$(2,2,2;0,0)_6$	$\dim C^6=4$, $\operatorname{rank} Q_0=3$, $h=1$, $\rho=0$ [$\mathbb{Q}$]	All-order survival [$\mathbb{Q}$]: closed multi-graviton completion (Theorem 9.1); incoming maps excluded by $n_+=n_-=0$ (Lemma 2.2)
All sectors through $E+J_L=\frac{23}{2}$	At $\frac{23}{2}$: $\sum_{\text{sectors}}h=2075$, $\sum_{\text{sectors}}(h-\rho)=170$	$d_1=0$; no d_2 candidate pairs; all d_3 candidate ranks zero [p]

2.3.4 $SU(3)$ Leigh–Strassler deformation

Deforming $\mathcal{N}=4$ $SU(3)$ Yang–Mills theory by the marginal $\mathcal{N}=1$ Leigh–Strassler superpotential [15] produces our first non-abelian two-loop lifts ($d_2\neq 0$). With three adjoint chirals and $\operatorname{tr}(t_At_B)=\frac{1}{2}\delta_{AB}$ ([14, results report, §6.1]),

$$\begin{aligned}\bar{W}(\gamma,\mathsf{w})=\;&\operatorname{tr}(\gamma^1\gamma^2\gamma^3)-q\operatorname{tr}(\gamma^1\gamma^3\gamma^2)\\&+\frac{h}{3}\left[\operatorname{tr}((\gamma^1)^3)+\operatorname{tr}((\gamma^2)^3)+\operatorname{tr}((\gamma^3)^3)\right]-\frac{1}{4}\mathsf{w}_{\dot{a}}^A\mathsf{w}_A{}^{\dot{a}}.\end{aligned} \tag{2.16}$$

Sectors $(\Theta, c; n_+, n_-)_Y$ are graded by total chiral weight $\Theta = \theta_1+\theta_2+\theta_3$ and $\mathbb{Z}_3$ charge $c = \theta_2+2\theta_3$ (mod 3) (for $\Theta \not\equiv 0 \pmod 3$, $c = 0$ represents three cyclic images). The census is evaluated at generic rational couplings $\rho = (1-q)/(1+q) = 12345/678$ and $h_e = 2h/(1+q) = 4321/97$, with multi-parameter and multi-prime checks in [14, supplement, Sections S6.1–S6.6]. Crucially, for the lowest two-loop lift $P_{14} : (7,0;0,0)_5 \to (1,0;2,2)_4$, we construct an exact rational lifting chain and dual witness certifying $\operatorname{rank} d_2 \geq 1$ unconditionally over $\mathbb{Q}$ (`ls_su3_P14_d2_cert.json.gz` in [14]):

$$\begin{aligned} P_{14}: \quad \dim E_1 = 5 \xrightarrow{\operatorname{rank} d_1=3} \dim E_2 = 2 \xrightarrow{\operatorname{rank} d_2=1\ [\mathbb{Q}]} \dim E_3 = 1, \\ Q_2\mathcal{O}_0 = 0, \qquad Q_0\mathcal{O}_1 = -Q_1\mathcal{O}_0, \qquad d_2[\mathcal{O}_0]_2 = [Q_1\mathcal{O}_1]_2 \neq 0 \quad [\mathbb{Q}]. \end{aligned} \tag{2.17}$$

Table 2.4: Sources of the two-loop lifts in the $SU(3)$ Leigh–Strassler theory: finite-rank cohomology, stable comparison, fortuitous dimension, and two-loop rank.

Source sector	**Finite-rank E_1**	**Stable rank**	**Fortuitous dim.**	$\operatorname{rank} d_2$
$P_{14} : (7,0;0,0)_5$	$h = 5$ $[\mathbb{Q}]$	$h_{SU(6)} = 0$ $[\mathbb{Q}]$	$f = 5$ $[\mathbb{Q}]$	1 $[\mathbb{Q}]$
$P_{16} : (8,0;0,0)_6$	$h = 4$ $[\mathbb{Q}]$	$h_{SU(7)} = 2$ $[p]$	$f \geq 2$ $[p]$	1 $[p, p']$
$\partial_{++} P_{14} : (7,0;1,0)_5$	$h = 5$ $[\mathbb{Q}]$	$h_{SU(6)} = 0$ $[\mathbb{Q}]$	$f = 5$ $[\mathbb{Q}]$	1 $[\mathbb{Q}]$
$\partial^2_{++} P_{14} : (7,0;2,0)_5$	$h = 5$ $[\mathbb{Q}]$	$h_{SU(6)} = 0$ $[\mathbb{Q}]$	$f = 5$ $[\mathbb{Q}]$	1 $[p]$
$(8,0;1,0)_6$	$h = 6$ $[p]$	$h_{SU(7)} = 4$ $[p]$	$f \geq 2$ $[p]$	1 $[p]$
$P_{19} : (8,0;1,0)_7$	$h = 19$ $[p]$	$h_{SU(8)} = 6$ $[p]$	$f \geq 13$ $[p]$	1 $[p]$
$P_{21} : (9,0;1,0)_7$	$h = 40$ $[p]$	$h_{SU(8)} = 5$ $[p]$	$f \geq 35$ $[p]$	6 $[p]$
$P_{21} : (9,1;1,0)_7$	$h = 36$ $[p]$	$h_{SU(8)} = 0$ $[p]$	$f = 36$ $[p]$	6 $[p]$

Table 2.5: Two-loop lifts in the $SU(3)$ Leigh–Strassler theory through $E + J_L = 11$.

Source	**Target**	$\operatorname{rank} d_2$	**Origin / certification**
$(7,0;0,0)_5$	$(1,0;2,2)_4$	1	P_{14} $[\mathbb{Q}]$ (`ls_su3_P14_d2_cert.json.gz`)
$(8,0;0,0)_6$	$(2,0;2,2)_5$	1	P_{16} $[p, p']$
$(7,0;1,0)_5$	$(1,0;3,2)_4$	1	$\partial_{++} P_{14}$ $[\mathbb{Q}]$
$(8,0;1,0)_6$	$(2,0;3,2)_5$	1	$\partial_{++} P_{16}$ $[p]$
$(8,0;1,0)_7$	$(2,0;3,2)_6$	1	P_{19} $[p]$
$(7,0;2,0)_5$	$(1,0;4,2)_4$	1	$\partial^2_{++} P_{14}$ $[p]$
$(9,c;1,0)_7$, $c = 0, 1$	$(3,c;3,2)_6$	6 each	P_{21} $[p]$
$(8,0;1,1)_7$	$(2,0;3,3)_6$	5	$E + J_L = 11$ $[p]$
$(8,0;2,0)_6$	$(2,0;4,2)_5$	1	$\partial^2_{++} P_{16}$ $[p]$
$(8,0;2,0)_7$	$(2,0;4,2)_6$	5	Includes $\partial_{++} P_{19}$ $[p]$
$(8,0;2,0)_8$	$(2,0;4,2)_7$	1	$E + J_L = 11$ $[p]$

2.3.5 $SU(3)$ β-deformation

Setting $h = 0$ in (2.16) gives the $SU(3)$ β-deformation,

$$\bar{W}(\gamma, \mathsf{w}) = \operatorname{tr}(\gamma^1\gamma^2\gamma^3) - q\operatorname{tr}(\gamma^1\gamma^3\gamma^2) - \frac{1}{4}\mathsf{w}^A_{\dot{a}}\mathsf{w}_A{}^{\dot{a}}, \qquad h = 0, \tag{2.18}$$

which conserves $(\theta_1, \theta_2, \theta_3)$ individually ([14, supplement, Section S6.7], evaluated at $\rho = (1-q)/(1+q) = 12345/678$). A two-loop lift appears at $E + J_L = \frac{21}{2}$ with $Q_2\mathcal{O}_0 = 0$ and

$$Q_2\mathcal{O}_0 = 0, \qquad d_2[\mathcal{O}_0]_2 = [Q_1\mathcal{O}_1]_2 \neq 0. \tag{2.19}$$

Table 2.6: Two-loop tests in the $SU(3)$ β-deformation.

Scope	**Source**	**Target**	**Lifting result**
All sectors through $E + J_L = 10$	All d_2 candidates	—	$\operatorname{rank} d_2 = 0$ $[p]$
$E + J_L = \frac{21}{2}$	$(5, 2, 2; 1, 0)_8$, $h = 1$	$(3, 0, 0; 3, 2)_7$, $\dim E_2 = 4$	$\operatorname{rank} d_2 = 1$ $[p, p']$

2.3.6 $\mathcal{N} = 2$ orbifold quiver: $SU(N) \times SU(N)$

At equal gauge couplings $g_1 = g_2 = g$, the twisted action is the $\mathbb{Z}_2$ orbifold projection of $SU(2N)$ $\mathcal{N} = 4$ Yang–Mills theory ([14, results report, §3.1]):

$$\begin{gathered} S_{\text{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \left[\int d^3\theta \left(\frac{1}{2} \delta_{AB} C^A \bar{\partial} C^B + \frac{i}{3!} C_{ABC} C^A C^B C^C \right) \right]_{(0,2)}, \\ \operatorname{diag}(\mathbf{1}_N, -\mathbf{1}_N)\, C(\theta_1, \theta_2, \theta_3)\, \operatorname{diag}(\mathbf{1}_N, -\mathbf{1}_N) = C(-\theta_1, -\theta_2, \theta_3), \end{gathered} \tag{2.20}$$

retaining block-diagonal traceless vector multiplets $(\mathsf{c}, \mathsf{b}, \gamma^3, \beta_3)$ and off-diagonal bifundamental hypermultiplets $(\gamma^{1,2}, \beta_{1,2})$. Rescaling fields shows that every nonzero coupling ratio $\kappa = g_2/g_1 \neq 0$ has isomorphic spectral-sequence pages:

$$\kappa = \frac{g_2}{g_1} \neq 0 \quad \Longrightarrow \quad E_r(\kappa) \cong E_r(1) \qquad (r \geq 1). \tag{2.21}$$

Table 2.7: Lifting tests in the $\mathcal{N} = 2$ $SU(N) \times SU(N)$ orbifold quiver ([14, supplement, Section S7.1]).

Gauge group	**Complete census**	**One-loop d_1**	**Higher-loop test**
$SU(2) \times SU(2)$	$E + J_L \leq \frac{21}{2}$	Nonzero in listed sectors $[p]$	Every d_2 candidate has rank 0 at $\kappa^2 = 1, 9$ $[p]$
$SU(3) \times SU(3)$	$E + J_L \leq 9$	Nonzero in listed sectors $[p]$	Every d_2 candidate has rank 0; d_3 candidates at levels $9, 10$ have rank 0 $[p]$

2.3.7 $\mathcal{N} = 2$ SQCD: $SU(N)$ with $N_f = 2N$

Freezing the second node of (2.20) gives conformal $\mathcal{N} = 2$ $SU(N)$ SQCD with $N_f = 2N$ fundamental hypermultiplets ([14, results report, §4.1]), where D, E run over off-diagonal fundamental blocks and A, F over the dynamical $\mathfrak{su}(N)$ block:

$$\begin{gathered} \bar{W}(\gamma, \mathsf{w}) = i C_{ABC} \delta^{AD} \delta^{BE} \delta^{CF} (\gamma^1)_D (\gamma^2)_E (\gamma^3)_F - \frac{1}{4} \mathsf{w}^A_{\dot{a}} \mathsf{w}_A{}^{\dot{a}}, \\ G = SU(N), \qquad N_f = 2N. \end{gathered} \tag{2.22}$$

Table 2.8: Flavour-singlet lifting tests in $\mathcal{N} = 2$ SQCD ([14, supplement, Section S7.1]).

Gauge group	**Complete census**	**One-loop d_1**	**Higher-loop test**
$SU(2)$, $N_f = 4$	$E + J_L \leq 12$	$d_1 = 0$ throughout census $[p]$	Every d_2 candidate has rank 0 $[p]$
$SU(3)$, $N_f = 6$	$E + J_L \leq 9$	Nonzero in listed sectors $[p]$	Every d_2 candidate has rank 0 $[p]$

2.3.8 Adjoint-mass deformations

Adding an $\mathcal{N}=1$-preserving adjoint mass term ([14, results report, §5.1]),

$$\Delta S_{\mathrm{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \left[\frac{M}{2}\delta^{AB}(\gamma^3)_A(\gamma^3)_B\right]_{(0,2)}, \qquad M \neq 0, \tag{2.23}$$

(or $(M,-M)$ on the two-node quiver) flows upon integrating out γ^3 to quartic $\mathcal{N}=1$ theories, including the Klebanov–Witten-type quartic potential ([14, supplement, Section S7.2]):

$$\begin{aligned} 0 &= \frac{\partial \bar{W}}{\partial(\gamma^3)_F} = M\delta^{FG}(\gamma^3)_G + iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E, \\ \bar{W}(\gamma,0) &= \frac{1}{2M}C_{ABC}C_{DEF}\delta^{CF}\delta^{AG}\delta^{BH}\delta^{DI}\delta^{EJ}(\gamma^1)_G(\gamma^2)_H(\gamma^1)_I(\gamma^2)_J. \end{aligned} \tag{2.24}$$

Sectors $(A,B;n_+,n_-;\mathrm{gh})$ carry infrared weight $w = A+B+2(n_++n_-)$. In the one-mass $SU(2)$ theory, all three benchmark classes of [31]—w^2, $\mathcal{O}_7$, and $\mathcal{O}_{16}$—are unconditionally certified to survive to all loop orders over $\mathbb{Q}$ (Table 2.9): incoming d_r maps are ruled out over $\mathbb{Q}$ by $n_+ = n_- = 0$ (Lemma 2.2), while explicit rational closed completions are constructed for w^2 and $\mathcal{O}_{16}$ using spider kernels in [14, supplement, Section S8.4] and for $\mathcal{O}_7$ in `mass_su2_O7_cert.json.gz` [14].

Table 2.9: Protected benchmark classes of the mass-deformed $SU(2)$ theory.

Class	Sector	Classification	Lifting result
w^2	$(6,6;0,0;-4)$	Product of graviton w [31]	Explicit closed completion through $\hbar^3$; incoming excluded ($n_\pm = 0$): all-order survival [$\mathbb{Q}$]
$\mathcal{O}_7$	$(5,2;0,0;-2)$	Class of [31]	26-term rational closed completion (`mass_su2_O7_cert.json.gz`); incoming excluded: all-order survival [$\mathbb{Q}$]
$\mathcal{O}_{16}$	$(8,8;0,0;-5)$	Non-graviton class of [31]	Explicit rational completion through $\hbar^4$ [14, supplement, S8.4]; $r \geq 5$ targets empty; incoming excluded: all-order survival [$\mathbb{Q}$]

Table 2.10: Census of the adjoint-mass-deformed theories.

UV theory and mass	Weight bound	Nonzero d_1 maps	Higher-loop results
$\mathcal{N}=4$ $SU(2)$, M	$w \leq 16$	$(6,6;1,1;-3)$ [p]	All tested d_2, d_3 candidate ranks zero [p]
$\mathcal{N}=4$ $SU(3)$, M	$w \leq 13$	None in census	All tested d_2, d_3 candidate ranks zero [p]
SQCD $SU(2)$, $N_f=4$, M	$w \leq 15$	None in census	All tested d_2, d_3 candidate ranks zero [p]
SQCD $SU(3)$, $N_f=6$, M	$w \leq 13$	None in census	All tested d_2 candidate ranks zero [p]
$SU(2)\times SU(2)$ quiver, $(M,-M)$	$w \leq 13$	$(2,2;k,1;1)$, $k=1,2,3$ [p]	All tested d_2, d_3 candidate ranks zero [p]

2.3.9 Massive SQED, higher-loop abelian lifts, and $\mathcal{N}=2$ component lifts

Our sharpest higher-loop obstructions arise in abelian gauge theories coupled to neutral singlets and in component-basis non-abelian $\mathcal{N}=2$ gauge theories. Starting with $U(1)$ massive SQED

with chiral multiplets $(\gamma_+, \gamma_-, \gamma_s)$ of charges $(+1, -1, 0)$ and $gmy \neq 0$ ([14, results report, §5.4]; [14, supplement, Section S8.1]),

$$S_{\mathrm{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \Big[\mathsf{b}\bar{\partial}\mathsf{c} + (\bar{\partial}\gamma_+)\beta^+ + (\bar{\partial}\gamma_-)\beta^- + (\bar{\partial}\gamma_s)\beta^s + ig\mathsf{c}(\gamma_+\beta^+ - \gamma_-\beta^-) + \frac{m}{2}\gamma_s^2 + y\gamma_s\gamma_+\gamma_- - \frac{1}{4}\mathsf{w}_{\dot{a}}\mathsf{w}^{\dot{a}} \Big]_{(0,2)}, \tag{2.25}$$

the unique classical class at weight 14 with derivatives $(n_+, n_-) = (1, 0)$ has $d_1 = 0$ and no incoming d_1, yet lifts at two loops over $\mathbb{Q}$:

$$d_2[\mathcal{O}_0]_2 = -\frac{2(ig)^2 y^4}{m^2(64\pi^2)^2} \Big[\partial_{+\dot{a}}(\gamma_+\gamma_-)\, \partial_{+\dot{b}}\partial_{+\dot{c}}(\gamma_+\gamma_-) \times \partial_+{}^{\dot{b}}\partial_+{}^{\dot{c}}(\gamma_+\gamma_-) \Big]_2 \neq 0. \tag{2.26}$$

Table 2.11: Exact two-loop lift in massive SQED ([14, supplement, Section S8.1]).

Source	**First page E_1**	**Second obstruction**	**Certificate**
Weight 14, $(n_+, n_-) = (1, 0)$, $\mathrm{gh} = -5$	$\dim E_1 = 18 - 17 = 1$; $d_1 = 0$; incoming excluded ($n_- = 0$)	$d_2[\mathcal{O}_0]_2 \neq 0$ (2.26)	Exact quotient rank $394 \to 395$ [$\mathbb{Q}$]

Extending the abelian scan to massless $\mathcal{N} = 2$ SQED with one hypermultiplet ($W = sxy$, `singlet`), its cubic neutral-singlet deformation ($W = sxy + \frac{1}{3}s^3$, `singlet_cubic_self`), and the $\mathcal{N} = 2$ $U(1)^2$ linear quiver (`n2_quiver2`) uncovers infinite towers of two-, three-, and four-loop lifts ([14, results report, §8]), summarized in Table 2.12. Notably, `singlet_cubic_self` exhibits $d_2 \neq 0$ already at weight 9, $d_3 \neq 0$ at weights 11 and 13, and a *four-loop lift* $d_4 \neq 0$ at weight 16, all certified over $\mathbb{Q}$ via machine-verified rational chains and dual witnesses in [14].

Table 2.12: Higher-loop lifts ($r = 2, 3, 4$) in abelian gauge theories with neutral singlets, graded by $(w; n_+, n_-; \mathrm{gh})$ ([14, results report, §8]).

Theory (W)	**First sector**	r	$\mathrm{rk}\, d_r$	**Status and scope**
Massive SQED ($\frac{m}{2}s^2 + ysxy$)	$(12; 1, 1; -4)$	2	1	[$\mathbb{Q}$]; tower through $w = 24$ (incl. $(14; 1, 0; -5)$ [$\mathbb{Q}$])
$\mathcal{N} = 2$ SQED (sxy)	$(11; 1, 1; -5)_{(0,-1)}$	3	1	[$\mathbb{Q}$]; $d_3 \neq 0$ tower through $w = 19$ [p]
Cubic-singlet SQED ($sxy + \frac{1}{3}s^3$)	$(9; 1, 1; -4)$	2	1	[$\mathbb{Q}$] (rational certificate in [14]); tower through $w = 18$
Cubic-singlet SQED ($sxy + \frac{1}{3}s^3$)	$(11; 1, 0; -5)$, $(13; 1, 1; -6)$	3	1	[$\mathbb{Q}$] (rational certificates in [14]); tower through $w = 17$
Cubic-singlet SQED ($sxy + \frac{1}{3}s^3$)	$(16; 1, 1; -8)$	4	1	[$\mathbb{Q}$] (rational certificate in [14]); first 4-loop lift
$\mathcal{N} = 2$ $U(1)^2$ quiver	$(10; 2, 1; -3)$, $(11; 1, 1; -4)$	2	4, 1	[p]; $(11; 1, 1; -4)$ traced to Massey triples $\langle a, Z, Y \rangle$

Finally, in non-abelian $\mathcal{N} = 2$ component-basis enumerations ([14, results report, §9]), Table 2.13 records genuine two-loop lifts $d_2 \neq 0$ in pure $\mathcal{N} = 2$ $SU(2)$ Yang–Mills (certified over $\mathbb{Q}$ at weight 10, and extending through weight 16) and pure $\mathcal{N} = 2$ $SU(3)$ Yang–Mills (weights 11, 12), together with both two-loop ($d_2 \neq 0$, certified over $\mathbb{Q}$ at weight 9, extending through weight 15) and *three-loop ($d_3 \neq 0$)* lifts in $\mathcal{N} = 2$ $SU(2)$ Yang–Mills coupled to a flipping singlet ($W = s\,\mathrm{tr}(\Phi^2)$), where both the weight-13 sector $(13; 1, 1; -5)_1$ and the weight-14 sector $(14; 1, 0; -6)_2$ are certified over $\mathbb{Q}$.

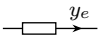
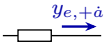
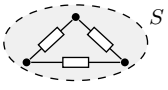
Table 2.13: Higher-loop lifts ($r = 2, 3$) in non-abelian $\mathcal{N} = 2$ gauge theories and singlet-flip deformations, graded by $(w; n_+, n_-; \mathrm{gh})_{\mathrm{flav}}$ ([14, results report, §9]).

Theory	**Sector**	r	$\operatorname{rk} d_r$	**Certification and census scope**
Pure $\mathcal{N} = 2$ $SU(2)$ YM	$(10; 0, 0; -5)_{-2}$	2	1	$[\mathbb{Q}]$ (rational certificate in [14]); $\dim E_1 = 1$, $d_1 = 0$
Pure $\mathcal{N} = 2$ $SU(2)$ YM	$11 \le w \le 16$ (15 sectors)	2	1–4	$[p]$; includes $\operatorname{rank} d_2 = 3$ at $(13; 1, 1; -6)_{-2}$, $\operatorname{rank} d_2 = 4$ at $(14; 1, 1; -7)_{-1}$, and $w = 15, 16$ lifts
Pure $\mathcal{N} = 2$ $SU(3)$ YM	$(11; 0, 0; -5)_{-1}$, $(12; 0, 0; -6)_0$	2	1, 1	$[p]$; complete through $w = 12$ ($d_2 = 0$ for $w \le 10$)
$\mathcal{N} = 2$ $SU(2)$ YM + flip singlet	$(9; 1, 1; -3)_0$	2	1	$[\mathbb{Q}]$ ($\dim E_1 = 5$, $\operatorname{rank} d_1 = 4 \Rightarrow \dim E_2 = 1$); further $d_2 \neq 0$ at $w = 11, 12, 15$ $[p]$
$\mathcal{N} = 2$ $SU(2)$ YM + flip singlet	$(13; 1, 1; -5)_1$	3	≥ 1	$[\mathbb{Q}]$ (22-prime CRT rational dual witness verified over $\mathbb{Q}$)
$\mathcal{N} = 2$ $SU(2)$ YM + flip singlet	$(14; 1, 0; -6)_2$	3	1	$[\mathbb{Q}]$ ($\dim E_1 = 1$, $d_1 = d_2 = 0$, incoming excluded by $n_- = 0$)

2.4 Electrical-circuit dictionary and local network reduction

We conclude our summary by explaining the physical mechanism that lets us evaluate n-loop holomorphic Feynman integrals in closed form. As reviewed in Section 3.9 ([5, 9]), passing to edge-current variables $y_{e,+\dot{a}} = (x_{e(0)} - x_{e(1)})^-_{\dot{a}} / (2t_e)$ turns the n-loop graph kernel (3.64) into the integral of an electrical resistor network on Γ (Table 2.14): the antiholomorphic vertex coordinate $\frac{1}{2}x^-_{v,\dot{a}}$ is the node electric potential, the Schwinger parameter $t_e > 0$ is the resistance of edge e, output momenta λ_v are external currents injected at node v, vertex delta functions enforce Kirchhoff's current law, and vanishing of $\sum_{e \in C} \pm t_e y_{e,+\dot{a}}$ around cycles is Kirchhoff's voltage law.

Table 2.14: Holomorphic graph kernels as electrical networks. $\Gamma_1[S]$ denotes the edges with both endpoints in $S \subseteq \Gamma_0$; the Laman bound is [9, Equations (2.22)–(2.23)].

Graph kernel	Circuit	
Schwinger time t_e	resistance of the resistor e	
antiholomorphic coordinate $\frac{1}{2}x^-_{v,\dot{a}}$ (3.58)	electric potential (voltage) at node v	
edge current $y_{e,+\dot{a}}$ (3.58), $\dot{a} = \dot{+}, \dot{-}$	current through e, two components	
output momentum λ_v (3.32)	current injected at the node v	
vertex delta functions (3.52), (3.63)	Kirchhoff's current law	
loop relation (3.53)	voltage law with resistances t_e	
weighted Laplacian [5, Remark 6]	conductance matrix	
$-\partial/\partial z_e^{+\dot{a}}$	insertion of the current $y_{e,+\dot{a}}$	
Laman bound $\lvert\Gamma_1[S]\rvert \le 2\lvert S\rvert - 3$ (4.3), $\lvert S\rvert \ge 2$	local bound on independent current directions	

Graph kernel	Circuit
kernel at $z = 0$	oriented volume swept by the independent currents 

Why does this electrical-network formulation allow an n-loop diagram to be reduced iteratively, one small piece at a time, rather than evaluated as a coupled $2n$-dimensional integral? Two structural properties work in tandem:

1. *Locality of Ohm's law (linear "black-box" elimination).* Because Ohm's law $t_e y_{e,+\dot{a}} = \frac{1}{2}(x_{e(0)} - x_{e(1)})^-_{\dot{a}}$ and Kirchhoff's current law (3.63) are linear and strictly local, any small two- or three-terminal subnetwork Γ_{sub} inside a large graph interacts with the rest of Γ *solely* through the potentials and net currents at its boundary terminals. Treating Γ_{sub} as a linear "black box" and solving for its internal node potentials replaces Γ_{sub} by its exact Norton equivalent—an effective resistance network on its boundary terminals together with current-division fractions that transfer internal injected currents λ_v onto those boundary nodes—leaving every current and voltage equation outside Γ_{sub} untouched.

2. *Global completeness of the projective resistance domain.* In (3.54), the Schwinger parameters range over the *entire* positive projective orthant $\mathbb{RP}^{|\Gamma_1|-1}_{>0}$, sweeping without gaps or ordering cuts across all positive resistance ratios and all relative resistance orderings. Consequently, the elementary network maps—series addition, parallel conductance addition, and the star–triangle (Y–Δ) transformation—are global real-analytic diffeomorphisms from the positive resistance orthant of Γ_{sub} onto the product of the effective boundary resistance orthant and a compact parameter fiber ($[0, 1]$ or a standard simplex) recording the internal current-division ratios r_j.

Together, these two properties turn classical circuit reductions into exact, local changes of variables on Schwinger space that peel off one loop per move while keeping all reconstructed edge currents polynomial in the fiber ratios r_j. Figure 2.1 displays the three local black-box moves proved in Sections 5 and 7: series–parallel reduction (Theorem 5.1, sufficient for all n-loop fans and spiders), sourced star–triangle reduction (Theorem 7.6), and two-star reduction (Theorem 7.7, reducing cubic product forests).

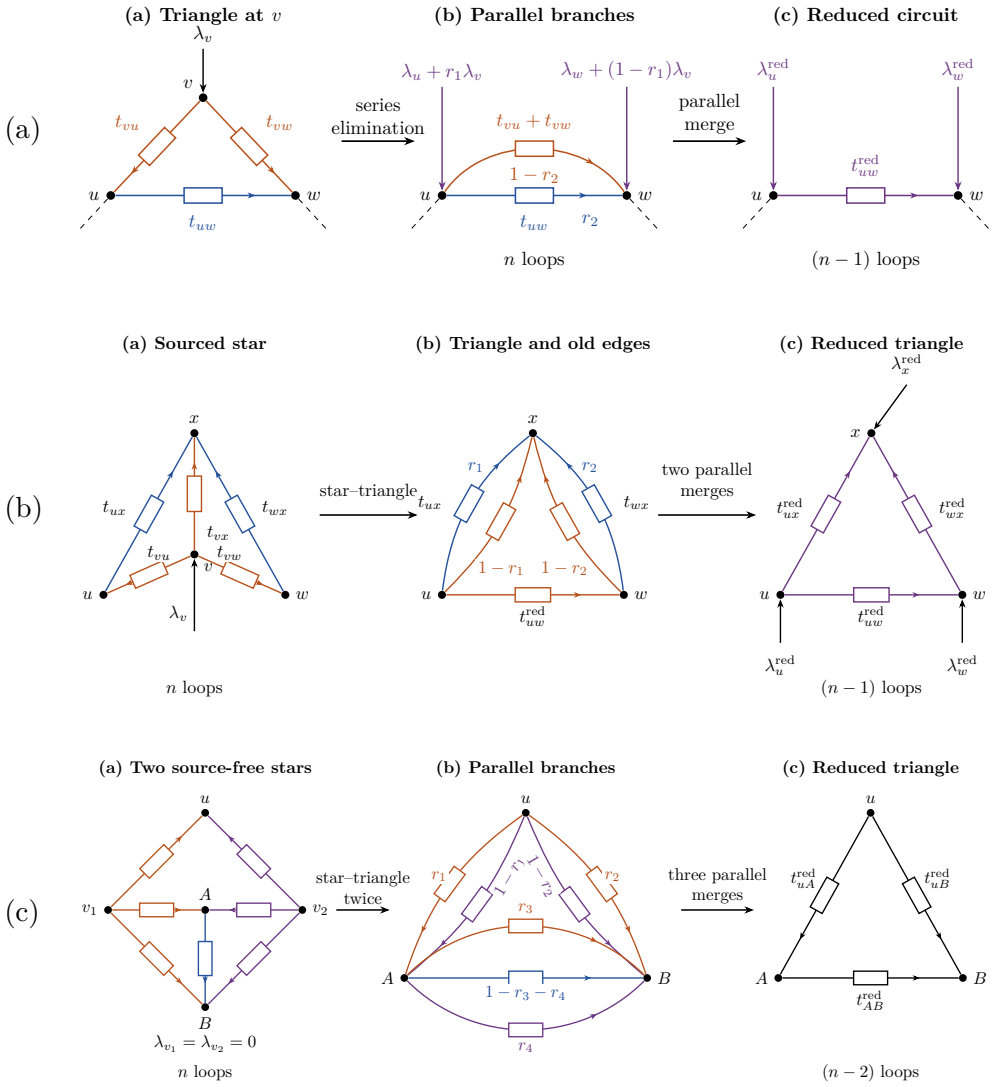


Figure 2.1: Local "black-box" network reduction moves (Sections 5 and 7). (a) Series–parallel reduction (Theorem 5.1): eliminating node v transfers λ_v to u and w with ratios $1-r$ and r, and merges the series branch (orange) in parallel with the existing uw resistor (blue), removing one loop. (b) Sourced star–triangle (Y–Δ) reduction (Theorem 7.6): eliminating trivalent node v transfers λ_v to u, w, x and merges the three new Δ-edges in parallel with existing edges. (c) Two source-free trivalent stars reduce simultaneously to a triangle via two star–triangle moves and three parallel merges (Theorem 7.7).

Explicitly, for the elementary series and parallel steps (5.7)–(5.9), the resistance, source, and current-reconstruction rules are

$$r = \frac{t_{vu}}{t_{vu}+t_{vw}}, \qquad t^{(1)}_{uw} = t_{vu}+t_{vw}, \qquad \lambda^{\mathrm{red}}_u = \lambda_u + (1-r)\lambda_v, \qquad \lambda^{\mathrm{red}}_w = \lambda_w + r\lambda_v,$$
$$\frac{1}{t^{\mathrm{red}}_{uw}} = \sum_{\substack{j \\ \text{parallel resistors}}} \frac{1}{t^{(j)}_{uw}}, \qquad r_j = \frac{t^{\mathrm{red}}_{uw}}{t^{(j)}_{uw}}, \qquad \sum_{\substack{j \\ \text{parallel resistors}}} r_j = 1, \tag{2.27}$$
$$y_{vu} = (1-r)\lambda_v - y^{(1)}_{uw}, \qquad y_{vw} = r\lambda_v + y^{(1)}_{uw}, \qquad y^{(j)}_{uw} = r_j y^{\mathrm{red}}_{uw},$$

while the corresponding star–triangle rules are given in Theorems 7.6 and 7.7. Iterating these local black-box moves evaluates every n-loop fan (Section 5), spider (Section 7.2), and cubic product forest (Section 7.3).

3 Review of holomorphic twist

This section reviews the holomorphic twist for readers who have not met it [3, 4], together with its higher brackets and current-integral representation [5, 6, 9]. The starting point is the BV formulation of the physical theory, in which the fields, the ghosts and their antifields carry the differential $(S_{\mathrm{BV}}, \cdot)$; its cohomology consists of the gauge-invariant operators modulo the equations of motion. Adding one supercharge Q_- to this differential gives a complex whose cohomology consists of the operators annihilated by Q_- modulo Q_--exact ones, that is, the BPS sector.

The twist simplifies this complex in three steps. First, a canonical change of BV variables removes the Q_--exact part of the action and exhibits most fields as doublets (u, v) with $(S_{\mathrm{BV}}, u) \propto v$; they do not contribute to the cohomology and are contracted (Section 3.2). Second, the remaining fields and antifields assemble into four Dolbeault forms $\mathsf{c}, \mathsf{b}, \gamma, \beta$ on $\mathbb{C}^2$ with a first-order action (Section 3.3); their zero-form components are the familiar BPS letters (Section 3.4). Third, the free differential $\bar\partial$ is contracted, so that local operators become polynomials in the zero-form fields and their holomorphic derivatives (Section 3.5). The interaction then acts on these operators through the higher brackets and current integrals reviewed in Sections 3.6–3.9.

3.1 Coordinates and fields

The index A labels the Lie algebra and i the matter representation. With the continuation $x^0 = ix^0_E$, we use Euclidean spinor coordinates and the correspondingly continued sigma matrices

$$x^{a\dot a} = x^\mu_E \bar\sigma^{\dot a a}_\mu . \tag{3.1}$$

The coordinates $x^{+\dot a}$ are holomorphic and $x^{-\dot a}$ antiholomorphic. In these coordinates,

$$\begin{aligned} &\frac{\partial}{\partial x^{a\dot a}} = -\frac{1}{2}\partial_{a\dot a}, \qquad \epsilon^{\dot+\dot-} = 1 = -\epsilon_{\dot+\dot-}, \qquad dx^-_{\dot a} = \epsilon_{\dot a\dot b} dx^{-\dot b}, \\ &\bar\partial = dx^{-\dot a}\frac{\partial}{\partial x^{-\dot a}} = -\frac{1}{2} dx^{-\dot a}\partial_{-\dot a}, \qquad \Omega = 2\, dx^{+\dot+} \wedge dx^{+\dot-}. \end{aligned} \tag{3.2}$$

Let c^A be the gauge ghost, V^A the gauge field, $\lambda^A, \bar\lambda^A$ the gauginos, and D^A the gauge auxiliary field. A chiral matter multiplet has components $\phi^i, \psi^i, \mathcal{F}^i$, with $\mathcal{F}^i$ the auxiliary field.

$$\begin{aligned} [t_B, t_C] &= iC^A{}_{BC} t_A, \qquad C_{ABC} = \delta_{AD} C^D{}_{BC} = C_{[ABC]}, \\ f^A_{\mu\nu} &= \partial_\mu V^A_\nu - \partial_\nu V^A_\mu + g C^A{}_{BC} V^B_\mu V^C_\nu , \\ f^A_{++} &= \partial_{+\dot a} V^{A\dot a}_+ + \frac{g}{2} C^A{}_{BC} V^B_{+\dot a} V^{C\dot a}_+ . \end{aligned} \tag{3.3}$$

Here $\partial_\mu = \partial/\partial x^\mu_E$ and $\partial_{a\dot a} = \sigma^\mu_{a\dot a}\partial_\mu$; undotted indices use $\epsilon^{+-} = 1 = -\epsilon_{+-}$. Write $\bar W^A_{\dot a}(x, \theta, \bar\theta)$ for the antichiral field-strength superfield, with $\theta^a, \bar\theta^{\dot a}$ the odd superspace coordinates [16, Eqs. (27.2.12), (27.3.16)]. Formula (3.35) follows from the identification $dx^{-\dot a} = 4\bar\theta^{\dot a}$ in (3.12).

3.2 Canonical changes of variables

In this subsection, we start from the original physical action and perform a canonical transformation ("adding Q-exact terms to the action").

We use the $N = 1$ off-shell superfield formulation in the gauge vector representation [16, Chapters 26–27].

Write $\Phi^{\mathcal{A}}$ for the collective array of fields and gauge ghosts (with $\mathcal{A}$ running over $c^A, V^A_\mu, \lambda^A_a, \bar\lambda^A_{\dot a}, D^A, \phi^i, \bar\phi_i, \psi^i_a, \bar\psi^{\dot a}_i, \mathcal{F}^i, \bar{\mathcal{F}}_i$), $\Phi^\ddagger_{\mathcal{A}}$ for their antifields, and s for the gauge BRST

differential. The BV bracket is

$$(F,G) = \int d^4x \left[\frac{\delta_R F}{\delta \Phi^\dagger_{\mathcal{A}}} \frac{\delta_L G}{\delta \Phi^{\mathcal{A}}} - \frac{\delta_R F}{\delta \Phi^{\mathcal{A}}} \frac{\delta_L G}{\delta \Phi^\dagger_{\mathcal{A}}} \right]. \tag{3.4}$$

Let S_{free} be the free quadratic kinetic action, and introduce the functional Ψ (which serves as a gauge-fixing fermion generating a BV canonical transformation),

$$\Psi = \int d^4x \left[\frac{1}{8} \nabla_+ \bar{\nabla}^2 K(\bar{\Phi}, \Phi) + \frac{1}{2} \nabla_+ W(\Phi, W_a) \right] \Big|_{\theta=\bar{\theta}=0}.$$

For the chosen nilpotent supercharge Q_- and with the gauge BRST transformation denoted by s, the physical and BV actions have the decomposition

$$\begin{aligned}
S_{\text{phys}} &= \int d^4x \Big[\frac{1}{8} \nabla_- \nabla_+ \bar{\nabla}^2 K(\bar{\Phi}, \Phi) + \frac{1}{2} \nabla_- \nabla_+ W(\Phi, W_a) \\
&\qquad + \frac{1}{4} \bar{\nabla}^2 \bar{W}(\bar{\Phi}, \bar{W}_{\dot{a}}) \Big] \Big|_{\theta=\bar{\theta}=0}, \\
Q_- \Psi &= \int d^4x \, \nabla_- \left[\frac{1}{8} \nabla_+ \bar{\nabla}^2 K(\bar{\Phi}, \Phi) + \frac{1}{2} \nabla_+ W(\Phi, W_a) \right] \Big|_{\theta=\bar{\theta}=0} \\
&= \int d^4x \left[\frac{1}{8} \nabla_- \nabla_+ \bar{\nabla}^2 K(\bar{\Phi}, \Phi) + \frac{1}{2} \nabla_- \nabla_+ W(\Phi, W_a) \right] \Big|_{\theta=\bar{\theta}=0} \\
&= S_{\text{phys}} - \frac{1}{4} \int d^4x \, \bar{\nabla}^2 \bar{W}(\bar{\Phi}, \bar{W}_{\dot{a}}) \Big|_{\theta=\bar{\theta}=0}.
\end{aligned}$$

$$\begin{aligned}
S_{\text{phys}} &= Q_- \Psi + \int d^4x \, [\bar{W}(\bar{\Phi}, \bar{W}_{\dot{a}})]_{\bar{\mathcal{F}}}, \\
\Psi &= \int d^4x \left[\frac{1}{8} \nabla_+ \bar{\nabla}^2 K(\bar{\Phi}, \Phi) + \frac{1}{2} \nabla_+ W(\Phi, W_a) \right] \Big|_{\theta=\bar{\theta}=0}, \\
S_{\text{BV}} &= S_{\text{phys}} + \int d^4x \, ((s + Q_-) \Phi^{\mathcal{A}}) \Phi^\dagger_{\mathcal{A}}, \\
Q_{\text{BV}} &= (S_{\text{free}}, \cdot), \qquad s\Psi = 0, \quad (s + Q_-)^2 = 0.
\end{aligned} \tag{3.5}$$

Here K is the Kähler potential, and $W(\Phi, W_a)$ is the chiral superpotential including its gauge kinetic term, with W_a the chiral field-strength superfield. The derivatives $\nabla_a, \bar{\nabla}_{\dot{a}}$ are gauge-covariant superspace derivatives, and $[\bar{W}]_{\bar{\mathcal{F}}} = \frac{1}{4} \bar{\nabla}^2 \bar{W}|_{\theta=\bar{\theta}=0}$ is the antichiral top component. The vertical bar $|_{\theta=\bar{\theta}=0}$ evaluates at $\theta = \bar{\theta} = 0$, taking the lowest component.

We now perform the BV canonical transformation generated by the gauge-fixing fermion Ψ:

$$\begin{aligned}
\Phi^\dagger_{\mathcal{A},\text{new}} &= \Phi^\dagger_{\mathcal{A},\text{old}} + \frac{\delta_L \Psi}{\delta \Phi^{\mathcal{A}}}, \qquad \Phi^{\mathcal{A}}_{\text{new}} = \Phi^{\mathcal{A}}_{\text{old}} \\
S_{\text{BV,new}} &= S_{\text{BV,old}} \left(\Phi, \Phi^\dagger_{\text{new}} - \frac{\delta_L \Psi}{\delta \Phi} \right) \\
&= \int d^4x \left([\bar{W}(\bar{\Phi}, \bar{W}_{\dot{a}})]_{\bar{\mathcal{F}}} + ((s + Q_-) \Phi^{\mathcal{A}}) \Phi^\dagger_{\mathcal{A},\text{new}} \right).
\end{aligned} \tag{3.6}$$

This removes the Q_--exact term from the action, which cancels almost all terms in the physical action, leaving only the antichiral superpotential.

Many of the fields can be arranged into the contractible pairs listed in the table below, which we will contract.

Table 3.1: Contractible pairs of the BV complex.

u	v	$(S_{\rm BV}, u) = av$
$V^A_{+\dot a}$	$\bar\lambda'^A_{\dot a} = \bar\lambda^A_{\dot a} - \frac{1}{2}\nabla_{+\dot a} c^A$	$a = -2$
λ^A_+	$D'^A = D^A + g C^A{}_{BC} c^B \lambda^C_+ - i f^A_{+-}$	$a = -1$
ϕ^i	$\psi'^i_- = \psi^i_- - \frac{g}{\sqrt{2}} c^A (t_A)^i{}_j \phi^j$	$a = -i\sqrt{2}$
ψ^i_+	$\mathcal{F}'^i = \mathcal{F}^i + \frac{g}{\sqrt{2}} c^A (t_A)^i{}_j \psi^j_+$	$a = i\sqrt{2}$

Nilpotency gives $(S_{\rm BV}, v) = 0$.

We perform an upper-triangular coordinate change, with $\bar\lambda \to \bar\lambda', D \to D', \psi_- \to \psi'_-, \mathcal{F} \to \mathcal{F}'$, whose Jacobian is equal to one.

Let $\mathcal{T}$ contain these pairs and their spacetime derivatives. The remaining fields $\mathcal{S} = \{c, V_-, \lambda_-; \bar\phi, \bar\psi, \bar{\mathcal{F}}\}$ are unchanged.

The key point is that the $\mathcal{S}$ letters transform only among themselves (without $\mathcal{T}$ dependence), while the $\mathcal{T}$ letters form contractible pairs and therefore do not contribute to the cohomology.

> We remind the reader that we will contract $\mathcal{T}$ and keep the $\mathcal{S}$ letters and their antifields; however, the $\mathcal{S}$ letters are not the familiar "BPS letters".

On field functionals (with $\partial_L/\partial v$ denoting the left partial derivative with respect to the graded variable v) the contraction is

$$\begin{aligned} h_{\mathcal{T}}\mathcal{O} &= \int_0^1 \frac{dt}{t} \left(\sum_{(u,v)} \frac{u}{a} \frac{\partial_L \mathcal{O}}{\partial v} \right)\Bigg|_{(u,v)\mapsto(tu,tv)}, \\ \Psi' &= h_{\mathcal{T}} \int d^4x \, [\bar W(\bar\Phi, \bar W_{\dot a})]_{\bar{\mathcal{F}}}, \\ (s + Q_-)\Psi' &= \int d^4x \, [\bar W]_{\bar{\mathcal{F}}} - \int d^4x \, [\bar W]_{\bar{\mathcal{F}}}\Big|_{\mathcal{T}=0}. \end{aligned} \tag{3.7}$$

The sum includes all spacetime derivatives of each pair. The second antifield translation, generated by Ψ' with the sign in (3.6), removes the $\mathcal{T}$ dependence of the antichiral action. It gives

$$\begin{aligned} S_{\rm BV,new} &= S_{\mathcal{S}} + S_{\mathcal{T}}, \qquad S_{\mathcal{T}} = \int d^4x \sum_{(u,v)} av \, u^\ddagger, \\ (S_{\mathcal{S}}, S_{\mathcal{T}}) &= 0, \qquad (S_{\mathcal{S}}, S_{\mathcal{S}}) = (S_{\mathcal{T}}, S_{\mathcal{T}}) = 0. \end{aligned} \tag{3.8}$$

Here $S_{\mathcal{S}}$ depends only on the retained fields and their antifields.

Including the antifields and all spacetime derivatives of the fields, the contraction satisfies

$$\begin{aligned} (S_{\rm BV}, v^\ddagger) &= (-1)^{|u|} a u^\ddagger, \qquad (S_{\rm BV}, u^\ddagger) = 0, \\ h_{\mathcal{T}}\mathcal{O} &= \int_0^1 \frac{dt}{t} \sum_{(u,v)} \left(\frac{u}{a}\frac{\partial_L}{\partial v} + \frac{(-1)^{|u|} v^\ddagger}{a} \frac{\partial_L}{\partial u^\ddagger} \right) \mathcal{O}\Big|_{\mathcal{T}\mapsto t\mathcal{T}}, \\ (S_{\rm BV}, h_{\mathcal{T}}\mathcal{O}) &+ h_{\mathcal{T}}(S_{\rm BV}, \mathcal{O}) = \mathcal{O} - \mathcal{O}|_{\mathcal{T}=0}, \qquad [h_{\mathcal{T}}, \partial_\mu] = 0. \end{aligned} \tag{3.9}$$

For $(S_{\rm BV}, \mathcal{O}) = 0$, this gives

$$\mathcal{O} = \mathcal{O}|_{\mathcal{T}=0} + (S_{\rm BV}, h_{\mathcal{T}}\mathcal{O}). \tag{3.10}$$

The representative on the right contains only retained fields and antifields in the complex

$\mathcal{S}$.

In particular,

$$\nabla_{+\dot{a}} \longmapsto \partial_{+\dot{a}}, \qquad \bar{\lambda}^A_{\dot{a}} \longmapsto \frac{1}{2}\partial_{+\dot{a}} c^A = -\frac{\partial c^A}{\partial x^{+\dot{a}}}. \tag{3.11}$$

3.3 Assemble the retained fields into Dolbeault forms

In the remaining complex $\mathcal{S}$, the fields and antifields assemble as

$$\begin{aligned}
\mathsf{c}^A &= c^A - 2\bar{\theta}_{\dot{a}} V_-^{A\dot{a}} + 2i\bar{\theta}_{\dot{a}}\bar{\theta}^{\dot{a}}\lambda_-^A, \\
\gamma_i &= \bar{\phi}_i - \sqrt{2}\bar{\theta}_{\dot{a}}\bar{\psi}_i^{\dot{a}} + \bar{\theta}_{\dot{a}}\bar{\theta}^{\dot{a}}\bar{\mathcal{F}}_i, \\
\beta^i &= \bar{\mathcal{F}}^{\ddagger i}_{+} + \sqrt{2}\bar{\theta}_{\dot{a}}\bar{\psi}^{\ddagger \dot{a} i} + \bar{\theta}_{\dot{a}}\bar{\theta}^{\dot{a}}\bar{\phi}^{\ddagger i}, \\
\mathsf{b}_A &= -\frac{i}{2}\lambda_A^{\ddagger -} + \bar{\theta}_{\dot{a}} V_A^{\ddagger -\dot{a}} + \bar{\theta}_{\dot{a}}\bar{\theta}^{\dot{a}} c_A^{\ddagger}, \\
\bar{\partial} &= 2\bar{\theta}_{\dot{a}}\partial_-{}^{\dot{a}}, \qquad dx^{-\dot{a}} = 4\bar{\theta}^{\dot{a}}.
\end{aligned} \tag{3.12}$$

Here $\bar{\psi}^{\ddagger i}_{\dot{a}}$ is BV-dual to $\bar{\psi}^{\dot{a}}_i$, and $\bar{\psi}^{\ddagger \dot{a} i}$ in β^i has its dotted index raised as in (3.2).

$$\bar{\psi}^{\ddagger i}_{\dot{a}}((s+Q_-)\bar{\psi}^{\dot{a}}_i) = -((s+Q_-)\bar{\psi}_{i\dot{a}})\bar{\psi}^{\ddagger \dot{a} i},$$
$$\sqrt{2}\bar{\theta}_{\dot{a}}\bar{\psi}^{\ddagger \dot{a} i} = -\sqrt{2}\bar{\psi}^{\ddagger i}_{\dot{a}}\bar{\theta}^{\dot{a}}.$$

The identification $dx^{-\dot{a}} = 4\bar{\theta}^{\dot{a}}$ and the volume relation

$$\Omega \wedge dx^{-\dot{+}} \wedge dx^{-\dot{-}} = -8\, d^4x_E, \qquad d^4x_E = dx^0_E \wedge dx^1_E \wedge dx^2_E \wedge dx^3_E \tag{3.13}$$

give the action S_{tw} in (3.37). Because (3.13) carries a minus sign when passing from $\int d^4x_E\, d^2\bar{\theta}$ to integration against Ω on $\mathbb{C}^2$, both $S_{\mathrm{tw}} = -S_{\mathcal{S}}$ and the twisted BV pairing $(\cdot,\cdot)$ on $\mathbb{C}^2$ acquire a minus sign relative to (3.4); these two signs cancel in the Hamiltonian vector field $(S_{\mathrm{tw}},\cdot)$.

Its free kinetic part is

$$\begin{aligned}
S_{\mathrm{free}} &= \int_{\mathbb{C}^2} \Omega \wedge [\mathsf{b}_A \bar{\partial}\mathsf{c}^A + (\bar{\partial}\gamma_i)\beta^i]_{(0,2)}, \\
Q_{\mathrm{BV}} &= (S_{\mathrm{free}},\cdot) = \bar{\partial}, \qquad (S_{\mathrm{tw}},\cdot) = Q_{\mathrm{BV}} + (I,\cdot).
\end{aligned} \tag{3.14}$$

The original-coordinate embedding keeps the antifield translations (3.6) (with superscript (0) denoting the Dolbeault zero-form component in (3.12)):

$$\begin{aligned}
\beta^{(0)i} &= \bar{\mathcal{F}}^{\ddagger i}_{\mathrm{old}} + \frac{\delta_L(\Psi+\Psi')}{\delta\bar{\mathcal{F}}_i}, \\
\mathsf{b}^{(0)}_A &= -\frac{i}{2}\left(\lambda^{\ddagger -}_{A,\mathrm{old}} + \frac{\delta_L(\Psi+\Psi')}{\delta\lambda^A_-}\right).
\end{aligned} \tag{3.15}$$

For canonical matter and quadratic gauge kinetic terms, the combined translation has generator

$$\begin{aligned}
\Psi + \Psi' = \int d^4x \Big[&-\frac{1}{\sqrt{2}}\phi^i \nabla_{+\dot{a}}\bar{\psi}^{\dot{a}}_i + g\bar{\phi}_j (t_A)^j{}_i \phi^i \lambda^A_+ \\
&- \frac{i}{\sqrt{2}}\bar{\mathcal{F}}_i \psi^i_+ - \frac{i}{\sqrt{2}}\psi^i_+ \frac{\partial W(\phi,0)}{\partial\phi^i} \\
&+ \frac{i}{2}\delta_{AB} f^A_{++}\lambda^B_- - \frac{1}{2}\delta_{AB}(D^A + i f^A_{+-})\lambda^B_+ \Big].
\end{aligned} \tag{3.16}$$

They give

$$\begin{aligned}\beta^{(0)i} &= \bar{\mathcal{F}}^{\ddagger i}_{\text{old}} - \frac{i}{\sqrt{2}}\psi^i_+,\\ \mathsf{b}^{(0)}_A &= -\frac{i}{2}\lambda^{\ddagger -}_{A,\text{old}} + \frac{1}{4}\delta_{AB} f^B_{++}.\end{aligned} \tag{3.17}$$

For the quadratic gauge kinetic term used below, using $\mathsf{w}^A_{\dot a} \mapsto \frac{1}{2}\partial_{+\dot a}\mathsf{c}^A$ after homotopy transfer (see (3.36) and [4, Section 3.3]),

$$-\frac{1}{4}\delta_{AB}\mathsf{w}^A_{\dot a}\mathsf{w}^{B\dot a} = -\frac{1}{16}\delta_{AB}\partial_{+\dot a}(\mathsf{c}^A\partial_+{}^{\dot a}\mathsf{c}^B), \tag{3.18}$$

so it is a total derivative.

3.4 BV completion of the Chang–Yin letters

For the canonical kinetic terms of (3.16), the Chang–Yin letters are the scalar $\bar\phi(z)$, chiral fermion $\psi_+(z)$, self-dual curvature $f_{++}(z)$ and antichiral gaugino $\bar\lambda(z)$. Their component complex uses the auxiliary $\bar{\mathcal{F}}$ and gaugino λ^- equations of motion (which are exactly the antifield parts of β and b). In the present conventions these equations are

$$\mathcal{F}^i + \frac{\partial \bar W(\bar\phi,0)}{\partial\bar\phi_i} = 0, \qquad \nabla_{+\dot a}\bar\lambda^{A\dot a} - \sqrt{2}g\delta^{AB}\bar\phi_j(t_B)^j{}_i\psi^i_+ = 0. \tag{3.19}$$

The original antifields implement these equations within the BV complex:

$$\begin{aligned}(S_{\text{BV}}, \bar{\mathcal{F}}^{\ddagger i}_{\text{old}}) &= igc^A(t_A)^i{}_j\bar{\mathcal{F}}^{\ddagger j}_{\text{old}} - \mathcal{F}^i - \frac{\partial\bar W(\bar\phi,0)}{\partial\bar\phi_i},\\ (S_{\text{BV}}, \lambda^{\ddagger -}_{A,\text{old}}) &= -gC^B{}_{AC}c^C\lambda^{\ddagger -}_{B,\text{old}} + i\delta_{AB}\nabla_{+\dot a}\bar\lambda^{B\dot a}\\ &\quad - i\sqrt{2}g\bar\phi_j(t_A)^j{}_i\left(\psi^i_+ + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\text{old}}\right).\end{aligned} \tag{3.20}$$

The completed fermion and curvature are precisely the retained letters of (3.17), with their fixed component factors:

$$\psi^i_+ + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\text{old}} = i\sqrt{2}\beta^{(0)i}, \qquad f^A_{++} - 2i\delta^{AB}\lambda^{\ddagger -}_{B,\text{old}} = 4\delta^{AB}\mathsf{b}^{(0)}_B. \tag{3.21}$$

Using $Q_-\psi^i_+ = i\sqrt{2}\mathcal{F}^i$ and $Q_- f^A_{++} = -2\nabla_{+\dot a}\bar\lambda^{A\dot a}$ gives

$$\begin{aligned}&(S_{\text{BV}}, \psi^i_+ + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\text{old}})\\ &\quad = igc^A(t_A)^i{}_j\left(\psi^j_+ + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger j}_{\text{old}}\right) - i\sqrt{2}\frac{\partial\bar W(\bar\phi,0)}{\partial\bar\phi_i},\\ &(S_{\text{BV}}, f^A_{++} - 2i\delta^{AB}\lambda^{\ddagger -}_{B,\text{old}})\\ &\quad = -gC^A{}_{BC}c^B\left(f^C_{++} - 2i\delta^{CD}\lambda^{\ddagger -}_{D,\text{old}}\right) - 2\nabla_{+\dot a}\bar\lambda^{A\dot a}\\ &\qquad - 2i\left[i\nabla_{+\dot a}\bar\lambda^{A\dot a} - i\sqrt{2}g\delta^{AB}\bar\phi_j(t_B)^j{}_i\left(\psi^i_+ + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\text{old}}\right)\right]\\ &\quad = -4g\delta^{AB}C^C{}_{BD}c^D\mathsf{b}^{(0)}_C - 4ig\delta^{AB}\bar\phi_j(t_B)^j{}_i\beta^{(0)i}.\end{aligned} \tag{3.22}$$

The auxiliary field and the gaugino divergence cancel, leaving the familiar classical terms and keeping the $\mathcal{S}$ complex closed under $(S_{\text{BV}}, \cdot)$.

At the insertion $x = 0$, physical operators are gauge-invariant words independent of the undifferentiated ghost $c^A(0)$.

We can now build the full dictionary between twisted letters and Chang–Yin letters (jets).

The symbol $\longleftrightarrow$ denotes the image after the doublet contraction and projection to holomorphic zero-form jets. Parentheses symmetrize the dotted indices with weight one, and $n \geq 0$.

$$\begin{aligned}
\nabla_{+(\dot a_1}\cdots\nabla_{+\dot a_n)}\bar\phi_i &\longleftrightarrow \partial_{+\dot a_1}\cdots\partial_{+\dot a_n}\gamma_i^{(0)},\\
\nabla_{+(\dot a_1}\cdots\nabla_{+\dot a_n)}\left(\psi_+^i + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\rm old}\right) &\longleftrightarrow i\sqrt{2}\partial_{+\dot a_1}\cdots\partial_{+\dot a_n}\beta^{(0)i},\\
\nabla_{+(\dot a_1}\cdots\nabla_{+\dot a_n)}\left(f^A_{++} - 2i\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}\right) &\longleftrightarrow 4\delta^{AB}\partial_{+\dot a_1}\cdots\partial_{+\dot a_n}\mathsf{b}^{(0)}_B,\\
2\nabla_{+(\dot a_1}\cdots\nabla_{+\dot a_n}\bar\lambda^A_{\dot a_0)} &\longleftrightarrow \partial_{+\dot a_0}\cdots\partial_{+\dot a_n}\mathsf{c}^{A(0)}.
\end{aligned} \tag{3.23}$$

Ordered covariant derivatives reduce to symmetrized derivatives and curvature terms. Completing each f^A_{++} to $f^A_{++} - 2i\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}$ gives representatives in the retained complex.

For ordered derivatives, the curvature commutator determines the additional b term. Since $\nabla_{+\dot a}\bar\phi_i = \partial_{+\dot a}\bar\phi_i + igV^A_{+\dot a}(t_A)^j{}_i\bar\phi_j$, the full BV variation is

$$\begin{aligned}
(S_{\rm BV}, \nabla_{+\dot a}\nabla_+{}^{\dot b}\bar\phi_i) = &-igc^A(t_A)^j{}_i\nabla_{+\dot a}\nabla_+{}^{\dot b}\bar\phi_j\\
&-2ig\Big[\frac{1}{2}\left(\nabla_{+\dot a}\bar\lambda^{A\dot b} + \nabla_+{}^{\dot b}\bar\lambda^A_{\dot a}\right)\bar\phi_j + \bar\lambda^A_{\dot a}\nabla_+{}^{\dot b}\bar\phi_j + \bar\lambda^{A\dot b}\nabla_{+\dot a}\bar\phi_j\Big](t_A)^j{}_i\\
&-ig\delta_{\dot a}{}^{\dot b}(\nabla_{+\dot c}\bar\lambda^{A\dot c})(t_A)^j{}_i\bar\phi_j.
\end{aligned} \tag{3.24}$$

The last term has the free gaugino equation as its linear field factor. The complementary antifield term has variation

$$\begin{aligned}
&\left(S_{\rm BV}, g\delta_{\dot a}{}^{\dot b}\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}(t_A)^j{}_i\bar\phi_j\right)\\
&\quad= -ig^2\delta_{\dot a}{}^{\dot b}\delta^{AB}c^C\lambda^{\ddagger-}_{B,\rm old}(t_C)^j{}_i(t_A)^k{}_j\bar\phi_k\\
&\qquad+ ig\delta_{\dot a}{}^{\dot b}(\nabla_{+\dot c}\bar\lambda^{A\dot c})(t_A)^j{}_i\bar\phi_j\\
&\qquad- i\sqrt{2}g^2\delta_{\dot a}{}^{\dot b}\delta^{AB}\bar\phi_k(t_B)^k{}_\ell\left(\psi_+^\ell + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger\ell}_{\rm old}\right)(t_A)^j{}_i\bar\phi_j.
\end{aligned} \tag{3.25}$$

The divergence cancels between these two equations.

$$\begin{aligned}
&\nabla_{+\dot a}\nabla_+{}^{\dot b}\bar\phi_i + g\delta_{\dot a}{}^{\dot b}\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}(t_A)^j{}_i\bar\phi_j\\
&\quad= \frac{1}{2}\left(\nabla_{+\dot a}\nabla_+{}^{\dot b} + \nabla_+{}^{\dot b}\nabla_{+\dot a}\right)\bar\phi_i + \frac{ig}{2}\delta_{\dot a}{}^{\dot b}\left(f^A_{++} - 2i\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}\right)(t_A)^j{}_i\bar\phi_j\\
&\quad\longleftrightarrow \partial_{+\dot a}\partial_+{}^{\dot b}\gamma_i^{(0)} + 2ig\delta_{\dot a}{}^{\dot b}\delta^{AB}\mathsf{b}^{(0)}_B(t_A)^j{}_i\gamma_j^{(0)}.
\end{aligned} \tag{3.26}$$

The antisymmetric gaugino jet gives the other relation used in [1, Eq. (2.4)]. Equation (3.20) gives the explicit BV boundary

$$\begin{aligned}
&\nabla_{+\dot a}\bar\lambda^{A\dot b} + \frac{ig}{2}\delta_{\dot a}{}^{\dot b}\delta^{AB}C^C{}_{BD}c^D\lambda^{\ddagger-}_{C,\rm old}\\
&\quad-\left[\frac{1}{2}\left(\nabla_{+\dot a}\bar\lambda^{A\dot b} + \nabla_+{}^{\dot b}\bar\lambda^A_{\dot a}\right) + \frac{g}{\sqrt{2}}\delta_{\dot a}{}^{\dot b}\delta^{AB}\bar\phi_j(t_B)^j{}_i\left(\psi_+^i + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\rm old}\right)\right]\\
&\qquad= \left(S_{\rm BV}, -\frac{i}{2}\delta_{\dot a}{}^{\dot b}\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}\right).
\end{aligned} \tag{3.27}$$

The representative in square brackets transfers as

$$\begin{aligned}
&\frac{1}{2}\left(\nabla_{+\dot a}\bar\lambda^{A\dot b} + \nabla_+{}^{\dot b}\bar\lambda^A_{\dot a}\right) + \frac{g}{\sqrt{2}}\delta_{\dot a}{}^{\dot b}\delta^{AB}\bar\phi_j(t_B)^j{}_i\left(\psi_+^i + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\rm old}\right)\\
&\qquad\longleftrightarrow \frac{1}{2}\partial_{+\dot a}\partial_+{}^{\dot b}\mathsf{c}^{A(0)} + ig\delta_{\dot a}{}^{\dot b}\delta^{AB}(t_B)^j{}_i\gamma_j^{(0)}\beta^{(0)i}.
\end{aligned} \tag{3.28}$$

The induced classical differential acts on the four zero-form fields by

$$\begin{aligned} Q_0\mathsf{c}^A &= -\frac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C, & Q_0\gamma_i &= -ig\mathsf{c}^A(t_A)^j{}_i\gamma_j, \\ Q_0\beta^i &= ig\mathsf{c}^A(t_A)^i{}_j\beta^j - \frac{\partial\bar{W}(\gamma,0)}{\partial\gamma_i}, & Q_0\mathsf{b}_A &= -gC^B{}_{AC}\mathsf{c}^C\mathsf{b}_B - ig(t_A)^j{}_i\gamma_j\beta^i. \end{aligned} \tag{3.29}$$

Here all fields denote their zero-form parts, and $[Q_0, \partial_{+\dot{a}}] = 0$ supplies the transformations of every jet.

3.5 Dolbeault contraction and Wick contraction

We now prove that, with the twisted action S_{tw}, the antiholomorphic parts of the letters are indeed contractible.

We use the perturbative homotopy transfer of Section 9.3, one step before its page E_0: we first take the $\bar{\partial}$-cohomology of the letters.

For polynomial Dolbeault jets, let ι include holomorphic representatives and π project onto them. Here $\iota_{-\dot{a}}$ contracts a differential form with $\partial/\partial x^{-\dot{a}}$; N counts antiholomorphic coordinate and form degree:

$$\begin{gathered} \{\bar{\partial}, \iota_{-\dot{a}}\} = \frac{\partial}{\partial x^{-\dot{a}}}, \qquad N = \{\bar{\partial}, \iota_{x^{-\dot{a}}\partial/\partial x^{-\dot{a}}}\} = x^{-\dot{a}}\frac{\partial}{\partial x^{-\dot{a}}} + dx^{-\dot{a}}\iota_{-\dot{a}}, \\ \pi X = X|_{x^{-\dot{a}}=dx^{-\dot{a}}=0}, \qquad h_{\bar{\partial}} = \begin{cases} \iota_{x^{-\dot{a}}\partial/\partial x^{-\dot{a}}}N^{-1}, & N>0, \\ 0, & N=0, \end{cases} \\ \bar{\partial}h_{\bar{\partial}} + h_{\bar{\partial}}\bar{\partial} = 1 - \iota\pi, \qquad h_{\bar{\partial}}^2 = h_{\bar{\partial}}\iota = \pi h_{\bar{\partial}} = 0. \end{gathered} \tag{3.30}$$

Indeed, since $[N, \bar{\partial}] = 0$ and $\{\bar{\partial}, \iota_{x^{-\dot{a}}\partial/\partial x^{-\dot{a}}}\} = N$, the left-hand side of $\bar{\partial}h_{\bar{\partial}} + h_{\bar{\partial}}\bar{\partial} = 1 - \iota\pi$ equals $NN^{-1} = 1$ on $N > 0$, while on $N = 0$ both $\bar{\partial}$ and $h_{\bar{\partial}}$ vanish and $\iota\pi = 1$.

The contraction removes antiholomorphic coordinates and forms, leaving the holomorphic jets of the zero-form fields. For a Dolbeault-closed output field, translation therefore acts as

$$\begin{aligned} \pi X(x_v + x) &= \pi \exp\left(x_v^{+\dot{a}}\frac{\partial}{\partial x^{+\dot{a}}} + x_v^{-\dot{a}}\frac{\partial}{\partial x^{-\dot{a}}}\right) X(x) \\ &= \exp\left(x_v^{+\dot{a}}\frac{\partial}{\partial x^{+\dot{a}}}\right)\pi X(x). \end{aligned} \tag{3.31}$$

This is the source of the momentum substitution in the higher bracket (3.43): the source at the root is eliminated by momentum conservation, and the sources at the integrated vertices become derivatives of their surviving letters,

$$\lambda_{k,+\dot{a}} = -\sum_{\substack{v=1,\dots,k-1 \\ \text{integrated vertices}}} \lambda_{v,+\dot{a}}, \qquad \lambda_{v,+\dot{a}} \longmapsto \frac{\partial}{\partial x_v^{+\dot{a}}} = -\frac{1}{2}\partial_{v,+\dot{a}} \qquad (v<k), \tag{3.32}$$

with the Leibniz rule when several letters survive at v. A fully contracted integrated vertex has $\lambda_v = 0$.

We now write the Wick contraction explicitly so that the graph kernels determine higher brackets on words and their holomorphic derivatives; following [5, 9], we sometimes refer to this field-space pairing at fixed spacetime arguments—with the spacetime differential-form factor F factored out—as the *fibre contraction*.

Let F be the spacetime form in (3.43) and $e_a = (v_a, w_a)$ its ordered edges. For operators with no input holomorphic derivatives, the Wick contraction is

$$W_\Gamma[F; \mathcal{O}_1, \dots, \mathcal{O}_k] \\ = F \prod_{a=1}^{|\Gamma_1|}{}^{\text{edge order}} \left(\begin{array}{c} \dfrac{\partial_L}{\partial \mathsf{b}_{A,v_a}} \dfrac{\partial_L}{\partial \mathsf{c}^A_{w_a}} + \dfrac{\partial_L}{\partial \mathsf{c}^A_{v_a}} \dfrac{\partial_L}{\partial \mathsf{b}_{A,w_a}} \\ - \dfrac{\partial_L}{\partial \beta^i_{v_a}} \dfrac{\partial_L}{\partial \gamma_{i,w_a}} - \dfrac{\partial_L}{\partial \gamma_{i,v_a}} \dfrac{\partial_L}{\partial \beta^i_{w_a}} \end{array} \right) \mathcal{O}_1 \cdots \mathcal{O}_k. \tag{3.33}$$

The field-space derivative operators act on the ordered product from right to left. The displayed formula uses the same separation of spacetime and field-space factors as (3.39). For holomorphic jets, let X, Y be the paired fields in a term and let r, s and p, q be their respective derivative orders. Each oriented edge $e = (v, w)$ replaces the two field-space derivatives in every term of (3.33) by

$$\frac{\partial_L}{\partial X_v} \frac{\partial_L}{\partial Y_w} \\ \longmapsto \sum_{r,s,p,q \geq 0} (-2)^{r+s} 2^{p+q} \frac{\partial^{r+p}}{\partial (z_e^{+\dot{+}})^{r+p}} \frac{\partial^{s+q}}{\partial (z_e^{+\dot{-}})^{s+q}} \frac{\partial_L}{\partial [(\partial^r_{+\dot{+}} \partial^s_{+\dot{-}} X)(x_v)]} \frac{\partial_L}{\partial [(\partial^p_{+\dot{+}} \partial^q_{+\dot{-}} Y)(x_w)]}. \tag{3.34}$$

Here $[P_e]_\Omega = [P_{\varepsilon,L;e}]_\Omega$ denotes the propagator one-form factor and $[K_{\varepsilon;e}]_\Omega$ denotes the cut-edge two-form factor defined below in (3.41)–(3.43). The z_e derivatives act only on the kernel of edge e, whose argument is $x_v - x_w + z_e$; set all $z_e = 0$ afterwards. Indeed,

$$\partial_{v,+\dot{a}} [P_e]_\Omega = -2 \frac{\partial}{\partial z_e^{+\dot{a}}} [P_e]_\Omega, \qquad \partial_{w,+\dot{a}} [P_e]_\Omega = 2 \frac{\partial}{\partial z_e^{+\dot{a}}} [P_e]_\Omega.$$

These identities also hold for the cut edge $[K_{\varepsilon;e}]_\Omega$. The derivatives with respect to the input jets remove the contracted letters; all remaining letters are translated by (3.31). For example,

$$W_{(v,w)}[F; (\partial_{+\dot{a}} \mathsf{b}_A)(x_v), \mathsf{c}^B(x_w)] = -2 \delta_A{}^B \left. \frac{\partial F}{\partial z_e^{+\dot{a}}} \right|_{z_e = 0}.$$

The projection to $H(\bar{\partial})$ is given in (3.30). Applying $\bar{\partial}$ to the ordered propagators gives the cut-edge signs $(-1)^{j-1}$ of (3.43).

For labelled vertices and a fixed edge order, every graph is included once. The derivatives supply the multiplicities of identical letters; the factor $1/(n+1)!$ in (8.1) accounts for the identical interaction insertions.

3.6 Fields and interaction

Collecting the results of Sections 3.1–3.3, we record the component dictionary and the twisted action that enter the graph operations [3–6, 9].

Proposition 3.1 (Component dictionary)**.** *For canonical matter and quadratic gauge kinetic*

terms, the map between twisted theory and physical components is

$$\begin{aligned}
\mathsf{c}^A &= c^A - \frac{1}{2}dx^-_{\dot a} V_-^{A\dot a} + \frac{i}{4}dx^{-\dot +}\wedge dx^{-\dot -}\lambda_-^A,\\
\gamma_i &= \bar\phi_i - \frac{1}{2\sqrt{2}}dx^-_{\dot a}\bar\psi_i^{\dot a} + \frac{1}{8}dx^{-\dot +}\wedge dx^{-\dot -}\bar{\mathcal{F}}_i,\\
\beta^i &= \bar{\mathcal{F}}^{\ddagger i} - \frac{i}{\sqrt{2}}\psi_+^i + \frac{1}{2\sqrt{2}}dx^-_{\dot a}\bar\psi^{\ddagger\dot a i} + \frac{1}{8}dx^{-\dot +}\wedge dx^{-\dot -}\bar\phi^{\ddagger i},\\
\mathsf{b}_A &= \frac{1}{4}\delta_{AB}f^B_{++} - \frac{i}{2}\lambda_A^{\ddagger -} + \frac{1}{4}dx^-_{\dot a}V_A^{\ddagger -\dot a} + \frac{1}{8}dx^{-\dot +}\wedge dx^{-\dot -}c_A^\ddagger,\\
\mathsf{w}^A_{\dot a} &= \bar W^A_{\dot a}\Big|_{\theta=0,\,\bar\theta^{\dot b}=dx^{-\dot b}/4}\\
&= \bar\lambda^A_{\dot a} + \frac{i}{4}dx^-_{\dot a}D^A + \frac{1}{8}dx^{-\dot b}\epsilon^{bc}\sigma^\mu_{b\dot a}\sigma^\nu_{c\dot b}f^A_{\mu\nu}\\
&\quad - \frac{i}{8}dx^{-\dot +}\wedge dx^{-\dot -}\nabla^b{}_{\dot a}\lambda_b^A.
\end{aligned}\tag{3.35}$$

where ‡ denotes the antifield.[14]

Proposition 3.2 (Twisted action and interaction density)**.** With $(t_A)^j{}_i$ the generators in the matter representation, the twisted action is

$$\begin{aligned}
S_{\mathrm{tw}} = \int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\left(\bar\partial\mathsf{c}^A - \frac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\right)\\
+\left(\bar\partial\gamma_i - ig\mathsf{c}^A(t_A)^j{}_i\gamma_j\right)\beta^i + \bar W(\gamma,\mathsf{w})\Big]_{(0,2)}.
\end{aligned}\tag{3.37}$$

where the projection $[\cdot]_{(0,2)}$ selects antiholomorphic form degree two and the function $\bar W$ contains the antichiral superpotential and gauge kinetic terms expressed in γ and w. Ω is the holomorphic form $2\,dx^{+\dot +}\wedge dx^{+\dot -}$.

Throughout, $\bar W$ is a polynomial or, more generally, a formal power series in γ and w, read coefficientwise, as perturbation theory requires.

The interaction density entering the higher operations is

$$\mathcal{I} = -\frac{g}{2}C^A{}_{BC}\mathsf{b}_A\mathsf{c}^B\mathsf{c}^C - ig\mathsf{c}^A(t_A)^j{}_i\gamma_j\beta^i + \bar W(\gamma,\mathsf{w}).\tag{3.38}$$

$\mathcal{I}$ in (3.38) is a polynomial in the full Dolbeault fields, retaining all antiholomorphic form degrees.

Its integrated interaction is $I = \int_{\mathbb{C}^2}\Omega\wedge[\mathcal{I}]_{(0,2)}$.

In (3.43), Wick contraction precedes the final $H(\bar\partial)$ projection: only the surviving fields are replaced by their holomorphic zero-form Taylor expansions.

The propagator forms remain in the integral and supply its antiholomorphic integration degree.

[14] Homotopy transfer sends the antichiral field strength to

$$\begin{aligned}
\mathsf{w}^A_{\dot a} &\xmapsto{\text{homotopy transfer}} \frac{1}{2}\partial_{+\dot a}\mathsf{c}^A = -\frac{\partial\mathsf{c}^A}{\partial x^{+\dot a}}\\
&= \frac{1}{2}\partial_{+\dot a}c^A - \frac{1}{4}dx^-_{\dot b}\partial_{+\dot a}V_-^{A\dot b} + \frac{i}{8}dx^{-\dot +}\wedge dx^{-\dot -}\partial_{+\dot a}\lambda_-^A.
\end{aligned}\tag{3.36}$$

$$[\bar\lambda^A_{\dot a}] = [\mathsf{w}^{A(0)}_{\dot a}] = \frac{1}{2}[\partial_{+\dot a}\mathsf{c}^{A(0)}].$$

The doublet $(S_{\mathrm{BV}}, V^A_{+\dot a}) = -2\bar\lambda^A_{\dot a} + \nabla_{+\dot a}c^A$ identifies the gaugino with the zero-form part of $\frac{1}{2}\partial_{+\dot a}\mathsf{c}^A$ after transfer.

3.7 Propagators and higher brackets

The graph operations are built from the BV pairing, the heat kernel and the propagator. Define Δ_{vw} as the inverse BV pairing contracting the fields at vertices v, w.

Its field-space part is

$$\begin{aligned}\Delta_{vw} = &\frac{\partial_L}{\partial \mathsf{b}_{A,v}}\frac{\partial_L}{\partial \mathsf{c}^A_w} + \frac{\partial_L}{\partial \mathsf{c}^A_v}\frac{\partial_L}{\partial \mathsf{b}_{A,w}} \\ &- \frac{\partial_L}{\partial \beta^i_v}\frac{\partial_L}{\partial \gamma_{i,w}} - \frac{\partial_L}{\partial \gamma_{i,v}}\frac{\partial_L}{\partial \beta^i_w}.\end{aligned} \tag{3.39}$$

Dolbeault gauge fixing (a choice of Lagrangian submanifold) uses the codifferential $\bar\partial^*$ and the Laplacian $H = \{\bar\partial, \bar\partial^*\}$.

With ω_E^{-1} the inverse BV pairing, the heat kernel and, for $0 < \varepsilon < L$, the propagator are [5, 17]

$$K_t = (e^{-tH} \otimes 1)\omega_E^{-1}. \tag{3.40}$$

$$[K_t(x_1, x_2)]_\Omega = -\frac{e^{-|x_1 - x_2|_E^2/(2t)}}{32\pi^2 t^2} d(x_1 - x_2)^{-\dot+} \wedge d(x_1 - x_2)^{-\dot-}. \tag{3.41}$$

$$\begin{aligned}P_{\varepsilon,L} &= \int_\varepsilon^L (\bar\partial^* \otimes 1) K_t \, dt, \\ (\bar\partial \otimes 1 + 1 \otimes \bar\partial) P_{\varepsilon,L} &= -\int_\varepsilon^L \partial_t K_t \, dt = K_\varepsilon - K_L.\end{aligned} \tag{3.42}$$

Here the subscript Ω denotes the antiholomorphic differential-form factor remaining after factoring out the holomorphic volume form Ω on one endpoint (explicitly written for $[P_{\varepsilon,L;e}]_\Omega$ in (3.57)).

The higher brackets are the basic objects in the computation of the anomalous supersymmetry transformation and of the quantum product.

Definition 3.3 (Brackets and the full quantum differential)**.** Let $\mathcal{O}_1, \dots, \mathcal{O}_k$ be local operators with fermion numbers $|\mathcal{O}_i|$, with $\mathcal{O}_k$ placed at the origin. Their higher bracket is

$$\begin{aligned}\{\mathcal{O}_1, \dots, \mathcal{O}_k\}_0 = (-1)^{\sum_{i=1}^k (k-i)|\mathcal{O}_i|} \sum_\Gamma \lim_{\varepsilon \downarrow 0} \sum_{j=1}^{|\Gamma_1|} (-1)^{j-1} \int_{(\mathbb{C}^2)^{k-1}} \left(\bigwedge_{v<k} \Omega_v \right) \\ \times W_\Gamma \Bigg[\left(\bigwedge_{a<j} [P_{\varepsilon,L;e_a}]_\Omega \right) \wedge [K_{\varepsilon;e_j}]_\Omega \wedge \left(\bigwedge_{a>j} [P_{\varepsilon,L;e_a}]_\Omega \right); \\ \mathcal{O}_1(x_1), \dots, \mathcal{O}_k(0) \Bigg] \Big|_{H(\bar\partial)}.\end{aligned} \tag{3.43}$$

Here Γ runs over connected contraction graphs through these insertions, with edges $\Gamma_1 = (e_1, \dots, e_{|\Gamma_1|})$ in their chosen contraction order, and Ω_v is $\Omega = 2\, dx_v^{++} \wedge dx_v^{+-}$ at vertex v. In the j-th term of the edge sum, the *cut edge* e_j carries the two-form heat kernel $[K_{\varepsilon;e_j}]_\Omega$ evaluated at the ultraviolet boundary $t_{e_j} = \varepsilon$ of its Schwinger-time interval, whereas every uncut edge e_a ($a \neq j$) carries the regulated Bochner–Martinelli propagator one-form $[P_{\varepsilon,L;e_a}]_\Omega$ integrated over $\varepsilon < t_{e_a} < L$. Finally, W_Γ denotes the ordered Wick contraction of Section 3.5 ((3.33)–(3.34)), and $|_{H(\bar\partial)}$ projects onto $\bar\partial$-cohomology via (3.30), keeping holomorphic local operators.

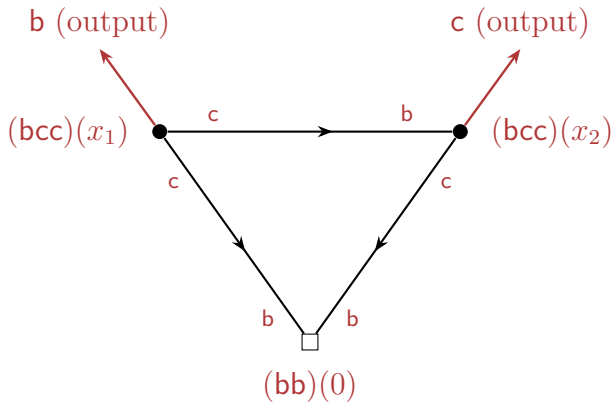


Figure 3.1: A contraction graph in $\{\mathsf{bcc}, \mathsf{bcc}, \mathsf{bb}\}_0$. The output letters are b at x_1 and c at x_2; internal arrows run from c to b.

For an interaction density $\mathcal{I}$ in the action and a holomorphic local operator $\mathcal{O}$, the full quantum BV differential is

$$\mathbf{Q} = \sum_{n\geq 0} \hbar^n Q_n, \qquad Q_n\mathcal{O} = \frac{1}{(n+1)!}\{\mathcal{I}^{\otimes(n+1)}, \mathcal{O}\}_0. \tag{3.44}$$

3.8 Where the higher brackets and cut edges come from

The cut-edge prescription in (3.43)–(3.44) is a BV Ward identity for a regularized composite-operator insertion [3, 4, 6, 17]. We explain this derivation in detail, emphasizing why two distinct mechanisms produce cut edges and why ultraviolet finiteness makes the holomorphically twisted theory far simpler than its untwisted counterpart; a reader already familiar with higher brackets may skip to Section 3.9.

Take the holomorphic interaction density $\mathcal{I}$ in (3.38) and a polynomial $\mathcal{O}$ in the Dolbeault zero-form fields $\mathsf{c}, \mathsf{b}, \gamma, \beta$ and their holomorphic ∂_+ jets:

$$I = \int_{\mathbb{C}^2} \Omega \wedge [\mathcal{I}]_{(0,2)}, \qquad F = \mathcal{O}(0).$$

Let $\partial_{P_{\varepsilon,L}}$ contract one unordered field pair with $P_{\varepsilon,L}$. At fixed infrared scale $L > 0$, evaluated on smooth compactly supported external fields, both the effective interaction $I[L]$ and the effective operator insertion $F[L]$ exist coefficientwise as $\varepsilon \downarrow 0$ [5, Theorem 5, Corollary 4 and Theorem 6]:

$$\begin{aligned} e^{I[L]/\hbar} &= \lim_{\varepsilon\downarrow 0} e^{\hbar\partial_{P_{\varepsilon,L}}} e^{I/\hbar}, \\ F[L]e^{I[L]/\hbar} &= \lim_{\varepsilon\downarrow 0} e^{\hbar\partial_{P_{\varepsilon,L}}} \left(\mathcal{O}(0)e^{I/\hbar}\right). \end{aligned} \tag{3.45}$$

Two separate ultraviolet-finiteness properties underlie (3.45):

1. *No bulk counterterms in I.* By [5, Theorem 5 and Corollary 4], connected bulk graphs of I with only $P_{\varepsilon,L}$ propagators have smooth $\varepsilon \downarrow 0$ limits, so the bare interaction I requires no $I^{\mathrm{CT}}(\varepsilon)$ counterterms.

2. *No composite-operator counterterms in $\mathcal{O}(0)$.* Because $\mathcal{O}(0)$ is built exclusively from Dolbeault zero-forms and their ∂_+ derivatives, it carries antiholomorphic form degree zero. In any subgraph collapsing m interaction vertices onto $\mathcal{O}(0)$, the m integrated vertices supply $2m$ antiholomorphic integration directions while each uncut propagator $[P_{\varepsilon,L;e}]_\Omega$ has antiholomorphic form degree one; as $\varepsilon \downarrow 0$ without a cut edge, no divergent contact counterterm $\mathcal{O}^{\mathrm{CT}}(\varepsilon)$ is needed [5, Theorem 6].

It is precisely for this twofold reason that computing anomalous supersymmetry transformations directly in the untwisted four-dimensional physical theory is much harder: there, rich ultraviolet divergences appear both in the bulk action and in composite-operator mixing, requiring coupled counterterm subtractions before taking $\varepsilon \downarrow 0$.

Now let $K_s = (e^{-sH} \otimes 1)\omega_E^{-1}$ be the scale-s heat kernel, let Δ_s contract one field pair within a single functional via K_s, and let $(\ ,\ )_s$ denote the scale-s BV bracket contracting one field pair between two functionals via K_s. Acting on effective observables at scale L, the quantum BV differential is

$$\mathbf{Q}_L F[L] = Q_{\mathrm{BV}} F[L] + (I[L], F[L])_L + \hbar \Delta_L F[L].$$

To express $\mathbf{Q}_L F[L]$ in terms of local operations at the ultraviolet cutoff ε, push $Q_{\mathrm{BV}} = \bar\partial$ through the propagator exponential $e^{\hbar \partial_{P_{\varepsilon,L}}}$ using $(\bar\partial \otimes 1 + 1 \otimes \bar\partial) P_{\varepsilon,L} = K_\varepsilon - K_L$ (3.42) [17], [4, §2.3]. For ordered edges $e_1, \dots, e_E$,

$$\begin{aligned} \bar\partial \bigwedge_{i=1}^{E} [P_{\varepsilon,L;e_i}]_\Omega = \sum_{\substack{j=1,\dots,E \\ \text{cut edges}}} (-1)^{j-1} \left(\bigwedge_{i<j} [P_{\varepsilon,L;e_i}]_\Omega \right) \\ \wedge [K_{\varepsilon;e_j} - K_{L;e_j}]_\Omega \wedge \left(\bigwedge_{i>j} [P_{\varepsilon,L;e_i}]_\Omega \right). \end{aligned} \tag{3.46}$$

All K_L contributions from (3.46) cancel against $(I[L], F[L])_L + \hbar \Delta_L F[L]$ (together with the bulk scale-L master equation on $I[L]$). At the ultraviolet scale ε, the integrated holomorphic interaction satisfies $Q_{\mathrm{BV}} I = 0$, and single-vertex contractions $\Delta_\varepsilon I = 0$ and $\Delta_\varepsilon \mathcal{O}(0) = 0$ vanish because $d(x_v - x_v)^{-\dot a} = 0$ kills self-loops. What remains is the exact BV Ward identity

$$(\mathbf{Q}_L F[L])\, e^{I[L]/\hbar} = \lim_{\varepsilon \downarrow 0} e^{\hbar \partial_{P_{\varepsilon,L}}} \left[\left(Q_{\mathrm{BV}} \mathcal{O}(0) + (I, \mathcal{O}(0))_\varepsilon + \frac{1}{\hbar}\, \mathcal{O}(0) \cdot \frac{1}{2} (I, I)_\varepsilon \right) e^{I/\hbar} \right]. \tag{3.47}$$

On holomorphic zero-form inputs we also have $Q_{\mathrm{BV}} \mathcal{O}(0) = \bar\partial \mathcal{O}(0) = 0$. Equation (3.47) shows that the quantum differential arises from combining two scale-ε mechanisms before taking $\varepsilon \downarrow 0$:

- *Regularized action–operator bracket $(I, \mathcal{O}(0))_\varepsilon$ (I–$\mathcal{O}$ spoke cuts).* Here the interaction I acts directly on $\mathcal{O}(0)$ through the regularized BV bracket at scale ε, placing the cut heat kernel $[K_{\varepsilon;e}]_\Omega$ on an edge connecting an interaction vertex to $\mathcal{O}(0)$.
- *Scale-ε QME defect insertion $\frac{1}{2}(I, I)_\varepsilon$ (I–I internal cuts).* Classically $(I, I) = 0$, and in the bulk theory $\lim_{\varepsilon \downarrow 0} e^{\hbar \partial_{P_{\varepsilon,L}}} [\frac{1}{2}(I, I)_\varepsilon e^{I/\hbar}] = 0$ whenever the anomaly-cancellation condition holds (Theorem 6.2), so disconnected insertions of $\frac{1}{2}(I, I)_\varepsilon$ drop out. At finite $\varepsilon > 0$, however, the point-split bracket $\mathcal{D}_{\mathrm{QME}}(\varepsilon) := \frac{1}{2}(I, I)_\varepsilon$ is a nonzero bilocal *quantum master equation (QME) defect* that places $[K_{\varepsilon;e}]_\Omega$ on an internal edge between two interaction vertices. When propagator contractions in $e^{\hbar \partial_{P_{\varepsilon,L}}}$ wire that defect into a connected cluster containing $\mathcal{O}(0)$, its explicit $1/\hbar$ prefactor combines with the propagators to yield a finite local contribution as $\varepsilon \downarrow 0$.

By [5, Theorem 6], each connected total-collision boundary cluster containing $\mathcal{O}(0)$ and m interaction vertices collapses as $\varepsilon \downarrow 0$ to a finite, constant-coefficient holomorphic differential operator at the origin, while the remaining uncollapsed vertices dress that output word into its scale-L effective representative:

$$\begin{aligned} &Q_{\mathrm{BV}} F[L] + (I[L], F[L])_L + \hbar \Delta_L F[L] \\ &\quad = (\mathbf{Q}\mathcal{O})[L] \\ &\quad = e^{-I[L]/\hbar} \lim_{\varepsilon \downarrow 0} e^{\hbar \partial_{P_{\varepsilon,L}}} \left((\mathbf{Q}\mathcal{O})(0) e^{I/\hbar} \right). \end{aligned} \tag{3.48}$$

For m interaction vertices and the root $\mathcal{O}(0)$, antiholomorphic form-degree saturation (one degree-two cut edge $[K_\varepsilon]_\Omega$ and $|\Gamma_1|-1$ degree-one propagators $[P_{\varepsilon,L}]_\Omega$ across m integrated vertices) fixes the loop order n:

$$|\Gamma_0| = m+1, \qquad 2+(|\Gamma_1|-1) = 2m, \qquad n = |\Gamma_1|-|\Gamma_0|+1 = m-1, \tag{3.49}$$

so $m = n+1$ and $|\Gamma_1| = 2n+1$. Extracting from (3.47) the connected total-collision graphs with $n+1$ insertions of I and $2n$ propagators $P_{\varepsilon,L}$ (and noting $Q_0\mathcal{O} = \lim_{\varepsilon\downarrow 0}(I,\mathcal{O}(0))_\varepsilon|_{H(\bar\partial)}$ at $n=0$) gives the explicit defining decomposition of $Q_n\mathcal{O}$ for $n \geq 1$:

$$\begin{aligned} Q_n\mathcal{O}(0) = \lim_{\varepsilon\downarrow 0} \Bigg[& \underbrace{\frac{1}{n!}\Big((\hbar\partial_{P_{\varepsilon,L}})^{2n}\Big((I,\mathcal{O}(0))_\varepsilon \frac{I^{\,n}}{\hbar^n}\Big)\Big)_{\mathrm{conn}}}_{\text{cuts on } I\text{–}\mathcal{O} \text{ edges}} \\ & + \underbrace{\frac{1}{(n-1)!}\Big((\hbar\partial_{P_{\varepsilon,L}})^{2n}\Big(\mathcal{O}(0)\,\frac{1}{\hbar}\cdot\frac{1}{2}(I,I)_\varepsilon \frac{I^{\,n-1}}{\hbar^{n-1}}\Big)\Big)_{\mathrm{conn}}}_{\text{cuts on } I\text{–}I \text{ edges from the QME defect}} \Bigg]\Bigg|_{H(\bar\partial)}, \end{aligned} \tag{3.50}$$

where $(\cdots)_{\mathrm{conn}}$ denotes the connected total-collision part (independent of L once $\varepsilon \downarrow 0$). Counting powers of $\hbar$ and symmetry factors,

$$\frac{\hbar^{|\Gamma_1|-m}}{m!} = \frac{\hbar^{m-1}}{m!} = \frac{\hbar^n}{(n+1)!}. \tag{3.51}$$

Indeed, among $n+1$ labelled interaction vertices, choosing which vertex enters $(I,\mathcal{O}(0))_\varepsilon$ gives $n+1$ choices with weight $\frac{1}{n!}\cdot\frac{1}{n+1} = \frac{1}{(n+1)!}$, while choosing which unordered pair enters $\frac{1}{2}(I,I)_\varepsilon$ gives $\binom{n+1}{2}$ choices with weight $\frac{1}{(n-1)!}\cdot\frac{1}{2}\cdot\frac{1}{\binom{n+1}{2}} = \frac{1}{(n+1)!}$. Thus the two terms in (3.50) assemble with uniform weight $\frac{1}{(n+1)!}$ and the Koszul signs $(-1)^{j-1}$ of (3.46) into the single higher bracket $Q_n\mathcal{O} = \frac{1}{(n+1)!}\{\mathcal{I}^{\otimes(n+1)},\mathcal{O}\}_0$ of (3.44).

At tree level ($n=0$), only the first term is present and the single cut edge gives $Q_0\mathcal{O} = \{\mathcal{I},\mathcal{O}\}_0$. At one loop ($n=1$), ordering the triangle edges on $\{1,2,\mathcal{O}\}$ as $(12,1\mathcal{O},2\mathcal{O})$, the three cut terms are

$$\begin{aligned} & [K_{\varepsilon;12}]_\Omega \wedge [P_{\varepsilon,L;1\mathcal{O}}]_\Omega \wedge [P_{\varepsilon,L;2\mathcal{O}}]_\Omega \\ -&[P_{\varepsilon,L;12}]_\Omega \wedge [K_{\varepsilon;1\mathcal{O}}]_\Omega \wedge [P_{\varepsilon,L;2\mathcal{O}}]_\Omega \\ +&[P_{\varepsilon,L;12}]_\Omega \wedge [P_{\varepsilon,L;1\mathcal{O}}]_\Omega \wedge [K_{\varepsilon;2\mathcal{O}}]_\Omega, \qquad Q_1\mathcal{O} = \frac{1}{2}\{\mathcal{I},\mathcal{I},\mathcal{O}\}_0, \end{aligned}$$

where the first cut (12) comes from the QME defect $\frac{1}{2}(I,I)_\varepsilon$ and the remaining two cuts $(1\mathcal{O}, 2\mathcal{O})$ come from $(I,\mathcal{O})_\varepsilon$. Likewise, at two loops ($n=2$), the fan graph on $\{1,2,3,\mathcal{O}\}$ has $2n+1=5$ edges $(12, 23, 1\mathcal{O}, 2\mathcal{O}, 3\mathcal{O})$: two cuts sit on the internal I–I path edges (12 and 23) from the QME defect $\frac{1}{2}(I,I)_\varepsilon$, and three cuts sit on the I–$\mathcal{O}$ spokes $(1\mathcal{O}, 2\mathcal{O}, 3\mathcal{O})$ from $(I,\mathcal{O})_\varepsilon$, summing to $Q_2\mathcal{O} = \frac{1}{6}\{\mathcal{I},\mathcal{I},\mathcal{I},\mathcal{O}\}_0$. Figure 3.2 displays both cases.

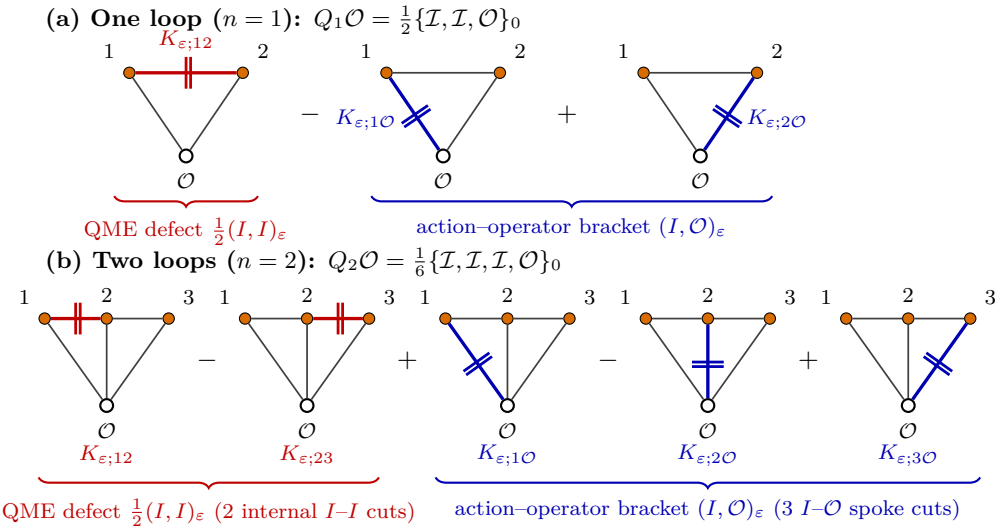


Figure 3.2: Origin of the $2n+1$ cut edges in $Q_n\mathcal{O}$ from the BV Ward identity (3.47)–(3.50) for edge order $(12, 1\mathcal{O}, 2\mathcal{O})$ at $n=1$ and $(12, 23, 1\mathcal{O}, 2\mathcal{O}, 3\mathcal{O})$ at $n=2$, with signs $(-1)^{j-1}$ as in (3.46). Double-slashed edges carry the UV heat kernel $[K_{\varepsilon;e}]_\Omega$ at $t_e=\varepsilon$; thin edges carry propagators $[P_{\varepsilon,L;e}]_\Omega$. Orange vertices represent interaction insertions I, and the open vertex represents $\mathcal{O}(0)$. Red internal I–I cuts originate from inserting the scale-ε QME defect $\frac{1}{2}(I,I)_\varepsilon$; blue I–$\mathcal{O}$ spoke cuts originate from $(I,\mathcal{O})_\varepsilon$.

3.9 Positive Schwinger times and the operatope

Computing a higher bracket amounts to computing the kernels of its graphs, with the external momenta supplied by the uncontracted output letters.

Proposition 3.4 (Current representation; external input [9, Section 2.7, Equations (2.32)–(2.33)], [5, Lemma 2, Proposition 9])**.** *Let Γ have vertices $\Gamma_0=\{1,\dots,k\}$, the insertion at $x_k=0$, ordered edges $\Gamma_1=(e_1,\dots,e_{2n+1})$ and Schwinger times $t_e>0$. With the source λ_v of the output letters at v, and the edge displacement z_e generating the holomorphic derivatives of the input letters (3.34), and the edge currents $y_{e,+\dot a}$ of (3.58), the kernel $\mathcal{I}_\Gamma(\lambda;z)$ of (3.55) is*

$$\begin{aligned}\mathcal{I}_\Gamma(\lambda;z) = (-1)^n \int_{\Delta_\Gamma} e^{-\sum_e z_e^{+\dot a} y_{e,+\dot a}} \bigwedge_{e\in\Gamma_1} d^2 y_e \\ \times \prod_{v<k}\prod_{\dot a=\dot+,\dot-} \delta\Big(\lambda_{v,+\dot a} - \sum_{e:e(0)=v} y_{e,+\dot a} + \sum_{e:e(1)=v} y_{e,+\dot a}\Big).\end{aligned} \tag{3.52}$$

Here the currents obey the loop law around every oriented cycle C,

$$\sum_{\substack{e\in C\\ \text{along } C}} t_e y_{e,+\dot a} - \sum_{\substack{e\in C\\ \text{against } C}} t_e y_{e,+\dot a} = 0. \tag{3.53}$$

for one common set of positive Schwinger times, the same for both dotted components, and Δ_Γ carries the orientation induced by the UV faces (3.55):

$$\Delta_\Gamma = \bigcup_{(t_e)\in\mathbb{RP}^{|\Gamma_1|-1}_{>0}} \Big\{(y_{e,+\dot a})_{e\in\Gamma_1} : \sum_{\substack{e\in C\\ \text{along } C}} t_e y_{e,+\dot a} - \sum_{\substack{e\in C\\ \text{against } C}} t_e y_{e,+\dot a} = 0 \ \text{ for every cycle } C\Big\}. \tag{3.54}$$

We now derive this representation, following [5, 9].

The heat edge and the current form

Let $\Gamma_0 = \{1, \dots, n+2\}$ be the vertex set of a general n-loop Feynman graph satisfying the Laman condition, and order the $2n+1$ edges as $\Gamma_1 = (e_1, \dots, e_{2n+1})$.

Fix $x_{n+2} = 0$ and $\lambda_{n+2} = -\sum_{v<n+2} \lambda_v$.

We compute the kernel by applying $\bar\partial$, which places a heat kernel on each edge in turn [5, 9].

$$\begin{aligned}
\mathcal{I}_\Gamma(\lambda; z) &= (-8\pi^2)^n \lim_{\varepsilon \downarrow 0} \sum_{j=1}^{2n+1} (-1)^{j-1} \int_{(\mathbb{C}^2)^{n+1}} \left(\bigwedge_{v<n+2} \Omega_v \right) e^{\sum_{v<n+2} \lambda_{v,+\dot a} x_v^{+\dot a}} \\
&\quad \times \left(\bigwedge_{i<j} [P_{\varepsilon,L;e_i}]_\Omega \right) \wedge [K_{\varepsilon;e_j}]_\Omega \wedge \left(\bigwedge_{i>j} [P_{\varepsilon,L;e_i}]_\Omega \right) \\
&= (-8\pi^2)^n \lim_{\varepsilon \downarrow 0} \int_{(\mathbb{C}^2)^{n+1}} \left(\bigwedge_{v<n+2} \Omega_v \right) e^{\sum_{v<n+2} \lambda_{v,+\dot a} x_v^{+\dot a}} \, \bar\partial \left(\bigwedge_{i=1}^{2n+1} [P_{\varepsilon,L;e_i}]_\Omega \right) \Big|_{\mathrm{UV}} \qquad (3.55) \\
&= (-1)^n (-8\pi^2)^n \lim_{\varepsilon \downarrow 0} \int_{(\mathbb{C}^2)^{n+1}} \left(\bigwedge_{v<n+2} \Omega_v \right) e^{\sum_{v<n+2} \lambda_{v,+\dot a} x_v^{+\dot a}} \\
&\quad \times \int_{\partial_{\mathrm{UV}}[\varepsilon,L]^{2n+1}} \bigwedge_{i=1}^{2n+1} \Big([K_{t_{e_i};e_i}]_\Omega - dt_{e_i} \wedge [(\bar\partial^* \otimes 1) K_{t_{e_i};e_i}]_\Omega \Big),
\end{aligned}$$

$$\partial_{\mathrm{UV}}[\varepsilon, L]^{2n+1} = \bigcup_{j=1}^{2n+1} \{ t_{e_j} = \varepsilon, \quad \varepsilon \le t_{e_i} \le L \ (i \neq j) \}. \qquad (3.56)$$

Here the facet $t_{e_j} = \varepsilon$ carries the orientation $(-1)^{j-1} \bigwedge_{i \neq j} dt_{e_i}$, matching the sign from passing $\bar\partial$ through $j-1$ propagator one-forms, while moving the $2n$ Schwinger differentials $-dt_{e_i}$ $(i \neq j)$ to the front crosses $0 + 1 + \cdots + (2n-1) = n(2n-1)$ one-forms and produces the overall sign $(-1)^{n(2n-1)} = (-1)^n$ (the infrared boundary $t_{e_j} = L$ cancels against $(I[L], \cdot)_L + \hbar \Delta_L$ as in Section 3.8).

We can further simplify the last line of (3.55) by combining the terms on each edge $e = (v, w)$ into a single current two-form:

$$\begin{aligned}
[P_{\varepsilon,L;e}]_\Omega = \; & - \frac{(x_v - x_w)^-_{\dot a} \, d(x_v - x_w)^{-\dot a}}{32\pi^2} \int_\varepsilon^L \frac{dt_e}{t_e^3} \\
& \times \exp\left[- \frac{(x_v - x_w)^-_{\dot a} (x_v - x_w + z_e)^{+\dot a}}{2t_e} \right].
\end{aligned} \qquad (3.57)$$

$$y_{e,+\dot a} = \frac{(x_{e(0)} - x_{e(1)})^-_{\dot a}}{2t_e}, \qquad d^2 y_e = dy_{e,+\dot 1} \wedge dy_{e,+\dot 2}. \qquad (3.58)$$

$$[K_{t_e;e}]_\Omega - dt_e \wedge [(\bar\partial^* \otimes 1) K_{t_e;e}]_\Omega = - \frac{e^{-y_{e,+\dot a}(x_{e(0)} - x_{e(1)} + z_e)^{+\dot a}}}{8\pi^2} \, d^2 y_e. \qquad (3.59)$$

$$\begin{aligned}
\mathcal{I}_\Gamma(\lambda; z) = \; & - \frac{1}{(8\pi^2)^{n+1}} \lim_{\varepsilon \downarrow 0} \int_{(\mathbb{C}^2)^{n+1}} \left(\bigwedge_{v<n+2} \Omega_v \right) e^{\sum_{v<n+2} \lambda_{v,+\dot a} x_v^{+\dot a}} \\
& \times \int_{\partial_{\mathrm{UV}}[\varepsilon,L]^{2n+1}} e^{-\sum_{e \in \Gamma_1} y_{e,+\dot a}(x_{e(0)} - x_{e(1)} + z_e)^{+\dot a}} \bigwedge_{e \in \Gamma_1} d^2 y_e.
\end{aligned} \qquad (3.60)$$

Deforming the integration contour to purely imaginary x_v^+ and real x_v^- (for real λ_v and imaginary z_e, then extending coefficientwise to formal power series in λ, z) regroups the exponent

by vertex:

$$\begin{aligned}&\sum_{v<n+2}\lambda_{v,+\dot a}x_v^{+\dot a}-\sum_e y_{e,+\dot a}(x_{e(0)}-x_{e(1)}+z_e)^{+\dot a}\\&\quad=-\sum_e y_{e,+\dot a}z_e^{+\dot a}+\sum_{v<n+2}\left(\lambda_{v,+\dot a}-\sum_{e:e(0)=v}y_{e,+\dot a}+\sum_{e:e(1)=v}y_{e,+\dot a}\right)x_v^{+\dot a},\end{aligned}\tag{3.61}$$

so integrating out $\Omega_v = 2\,dx_v^{+\dot+}\wedge dx_v^{+\dot-}$ over $(i\mathbb{R})^2$ at each vertex $v<n+2$ yields $2(2\pi i)^2=-8\pi^2$ times the current-conservation delta function

$$\begin{aligned}&\int_{(i\mathbb{R})^2}\frac{dx_v^{+\dot+}\wedge dx_v^{+\dot-}}{(2\pi i)^2}\exp\left[\left(\lambda_{v,+\dot a}-\sum_{e:e(0)=v}y_{e,+\dot a}+\sum_{e:e(1)=v}y_{e,+\dot a}\right)x_v^{+\dot a}\right]\\&\quad=\prod_{\dot a=\dot+,\dot-}\delta\left(\lambda_{v,+\dot a}-\sum_{e:e(0)=v}y_{e,+\dot a}+\sum_{e:e(1)=v}y_{e,+\dot a}\right),\end{aligned}\tag{3.62}$$

enforcing Kirchhoff's current law at every node:

$$\sum_{e:e(0)=v}y_{e,+\dot a}-\sum_{e:e(1)=v}y_{e,+\dot a}=\lambda_{v,+\dot a}.\tag{3.63}$$

Since $(-8\pi^2)^n(-1/8\pi^2)^{2n+1}(-8\pi^2)^{n+1}=1$, only the sign $(-1)^n$ remains, giving (3.52).

Independent currents

Choose a spanning tree T of Γ. Deleting any tree edge $e'\in T$ splits T into two connected components, so summing (3.63) over either component expresses $y_{e'}$ as a linear combination with coefficients ±1 of the external sources λ_v and the chord currents $(y_e)_{e\notin T}$. Eliminating the tree currents therefore has unit Jacobian for both dotted components and gives

$$\mathcal{I}_\Gamma(\lambda;z)=\int_{\substack{y\in\Delta_\Gamma\\ \sum_{e:e(0)=v}y_e-\sum_{e:e(1)=v}y_e=\lambda_v}}e^{-\sum_e y_{e,+\dot a}z_e^{+\dot a}}\bigwedge_{e\notin T}(-dy_{e,+\dot+}\wedge dy_{e,+\dot-}).\tag{3.64}$$

The domain Δ_Γ consists of the current solutions for positive resistance ratios, with the heat-boundary orientation.

Remark 3.5 (Circuit dictionary)**.** Equation (3.52) describes an electrical network on Γ, summarized in Table 2.14 of Section 2.4. Here $\frac12 x_{v,\dot a}^-$ acts as the node electric potential at v, t_e is the resistance of edge e, $y_{e,+\dot a}=(x_{e(0)}-x_{e(1)})_{\dot a}^-/(2t_e)$ is the edge current obeying Ohm's law, and λ_v is the external current injected at v. The delta functions in (3.52) impose Kirchhoff's current law (3.63) at every node, while (3.53) is Kirchhoff's voltage law around every loop. Finally, each $-\partial/\partial z_e^{+\dot a}$ inserts $y_{e,+\dot a}$, so at $z=0$ the kernel is the oriented volume swept out by the independent chord currents as (t_e) ranges over $\mathbb{RP}^{|\Gamma_1|-1}_{>0}$, and z_e is the generating source recording current moments. As summarized in Section 2.4, this dictionary brings series, parallel, and star–triangle network reductions to holomorphic graph kernels as iterative changes of variables on Schwinger space.

4 Constraints and classification of Q_n graphs

Only a few graphs can have a nonzero contribution to (3.52); they must satisfy the so-called Laman condition. In addition, the diagrams in 4d separate further into a few classes.

Write $\Gamma_1[S]$ for the edges with both ends in a vertex set $S \subseteq \Gamma_0$.

Some of the proofs are easier to understand in terms of a circuit diagram, so we sometimes provide more than one proof to illustrate the idea.

Lemma 4.1 (Laman count)**.** *Every nonzero graph satisfies*

$$|\Gamma_1| = 2|\Gamma_0| - 3. \tag{4.1}$$

Proof (graph integral). The heat edge has antiholomorphic form degree two and every other propagator degree one; together they saturate the relative antiholomorphic integration directions:

$$2 + (|\Gamma_1| - 1) = 2(|\Gamma_0| - 1).$$

□

Proof (circuit). The resistance ratios sweep out the real components of the independent loop currents in (3.52):

$$\underbrace{|\Gamma_1| - 1}_{\text{independent resistance ratios}} = \underbrace{2}_{\text{holomorphic dimension}} \underbrace{(|\Gamma_1| - |\Gamma_0| + 1)}_{\text{loops}}.$$

□

Lemma 4.2 (Laman bound on subgraphs)**.** *Every $S \subseteq \Gamma_0$ with $\Gamma_1[S] \neq \emptyset$ satisfies*

$$|\Gamma_1[S]| \leq 2|S| - 3.$$

Proof (graph integral). Fix $v_0 \in S$; the coordinates $(x_v - x_{v_0})^{-\dot{a}}$, $v \in S \setminus \{v_0\}$, give $2(|S| - 1)$ one-form directions in the numerators of the propagators. The total form degree cannot exceed the number of one-form directions. If the heat kernel edge K lies in S,

$$|\Gamma_1[S]| + 1 \leq 2(|S| - 1).$$

If the heat edge lies outside S, the radial vector annihilates every internal propagator one-form [5, Proposition 5, Lemma 2], [9, Equation (2.15)]:

$$R = \sum_{u \in S \setminus \{v_0\}} (x_u - x_{v_0})^{-\dot{c}} \frac{\partial}{\partial x_u^{-\dot{c}}},$$
$$\iota_R\big[-(x_v - x_w)^-_{\dot{a}} d(x_v - x_w)^{-\dot{a}}\big] = -(x_v - x_w)^-_{\dot{a}} (x_v - x_w)^{-\dot{a}} = 0,$$

so the number of holomorphic directions drops by one.

$$|\Gamma_1[S]| \leq 2(|S| - 1) - 1.$$

□

Proof (circuit). Ohm's law is local: the $2|\Gamma_1[S]|$ current components inside S are fixed by the $2(|S| - 1)$ relative potentials and by $|\Gamma_1[S]| - 1$ independent resistances, a common rescaling of potentials and resistances, or the fixed resistance on the UV boundary, removing one. They vary in all directions only if

$$2|\Gamma_1[S]| \leq 2(|S| - 1) + |\Gamma_1[S]| - 1.$$

□

Lemma 4.2 immediately leads to:

Corollary 4.3 (No self-loops or parallel edges)**.** *A nonzero graph has no self-loop and at most one edge between any two vertices.*

Proof (propagators). A self-loop carries $d(x_v - x_v)^{-\dot{a}} = 0$ [5, Section 2.3]. Two edges between v and w, with holomorphic shifts z, z', share the one-form direction of $x_v - x_w$ [9, Section 2.1]:

$$\begin{aligned}[P_{\varepsilon,L;vw}(x_v - x_w + z)]_\Omega \wedge [P_{\varepsilon,L;vw}(x_v - x_w + z')]_\Omega = 0,\\ [K_{\varepsilon;vw}(x_v - x_w + z)]_\Omega \wedge [P_{\varepsilon,L;vw}(x_v - x_w + z')]_\Omega = 0.\end{aligned} \tag{4.2}$$

□

Proof (circuit). A self-loop is shorted: $t_e y_{e,+\dot{a}} = 0$ with $t_e > 0$ gives $y_e = 0$. Two resistors between the same nodes are in parallel: $t_e y_e = t_{e'} y_{e'}$, so their four current components vary in only three directions. □

Corollary 4.3 is the case $|S| \le 2$ of Lemma 4.2.

These are the Laman constraints in two holomorphic dimensions [9, Equations (2.22)–(2.23)], [6, Section 3.3], [5, Definition 16, Proposition 5, Theorem 6].

Lemma 4.4 (Counts at n loops)**.** *A nonzero connected n-loop graph satisfies*

$$\begin{gathered}|\Gamma_0| = n + 2, \qquad |\Gamma_1| = 2n + 1, \qquad |\Gamma_1| - |\Gamma_0| + 1 = n,\\ |\Gamma_1[S]| \le 2|S| - 3 \qquad (S \subseteq \Gamma_0,\ |S| \ge 2).\end{gathered} \tag{4.3}$$

Proof. Lemma 4.1 and $n = |\Gamma_1| - |\Gamma_0| + 1$ give the first line; Lemma 4.2 the second. □

We now turn to the condition special to $d = 4$.

By (3.44), the Feynman graphs of the loop-corrected supercharge Q_n have $n + 1$ interaction vertices and one insertion $\mathcal{O}$.

Proposition 4.5 (Fan diagrams for interactions of degree at most three)**.** *For a translation-invariant interaction $\mathcal{I}$ whose vertices have degree at most three, every nonzero connected n-loop kernel with one local-operator insertion is a fan of cubic vertices (Figure 4.1), for $n \ge 1$. (The $n = 0$ case is a single edge connecting two vertices, representing the classical tree-level SUSY transformation.) The interaction vertices form a path; each is joined to the insertion, and one output field remains at each endpoint of the path.*

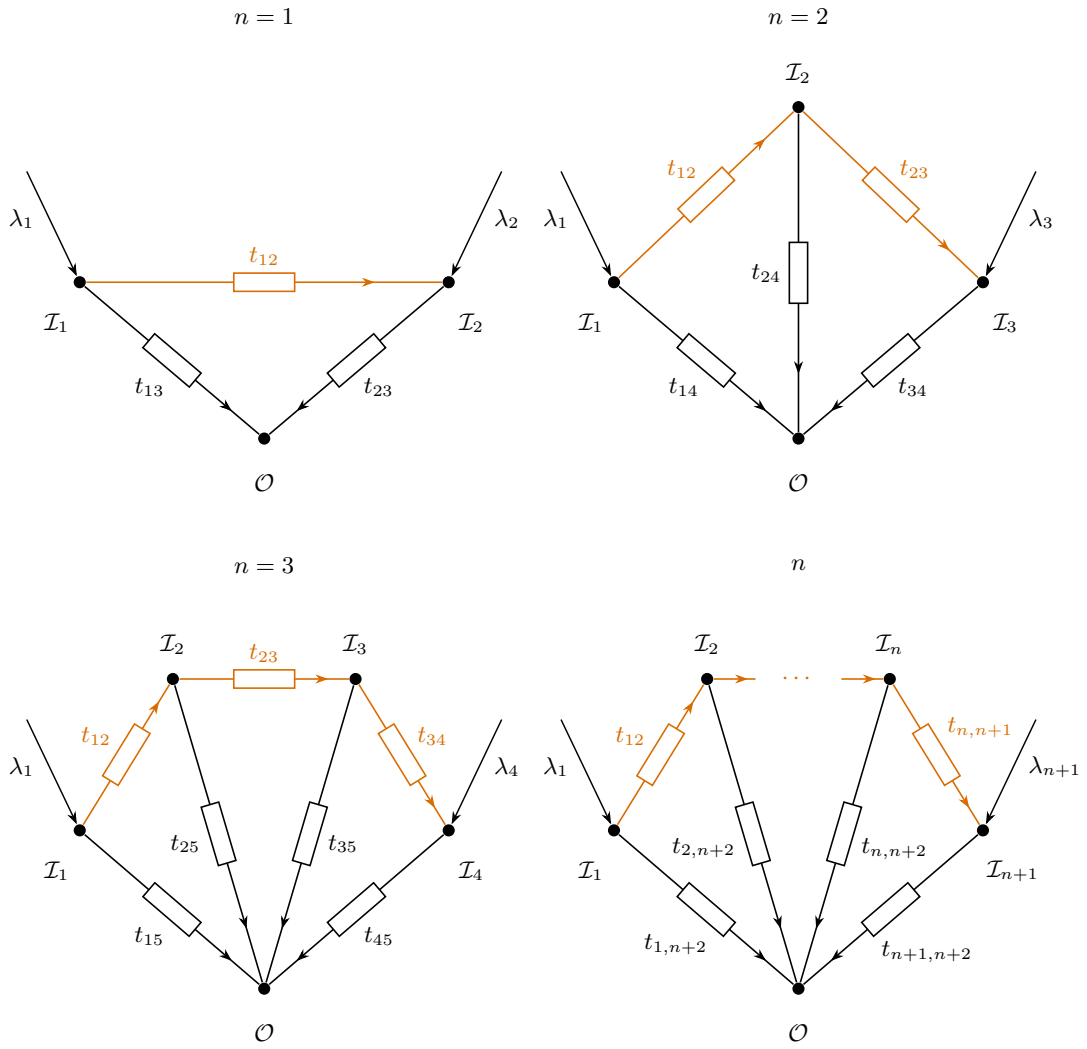


Figure 4.1: Fans at one, two, three and n loops as circuits, with the interaction path in orange, resistances t_{vw} and injected currents λ_1, λ_{n+1}.

By Lemma 4.4, there are $n+1$ interaction vertices besides the insertion $\mathcal{O}$.

The proof of Proposition 4.5 uses two lemmas:

Lemma 4.6 (Spokes)**.** *Let* $\deg_\Gamma \mathcal{O}$ *be the number of edges incident to* $\mathcal{O}$ *in a graph of Proposition 4.5. Then* $\deg_\Gamma \mathcal{O} = n+1$*: every interaction vertex is joined to* $\mathcal{O}$ *by one edge (called a* spoke *below) and exactly two interaction fields remain uncontracted.*

Proof (circuit). The uncontracted interaction fields inject the external currents. By Corollary 4.3 and Lemma 4.4, with equality in the first step only for cubic vertices,

$$\begin{aligned} \#(\text{sources}) &\le \underbrace{3}_{\text{degree bound}} \underbrace{(n+1)}_{\mathcal{I}\text{ vertices}} - \underbrace{2(2n+1-\deg_\Gamma \mathcal{O})}_{\substack{\text{letters contracted between} \\ \mathcal{I}\text{ vertices}}} - \underbrace{\deg_\Gamma \mathcal{O}}_{\substack{\text{edges from } \mathcal{O} \\ \text{to } \mathcal{I}\text{ vertices}}} \\ &= \underbrace{\deg_\Gamma \mathcal{O}}_{\substack{\le n+1 \\ \text{by Corollary 4.3}}} \quad -n+1 \le 2. \end{aligned} \tag{4.4}$$

If there is only one source, every current is a real multiple of the source current and sweeps at most $n < 2n$ dimensions; a nonzero kernel therefore has at least two sources, and both inequalities are then equalities. □

Lemma 4.7 (No interaction cycles)**.** *If every interaction vertex is joined to $\mathcal{O}$, no set S of interaction vertices carries a cycle.*

Proof. A loop of resistors on the interaction nodes S, together with their $|S|$ spokes, gives a subcircuit on $S \cup \{\mathcal{O}\}$ with more resistors than the local count of Ohm's law allows (Lemma 4.2):

$$|\Gamma_1[S \cup \{\mathcal{O}\}]| \geq 2|S| > 2(|S| + 1) - 3.$$

□

Proof of Proposition 4.5. By Lemma 4.6, each of the $n + 1$ cubic interaction nodes is wired to $\mathcal{O}$ by one spoke, which leaves $2n + 1 - (n + 1) = n$ internal resistors; they close no loop (Lemma 4.7) and form a tree. Each node spends one terminal on its spoke and keeps at most two for internal resistors, so the tree is a chain: its interior nodes wire all three terminals, and each end keeps one free terminal, the two sources of Lemma 4.6. □

Corollary 4.8 (Renormalizable theories)**.** *For every renormalizable $\mathcal{N} = 1$ gauge theory, massless or massive, every nonzero connected graph of Q_n, $n \geq 1$, with one insertion $\mathcal{O}$ is a fan of cubic vertices.*

Proof. The superpotential has degree at most three, and Proposition 4.5 applies. □

The fan kernel is the simplest one to evaluate. If the theory is non-renormalizable (with $\mathcal{I}$ of degree greater than three), other kinds of kernels can also appear; they are, however, also highly constrained.

Proposition 4.9 (Superpotential vertices and antifield count)**.** *For the interaction (3.38), allow $\bar{W}(\gamma, \mathsf{w})$ to have arbitrary degree. Every nonzero connected n-loop graph of $Q_n\mathcal{O}$ with one insertion $\mathcal{O}$, $n \geq 1$, contains at most one $\bar{W}$ vertex. Each graph term lowers the number of b, β factors by n plus the number of $\bar{W}$ vertices.*

We prove Proposition 4.9 with the help of two lemmas.

Lemma 4.10 (Antifield terminals)**.**

Every contraction pairs a b or β factor with a c or γ factor. A gauge or matter vertex of (3.38) supplies one b or β factor, a $\bar{W}$ vertex none.

Proof. A gauge or matter node, bcc or $\mathsf{c}\gamma\beta$ in (3.38), has one antifield terminal, a node of $\bar{W}(\gamma, \mathsf{w})$ none. □

Lemma 4.11 (Internal edges)**.** *A nonzero connected n-loop graph of Q_n has at least n edges between interaction vertices.*

Proof. Two resistors from $\mathcal{O}$ to the same interaction node would be in parallel (Corollary 4.3), so at most $n+1$ of the $2n+1$ resistors (Lemma 4.4) end at $\mathcal{O}$, and $2n+1-\deg_\Gamma \mathcal{O} \geq 2n+1-(n+1) = n$ join interaction nodes.

This is a weaker version of Lemma 4.6, which does not use the cubic condition. □

Proof of Proposition 4.9. By Lemmas 4.10 and 4.11,

$$\begin{gathered} \underbrace{n}_{\substack{\deg_\Gamma \mathcal{O} \leq n+1 \\ \text{at most } n+1 \text{ spokes}}} \leq \underbrace{2n + 1 - \deg_\Gamma \mathcal{O}}_{\text{internal resistors}} \leq \underbrace{n + 1 - \#(\bar{W} \text{ vertices})}_{\substack{\text{antifield terminals,} \\ \text{one per internal resistor}}} \\ \Longrightarrow \quad \#(\bar{W} \text{ vertices}) \leq 1, \end{gathered} \tag{4.5}$$

and each of the $2n + 1$ resistors takes one antifield terminal, a b or β factor:

$$\begin{aligned} \#(\mathsf{b}, \beta \text{ in the output}) &= \#(\mathsf{b}, \beta \text{ in } \mathcal{O}) + \big[n + 1 - \#(\bar{W} \text{ vertices})\big] - (2n + 1) \\ &= \#(\mathsf{b}, \beta \text{ in } \mathcal{O}) - n - \#(\bar{W} \text{ vertices}). \end{aligned} \tag{4.6}$$

□

Corollary 4.12 (Finite loop order)**.** *For $\bar{W}$ of any degree, or a formal power series read coefficientwise, $Q_n\mathcal{O} = 0$ for $n > \#(\mathsf{b}, \beta \text{ in } \mathcal{O})$, and terms with a $\bar{W}$ vertex vanish already for $n + 1 > \#(\mathsf{b}, \beta \text{ in } \mathcal{O})$.*

Proof. The left side of (4.6) is nonnegative. □

Corollary 4.12 and dimensional counting make the perturbative expansion truncate at a finite order, so that the perturbative cohomology is well defined.

Proposition 4.13 (Graphs with one superpotential vertex)**.** *For the interaction (3.38), a nonzero connected n-loop graph of Q_n, $n \geq 1$, with one insertion $\mathcal{O}$ and one $\bar{W}$ vertex joins $\mathcal{O}$ to all $n + 1$ interaction vertices, for any degree of $\bar{W}$. Removing $\mathcal{O}$ leaves a tree whose arms meet at $\bar{W}$.*

This type of kernel is called a spider.

Proof. With one $\bar{W}$ vertex the two bounds in (4.5) coincide:

$$n \leq 2n + 1 - \deg_\Gamma \mathcal{O} \leq n \quad \Longrightarrow \quad \deg_\Gamma \mathcal{O} = n + 1, \tag{4.7}$$

so every interaction node is wired to $\mathcal{O}$, and the n internal resistors close no loop (Lemma 4.7) and form a tree. A cubic node keeps at most two internal resistors besides its spoke, so the tree branches only at the $\bar{W}$ node; with a arms of ℓ_i cubic nodes,

$$\begin{gathered} \Gamma \setminus \{\mathcal{O}\} = \text{arms of lengths } \ell_1, \dots, \ell_a \text{ meeting at } \bar{W}, \\ \ell_1 + \cdots + \ell_a = n, \qquad \ell_i \geq 1, \qquad \deg \bar{W} \geq a + 1, \end{gathered} \tag{4.8}$$

where $\deg \bar{W}$, the number of field factors of the contributing monomial, covers the a internal resistors at the $\bar{W}$ node and its spoke. □

Remark 4.14. With at most two arms the graph of Proposition 4.13 is a fan; the bound $a + 1 \leq \deg \bar{W}$ gives this case for $\deg \bar{W} \leq 3$.

Corollary 4.15 (Graphs of Q_n)**.** *For the interaction (3.38), every nonzero connected graph of Q_n, $n \geq 1$, with one insertion $\mathcal{O}$ is a fan, or a spider whose arms meet at its $\bar{W}$ vertex, a book when all its arms have length one (Definition 7.4); for $\deg \bar{W} \leq 3$ it is a fan.*

Proof. Propositions 4.5, 4.9 and 4.13. □

This completes the classification of the diagrams of Q_n.

The calculation of the quantum product in Section 6 involves other kinds of kernels.

Proposition 4.16 (Superpotential vertices in quantum product diagrams)**.** *Let Γ be a nonzero connected marked graph of the quantum product (6.11) at order $\hbar^n$, $n \geq 1$, with insertions A, B, two arbitrary composite operators.*

(i) *Every connected component of $\Gamma \setminus \{A, B\}$ contains at most one $\bar{W}$ vertex.*

(ii) *Each graph term satisfies*

$$\#(\mathsf{b}, \beta \text{ in the output}) = \#(\mathsf{b}, \beta \text{ in } A) + \#(\mathsf{b}, \beta \text{ in } B) - n - \#(\bar{W} \text{ vertices}), \tag{4.9}$$

so the order-$\hbar^n$ term vanishes unless $\#(\mathsf{b}, \beta \text{ in } A) + \#(\mathsf{b}, \beta \text{ in } B) \geq n + \#(\bar{W} \text{ vertices})$.

Proof. (i) The $|T_j|$ nodes of a component T_j are connected through its internal resistors I_j, each of which takes its antifield terminal from a node of T_j (Lemma 4.10):

$$|T_j| - 1 \leq |I_j| \leq |T_j| - \#(\bar{W} \text{ vertices in } T_j).$$

(ii) Each of the $2n$ physical resistors takes one antifield terminal; the n interaction nodes have $n - \#(\bar{W} \text{ vertices})$ of them. □

The graphs of the quantum product are forests of interaction trees attached to both insertions, classified in Proposition 7.2.

5 Fan kernels

In Section 3, we saw that only the antichiral superpotential $\bar{W}$ of the physical action enters the holomorphically twisted interaction, while all other vertices arise from the first-order BV kinetic terms. In four dimensions, any renormalizable $\mathcal{N} = 1$ supersymmetric gauge theory has $\deg \bar{W} \leq 3$, so all interaction vertices are at most cubic and the n-loop supercharge Q_n is determined entirely by the n-loop fan kernel (Corollary 4.8). In this section we establish the general local series–parallel reduction operator on Schwinger space and iterate it to evaluate the n-loop fan kernel and its full displacement Taylor expansion in closed form.

5.1 The one-loop triangle diagram

We first review the one-loop triangle in the circuit picture (Figure 5.1).

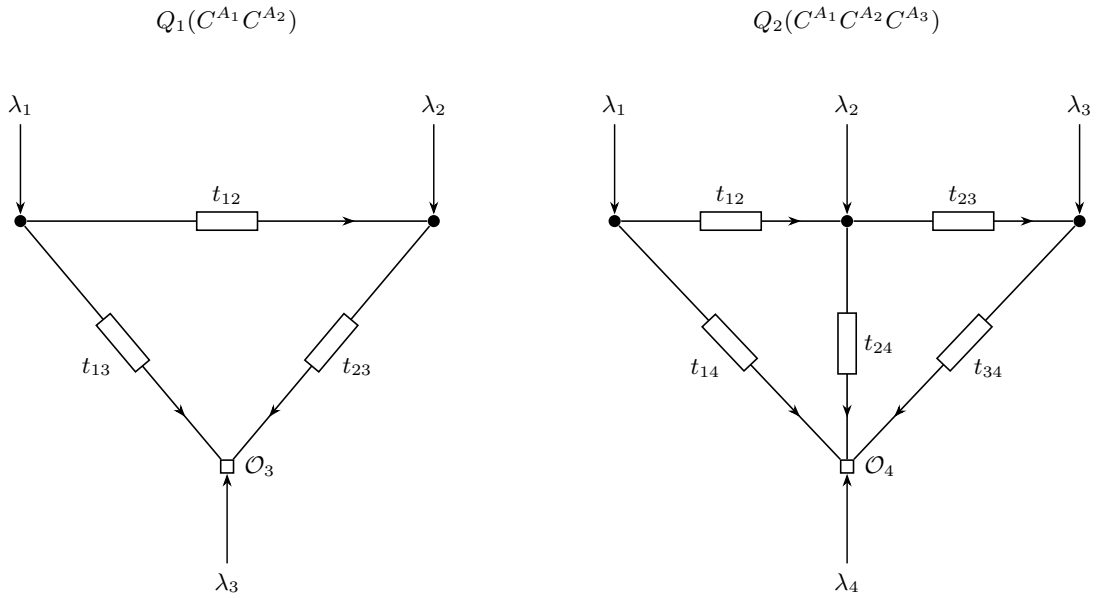


Figure 5.1: The one- and two-loop fan circuits. Dots are interaction vertices and the square is the local insertion. Each edge carries a resistor t_{vw}; λ_v is the current injected at vertex v.

The vertices of the triangle are $1, 2, 3$, with 3 at the origin, the insertion. With the gauge $t_{12} + t_{13} + t_{23} = 1$, current conservation and the cycle equation give[15]

$$\begin{aligned} y_{12,+\dot{a}} &= t_{13}\lambda_{1,+\dot{a}} - t_{23}\lambda_{2,+\dot{a}}, \\ y_{13,+\dot{a}} &= \lambda_{1,+\dot{a}} - y_{12,+\dot{a}}, \\ y_{23,+\dot{a}} &= \lambda_{2,+\dot{a}} + y_{12,+\dot{a}}. \end{aligned} \tag{5.1}$$

Using simplex coordinates r_1, r_2 with $(t_{12}, t_{13}, t_{23}) = (1 - r_1 - r_2, r_1, r_2)$, the Jacobian is

$$-dy_{12,++} \wedge dy_{12,+-} = (\lambda_{1,+\dot{a}}\lambda_{2,+}{}^{\dot{a}})\, dr_1 \wedge dr_2, \tag{5.2}$$

so

$$\mathcal{I}_{\triangle}(\lambda; z) = (\lambda_{1,+\dot{a}}\lambda_{2,+}{}^{\dot{a}}) \int_0^1 dr_1 \int_0^{1-r_1} dr_2\, e^{-\sum_e y_{e,+\dot{b}} z_e^{+\dot{b}}}. \tag{5.3}$$

[15] The first line of (5.1) is read off from $\partial y_{12}/\partial\lambda_1 = t_{13}$ and $\partial y_{12}/\partial\lambda_2 = -t_{23}$. With λ_2 switched off, the current λ_1 enters at vertex 1 and leaves at vertex 3 through the two parallel paths $1 \to 3$ and $1 \to 2 \to 3$; the path through edge 12, of resistance $t_{12} + t_{23}$, carries the fraction $t_{13}/(t_{12} + t_{13} + t_{23}) = t_{13}$. The other two lines are current conservation at vertices 1 and 2.

Expanding the exponential and integrating each monomial yields the complete displacement dependence, as in [4, Equation (2.47)]:[16]

$$\begin{aligned}\mathcal{I}_\Gamma(\lambda;z) = (\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a})\, e^{-\lambda_{1,+\dot a}z_{13}^{+\dot a}-\lambda_{2,+\dot a}z_{23}^{+\dot a}} \\ \times \sum_{i,j\ge 0} \frac{[-\lambda_{1,+\dot b}(z_{12}+z_{23}-z_{13})^{+\dot b}]^i[\lambda_{2,+\dot c}(z_{12}+z_{23}-z_{13})^{+\dot c}]^j}{(i+j+2)!}.\end{aligned} \tag{5.4}$$

5.2 Spanning trees and local series–parallel operators

In principle, one can solve Kirchhoff's equations on any graph Γ globally at fixed resistances via Kirchhoff's spanning-tree theorem [5, Theorem 4 and Corollary 2]: solving current conservation on each spanning tree T and averaging over all spanning trees,

$$\begin{aligned} y^T_{e,+\dot a} &= \begin{cases} \displaystyle\sum_{\substack{v \text{ connected to } e(0)\\ \text{in } T\setminus\{e\}}} \lambda_{v,+\dot a}, & e\in T,\\ 0, & e\notin T, \end{cases} \\ y_{e,+\dot a}(t) &= \frac{\displaystyle\sum_{\substack{T\\ \text{spanning trees}}} \left(\prod_{h\in\Gamma_1\setminus T} t_h\right) y^T_{e,+\dot a}}{\displaystyle\sum_{\substack{T\\ \text{spanning trees}}} \prod_{h\in\Gamma_1\setminus T} t_h}, \end{aligned} \tag{5.5}$$

satisfies both current conservation (since the normalized tree weights sum to one) and the loop voltage law.[17] While scale-invariant (allowing the simplex gauge $\sum_{e\in\Gamma_1} t_e = 1$), (5.5) introduces rational spanning-tree denominators whose complexity grows exponentially with loop order.

Instead, we eliminate vertices locally, turning elementary circuit reductions into differential-integral operators on the reduced graph kernel (indicated by superscript red).

Series elimination and parallel merge

Suppose a vertex v has only two neighbours u and w (Figure 5.2). Current conservation at v and Ohm's law between u and w give

$$\begin{aligned} y_{vu,+\dot a} + y_{vw,+\dot a} &= \lambda_{v,+\dot a}, \\ \frac{(x_u - x_w)^-_{\dot a}}{2} &= -t_{vu}y_{vu,+\dot a} + t_{vw}y_{vw,+\dot a} = t_{vw}\lambda_{v,+\dot a} - (t_{vu}+t_{vw})y_{vu,+\dot a}. \end{aligned} \tag{5.6}$$

[16] As in (3.34), the displacements generate the holomorphic derivatives of the letters of $\mathcal{O}$.

[17] Indeed, orienting any cycle L along its traversal and grouping $\sum_{e\in L} t_e(\prod_{h\notin T} t_h)y^T_{e,+\dot a}$ by the two-component spanning forest $F = T\setminus\{e\}$ (with common weight $\prod_{h\notin F} t_h$), each exit of L from one component of F pairs with the next entry f into it; the resulting trees T, T' satisfy $y^T_{e,+\dot a} + y^{T'}_{f,+\dot a} = \sum_{v\in\Gamma_0}\lambda_{v,+\dot a} = 0$.

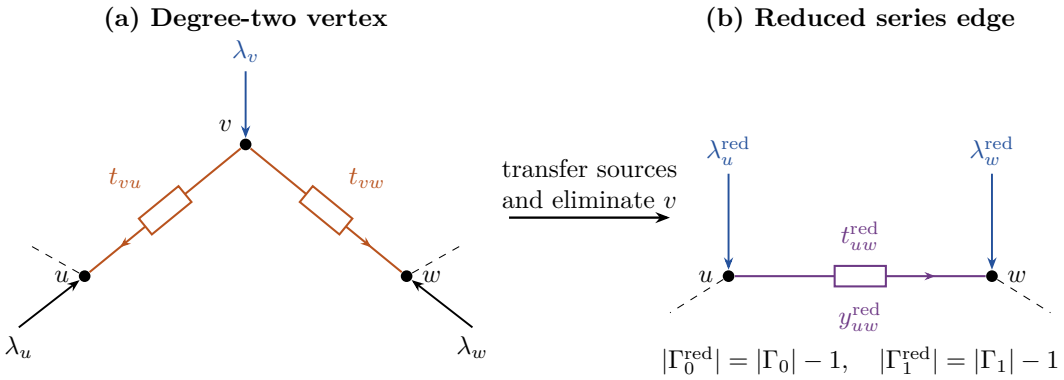


Figure 5.2: Series elimination: transferring the source λ_v to u, w via (5.7) replaces vu, vw by a single series resistor $t_{uw}^{\mathrm{red}} = t_{vu} + t_{vw}$ without changing the loop number.

Setting $r_j = t_{vu}/(t_{vu} + t_{vw})$ replaces the pair (vu, vw) by a single edge $u \to w$ and transfers λ_v to u and w:

$$\begin{aligned} r_j &= \frac{t_{vu}}{t_{vu} + t_{vw}}, & t_{uw}^{\mathrm{red}} &= t_{vu} + t_{vw}, \\ \lambda_{u,+\dot{a}}^{\mathrm{red}} &= \lambda_{u,+\dot{a}} + (1 - r_j)\lambda_{v,+\dot{a}}, & \lambda_{w,+\dot{a}}^{\mathrm{red}} &= \lambda_{w,+\dot{a}} + r_j\lambda_{v,+\dot{a}}, \\ y_{vu,+\dot{a}} &= (1 - r_j)\lambda_{v,+\dot{a}} - y_{uw,+\dot{a}}^{\mathrm{red}}, & y_{vw,+\dot{a}} &= r_j\lambda_{v,+\dot{a}} + y_{uw,+\dot{a}}^{\mathrm{red}}. \end{aligned} \tag{5.7}$$

On an n-loop Laman graph, series elimination preserves the loop number $h_1(\Gamma^{\mathrm{red}}) = n$ while reducing $|\Gamma_1|$ from $2n+1$ to $2n$: on the $(2n-1)$-dimensional projective space of reduced resistances, the $2n$-form $\bigwedge_{e \notin T}(-dy_{e,++} \wedge dy_{e,+\dot{-}})$ acquires one leg from the r_j-dependence of the transferred sources in (5.7), accompanied by the orientation sign $dt_{vw} \wedge \bigwedge_{e \neq vu,vw} dt_e = -t_{uw}^{\mathrm{red}}\, dr_j \wedge \bigwedge_{e \neq vu,vw} dt_e$ when (vu, vw) are ordered first.

Next, for q parallel edges $(uw)^{(1)}, \ldots, (uw)^{(q)}$ oriented from u to w, their common voltage difference gives

$$t_{uw}^{(j)} y_{uw,+\dot{a}}^{(j)} = \frac{(x_u - x_w)_{\dot{a}}^{-}}{2}, \qquad y_{uw,+\dot{a}}^{\mathrm{red}} = \sum_{j=1}^{q} y_{uw,+\dot{a}}^{(j)} = \left(\sum_{j=1}^{q} \frac{1}{t_{uw}^{(j)}}\right) \frac{(x_u - x_w)_{\dot{a}}^{-}}{2}, \tag{5.8}$$

so their conductances add and the merged current divides proportionally:

$$\frac{1}{t_{uw}^{\mathrm{red}}} = \sum_{j=1}^{q} \frac{1}{t_{uw}^{(j)}}, \qquad r_j = \frac{t_{uw}^{\mathrm{red}}}{t_{uw}^{(j)}}, \qquad \sum_{j=1}^{q} r_j = 1, \qquad y_{uw,+\dot{a}}^{(j)} = r_j y_{uw,+\dot{a}}^{\mathrm{red}}. \tag{5.9}$$

Combining a series elimination with a two-edge parallel merge removes one loop and produces a Laman graph again (Figure 2.1(a)).

Theorem 5.1 (Series–parallel reduction operator)**.** *Let Γ be a Laman circuit with a degree-two vertex v, incident edges vu, vw, and an existing edge uw. Order vu, vw first and retain the relative order of all remaining edges, including the merged edge uw. Setting*

$$\begin{aligned} r_1 &= \frac{t_{vw}}{t_{vu} + t_{vw}}, & r_2 &= \frac{t_{vu} + t_{vw}}{t_{vu} + t_{vw} + t_{uw}} \qquad (0 < r_1, r_2 < 1), \\ \lambda_u^{\mathrm{red}} &= \lambda_u + r_1\lambda_v, & \lambda_w^{\mathrm{red}} &= \lambda_w + (1 - r_1)\lambda_v, \end{aligned}$$

the reduced graph Γ^{red} *is Laman and the kernel satisfies*

$$\mathcal{I}_\Gamma(\lambda; z) = \int_0^1 dr_1 \int_0^1 dr_2\, e^{-\lambda_{v,+\dot{a}}(r_1 z_{vu} + (1-r_1) z_{vw})^{+\dot{a}}} \times \underbrace{r_2 \lambda_{v,+\dot{a}} \frac{\partial}{\partial z^{\mathrm{red},+}_{uw,\dot{a}}}}_{\text{local current Jacobian}} \mathcal{I}_{\Gamma^{\mathrm{red}}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}}), \tag{5.10}$$

where the derivative acts before substituting

$$z^{\mathrm{red}}_{uw} = r_2 z_{uw} + (1-r_2)(z_{vw} - z_{vu}), \qquad z^{\mathrm{red}}_e = z_e \quad (e \neq uw).$$

Proof. Inverting the resistance ratios gives

$$t_{vu} = \frac{(1-r_1) t^{\mathrm{red}}_{uw}}{1-r_2}, \quad t_{vw} = \frac{r_1 t^{\mathrm{red}}_{uw}}{1-r_2}, \quad t_{uw} = \frac{t^{\mathrm{red}}_{uw}}{r_2}, \quad d\Big(\frac{t_{vu}}{t_{uw}}\Big) \wedge d\Big(\frac{t_{vw}}{t_{uw}}\Big) = -\frac{r_2}{(1-r_2)^3}\, dr_1 \wedge dr_2,$$

and the original edge currents are reconstructed from y^{red}_{uw} via (5.7)–(5.9):

$$y_{uw} = r_2 y^{\mathrm{red}}_{uw}, \qquad y_{vu} = r_1 \lambda_v - (1-r_2) y^{\mathrm{red}}_{uw}, \qquad y_{vw} = \lambda_v - y_{vu}, \tag{5.11}$$

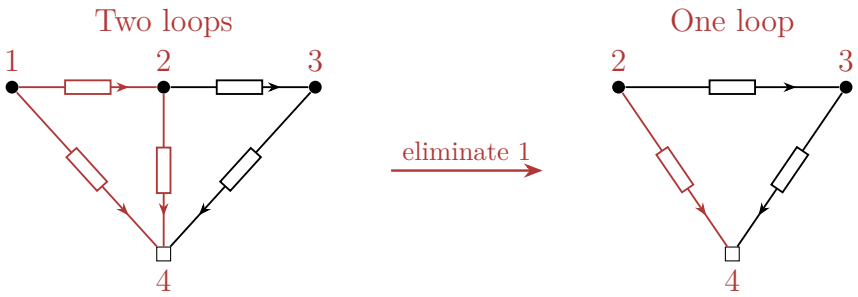
so $\sum_{e\in\Gamma_1} y_{e,+\dot{a}} z_e^{+\dot{a}} = \lambda_{v,+\dot{a}}(r_1 z_{vu} + (1-r_1) z_{vw})^{+\dot{a}} + \sum_{e\in\Gamma_1^{\mathrm{red}}} y^{\mathrm{red}}_{e,+\dot{a}} z^{\mathrm{red},+\dot{a}}_e$. Choosing a spanning tree of Γ^{red} containing uw (extended to Γ by vw), current conservation at fixed λ makes $d(y^{\mathrm{red}}_{uw,+\dot{a}} - r_1 \lambda_{v,+\dot{a}})$ a linear combination of the retained chord-current differentials. Working modulo those retained chord differentials ($\equiv$),

$$\begin{aligned} dy_{vu,+\dot{a}} &= r_2 \lambda_{v,+\dot{a}} dr_1 + y^{\mathrm{red}}_{uw,+\dot{a}} dr_2 - (1-r_2) d\Big(y^{\mathrm{red}}_{uw,+\dot{a}} - r_1 \lambda_{v,+\dot{a}}\Big), \\ -dy_{vu,+\dot{+}} \wedge dy_{vu,+\dot{-}} &\equiv -r_2 \left(\lambda_{v,+\dot{a}} y^{\mathrm{red}}_{uw,+}{}^{\dot{a}}\right) dr_1 \wedge dr_2. \end{aligned}$$

This minus sign cancels the orientation sign from $d(t_{vu}/t_{uw}) \wedge d(t_{vw}/t_{uw})$, while $\lambda_{v,+\dot{a}} \partial/\partial z^{\mathrm{red},+}_{uw,\dot{a}}$ brings down $(\lambda_{v,+\dot{a}} y^{\mathrm{red}}_{uw,+}{}^{\dot{a}})$ from $e^{-\sum_{e\in\Gamma_1^{\mathrm{red}}} y^{\mathrm{red}}_{e,+\dot{b}} z^{\mathrm{red},+\dot{b}}_e}$, yielding (5.10). □

5.3 One elimination step on the fan

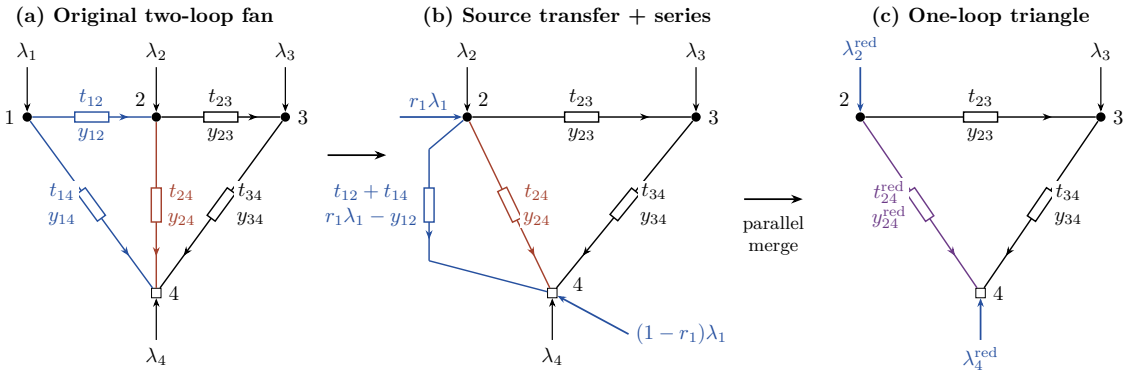
The n-loop fan of Figure 4.1 (with path $1 \to 2 \to \cdots \to n+1$, spokes oriented towards the insertion $n+2$, and $n \geq 2$) is reduced recursively from the left: each step applies Theorem 5.1 to eliminate the leftmost path vertex (Figures 5.3 and 5.4).



$$\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda;z) = \int_0^1 dr_1 \int_0^1 dr_2\; r_2\, e^{-\lambda_{1,+\dot{a}}[r_1 z_{12}+(1-r_1)z_{14}]^{+\dot{a}}} \times \lambda_{1,+}^{\dot{b}} \frac{\partial}{\partial z_{24}^{\mathrm{red},+\dot{b}}} \mathcal{I}_{\triangle}^{\text{raw cut}}\Big(\lambda_2 + r_1\lambda_1, \lambda_3, \lambda_4 + (1-r_1)\lambda_1; z^{\mathrm{red}}\Big).$$

$$z_{24}^{\mathrm{red}} = r_2 z_{24} + (1-r_2)(z_{14}-z_{12}), \qquad z_{23}^{\mathrm{red}} = z_{23}, \quad z_{34}^{\mathrm{red}} = z_{34}.$$

Figure 5.3: The two-loop fan on vertices $1,2,3,4$ reduces to the triangle on $2,3,4$ (Proposition 5.2); vertex 4 is the insertion. The boxed formula integrates the one-loop kernel over r_1, r_2.



$$r_1 = \frac{t_{14}}{t_{12}+t_{14}}, \qquad r_2 = \frac{t_{12}+t_{14}}{t_{12}+t_{14}+t_{24}}, \qquad \lambda_2^{\mathrm{red}} = \lambda_2 + r_1\lambda_1, \quad \lambda_4^{\mathrm{red}} = \lambda_4 + (1-r_1)\lambda_1.$$

$$-[y_{12} - r_1\lambda_1] = y_{14} - (1-r_1)\lambda_1 = (1-r_2)y_{24}^{\mathrm{red}}, \qquad y_{24} = r_2 y_{24}^{\mathrm{red}}.$$

$$t_{24}^{\mathrm{red}} = \left[\frac{1}{t_{12}+t_{14}} + \frac{1}{t_{24}}\right]^{-1}, \qquad y_{24}^{\mathrm{red}} = \underbrace{r_1\lambda_1 - y_{12}}_{\text{series branch}} + \underbrace{y_{24}}_{\text{original spoke}} = \lambda_2 + r_1\lambda_1 - y_{23}.$$

$$y_{12} = r_1\lambda_1 - (1-r_2)y_{24}^{\mathrm{red}}, \qquad y_{14} = \lambda_1 - y_{12}, \qquad y_{24} = r_2 y_{24}^{\mathrm{red}}.$$

Figure 5.4: Two-loop series–parallel reduction, supplementing Figure 5.3. Blue source arrows inject $r_1\lambda_1$ at 2 and $(1-r_1)\lambda_1$ at insertion 4; the series branch carries $r_1\lambda_1 - y_{12}$. The parallel merge gives the triangle on $2,3,4$; the formulas beneath the circuits reconstruct the original currents.

Proposition 5.2 (One elimination step)**.** *Specializing Theorem 5.1 to* $(v,u,w) = (1,2,n+2)$*, set*

$$r_1 = \frac{t_{1,n+2}}{t_{12}+t_{1,n+2}}, \qquad r_2 = \frac{t_{12}+t_{1,n+2}}{t_{12}+t_{1,n+2}+t_{2,n+2}},$$

so r_1 *is the fraction of* λ_1 *passed to vertex* 2 *and* r_2 *is the fraction of the merged spoke current*

carried by the original spoke $(2, n+2)$*, with*

$$t^{\mathrm{red}}_{2,n+2} = \left[\frac{1}{t_{12}+t_{1,n+2}} + \frac{1}{t_{2,n+2}}\right]^{-1}, \qquad \lambda^{\mathrm{red}}_2 = \lambda_2 + r_1\lambda_1, \qquad \lambda^{\mathrm{red}}_{n+2} = \lambda_{n+2} + (1-r_1)\lambda_1.$$

The removed currents follow from $y^{\mathrm{red}}_{2,n+2} = \lambda_2 + r_1\lambda_1 - y_{23}$ *via* (5.11)*:*

$$y_{2,n+2} = r_2 y^{\mathrm{red}}_{2,n+2}, \qquad y_{12} = r_1\lambda_1 - (1-r_2)y^{\mathrm{red}}_{2,n+2} = r_1 r_2\lambda_1 - (1-r_2)\lambda_2 + (1-r_2)y_{23}, \qquad y_{1,n+2} = \lambda_1 - y_{12}.$$

With Γ^{red} *the reduced* $(n-1)$*-loop fan on vertices* $2,\dots,n+2$*, the kernel satisfies*

$$\mathcal{I}_\Gamma(\lambda; z) = \int_0^1 dr_1 \int_0^1 dr_2 \; e^{-\lambda_{1,+\dot a}(r_1 z_{12} + (1-r_1) z_{1,n+2})^{+\dot a}} \times \underbrace{r_2 \lambda_{1,+}{}^{\dot b} \frac{\partial}{\partial z^{\mathrm{red},+\dot b}_{2,n+2}}}_{\substack{\text{inserts the current-area Jacobian} \\ r_2(\lambda_{1,+\dot b} y^{\mathrm{red}}_{2,n+2,+}{}^{\dot b})}} \mathcal{I}_{\Gamma^{\mathrm{red}}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}}), \tag{5.12}$$

where the derivative acts before substituting $z^{\mathrm{red}}_{2,n+2} = r_2 z_{2,n+2} + (1-r_2)(z_{1,n+2} - z_{12})$ *and* $z^{\mathrm{red}}_e = z_e$ *for* $e \neq (2, n+2)$*.*

Proof. Immediate from Theorem 5.1 with $(v,u,w) = (1,2,n+2)$ and $\lambda_{1,+}{}^{\dot b} y^{\mathrm{red}}_{2,n+2,+\dot b} = -\lambda_{1,+\dot b} y^{\mathrm{red}}_{2,n+2,+}{}^{\dot b}$ balanced by $\partial/\partial z^{\mathrm{red},+\dot b}_{2,n+2} = -\partial/\partial z^{\mathrm{red},+}_{2,n+2,\dot b}$. □

5.4 Final result at arbitrary loop order

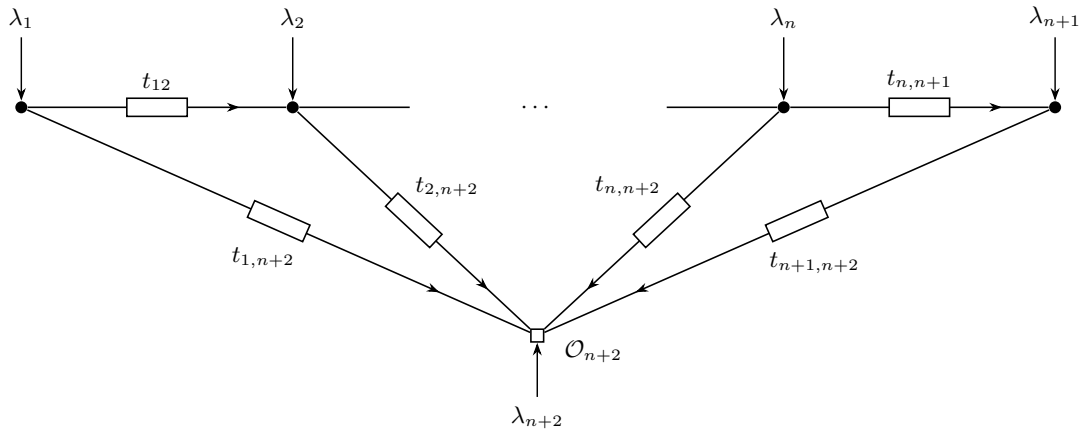


Figure 5.5: The n-loop fan circuit: a path of $n+1$ cubic interactions joined to the insertion. The resistors have resistances t_{vw}, and the external arrows carry the currents λ_v. The short stubs and central ellipsis indicate omitted intermediate vertices.

We keep the intermediate sources $\lambda_2,\dots,\lambda_n$ arbitrary during the reduction; in a renormalizable theory they vanish in Q_n because all three fields at interior path vertices are contracted, leaving only λ_1 and λ_{n+1} (and $\lambda_{n+2} = -\sum_{v<n+2}\lambda_v$), while in non-renormalizable theories at most one interior $\bar W$ vertex carries a source (Propositions 4.9 and 4.13).

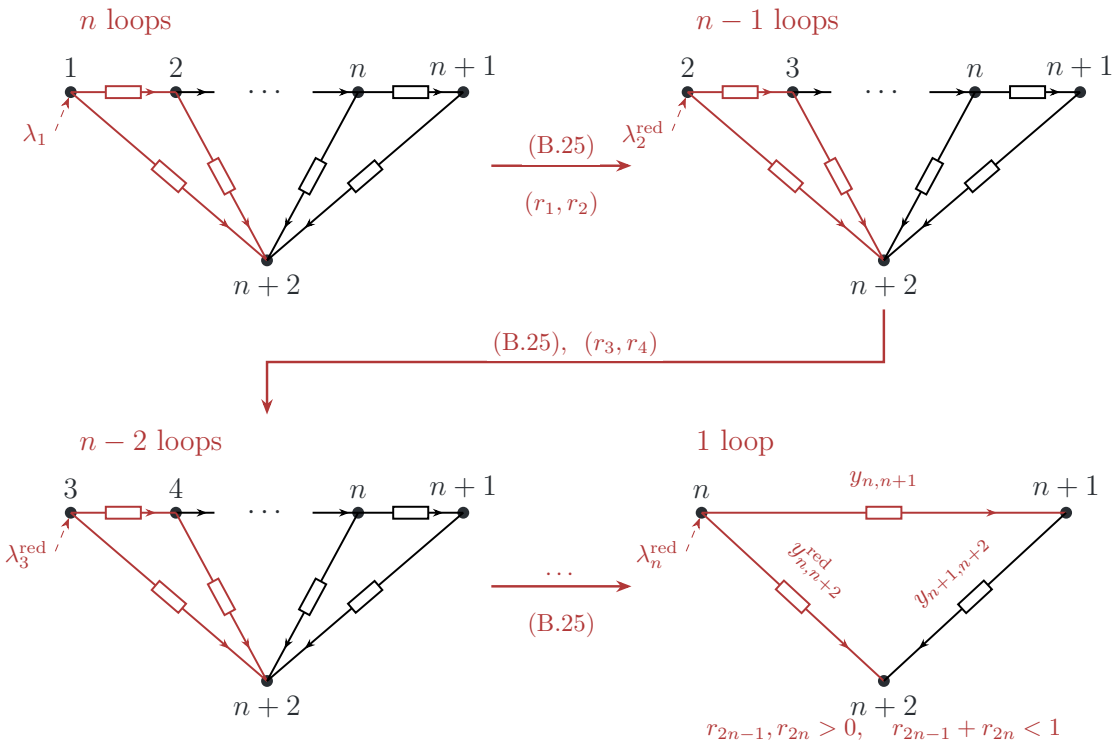


Figure 5.6: Iterating (5.10) removes vertices $1, 2, \ldots, n-1$ in succession, leaving the one-loop triangle on $n, n+1, n+2$. Source transfer proceeds left to right; operator evaluation and current reconstruction proceed backwards from the final triangle.

For $n \geq 2$, iterating Proposition 5.2 at vertices $i = 1, 2, \ldots, n-1$ (Figure 5.6) expresses the n-loop fan kernel as an ordered composition of $n-1$ integral–differential operators acting on the final one-loop triangle kernel (5.3) on vertices $(n, n+1, n+2)$:

$$\mathcal{I}_\Gamma(\lambda; z) = \left[\prod_{i=1}^{n-1} \left(\int_0^1 dr_{2i-1} \int_0^1 dr_{2i}\, e^{-\lambda^{\mathrm{red}}_{i,+\dot{a}}\left[r_{2i-1} z_{i,i+1} + (1-r_{2i-1}) z^{\mathrm{red}}_{i,n+2}\right]^{+\dot{a}}} r_{2i} \lambda^{\mathrm{red}\,\dot{b}}_{i,+} \frac{\partial}{\partial z^{\mathrm{red},+\dot{b}}_{i+1,n+2}} \right)\right] \times \mathcal{I}_\triangle(\lambda^{\mathrm{red}}; z^{\mathrm{red}}),$$

over the $2n$-dimensional domain

$$0 < r_j < 1 \quad (1 \leq j \leq 2n-2), \qquad r_{2n-1} > 0, \quad r_{2n} > 0, \quad r_{2n-1} + r_{2n} < 1. \tag{5.13}$$

Here $i = 1$ acts outermost, each derivative acts on everything to its right before applying the displacement substitution $z^{\mathrm{red}}_{i+1,n+2} = r_{2i} z_{i+1,n+2} + (1 - r_{2i})(z^{\mathrm{red}}_{i,n+2} - z_{i,i+1})$, and the accumulated source entering vertex i obeys

$$\lambda^{\mathrm{red}}_1 = \lambda_1, \qquad \lambda^{\mathrm{red}}_i = \lambda_i + r_{2i-3} \lambda^{\mathrm{red}}_{i-1} = \sum_{\substack{1 \leq j \leq i \\ \text{vertices}}} \left(\prod_{k=j}^{i-1} r_{2k-1} \right) \lambda_j \qquad (1 \leq i \leq n). \tag{5.14}$$

Each r_{2i-1} $(1 \leq i < n)$ is the fraction of λ^{red}_i passed along the path to $i+1$, each r_{2i} is the fraction of the merged spoke current carried by the original spoke $(i+1, n+2)$, and (r_{2n-1}, r_{2n}) are the simplex coordinates of the final triangle:

$$\mathcal{I}_\triangle(\lambda^{\mathrm{red}}; z^{\mathrm{red}}) = (\lambda^{\mathrm{red}}_{n,+\dot{a}} \lambda_{n+1,+}{}^{\dot{a}}) \int_0^1 dr_{2n-1} \int_0^{1-r_{2n-1}} dr_{2n}\, e^{-\sum_{e \in \triangle_1} y^{\mathrm{red}}_{e,+\dot{b}} z_e^{\mathrm{red},+\dot{b}}}, \tag{5.15}$$

with $y_{n,n+1,+\dot{a}} = r_{2n-1}\lambda^{\mathrm{red}}_{n,+\dot{a}} - r_{2n}\lambda_{n+1,+\dot{a}}$.

For $n \geq 3$, note how the $i = 1$ derivative acts across the $i = 2$ operator via the Leibniz rule:

$$r_2\lambda_{1,+}{}^{\dot{a}}\frac{\partial}{\partial z^{\mathrm{red},+\dot{a}}_{2,n+2}}\Big(e^{-\lambda^{\mathrm{red}}_{2,+\dot{b}}(r_3 z_{23}+(1-r_3)z^{\mathrm{red}}_{2,n+2})^{+\dot{b}}}\mathcal{F}\Big) = e^{-\lambda^{\mathrm{red}}_{2,+\dot{b}}(r_3 z_{23}+(1-r_3)z^{\mathrm{red}}_{2,n+2})^{+\dot{b}}} \times r_2\left[(1-r_3)\lambda_{1,+\dot{a}}\lambda^{\mathrm{red}\,\dot{a}}_{2,+} + (1-r_4)\lambda_{1,+}{}^{\dot{a}}\frac{\partial}{\partial z^{\mathrm{red},+\dot{a}}_{3,n+2}}\right]\mathcal{F}. \tag{5.16}$$

Evaluating the operators from the innermost ($i = n-1$) outward to $i = 1$ avoids nested chain rules completely:

Outward evaluation of each operator. Suppose the inner operators $n-1,\dots,i+1$ have already been evaluated on (5.15), producing a prefactor independent of the remaining displacements times $e^{-\sum_{e\in\Gamma_1^{\mathrm{red}}} y^{\mathrm{red}}_{e,+\dot{b}} z_e^{\mathrm{red},+\dot{b}}}$ on the $(n-i)$-loop fan. Applying the i-th operator gives

$$r_{2i}\lambda^{\mathrm{red}\,\dot{a}}_{i,+}\frac{\partial}{\partial z^{\mathrm{red},+\dot{a}}_{i+1,n+2}}e^{-\sum_{e\in\Gamma_1^{\mathrm{red}}} y^{\mathrm{red}}_{e,+\dot{b}} z_e^{\mathrm{red},+\dot{b}}} = r_{2i}(\lambda^{\mathrm{red}}_{i,+\dot{a}} y^{\mathrm{red}}_{i+1,n+2,+}{}^{\dot{a}})e^{-\sum_{e\in\Gamma_1^{\mathrm{red}}} y^{\mathrm{red}}_{e,+\dot{b}} z_e^{\mathrm{red},+\dot{b}}}. \tag{5.17}$$

Multiplying by $e^{-\lambda^{\mathrm{red}}_{i,+\dot{a}}[r_{2i-1}z_{i,i+1}+(1-r_{2i-1})z^{\mathrm{red}}_{i,n+2}]^{+\dot{a}}}$ and substituting $z^{\mathrm{red}}_{i+1,n+2} = r_{2i}z_{i+1,n+2} + (1-r_{2i})(z^{\mathrm{red}}_{i,n+2} - z_{i,i+1})$ converts the exponent into the exact current sum at stage $i-1$, because

$$\begin{aligned}
&\lambda^{\mathrm{red}}_{i,+\dot{a}}[r_{2i-1}z_{i,i+1} + (1-r_{2i-1})z^{\mathrm{red}}_{i,n+2}]^{+\dot{a}} + y^{\mathrm{red}}_{i+1,n+2,+\dot{a}}[r_{2i}z_{i+1,n+2} + (1-r_{2i})(z^{\mathrm{red}}_{i,n+2} - z_{i,i+1})]^{+\dot{a}} \\
&\quad = \big[r_{2i-1}\lambda^{\mathrm{red}}_{i,+\dot{a}} - (1-r_{2i})y^{\mathrm{red}}_{i+1,n+2,+\dot{a}}\big]z^{+\dot{a}}_{i,i+1} + \big[(1-r_{2i-1})\lambda^{\mathrm{red}}_{i,+\dot{a}} + (1-r_{2i})y^{\mathrm{red}}_{i+1,n+2,+\dot{a}}\big]z^{\mathrm{red},+\dot{a}}_{i,n+2} \\
&\qquad + r_{2i}y^{\mathrm{red}}_{i+1,n+2,+\dot{a}}z^{+\dot{a}}_{i+1,n+2}.
\end{aligned}$$

Meanwhile, current conservation at vertex $i+1$ in the reduced circuit gives

$$y^{\mathrm{red}}_{i+1,n+2,+\dot{a}} = \lambda_{i+1,+\dot{a}} + r_{2i-1}\lambda^{\mathrm{red}}_{i,+\dot{a}} - y_{i+1,i+2,+\dot{a}}, \qquad r_{2i}\lambda^{\mathrm{red}}_{i,+\dot{a}}y^{\mathrm{red}}_{i+1,n+2,+}{}^{\dot{a}} = r_{2i}\lambda^{\mathrm{red}}_{i,+\dot{a}}(\lambda_{i+1,+}{}^{\dot{a}} - y_{i+1,i+2,+}{}^{\dot{a}}), \tag{5.18}$$

which is independent of displacements, allowing the induction to proceed to $i-1$. □

Combining (5.16) and the outward induction (5.17)–(5.18) yields the complete reduction chain

(for $n=2$, the first reduced kernel is already the triangle (5.3), so the second step is omitted):

$$
\begin{aligned}
&\mathcal{I}_\Gamma(\lambda;z)\\
&= \int_0^1 dr_1 \int_0^1 dr_2\; e^{-\lambda_{1,+\dot a}(r_1 z_{12}+(1-r_1)z_{1,n+2})^{+\dot a}} \underbrace{r_2 \lambda_{1,+}{}^{\dot b} \frac{\partial}{\partial z^{\mathrm{red},+\dot b}_{2,n+2}}}_{\substack{\text{inserts the current-area Jacobian}\\ r_2(\lambda_{1,+\dot b} y^{\mathrm{red}}_{2,n+2,+}{}^{\dot b})}} \underbrace{\mathcal{I}_{\Gamma^{\mathrm{red}}}(\lambda^{\mathrm{red}};z^{\mathrm{red}})}_{(n-1)\text{-loop kernel}}\\
&= \int_0^1 dr_1 \int_0^1 dr_2 \int_0^1 dr_3 \int_0^1 dr_4\; e^{-\lambda_{1,+\dot a}(r_1 z_{12}+(1-r_1)z_{1,n+2})^{+\dot a}}\, e^{-\lambda^{\mathrm{red}}_{2,+\dot a}(r_3 z_{23}+(1-r_3)z^{\mathrm{red}}_{2,n+2})^{+\dot a}}\\
&\quad\times \underbrace{r_2\left[(1-r_3)\lambda_{1,+\dot a}\lambda^{\mathrm{red}\,\dot a}_{2,+} + (1-r_4)\lambda_{1,+}{}^{\dot b}\frac{\partial}{\partial z^{\mathrm{red},+\dot b}_{3,n+2}}\right]}_{\substack{\text{inserts the first current-area Jacobian}\\ r_2(\lambda_{1,+\dot b} y^{\mathrm{red}}_{2,n+2,+}{}^{\dot b})}} \underbrace{r_4 \lambda^{\mathrm{red}\,\dot c}_{2,+}\frac{\partial}{\partial z^{\mathrm{red},+\dot c}_{3,n+2}}}_{\substack{\text{inserts the current-area Jacobian}\\ r_4(\lambda^{\mathrm{red}}_{2,+\dot c} y^{\mathrm{red}}_{3,n+2,+}{}^{\dot c})}} \underbrace{\mathcal{I}_{(\Gamma^{\mathrm{red}})^{\mathrm{red}}}\begin{pmatrix}\lambda_3+r_3\lambda_2^{\mathrm{red}},\lambda_4,\dots,\lambda\\ \lambda_{n+2}+(1-r_1)\lambda_1\\ +(1-r_3)\lambda_2^{\mathrm{red}};z^{\mathrm{red}}\end{pmatrix}}_{(n-2)\text{-loop kernel}}\\
&= \cdots\\
&= \int_{(5.13)} d^{2n}r \underbrace{\prod_{i=1}^{n-1} r_{2i}\left[\sum_{j=1}^{i}\left(\prod_{k=j}^{i-1} r_{2k-1}\right)\lambda_{j,+\dot a_i}\right](\lambda_{i+1,+}{}^{\dot a_i} - y_{i+1,i+2,+}{}^{\dot a_i}(r))}_{\text{kernel factor from eliminating vertex } i}\\
&\quad\times \underbrace{\left[\sum_{j=1}^{n}\left(\prod_{k=j}^{n-1} r_{2k-1}\right)\lambda_{j,+\dot b}\right]\lambda_{n+1,+}{}^{\dot b}}_{\text{final triangle factor}}\, e^{-\sum_e y_{e,+\dot a}(r) z_e^{+\dot a}}.
\end{aligned}
\tag{5.19}
$$

Reading off the coefficients of $z_{i,i+1}$ from the outward restoration reconstructs all path currents backwards from $y_{n,n+1}$ to y_{12}:

$$
\begin{aligned}
y_{n,n+1,+\dot a} &= r_{2n-1}\sum_{j=1}^{n}\left(\prod_{k=j}^{n-1} r_{2k-1}\right)\lambda_{j,+\dot a} - r_{2n}\lambda_{n+1,+\dot a},\\
y_{i,i+1,+\dot a} &= r_{2i-1}r_{2i}\sum_{j=1}^{i}\left(\prod_{k=j}^{i-1} r_{2k-1}\right)\lambda_{j,+\dot a} - (1-r_{2i})\lambda_{i+1,+\dot a} + (1-r_{2i})y_{i+1,i+2,+\dot a}, \qquad 1\le i<n,
\end{aligned}
\tag{5.20}
$$

and current conservation at each path vertex determines every spoke current:

$$
\begin{aligned}
y_{1,n+2,+\dot a} &= \lambda_{1,+\dot a} - y_{12,+\dot a},\\
y_{i,n+2,+\dot a} &= \lambda_{i,+\dot a} + y_{i-1,i,+\dot a} - y_{i,i+1,+\dot a}, \qquad 2\le i\le n,\\
y_{n+1,n+2,+\dot a} &= \lambda_{n+1,+\dot a} + y_{n,n+1,+\dot a}.
\end{aligned}
\tag{5.21}
$$

All edge currents are polynomial in $r_1,\dots,r_{2n}$, free of the spanning-tree denominators of (5.5), so every Taylor coefficient is an elementary polynomial integral over (5.13).

5.5 Endpoint sources and final Taylor expansion

In renormalizable theories, $\lambda_2=\cdots=\lambda_n=0$ in Q_n, and derivatives of input letters in $\mathcal{O}$ sit solely on the spokes: $z_{i,i+1}=0$ and $z_{i,n+2}=-z_i$. Substituting (5.20) into (5.19) factorizes the

integration measure:

$$\begin{aligned}
&\mathcal{I}_\Gamma(\lambda;z)\big|_{\lambda_2=\cdots=\lambda_n=0,\ z_{i,i+1}=0,\ z_{i,n+2}=-z_i}\\
&=(\lambda_{1,+\dot a}\lambda_{n+1,+}{}^{\dot a})^n\int_{(5.13)}d^{2n}r\,\left[\prod_{j=1}^{n-1}r_{2j-1}^{n-j}r_{2j}(1-r_{2j})^{j-1}\right]r_{2n}^{n-1}e^{\sum_j y_{j,n+2,+\dot a}z_j^{+\dot a}}.
\end{aligned}\tag{5.22}$$

The exponential in (5.22) translates the two endpoint output fields according to

$$\sum_{j=1}^{n+1}y_{j,n+2,+\dot a}(r)z_j^{+\dot a}=\lambda_{1,+\dot a}x_1^{+\dot a}(r,z)+\lambda_{n+1,+\dot a}x_{n+1}^{+\dot a}(r,z),\tag{5.23}$$

with translated positions

$$\begin{aligned}
x_1^{+\dot a}(r,z)=&\sum_{j=1}^{n}(1-r_{2j-1})\left(\prod_{k=1}^{j-1}r_{2k-1}\right)\\
&\times\left[\left(\prod_{k=1}^{j-1}(1-r_{2k})\right)z_1^{+\dot a}+\sum_{i=2}^{j}r_{2i-2}\left(\prod_{k=i}^{j-1}(1-r_{2k})\right)z_i^{+\dot a}\right]\\
&+\left(\prod_{k=1}^{n}r_{2k-1}\right)z_{n+1}^{+\dot a},\\
x_{n+1}^{+\dot a}(r,z)=&\,r_{2n}\left[\left(\prod_{k=1}^{n-1}(1-r_{2k})\right)z_1^{+\dot a}+\sum_{i=2}^{n}r_{2i-2}\left(\prod_{k=i}^{n-1}(1-r_{2k})\right)z_i^{+\dot a}\right]\\
&+(1-r_{2n})z_{n+1}^{+\dot a}.
\end{aligned}\tag{5.24}$$

Using the elementary moments on (5.13),

$$\int_{(5.13)}dr_1\cdots dr_{2n}\prod_{j=1}^{2n}r_j^{m_j}=\frac{m_{2n-1}!\,m_{2n}!}{(m_{2n-1}+m_{2n}+2)!}\prod_{j=1}^{2n-2}\frac{1}{m_j+1},\tag{5.25}$$

and grouping by total displacement degree m, we obtain the complete Taylor expansion for arbitrary edge displacements:

$$\begin{aligned}
&\mathcal{I}_\Gamma(\lambda;z)\big|_{\lambda_2=\cdots=\lambda_n=0}\\
&=(\lambda_{1,+\dot a}\lambda_{n+1,+}{}^{\dot a})^n\sum_{m=0}^{\infty}\sum_{\substack{r_0,s_0,r_{ik},s_i\geq 0\\ r_0+s_0+\sum_{i=1}^n s_i+\sum_{k=1}^n\sum_{i=1}^k r_{ik}=m}}\\
&\quad\times\frac{[-\lambda_{1,+\dot b}z_{1,n+2}^{+\dot b}]^{r_0}[-\lambda_{n+1,+\dot c}z_{n+1,n+2}^{+\dot c}]^{s_0}}{r_0!s_0!}\\
&\quad\times\prod_{1\leq i\leq k\leq n}\frac{\left[-\lambda_{1,+\dot d}(z_{i,i+1}+z_{i+1,n+2}-z_{i,n+2})^{+\dot d}\right]^{r_{ik}}}{r_{ik}!}\\
&\quad\times\prod_{i=1}^{n}\frac{[\lambda_{n+1,+\dot e}(z_{i,i+1}+z_{i+1,n+2}-z_{i,n+2})^{+\dot e}]^{s_i}}{s_i!}\\
&\quad\times\frac{(\sum_{i=1}^n r_{in})!\,(n-1+\sum_{i=1}^n s_i)!}{(n+1+\sum_{i=1}^n r_{in}+\sum_{i=1}^n s_i)!}\\
&\quad\times\prod_{j=1}^{n-1}\frac{1}{n-j+1+\sum_{k=j}^n\sum_{i=1}^k r_{ik}}\\
&\quad\times\prod_{j=1}^{n-1}\frac{\left(1+\sum_{i=1}^j r_{ij}\right)!\left(j-1+\sum_{i=1}^j\left[s_i+\sum_{k=j+1}^n r_{ik}\right]\right)!}{\left(j+1+\sum_{i=1}^j\left[s_i+\sum_{k=j}^n r_{ik}\right]\right)!}.
\end{aligned}\tag{5.26}$$

At zero displacement ($m = 0$), (5.26) evaluates to

$$\mathcal{I}_\Gamma(\lambda;0)|_{\lambda_{2,+\dot a}=\cdots=\lambda_{n,+\dot a}=0} = \frac{(\lambda_{1,+\dot a}\lambda_{n+1,+}{}^{\dot a})^n}{(n!)^3(n+1)}. \tag{5.27}$$

For $n = 1, 2, 3$, (5.27) agrees with the operatope values [9, Equations (4.20), (4.44), (4.67)] (and public code [18]), [4, Equation (2.48)], and [13, Sections 4.4–4.6].

6 Quantum multiplication and the Leibniz rule

As we have seen, the differential Q_1 contains two input derivatives [4].

$$\begin{aligned} Q_1 = &-\frac{1}{64\pi^2}\int_0^1 dr_2 \int_0^{1-r_2} dr_1 \int d^3\theta \int d^4z_1\, d^4z_2\, C_{ARD}C_{BRE} \\ &\times (\partial_{+\dot a}C^D)(x+(1-r_1)z_1+r_1z_2,\theta)\,(\partial_+{}^{\dot a}C^E)(x+r_2z_1+(1-r_2)z_2,\theta) \\ &\times \frac{\delta}{\delta C^A(x+z_1,\theta)}\frac{\delta}{\delta C^B(x+z_2,\theta)}. \end{aligned} \tag{6.1}$$

Its two functional derivatives contract two input letters, which can belong to different factors (equation (6.2) below).

Acting on a product of two composite operators $A \times B$, these derivatives can land on different factors, producing a cross-contraction that explicitly violates the Leibniz rule. For example, in the zero-input-jet part of $\mathcal{N} = 4$ SYM, the Leibniz rule defect reads:

$$\begin{aligned} &Q_1(AB) - (Q_1A)B - (-1)^{|A|}A(Q_1B) \\ &\qquad = -\frac{1}{64\pi^2}\int d^3\theta\, C_{XRD}C_{YRE}(\partial_{+\dot a}C^D)(\partial_+{}^{\dot a}C^E)\frac{\delta A}{\delta C^X}\frac{\delta B}{\delta C^Y}. \end{aligned} \tag{6.2}$$

Here $C(z^{+\dot a}, \theta_i)$ is the superfield of Appendix A.1.

Even when both A and B are annihilated by Q_0 and Q_1 (so that they remain quantum cocycles), this cross-contraction can be nonzero; in fact, it is generically nonzero.

This raises an apparent tension with the ring structure of supercharge cohomology: supersymmetry acts as a graded derivation on separated operator products, the product of two closed operators should again define a closed class, and the Q-cohomology, or BPS ring, is expected to have a ring structure.

This raises a question: given two classes $[A]$ and $[B]$ of Q-closed operators, we expect a class, denoted $[AB]$, with the physical meaning of the product of A and B.

The failure of the Leibniz rule for Q_1, Q_2, ...[18] shows that this is not trivial: at the least, $[AB]$ cannot be the naive Wick product of A and B.

The quantum multiplication, defined in previous work [6, Sections 4.3 and 4.5], resolves this tension at every perturbative order [3, footnote 28]: $\mathbf{Q}$ obeys the Leibniz rule for the quantum "normal-ordering product".

The definition of the quantum product also uses the higher brackets, together with two auxiliary free fields.

The quadratic identities of the higher brackets, together with the quantum master equation, give the corrected Leibniz identity (6.13). Thus $\mathbf{Q}A = \mathbf{Q}B = 0$ implies $\mathbf{Q}(A \star B) = 0$. We first prove this identity and, from the marked quadratic identities with three insertions, the ring laws

[18] The higher corrections Q_n likewise act on $n+1$ input fields and produce contractions shared between at most $n+1$ factors, violating the ordinary Leibniz rule more severely.

on cohomology; we then give the graph formula and, in Section 7, the way to calculate these graphs.

For the enumeration of $\mathbf{Q}$-cohomology representatives, the quantum product usually need not be computed explicitly. It is an explicit construction showing that the product of two quantum cocycles is a quantum cocycle: for quantum cocycles A and B, $A \star B$ is a quantum cocycle completing $A_0 B_0$,[19] and it is lifted exactly when it becomes a quantum coboundary. In the successive homotopy transfer of Section 9.3, which is the algorithm used for the lifting tests, the quantum product induces the ordinary product of leading representatives on every page (Theorem 6.4).

6.1 The Leibniz rule of the quantum supercharge

Here $\{\cdot\}_0$ denotes the brackets of Section 3.7. The quantum differential and the interacting bracket of two operators are

$$\mathbf{Q}\mathcal{O} = \sum_{s\geq 1} \frac{\hbar^{s-1}}{s!}\{\mathcal{I}^{\otimes s}, \mathcal{O}\}_0, \qquad \{X, Y\}_{\mathcal{I}} = \sum_{r\geq 0} \frac{\hbar^r}{r!}\{\mathcal{I}^{\otimes r}, X, Y\}_0. \tag{6.3}$$

Lemma 6.1. *Suppose that the interaction satisfies*

$$\sum_{k\geq 2} \frac{\hbar^{k-2}}{k!}\{\mathcal{I}^{\otimes k}\}_0 = \sum_{n\geq 0} \frac{\hbar^n}{n+2} Q_n \mathcal{I} = 0. \tag{6.4}$$

Then $\mathbf{Q}^2 = 0$; at the first orders,

$$\begin{aligned} Q_0^2 = 0, \qquad & Q_0 Q_1 + Q_1 Q_0 = 0, \\ Q_0 Q_2 + Q_1^2 + Q_2 Q_0 = 0, \quad & Q_0 Q_3 + Q_1 Q_2 + Q_2 Q_1 + Q_3 Q_0 = 0. \end{aligned} \tag{6.5}$$

Proof. The quadratic identity of [6], with N interaction entries and one operator, reads

$$\begin{aligned} 0 = & \sum_{r=1}^{N-1} \binom{N}{r} \{\mathcal{I}^{\otimes(N-r)}, \{\mathcal{I}^{\otimes r}, \mathcal{O}\}_0\}_0 \\ & - \sum_{k=2}^{N} \binom{N}{k} \{\mathcal{I}^{\otimes(N-k)}, \{\mathcal{I}^{\otimes k}\}_0, \mathcal{O}\}_0. \end{aligned} \tag{6.6}$$

Multiplying by $\hbar^{N-2}/N!$ and summing over N gives

$$\begin{aligned} 0 = & \sum_{a,b\geq 0} \frac{\hbar^{a+b}}{(a+1)!\,(b+1)!} \{\mathcal{I}^{\otimes(a+1)}, \{\mathcal{I}^{\otimes(b+1)}, \mathcal{O}\}_0\}_0 \\ & - \sum_{m\geq 0,\ k\geq 2} \frac{\hbar^{m+k-2}}{m!\,k!} \{\mathcal{I}^{\otimes m}, \{\mathcal{I}^{\otimes k}\}_0, \mathcal{O}\}_0. \end{aligned} \tag{6.7}$$

The first sum is $\mathbf{Q}^2\mathcal{O}$ and the second vanishes by (6.4). The first coefficients agree with [4, Equations (1.12) and (2.25)–(2.26)]. □

Theorem 6.2. *Suppose that $\bar{W}$ is gauge invariant and the matter representation obeys*

$$\mathrm{Tr}_R(t_A\{t_B, t_C\}) = 0. \tag{6.8}$$

[19] Here, the completion of a Q_0 cocycle O_0 is a series $O = \sum_{\substack{i\geq 0 \\ \text{loop orders}}} \hbar^i O_i = O_0 + \hbar O_1 + \hbar^2 O_2 + \cdots$ (a finite sum) satisfying $\mathbf{Q}\sum_{\substack{i\geq 0 \\ \text{loop orders}}} \hbar^i O_i = 0$. Not all Q_0 cocycles have a completion; those without one are the "lifted" operators.

For the interaction (3.38), the integrated interaction satisfies

$$\bar\partial\mathcal{I}=0,\qquad Q_0\mathcal{I}=0,\qquad Q_1\mathcal{I}=0,\qquad Q_2\mathcal{I}=Q_3\mathcal{I}=\cdots=0. \tag{6.9}$$

Consequently, $\mathcal{I}$ satisfies (6.4), $(\bar\partial+Q_0+\sum_{n\geq1}\hbar^nQ_n)\mathcal{I}=0$, and $\mathbf{Q}^2=0$ by Lemma 6.1.

Proof. The free differential $\bar\partial$ maps the integrated Dolbeault density of $\mathcal{I}$ to a total derivative. The gauge and matter vertices satisfy the classical master equation by

$$C^A{}_{B[C}C^B{}_{DE]}=0,\qquad [t_A,t_B]=iC^C{}_{AB}t_C.$$

Since $\bar W$ contains no b or β, its self-bracket vanishes, and, up to holomorphic total derivatives,

$$\begin{aligned}Q_0\mathcal{I}=\{\mathcal{I},\mathcal{I}\}_0=&-2ig\mathsf{c}^A(t_A)^j{}_i\gamma_j\frac{\partial\bar W}{\partial\gamma_i}\\&-2g\mathsf{c}^AC^B{}_{AC}\mathsf{w}^C_{\dot a}\frac{\partial_L\bar W}{\partial\mathsf{w}^B_{\dot a}}=0.\end{aligned}$$

The last equality is gauge invariance of $\bar W$.

At one loop, the triangle graphs give

$$Q_1\mathcal{I}=\frac{ig^3}{128\pi^2}\operatorname{Tr}_R(t_A\{t_B,t_C\})\,\mathsf{c}^A(\partial_{+\dot a}\mathsf{c}^B)(\partial_+{}^{\dot a}\mathsf{c}^C)=0. \tag{6.10}$$

A circuit of $Q_n\mathcal{I}$ has $n+2$ interaction nodes and $2n+1$ resistors, and each resistor takes its own antifield terminal (Lemma 4.10): a gauge or matter node has one, a superpotential node none. Thus

$$2n+1\leq n+2-\#(\bar W\text{ nodes}),$$

which is impossible for $n\geq2$. Finally, $Q_n\mathcal{I}=\{\mathcal{I}^{\otimes(n+2)}\}_0/(n+1)!$ gives (6.4). □

Introduce a decoupled auxiliary pair, with action $\int\Omega\wedge[a\bar\partial b_{\mathrm{aux}}]_{(0,2)}$ and $|a|=0$, $|b_{\mathrm{aux}}|=1$. For every $n\geq0$, $Q_na=Q_nb_{\mathrm{aux}}=0$; the free differential is still $Q_{\mathrm{BV}}=\bar\partial$, which vanishes on the holomorphic auxiliary jets. Thus

$$\mathbf{Q}(Aa)=(\mathbf{Q}A)a,\qquad \mathbf{Q}(b_{\mathrm{aux}}B)=-b_{\mathrm{aux}}\mathbf{Q}B.$$

The quantum product is defined by the unique auxiliary contraction; $\gamma=\{(A,B)\}$ marks this edge, and $\mathrm{aux}=0$ sets the remaining auxiliary fields to zero after contraction:

$$\begin{aligned}A\star B&=\sum_{n\geq0}\frac{\hbar^n}{n!}\{\mathcal{I}^{\otimes n},A,B\}^{\gamma,0}=\{Aa,b_{\mathrm{aux}}B\}_{\mathcal{I}}\big|_{\mathrm{aux}=0}\,,\\\{A,B\}^{\gamma,0}&=(-1)^{|A|}\frac{\partial_L}{\partial a}\frac{\partial_L}{\partial b_{\mathrm{aux}}}(Aa\,b_{\mathrm{aux}}B)=AB.\end{aligned} \tag{6.11}$$

We will show later that these auxiliary fields serve precisely to define an integration contour, analogous to normal ordering in 2d CFT.

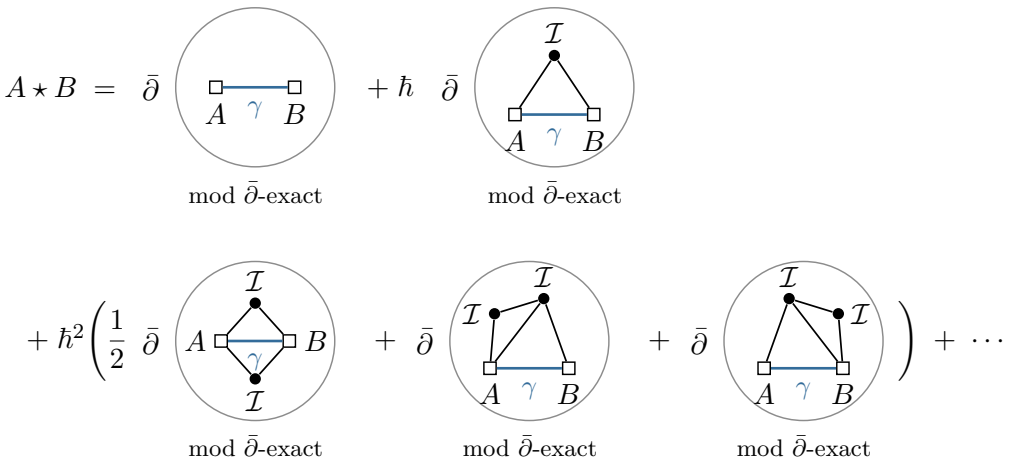


Figure 6.1: The quantum product (6.11) through order $\hbar^2$: the free differential $\bar\partial$ acting on the marked graphs, as in (3.55). Squares are the insertions A and B, dots the interaction vertices $\mathcal{I}$. The blue edge γ is the propagator of the pair a, b_{aux} in (6.26), which acts as the integration contour of (6.41); at order zero $\bar\partial$ turns it into the heat kernel and gives AB (6.11). Black edges contract fields of $\mathcal{I}$, A and B. Each coefficient is $\hbar^n/n!$ times the number of labellings of the interaction vertices.

Theorem 6.3 (Leibniz rule)**.**

Suppose that the interaction satisfies (6.4)*, which is the quantum master equation. (Explicitly, it is the gauge anomaly cancellation condition.) For operators X, Y, with $|X|$ the fermion number of X,*[20]

$$\mathbf{Q}\{X,Y\}_{\mathcal{I}} - \{\mathbf{Q}X, Y\}_{\mathcal{I}} + (-1)^{|X|}\{X, \mathbf{Q}Y\}_{\mathcal{I}} = 0, \tag{6.12}$$

and the quantum product obeys

$$\mathbf{Q}(A \star B) = (\mathbf{Q}A) \star B + (-1)^{|A|} A \star (\mathbf{Q}B). \tag{6.13}$$

In particular, quantum multiplication takes quantum cocycles to quantum cocycles and descends to $\mathbf{Q}$*-cohomology.*

Proof. For $N \geq 1$ interaction entries and the entries X, Y, the quadratic identity of the higher brackets [9, Section 3.2], [5, Theorem 7], [6, Equation (4.19)], in the conventions of (6.6), sorts its terms by the content of the inner bracket:

$$\begin{aligned} 0 = {} & \underbrace{\sum_{r=0}^{N-1} \binom{N}{r} \{\mathcal{I}^{\otimes(N-r)}, \{\mathcal{I}^{\otimes r}, X, Y\}_0\}_0}_{\text{inner bracket contains } X \text{ and } Y} - \underbrace{\sum_{r=1}^{N} \binom{N}{r} \{\mathcal{I}^{\otimes(N-r)}, \{\mathcal{I}^{\otimes r}, X\}_0, Y\}_0}_{\text{inner bracket contains } X} \\ & + (-1)^{|X|} \underbrace{\sum_{r=1}^{N} \binom{N}{r} \{\mathcal{I}^{\otimes(N-r)}, X, \{\mathcal{I}^{\otimes r}, Y\}_0\}_0}_{\text{inner bracket contains } Y} + \underbrace{\sum_{k=2}^{N} \binom{N}{k} \{\mathcal{I}^{\otimes(N-k)}, \{\mathcal{I}^{\otimes k}\}_0, X, Y\}_0}_{\text{inner bracket contains only interactions}} . \end{aligned} \tag{6.14}$$

Multiply by $\hbar^{N-1}/N!$ and sum over $N \geq 1$. Since $\binom{N}{r}/N! = 1/(r!\,(N-r)!)$, each double sum

[20] Equation (6.14) is the quadratic identity of the higher brackets [9, Section 3.2], [5, Theorem 7], [6, Equation (4.19)]: in its first three terms the inner bracket contains both X and Y, only X, and only Y, and the terms whose inner bracket contains interactions alone cancel by (6.4).

factorizes into an outer series in $s = N - r$ and an inner series:

$$\begin{aligned} 0 = {} & \underbrace{\sum_{s\geq 1} \frac{\hbar^{s-1}}{s!}\Big\{\mathcal{I}^{\otimes s}, \sum_{r\geq 0}\frac{\hbar^r}{r!}\{\mathcal{I}^{\otimes r}, X, Y\}_0\Big\}_0}_{\mathbf{Q}\{X,Y\}_{\mathcal{I}}} - \underbrace{\sum_{s\geq 0} \frac{\hbar^{s}}{s!}\Big\{\mathcal{I}^{\otimes s}, \sum_{r\geq 1}\frac{\hbar^{r-1}}{r!}\{\mathcal{I}^{\otimes r}, X\}_0, Y\Big\}_0}_{\{\mathbf{Q}X,Y\}_{\mathcal{I}}} \\ & + (-1)^{|X|}\underbrace{\sum_{s\geq 0} \frac{\hbar^{s}}{s!}\Big\{\mathcal{I}^{\otimes s}, X, \sum_{r\geq 1}\frac{\hbar^{r-1}}{r!}\{\mathcal{I}^{\otimes r}, Y\}_0\Big\}_0}_{\{X,\mathbf{Q}Y\}_{\mathcal{I}}} \\ & + \underbrace{\sum_{s\geq 0} \frac{\hbar^{s+1}}{s!}\Big\{\mathcal{I}^{\otimes s}, \sum_{k\geq 2}\frac{\hbar^{k-2}}{k!}\{\mathcal{I}^{\otimes k}\}_0, X, Y\Big\}_0}_{0 \text{ by (6.4)}}. \end{aligned} \tag{6.15}$$

The inner series are (6.3) and the left side of (6.4); this is (6.12). The Jacobi identity expresses the Leibniz rule for an adjoint action on a bracket, and (6.12) is its higher L_∞ form, with the interaction insertions supplying every perturbative order [6, Sections 4.3–4.5].

Set $X = Aa$ and $Y = b_{\mathrm{aux}}B$ in (6.12), and extract the auxiliary contraction:

$$0 = \mathbf{Q}(A \star B) - (\mathbf{Q}A) \star B - (-1)^{|A|} A \star (\mathbf{Q}B).$$

Consequently, $\mathbf{Q}A = \mathbf{Q}B = 0$ gives $\mathbf{Q}(A \star B) = 0$. For homogeneous X, Y with $|X| = |A| + 1$ and $|Y| = |B| + 1$, the same identity and $\mathbf{Q}^2 = 0$ (Lemma 6.1) give

$$\begin{aligned} &(A + \mathbf{Q}X) \star (B + \mathbf{Q}Y) - A \star B \\ &\qquad = \mathbf{Q}(X \star B + (-1)^{|A|} A \star Y + X \star \mathbf{Q}Y). \end{aligned} \tag{6.16}$$

The resulting cohomology class is therefore independent of representatives. □

6.2 The quantum BPS ring

Write $[\cdot]_{\mathbf{Q}}$ for the class of a quantum cocycle.

Theorem 6.4. *Suppose the interaction satisfies (6.4), the quantum master equation. The quantum product (6.11) induces an associative, graded-commutative, unital multiplication on $H(\mathbf{Q})$. For quantum cocycles A, B, D,*

$$[(A \star B) \star D]_{\mathbf{Q}} = [A \star (B \star D)]_{\mathbf{Q}}, \tag{6.17}$$

$$[A \star B]_{\mathbf{Q}} = (-1)^{|A||B|}[B \star A]_{\mathbf{Q}}. \tag{6.18}$$

The unit is strict:

$$A \star 1 = 1 \star A = A. \tag{6.19}$$

In the successive homotopy transfer of Section 9.3, the quantum product induces on every page E_r the ordinary product of leading representatives, and every d_r is a derivation of it.

Proof. The chain identity (6.13) makes the product well-defined on cohomology. A class of E_r has a representative $a + \hbar a_1 + \cdots$ whose $\mathbf{Q}$-variation starts at order $\hbar^r$ with $d_r a$. Since $A \star B = AB + O(\hbar)$, the product of two such representatives has leading term ab, and by (6.13) its $\mathbf{Q}$-variation starts at order $\hbar^r$ with $(d_r a)b + (-1)^{|a|} a\, d_r b$. For the unit, the identity insertion has no fields, so in a marked circuit with $n \geq 1$ interaction nodes it is wired only by the marked resistor. It is a dead end: the marked resistor carries the current injected at the identity insertion,

whatever its resistance, so this resistance moves no current and the current form vanishes. So only the zero-interaction marked segment contributes.

For three closed insertions X, Y, Z, sum the quadratic identities with all interaction entries, as in [6, Sections 4.3 and 4.5]. The pure-interaction terms vanish by (6.4), and the single-input terms vanish by closure. With the ordered-bracket sign of (3.43), the remaining identity is

$$\begin{aligned} 0 = {} & \{X, \{Y, Z\}_{\mathcal{I}}\}_{\mathcal{I}} - \{\{X, Y\}_{\mathcal{I}}, Z\}_{\mathcal{I}} \\ & - (-1)^{(|X|+1)(|Y|+1)} \{Y, \{X, Z\}_{\mathcal{I}}\}_{\mathcal{I}} \\ & + \mathbf{Q} \sum_{n \geq 0} \frac{\hbar^{n+1}}{n!} \{\mathcal{I}^{\otimes n}, X, Y, Z\}_0. \end{aligned} \tag{6.20}$$

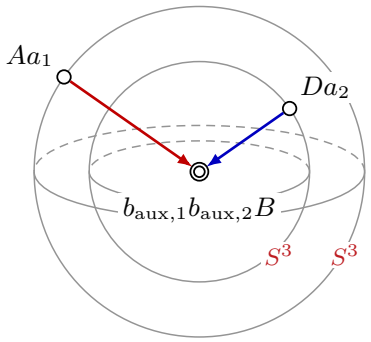
A ternary graph with n interactions has $n+3$ vertices and $2n+3$ edges, hence loop number $n+1$. This fixes the power of $\hbar$ in the last line.

Take two independent copies $(a_1, b_{\mathrm{aux},1})$ and $(a_2, b_{\mathrm{aux},2})$ of the free auxiliary pair defined above. Each a_i is even and each $b_{\mathrm{aux},i}$ is odd. Substitute

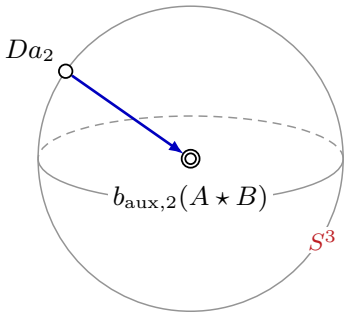
$$X = Aa_1, \qquad Y = Da_2, \qquad Z = b_{\mathrm{aux},1} b_{\mathrm{aux},2} B.$$

The last insertion is the root. This marks the edges AB and DB, as in Figure 6.2.

(a) Both marks end at the root

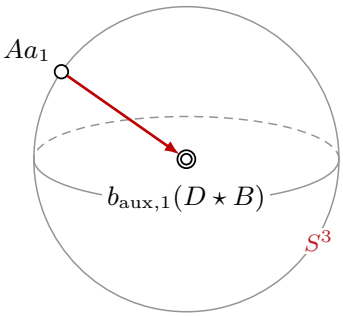


(b) Collapse A, B at B



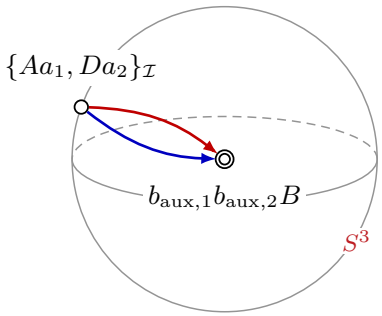
The second auxiliary field stays at the root.

(c) Collapse D, B at B



The first auxiliary field stays at the root.

(d) Collapse A, D



$P \wedge P = 0, \qquad K \wedge P = 0$

Both marks now join the same two vertices.

Figure 6.2: The common-root contraction and its three pair-collapse channels. Red and blue arrows mark the first and second auxiliary pairs; double circles mark the root. Each non-root insertion lies on its own integration sphere S^3 centered at the root, shown schematically in gray. Only auxiliary contractions are shown. Each component may also contain interaction vertices and physical contractions. The labels in (b) and (c) display the surviving fields; their Koszul signs are given in the proof.

The ordered-bracket sign gives

$$\begin{aligned}\{Aa_1, b_{\mathrm{aux},1}b_{\mathrm{aux},2}B\}_{\mathcal{I}}|_{a_1=b_{\mathrm{aux},1}=0} &= (-1)^{|A|}b_{\mathrm{aux},2}(A\star B),\\ \{Da_2, b_{\mathrm{aux},1}b_{\mathrm{aux},2}B\}_{\mathcal{I}}|_{a_2=b_{\mathrm{aux},2}=0} &= (-1)^{|D|+1}b_{\mathrm{aux},1}(D\star B).\end{aligned} \tag{6.21}$$

In the second line the odd derivative passes through $b_{\mathrm{aux},1}$, contributing the extra minus sign.

The third binary channel first collapses A, D. Both marked resistors then join that collapsed node to B: they are in parallel, and their proportional currents kill the measure for every holomorphic derivative (Corollary 4.3, (4.2)). Consequently,

$$\{\{Aa_1, Da_2\}_{\mathcal{I}}, b_{\mathrm{aux},1}b_{\mathrm{aux},2}B\}_{\mathcal{I}}|_{\mathrm{aux}=0} = 0. \tag{6.22}$$

On substituting (6.21) and (6.22) into (6.20), each of its three nested brackets becomes a product of three operators:

$$0 = \underbrace{(-1)^{|D|+1}A\star(D\star B)}_{\{X,\{Y,Z\}_{\mathcal{I}}\}_{\mathcal{I}}} - \underbrace{0}_{\substack{\{\{X,Y\}_{\mathcal{I}},Z\}_{\mathcal{I}}:\\ \text{parallel marks}}} \underbrace{-(-1)^{(|A|+1)(|D|+1)+|A|}D\star(A\star B)}_{-(-1)^{(|X|+1)(|Y|+1)}\{Y,\{X,Z\}_{\mathcal{I}}\}_{\mathcal{I}}} + \underbrace{\mathbf{Q}\,H}_{\mathbf{Q}\text{-exact}}, \tag{6.23}$$

with H the bracket in the last line of (6.20) at these inputs. The middle term vanishes, so only the two bracketings $A\star(D\star B)$ and $D\star(A\star B)$ remain. Multiplying by $(-1)^{|D|+1}$ gives the local homotopy

$$\begin{aligned}&A\star(D\star B) - (-1)^{|A||D|}D\star(A\star B)\\ &\quad= (-1)^{|D|}\mathbf{Q}\left[\sum_{n\geq 0}\frac{\hbar^{n+1}}{n!}\{\mathcal{I}^{\otimes n}, Aa_1, Da_2, b_{\mathrm{aux},1}b_{\mathrm{aux},2}B\}_0\Big|_{\mathrm{aux}=0}\right].\end{aligned} \tag{6.24}$$

Moreover, its n-interaction term has $2n+1$ physical contractions. The interactions supply at most n fields of type b or β, so at least $n+1$ must come from the inputs. The homotopy is therefore a finite sum of local operators for finite polynomial inputs.

Setting $B=1$ proves graded commutativity. For three closed inputs, (6.13), graded commutativity, and (6.24) give

$$\begin{aligned}[(A\star B)\star D]_{\mathbf{Q}} &\underset{(6.18)}{=} (-1)^{(|A|+|B|)|D|}[D\star(A\star B)]_{\mathbf{Q}}\\ &\underset{(6.24)}{=} (-1)^{|B||D|}[A\star(D\star B)]_{\mathbf{Q}}\\ &\underset{(6.18)}{=} [A\star(B\star D)]_{\mathbf{Q}}.\end{aligned} \tag{6.25}$$

The $\mathbf{Q}$-exact term of (6.24) drops out of the classes; this is the potential $\mathbf{Q}$-exact difference between the two bracketings in (6.17). This proves the stated ring laws. □

6.3 The graph formula of the quantum product

For a physical edge (v,w), the operator Δ_{vw} contracts one field at vertex v with its BV conjugate at vertex w, using the pairing (3.39). For example, in the $\mathcal{N}=4$ superfield variables, at fixed vertex positions,

$$\Delta_{vw} = -\delta^{DE}\int d^3\theta\,\frac{\delta}{\delta C_v^D(\theta)}\frac{\delta}{\delta C_w^E(\theta)}.$$

Here C_v^D is the field at vertex v, and the fibre derivatives are given by (A.4). Composing these contractions in edge order gives the graph coefficient

$$\begin{aligned}\frac{1}{n!}\{\mathcal{I}^{\otimes n},A,B\}^{\gamma,0} &= \frac{1}{n!}\{\mathcal{I}^{\otimes n},Aa,b_{\mathrm{aux}}B\}_0\Big|_{\mathrm{aux}=0}\\ &= \frac{(-1)^n}{n!(8\pi^2)^n}\sum_{\substack{\Gamma\text{ labelled, Laman}\\ |\Gamma_0|=n+2,\ |\Gamma_1|=2n+1\\ (n+1,n+2)\in\Gamma_1}}\Bigg[\mathcal{I}_\Gamma(\lambda;z)\\ &\qquad\times\prod_{(v,w)\in\Gamma_1\setminus\{(n+1,n+2)\}}^{\text{edge order}}\Delta_{vw}(\mathcal{I}_1\cdots\mathcal{I}_nA_{n+1}B_{n+2})\Bigg]_{\text{jets},\,x_v=x,\,z=0}.\end{aligned}\tag{6.26}$$

Here $\gamma=\{(n+1,n+2)\}$, with aux $=0$ imposed after contraction.

In the subscript jets, each derivative on a contracted input field differentiates the kernel through its edge displacement, while λ_v differentiates the surviving fields at vertex v.

The displacements are then set to zero and all vertex positions to the common insertion point x. The graph restrictions of Section 7 reduce this sum to product forests.

Corollary 6.5 (Finiteness of the quantum product)**.**

For polynomial local inputs and a polynomial superpotential in the interaction (3.38), the quantum product has finitely many powers of $\hbar$.

Proof. Each order-n product graph has n interaction vertices and $2n$ physical contractions, so Proposition 4.16 gives

$$\begin{aligned}\#(\mathsf{b},\beta\text{ in output}) &= \#(\mathsf{b},\beta\text{ in }A)+\#(\mathsf{b},\beta\text{ in }B)\\ &\quad+\underbrace{n-\#(\bar W\text{ vertices})}_{\text{interaction terminals}}-\underbrace{2n}_{\text{physical contractions}}\\ &= \#(\mathsf{b},\beta\text{ in }A)+\#(\mathsf{b},\beta\text{ in }B)-n-\#(\bar W\text{ vertices})\geq 0.\end{aligned}$$

Writing $[\hbar^n]$ for the coefficient of $\hbar^n$, we obtain

$$[\hbar^n](A\star B)=0\quad\text{if}\quad n>\#(\mathsf{b},\beta\text{ in }A)+\#(\mathsf{b},\beta\text{ in }B).\tag{6.27}$$

For polynomial inputs the counts are the maxima over their monomials; the same loss adds at the binary vertices of any fixed bracketing of finitely many inputs.

For a formal superpotential the bound holds coefficientwise in its coefficients. □

6.4 One- and two-loop examples in $\mathcal{N}=4$ Yang–Mills theory

The one-loop triangle has vertices $1:\mathcal{I}$, $2:A$ and $3:B$, and marked edge $\gamma=\{(2,3)\}$. Its kernel is exactly the same as the fan kernel. Pairing its displacement exponential with the spatial functional derivatives of (8.2) gives

$$\begin{aligned}\{\mathcal{I},A,B\}^{\{(2,3)\},0}(x) &= \frac{i}{32\pi^2}\int_0^1 dr_1\int_0^{1-r_1}dr_2\int d^3\theta\,C_{XYZ}\\ &\quad\times\int d^4z_1\,d^4z_2\\ &\quad\times(\partial_{+\dot a}C^X)(x+r_1z_1+(1-r_1)z_2,\theta)\\ &\quad\times\partial_{2,+}{}^{\dot a}\left(\frac{\delta A(x_2)}{\delta C^Y(x_2+z_1,\theta)}\right)\Big|_{x_2=x+r_2(z_2-z_1)}\frac{\delta B(x)}{\delta C^Z(x+z_2,\theta)}.\end{aligned}\tag{6.28}$$

Here x is the arbitrary common position of the local product.

At two loops the physical graphs have four edges (plus, as always, one edge for the auxiliary fields). They consist of a book and two triangles attached to opposite inputs (Figure 6.3). Their ordered edges, for $3 = A$ and $4 = B$, are

$$\begin{aligned} \Gamma_{\text{book}} &: (13, 14, 23, 24, 34), \\ \Gamma_A &: (12, 13, 23, 24, 34), \\ \Gamma_B &: (12, 14, 23, 24, 34). \end{aligned} \tag{6.29}$$

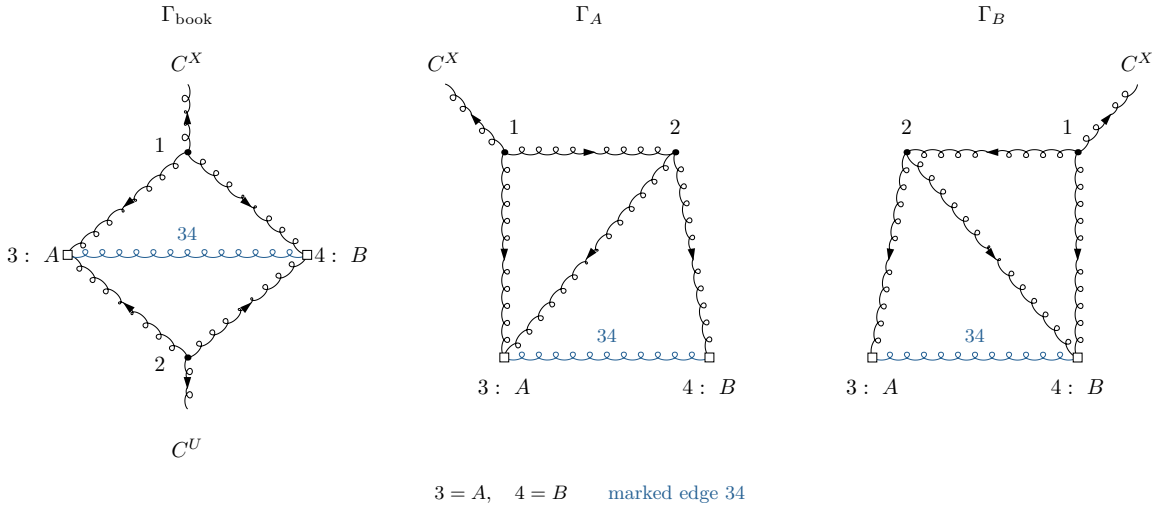


Figure 6.3: The two-loop quantum-product graphs Γ_{book}, Γ_A and Γ_B. Squares are the insertions $3 : A$ and $4 : B$; dots are cubic interactions. Black arrows orient the physical contractions, and the blue edge $(3, 4)$ carries the auxiliary pairing. The book leaves C^X, C^U uncontracted; each triangle leaves C^X.

The last edge is the auxiliary edge. These are still fan kernels (but with more free injected momenta).

$$\begin{aligned} \mathcal{I}_{\Gamma_{\text{book}}}(\lambda; 0) &= \frac{1}{24}\left[\lambda_{1,+\dot{a}}(\lambda_{2,+}{}^{\dot{a}} + 2\lambda_{3,+}{}^{\dot{a}})\right]\left[\lambda_{2,+\dot{b}}(\lambda_{1,+}{}^{\dot{b}} + 2\lambda_{3,+}{}^{\dot{b}})\right], \\ \mathcal{I}_{\Gamma_A}(\lambda; 0) &= -\frac{1}{24}(\lambda_{1,+\dot{a}}\lambda_{3,+}{}^{\dot{a}})^2, \\ \mathcal{I}_{\Gamma_B}(\lambda; 0) &= \frac{1}{24}(\lambda_{1,+\dot{a}}\lambda_{3,+}{}^{\dot{a}})^2. \end{aligned} \tag{6.30}$$

For the full result, the book kernel gives

$$\begin{aligned} &\frac{1}{2}\{\mathcal{I}, \mathcal{I}, A, B\}^{\gamma,0}(x)\Big|_{\Gamma_{\text{book}}} \\ &= \frac{C_{XYZ}C_{UVW}}{2(32\pi^2)^2}\int_0^1 dr_1 \int_0^1 dr_3 \int_0^1 dr_2 \int_0^{1-r_2} dr_4\,(1 - r_2 - r_4) \\ &\times \int d^3\theta^{(1)} d^3\theta^{(2)} \int \prod_{e\in\{13,14,23,24\}} d^4 z_e \\ &\times \partial_{1,+\dot{a}}(\partial_{3,+}{}^{\dot{a}} + r_3\partial_{2,+}{}^{\dot{a}})\partial_{2,+\dot{b}}(\partial_{3,+}{}^{\dot{b}} + r_1\partial_{1,+}{}^{\dot{b}}) \\ &\times \Bigg[C^X(x_1, \theta^{(1)})C^U(x_2, \theta^{(2)}) \\ &\quad \times \frac{\delta^2 A(x_3)}{\delta C^Y(x_3 + z_{13}, \theta^{(1)})\delta C^V(x_3 + z_{23}, \theta^{(2)})}\,\frac{\delta^2 B(x)}{\delta C^Z(x + z_{14}, \theta^{(1)})\delta C^W(x + z_{24}, \theta^{(2)})}\Bigg]_{x_v = x_v(x;r,z),\, v=1,2,3}. \end{aligned} \tag{6.31}$$

The translated positions are

$$\begin{aligned}
x_1(x;r,z) &= x + r_1 z_{13} + (1-r_1) z_{14} + r_1\big[r_2(z_{14}-z_{13}) + r_4(z_{24}-z_{23})\big],\\
x_2(x;r,z) &= x + r_3 z_{23} + (1-r_3) z_{24} + r_3\big[r_2(z_{14}-z_{13}) + r_4(z_{24}-z_{23})\big],\\
x_3(x;r,z) &= x + r_2(z_{14}-z_{13}) + r_4(z_{24}-z_{23}).
\end{aligned}$$

For Γ_A, the translated positions are

$$\begin{aligned}
x_1^{+\dot a}(x;r,z) &= x^{+\dot a} + \big[r_1 r_2 + (1-r_2)(r_1 r_3 + r_4)\big] z_{13}^{+\dot a} - r_2\big[r_1(1-r_3) - r_4\big] z_{23}^{+\dot a}\\
&\quad + (1 - r_1 r_3 - r_4) z_{24}^{+\dot a},\\
x_3^{+\dot a}(x;r,z) &= x^{+\dot a} + (1-r_3-r_4)\big[z_{24}^{+\dot a} - (1-r_2) z_{13}^{+\dot a} - r_2 z_{23}^{+\dot a}\big].
\end{aligned}$$

The full contribution is

$$\begin{aligned}
&\frac{1}{2}\{\mathcal{I},\mathcal{I},A,B\}^{\gamma,0}(x)\Big|_{\Gamma_A}\\
&\quad = \frac{C_{XYR}C_{RZW}}{(32\pi^2)^2}\int_0^1 dr_1\int_0^1 dr_2\int_0^1 dr_3\int_0^{1-r_3} dr_4\\
&\times r_2(1-r_1)(1-r_3-r_4)\int d^3\theta\int \prod_{e\in\{13,23,24\}} d^4 z_e\\
&\times (\partial_{+\dot a}\partial_{+\dot b} C^X)(x_1(x;r,z),\theta)\\
&\times \partial_{3,+}{}^{\dot a}\partial_{3,+}{}^{\dot b}\left(\frac{\delta^2 A(x_3)}{\delta C^Y(x_3+z_{13},\theta)\,\delta C^Z(x_3+z_{23},\theta)}\right)\Big|_{x_3=x_3(x;r,z)}\frac{\delta B(x)}{\delta C^W(x+z_{24},\theta)}.
\end{aligned}\tag{6.32}$$

For Γ_B, the translated positions are

$$\begin{aligned}
x_1^{+\dot a}(x;r,z) &= x^{+\dot a} + \big[1 - r_1 r_2 - r_1(1-r_2) r_3\big] z_{14}^{+\dot a} + r_1 r_3 z_{23}^{+\dot a} + r_1 r_2 (1-r_3) z_{24}^{+\dot a},\\
x_3^{+\dot a}(x;r,z) &= x^{+\dot a} + r_4\big[(1-r_2) z_{14}^{+\dot a} - z_{23}^{+\dot a} + r_2 z_{24}^{+\dot a}\big].
\end{aligned}$$

Thus

$$\begin{aligned}
&\frac{1}{2}\{\mathcal{I},\mathcal{I},A,B\}^{\gamma,0}(x)\Big|_{\Gamma_B}\\
&\quad = -\frac{C_{XZR}C_{RYW}}{(32\pi^2)^2}\int_0^1 dr_1\int_0^1 dr_2\int_0^1 dr_3\int_0^{1-r_3} dr_4\\
&\times r_1 r_2 r_4\int d^3\theta\int \prod_{e\in\{14,23,24\}} d^4 z_e\\
&\times (\partial_{+\dot a}\partial_{+\dot b} C^X)(x_1(x;r,z),\theta)\\
&\times \partial_{3,+}{}^{\dot a}\partial_{3,+}{}^{\dot b}\left(\frac{\delta A(x_3)}{\delta C^Y(x_3+z_{23},\theta)}\right)\Big|_{x_3=x_3(x;r,z)}\frac{\delta^2 B(x)}{\delta C^Z(x+z_{14},\theta)\,\delta C^W(x+z_{24},\theta)}.
\end{aligned}\tag{6.33}$$

The exchange $3 \leftrightarrow 4$ maps Γ_A to Γ_B: after restoring the displayed edge order, $\mathcal{I}_{\Gamma_B}(\lambda;0) = -\mathcal{I}_{\Gamma_A}(\lambda;0)$ in (6.30); the full displaced formulas also require exchanging the inputs, their edge displacements, and the root.

6.5 Order-by-order representatives

For $Q_0A = Q_0B = 0$, the first two orders of (6.13) are

$$\begin{aligned} Q_0\{\mathcal{I},A,B\}^{\gamma,0} + Q_1(AB) &= (Q_1A)B + (-1)^{|A|}A(Q_1B),\\ \frac{1}{2}Q_0\{\mathcal{I},\mathcal{I},A,B\}^{\gamma,0} + Q_1\{\mathcal{I},A,B\}^{\gamma,0} + Q_2(AB) &= (Q_2A)B + (-1)^{|A|}A(Q_2B)\\ &\quad + \{\mathcal{I},Q_1A,B\}^{\gamma,0} + (-1)^{|A|}\{\mathcal{I},A,Q_1B\}^{\gamma,0}. \end{aligned} \tag{6.34}$$

Thus, the classical variation of the one-loop term of the quantum product cancels the one-loop cross-contraction. Its two-loop term absorbs the remaining two-loop defect together with the action of Q_1 on the one-loop term.

If $A = \sum_i \hbar^i A_i$ and $B = \sum_j \hbar^j B_j$ are quantum cocycles, their product has coefficients

$$\mathcal{O}_r = \sum_{i+j+n=r} \frac{1}{n!}\{\mathcal{I}^{\otimes n}, A_i, B_j\}^{\gamma,0}, \qquad \mathcal{O}_0 = A_0B_0 \text{ (6.11)}, \tag{6.35}$$

$$\begin{aligned} \mathcal{O}_1 &= A_1B_0 + A_0B_1 + \{\mathcal{I},A_0,B_0\}^{\gamma,0},\\ \mathcal{O}_2 &= A_2B_0 + A_1B_1 + A_0B_2\\ &\quad + \{\mathcal{I},A_1,B_0\}^{\gamma,0} + \{\mathcal{I},A_0,B_1\}^{\gamma,0} + \frac{1}{2}\{\mathcal{I},\mathcal{I},A_0,B_0\}^{\gamma,0}. \end{aligned} \tag{6.36}$$

Equation (6.13) gives $\sum_{k=0}^{r} Q_k\mathcal{O}_{r-k} = 0$ at every order. These are the lifting equations (9.17) for the product A_0B_0: the quantum product gives in closed form the completion that the homotopy transfer of Section 9.3 constructs order by order.

6.6 Normal ordering and an explicit associator

Normal ordering. The quantum normal-ordering product is represented by the Bochner–Martinelli contour (6.41) [3, Section 2.3].

We now prove that it agrees, up to exact terms, with the quantum product of Sections 6.1–6.5.

Here $(A(x)B(0))[L]$ denotes the joint renormalized two-insertion expansion at scale L, defined for local insertions X by $X[L]\,e^{I[L]/\hbar} = \lim_{\varepsilon\downarrow 0} e^{\hbar\partial_{P_{\varepsilon,L}}}(X\,e^{I/\hbar})$ with $I = \int_{\mathbb{C}^2}\Omega\wedge[\mathcal{I}]_{(0,2)}$ [17]. (The scale L belongs to Costello's formalism of BV renormalization theory; it never affects our calculations.)

The contour argument uses renormalized separated insertions satisfying the supersymmetry descent Ward identity [3, Sections 2.2–2.3].

Let $A(x,\bar\theta)[L]$ be the renormalized supersymmetry multiplet of the effective insertion $A[L]$. In the coordinates of (3.12),

$$\begin{aligned} A^{(1)}(x)[L] &= \frac{1}{4}dx^{-\dot a}\,\frac{\partial_L A(x,\bar\theta)[L]}{\partial\bar\theta^{\dot a}}\bigg|_{\bar\theta=0},\\ (\bar\partial + (I[L],\,\cdot\,)_L + \hbar\Delta_L)A^{(1)}[L] + \bar\partial_x A[L] &= (\mathbf{Q}A)^{(1)}[L]. \end{aligned} \tag{6.37}$$

Taking the one-form component of the full composite multiplet gives its descendant [3, Section 2.2].

For $\mathbf{Q}A = 0$, write the form to the left of its coefficient:

$$\begin{aligned} A^{(1)}[L] &= dx^{-\dot{a}} A^{(1)}_{-\dot{a}}[L], \\ \frac{\partial A[L]}{\partial x^{-\dot{a}}} &= (\bar{\partial} + (I[L], \cdot)_L + \hbar\Delta_L) A^{(1)}_{-\dot{a}}[L]. \end{aligned} \tag{6.38}$$

This gives the primitive of an antiholomorphic translation in the effective-observable complex.

At zero displacement, (3.57) gives

$$\begin{gathered} [P_{0,\infty}(x)]_\Omega = -\frac{x^-_{\dot{a}}\, dx^{-\dot{a}}}{8\pi^2 |x|^4_E}, \qquad d(\Omega \wedge [P_{0,\infty}]_\Omega) = \delta^{(4)}_E(x) d^4x_E, \\ \int_{S^3} \Omega \wedge [P_{0,\infty}]_\Omega = 1. \end{gathered} \tag{6.39}$$

The last integral holds for every positively oriented embedded three-sphere enclosing the origin once; its normalization is that of the heat kernel,

$$\int_{|x|_E \le r} \Omega \wedge [K_\varepsilon]_\Omega = 1 - \left(1 + \frac{r^2}{2\varepsilon}\right) e^{-r^2/(2\varepsilon)} \xrightarrow{\varepsilon \downarrow 0} 1.$$

Let $(A(x)B(0))^{(1)}[L]$ be the first descendant of the joint insertion with respect to the position x.

For closed inputs and $S^3_1 - S^3_0 = \partial V$, with V disjoint from all insertions,

$$\begin{aligned} &\int_{S^3_1} \Omega \wedge [P_{0,\infty}]_\Omega (A(x)B(0))[L] - \int_{S^3_0} \Omega \wedge [P_{0,\infty}]_\Omega (A(x)B(0))[L] \\ &\quad = -(\bar{\partial} + (I[L], \cdot)_L + \hbar\Delta_L) \int_V \Omega \wedge [P_{0,\infty}]_\Omega \wedge (A(x)B(0))^{(1)}[L]. \end{aligned} \tag{6.40}$$

For $\mathbf{Q}A = \mathbf{Q}B = 0$, descent gives $\bar{\partial}_x (A(x)B(0))[L] = -(\bar{\partial} + (I[L], \cdot)_L + \hbar\Delta_L)(A(x)B(0))^{(1)}[L]$; the propagator is closed on V, so

$$\begin{aligned} \int_{\partial V} \Omega \wedge [P_{0,\infty}]_\Omega (A(x)B(0))[L] &= -\int_V \Omega \wedge [P_{0,\infty}]_\Omega \wedge \bar{\partial}_x (A(x)B(0))[L] \\ &= -(\bar{\partial} + (I[L], \cdot)_L + \hbar\Delta_L) \int_V \Omega \wedge [P_{0,\infty}]_\Omega \wedge (A(x)B(0))^{(1)}[L]. \end{aligned}$$

Equation (6.40) proves equality of the contour classes in the effective-observable complex.

Theorem 6.6 (Normal-ordering comparison)**.** *Let A and B be homogeneous quantum cocycles and $L > 0$. In the effective-observable complex at scale L, with differential $\mathbf{Q}_L = Q_{\mathrm{BV}} + (I[L], \cdot)_L + \hbar\Delta_L$, let B_R be the ball of radius R centred at the insertion of B and $S^3_R = \partial B_R$. The contour product and the effective representative of (6.11) satisfy*

$$\begin{aligned} &\int_{S^3_R} \Omega \wedge [P_{0,\infty}(x)]_\Omega \, (A(x)B(0))[L] \\ &\quad = (A \star B)[L] - \mathbf{Q}_L \int_{B_R} \Omega \wedge [P_{0,\infty}(x)]_\Omega \wedge (A(x)B(0))^{(1)}[L]. \end{aligned} \tag{6.41}$$

Proof.

Write $F(x) = (A(x)B(0))[L]$ and $F^{(1)}(x) = (A(x)B(0))^{(1)}[L]$. The volume integral uses the full joint descendant and has type $(2,2)$. Evaluate it as one coefficientwise graph limit in the decoupled auxiliary theory: physical edges carry $P_{\varepsilon,L}$ and the marked edge carries $P_{\varepsilon,\infty}$. Take the limit only after combining the ordered graphs. The marked edge is last, has zero displacement, and its single Wick factor $\hbar$ is divided out.

$$\begin{aligned} \bar\partial P_{\varepsilon,L} &= K_\varepsilon - K_L, \qquad \bar\partial P_{\varepsilon,\infty} = K_\varepsilon, \\ |\Gamma_0| &= n+2, \qquad |\Gamma_1| = 2n+1, \qquad \frac{1}{\hbar}\frac{\hbar^{2n+1}}{n!\hbar^n} = \frac{\hbar^n}{n!}. \end{aligned} \tag{6.42}$$

Apply (6.42) to every edge, with the signs $(-1)^{j-1}$ of (3.43). The upper physical-edge faces assemble into $\mathbf{Q}_L$; pure-interaction collision faces cancel by the quantum master equation, and faces with only one insertion vanish by $\mathbf{Q}A = \mathbf{Q}B = 0$. A remaining cluster contains both insertions and n interactions. The rescaling $t_e \mapsto \varepsilon t_e$, $x_v \mapsto \sqrt{\varepsilon}\, x_v$ sends the physical upper endpoint to $L/\varepsilon \to \infty$, yielding exactly the projective marked kernel (6.26). The ordered Wick signs are those of (3.43); the zero-interaction identity in (6.11) fixes the collision sign to $+AB$.

$$\left[\sum_{n\ge 0}\frac{\hbar^n}{n!}\{\mathcal{I}^{\otimes n}, Aa, b_{\mathrm{aux}}B\}_0\Big|_{\mathrm{aux}=0}\right][L] = (A\star B)[L]. \tag{6.43}$$

The graphs outside each collapsed cluster reconstruct its effective representative by the Ward identity $\mathbf{Q}_L(\mathcal{O}[L]) = (\mathbf{Q}\mathcal{O})[L]$ of a local insertion, the zero-form counterpart of (6.37). The same ultraviolet-finite holomorphic graph prescription applies to the decoupled auxiliary extension. Away from the collision faces, the closed-input descent identity is $\bar\partial_x F = -\mathbf{Q}_L F^{(1)}$. The remaining boundary is S^3_R. Since $\Omega \wedge [P_{0,\infty}]_\Omega$ has odd degree, the regulated Stokes identity gives

$$\begin{aligned} \mathbf{Q}_L \int_{B_R} \Omega\wedge[P_{0,\infty}]_\Omega \wedge F^{(1)} &= -\int_{B_R} \Omega\wedge[P_{0,\infty}]_\Omega \wedge \mathbf{Q}_L F^{(1)} \\ &= \int_{B_R} \Omega\wedge[P_{0,\infty}]_\Omega \wedge \bar\partial_x F \\ &= (A\star B)[L] - \int_{S^3_R} \Omega\wedge[P_{0,\infty}]_\Omega F. \end{aligned} \tag{6.44}$$

This is (6.41). The collision coefficient is independent of L by this rescaling [5, Theorem 6]; L remains in the effective representatives and their differential. □

An explicit associator example. **Example 6.7.** In $\mathcal{N}=4$ Yang–Mills theory, let A and B be gauge-invariant polynomials in the scalar fields $(\gamma^I)_X$ (a subsector of the BMN sector), without holomorphic derivatives, and set $D = (\gamma^3)_X(\beta_2)^X$. Then

$$\begin{aligned} &(A\star B)\star D - A\star(B\star D) \\ &\quad = \mathbf{Q}\left\{\frac{\hbar}{64\pi^2}\left[(\partial_{+\dot a}B)\partial_+{}^{\dot a}\frac{\partial A}{\partial(\gamma^2)_X} + (\partial_{+\dot a}A)\partial_+{}^{\dot a}\frac{\partial B}{\partial(\gamma^2)_X}\right](\gamma^3)_X\right\}. \end{aligned} \tag{6.45}$$

Proof. Gauge invariance gives $Q_0A = Q_0B = 0$. Since $C^{XYZ}(\gamma^3)_X(\gamma^3)_Y(\gamma^1)_Z$ is zero, Q_0D is also zero. The possible one-loop action on D contains $(\theta_1\theta_2)\theta_2 = 0$, and $Q_{n\ge 2}$ requires at least three input fields. Thus $\mathbf{Q}D = 0$. Equation (6.28) gives

$$\{\mathcal{I}, A, D\}^{\gamma,0} = \frac{g}{64\pi^2} C_{EX}{}^Y(\partial_{+\dot a}\mathsf{c}^E)\partial_+{}^{\dot a}\left(\frac{\partial A}{\partial(\gamma^2)_X}\right)(\gamma^3)_Y.$$

Each physical contraction uses one b or β. For an order-n product, the n interactions supply at most n such factors, and D supplies one; its $2n$ physical contractions therefore require $n \leq 1$ (one-loop exact). Products between operators containing only c and γ have no quantum correction. Expanding the two bracketings consequently gives, exactly,

$$\begin{aligned}&(A \star B) \star D - A \star (B \star D)\\ &\quad = \frac{\hbar g}{64\pi^2} C_{EX}{}^{Y} (\partial_{+\dot{a}} \mathsf{c}^{E}) \left[\frac{\partial A}{\partial(\gamma^2)_X} \partial_+{}^{\dot{a}} B + (\partial_+{}^{\dot{a}} A) \frac{\partial B}{\partial(\gamma^2)_X} \right] (\gamma^3)_Y .\end{aligned}$$

We have

$$Q_0 \left[\partial_+{}^{\dot{a}} \left(\frac{\partial A}{\partial(\gamma^2)_X} \right) (\gamma^3)_X \right] = -g C_{EX}{}^{Y} (\partial_+{}^{\dot{a}} \mathsf{c}^{E}) \frac{\partial A}{\partial(\gamma^2)_X} (\gamma^3)_Y .$$

The same identity holds for B. Together with the antisymmetry of the dotted-index contraction, it evaluates the displayed $\mathbf{Q}$-variation to the associator. For an operator containing only γ and its derivatives, an n-loop supercharge graph would require $2n + 1 \leq n + 1$, so all its positive-loop variations vanish. This also gives $\mathbf{Q}A = \mathbf{Q}B = 0$ and shows that no further terms enter the primitive. □

For example, take $A = B = \delta^{XY} (\gamma^2)_X (\gamma^2)_Y$. The two gradient terms coincide, and the formula becomes

$$\begin{aligned}&(A \star B) \star D - A \star (B \star D)\\ &\quad = \frac{\hbar g}{16\pi^2} C_E{}^{XY} (\partial_{+\dot{a}} \mathsf{c}^{E}) (\partial_+{}^{\dot{a}} A) (\gamma^2)_X (\gamma^3)_Y\\ &\quad = \mathbf{Q} \left[\frac{\hbar}{16\pi^2} \delta^{XY} (\partial_{+\dot{a}} A) (\partial_+{}^{\dot{a}} (\gamma^2)_X) (\gamma^3)_Y \right] .\end{aligned} \tag{6.46}$$

The anomaly brackets, with their holomorphic spectral parameters, form a homotopical Lie conformal algebra: their Jacobi identities hold through the higher brackets [6, Sections 4.2–4.5]. The normal-ordering product and its associator homotopies are additional operations [3, Section 2.3]. The corrected Leibniz rule also permits a normal-ordering calculation of the supercharge on composites; [3, Section 3.2] illustrates this with the one-loop supersymmetry anomaly of the stress tensor.

7 Other kernels and product forests

As derived in Section 4, kernels beyond the fans arise in two cases:

1. The quantum product has general product forests: the marked edge AB and interaction trees, each joined to both insertions by $|T_j| + 1$ spokes. Marked books are the special case in which every interaction tree is a single vertex.
2. For a superpotential of degree greater than three (non-renormalizable), the loop corrections to the supersymmetry transformation themselves contain spiders (Proposition 4.13).

7.1 Restrictions on quantum-product graphs

Proposition 7.1 (Edge counts of the product graphs)**.** *Let Γ be a nonzero connected marked graph of (6.11) at order $\hbar^n$, $n \geq 1$. It has n interaction vertices, the two insertions A, B, and $2n + 1$ edges: the auxiliary edge AB, which contributes its kernel, and $2n$ edges contracting fields in $\mathcal{I}$, A and B. Removing A and B leaves the interaction subgraph (Figure 7.1). For one of its components T_j, with $|T_j|$ vertices, I_j denotes its internal edges and E_j its edges to A or B, the spokes. Every edge other than AB is an internal edge or a spoke of exactly one component, and*

$$|T_j| - 1 \;\leq\; |I_j| \;\leq\; |T_j| - \#(\bar{W} \text{ vertices in } T_j), \qquad |I_j| + |E_j| = 2|T_j|. \tag{7.1}$$

Proof. In the circuit of Γ (Remark 3.5) every edge is a resistor, and the marked resistor joins the insertions A and B.

Internal resistors.

$$\underbrace{|T_j| - 1}_{T_j \text{ connected}} \;\le\; |I_j| \;\le\; \underbrace{|T_j| - \#(\bar{W} \text{ vertices in } T_j)}_{\text{antifield terminals in } T_j}.$$

It is the antifield-terminal count of Lemma 4.10.

Resistors at the insertions. A physical resistor between A and B would be in parallel with the marked resistor (Corollary 4.3). Every physical resistor therefore ends at an interaction node, and it is an internal resistor or a spoke of the component of that node. Summing over the components,

$$\sum_j (|I_j| + |E_j|) = 2n = 2\sum_j |T_j|. \tag{7.2}$$

Local Ohm's law on $T_j \cup \{A, B\}$. The subcircuit formed by the nodes of T_j, the two insertions and the marked resistor obeys the local count of Local Ohm's law (Lemma 4.2):

$$\begin{aligned} & |I_j| + |E_j| + \underbrace{1}_{AB} \;\le\; 2(|T_j| + 2) - 3 \\ \Longrightarrow\quad & |I_j| + |E_j| \;\le\; 2|T_j|. \end{aligned}$$

By (7.2), equality holds for every component. □

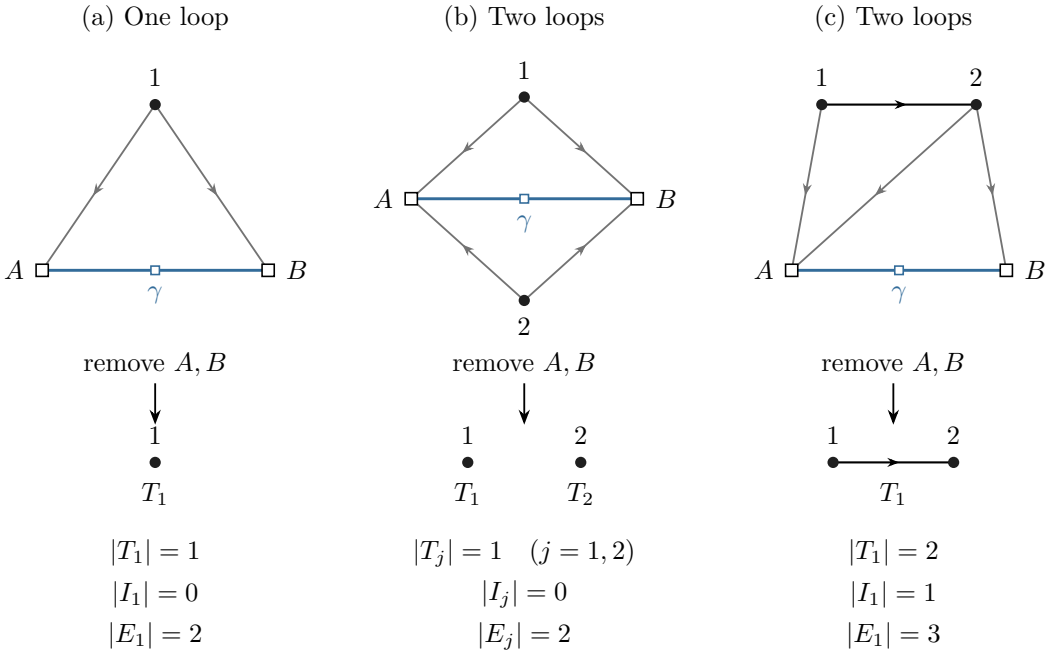


Figure 7.1: Marked graph topologies through two loops and their internal forests. The squares $A = n+1$ and $B = n+2$ are the two operator insertions; the blue edge carries the auxiliary mark γ. Deleting A, B and all incident edges leaves (a) one isolated vertex, (b) two isolated vertices, and (c) a two-vertex tree. The black edge is internal; the grey edges are spokes to the insertions. The displayed spoke counts refer to the full graphs in the top row.

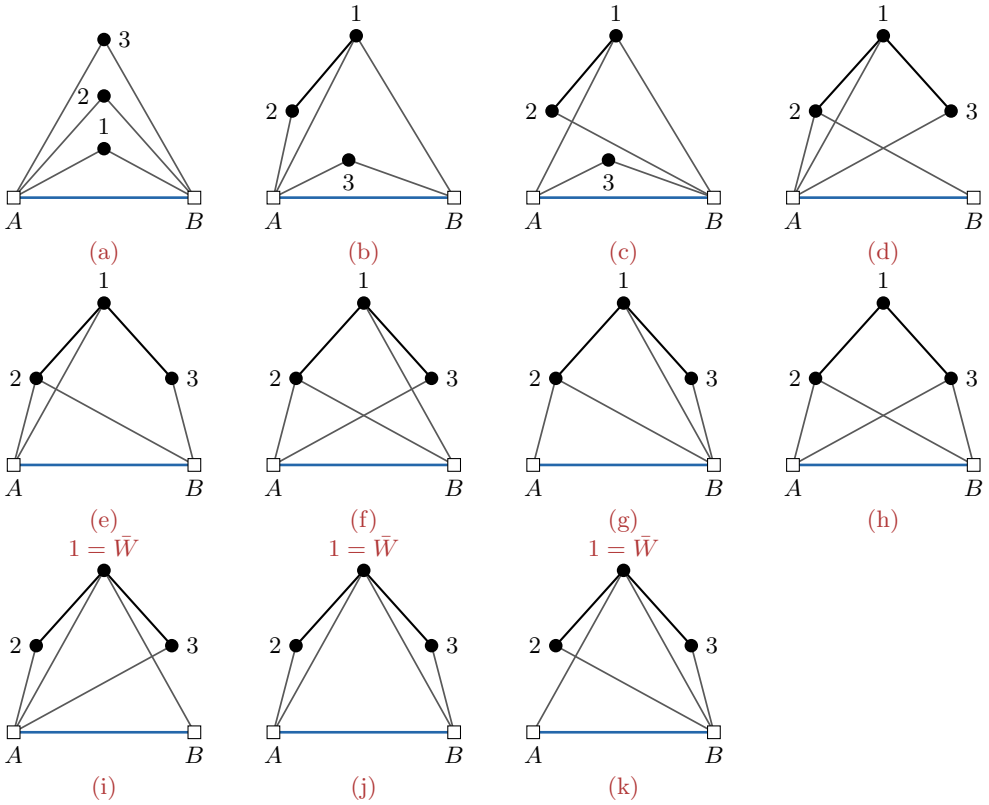


Figure 7.2: All eleven marked three-loop topologies up to permutations of the interaction vertices, fixing A and B. Panels (a)–(h) allow cubic interactions; in (i)–(k), vertex 1 is a superpotential vertex with four resistors. Blue: the marked edge AB; black: internal edges; gray: spokes.

Proposition 7.2 (Graphs of the quantum product)**.** *For the interaction* (3.38)*, with* $\bar{W}$ *of any degree, let* Γ *be a nonzero connected marked graph of* (6.11) *at order* $\hbar^n$*,* $n \geq 1$*. Then*

$$\Gamma \setminus \{A, B\} = \coprod_j T_j, \qquad T_j \text{ a tree}, \qquad \#(\bar{W} \text{ vertices in } T_j) \leq 1, \qquad |E_j| = |T_j| + 1. \tag{7.3}$$

Every T_j *is joined to both* A *and* B*, and loop diagrams contain no quadratic term of the superpotential* $\bar{W}$ *(only terms of at least cubic degree enter the theory). Each tree has one uncontracted field if all its vertices are cubic, and* $\deg \bar{W} - 2$ *if it contains a superpotential monomial; its number of source-bearing vertices is bounded by that number.*

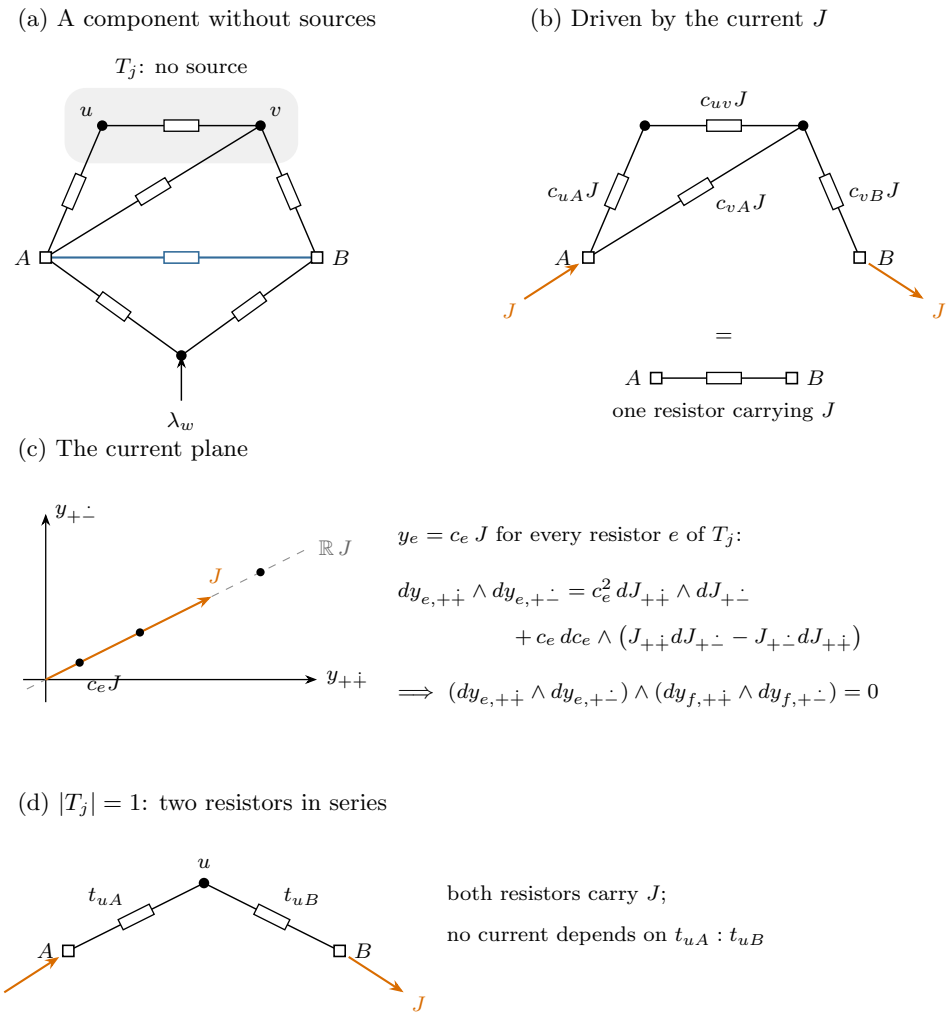


Figure 7.3: A component without sources. (a) A component T_j with no injected current meets the rest of the circuit only at the insertions A and B; here the rest is the marked resistor (blue) and a node fed by λ_w. (b) The current J entering T_j at A leaves it at B, and every current of T_j is c_eJ with a real current-division ratio c_e: the component acts as one resistor carrying J. (c) All currents of T_j lie on the line $\mathbb{R}J$ of the current plane, so each of their area forms is (7.4), and the product of two of them vanishes. (d) For $|T_j| = 1$ the two resistors are in series, and no current depends on the ratio of their resistances.

Proof.

The proof parallels that of Proposition 4.5.

Components without sources. Suppose that no current is injected at the nodes of T_j (Figure 7.3(a)). The component touches the rest of the circuit only at the insertions A and B, so the current J that enters it at A leaves it at B, and inside it Kirchhoff's laws are those of a passive network driven by J. Every current of the component is therefore a real multiple c_eJ of J, with a current-division ratio c_e fixed by its resistances: the component acts as one resistor carrying J (Figure 7.3(b)). As the resistances vary, its currents move along the single direction of J in the current plane (Figure 7.3(c)):

$$dy_{e,+\dot{+}} \wedge dy_{e,+\dot{-}} = c_e^2\,dJ_{+\dot{+}} \wedge dJ_{+\dot{-}} + c_e\,dc_e \wedge (J_{+\dot{+}}dJ_{+\dot{-}} - J_{+\dot{-}}dJ_{+\dot{+}}). \tag{7.4}$$

Replace the source-free component by its effective resistor between A, B and merge it with the marked resistor; call the resulting circuit Γ^{red}. If $|T_j| = 1$, its vertex v has $\lambda_v = 0$, and Theorem 5.1 gives

$$\mathcal{I}_\Gamma(\lambda;z)=\int_0^1 dr_1\int_0^1 dr_2\ \underbrace{r_2\lambda_{v,+}{}^{\dot a}\frac{\partial}{\partial z_{AB}^{\mathrm{red},+\dot a}}\Big|_{\lambda_v=0}}_{\text{zero current-area operator}}\mathcal{I}_{\Gamma^{\mathrm{red}}}(\lambda;z^{\mathrm{red}})=0.$$

If $|T_j|\geq 2$, choose a spanning tree containing AB and $|T_j|$ resistors of the component. There are $|T_j|$ complementary component resistors; for any two, e,f, (7.4) gives

$$\begin{aligned}
&(dy_{e,++}\wedge dy_{e,+-})\wedge(dy_{f,++}\wedge dy_{f,+-})\\
&=\Big[c_e^2 dJ_{++}\wedge dJ_{+-}+c_e\,dc_e\wedge(J_{++}dJ_{+-}-J_{+-}dJ_{++})\Big]\\
&\quad\wedge\Big[c_f^2 dJ_{++}\wedge dJ_{+-}+c_f\,dc_f\wedge(J_{++}dJ_{+-}-J_{+-}dJ_{++})\Big]=0,\\
\mathcal{I}_\Gamma(\lambda;z)&=\underbrace{0}_{\text{current-area Jacobian of the component}}\mathcal{I}_{\Gamma^{\mathrm{red}}}(\lambda;z^{\mathrm{red}})=0.
\end{aligned}$$

Trees. If the internal resistors of T_j close a loop, they number at least $|T_j|$; by (7.1) they number exactly $|T_j|$, T_j has no $\bar W$ node, and $|E_j|=|T_j|$. The $2|T_j|$ ends of the internal resistors and the $|T_j|$ spokes then wire all $3|T_j|$ terminals of its cubic nodes. No current is injected in T_j, and it becomes a component without sources, which gives zero. Every T_j is therefore a tree, $|I_j|=|T_j|-1$, and (7.1) gives at most one $\bar W$ node and $|E_j|=|T_j|+1$, which is (7.3).

Spokes and degrees. Since $|E_j|=|T_j|+1$ and each node can be joined to each of A and B at most once, each component must have spokes to both A and B.

Quadratic terms of $\bar W$. A quadratic term of $\bar W$ is a node with two terminals, both wired by the previous step, so it injects no current. It is the only $\bar W$ node of its tree, whose $3(|T_j|-1)+2=3|T_j|-1$ terminals are then all wired by the $|T_j|-1$ internal resistors and the $|T_j|+1$ spokes. The tree carries no source, and the first step gives zero.

A source at a vertex is supplied by its uncontracted fields. For cubic vertices the count is one; if T_j contains a superpotential monomial,

$$\begin{aligned}
\#(\text{uncontracted fields in } T_j)&=3(|T_j|-1)+\deg\bar W-2(|T_j|-1)-(|T_j|+1)\\
&=\deg\bar W-2,\\
\#\{v\in T_j:\lambda_v\neq 0\}&\leq\#(\text{uncontracted fields in } T_j)=\deg\bar W-2.
\end{aligned}$$

Thus the bound is one for cubic interactions, and is not uniformly one for arbitrary $\bar W$. □

Proposition 7.3 (Cubic interactions have a starting point for the iteration: a vertex of degree two)**.** *Let every vertex of $\mathcal{I}$ be cubic, as are the gauge and matter vertices of (3.38) together with a cubic $\bar W$. In every tree T_j of (7.3), exactly one vertex has degree two, and its uncontracted field is the only output field of T_j; all other vertices have degree three. The tree degree of a vertex is its number of internal edges, and the number of spokes at a vertex of given tree degree is*

tree degree	0	1	2	3
spokes if $\deg_\Gamma v=3$	–	2	1	0
spokes if $\deg_\Gamma v=2$	2	1	0	–

(7.5)

and a vertex with two spokes is joined to both A and B (Figure 7.4).

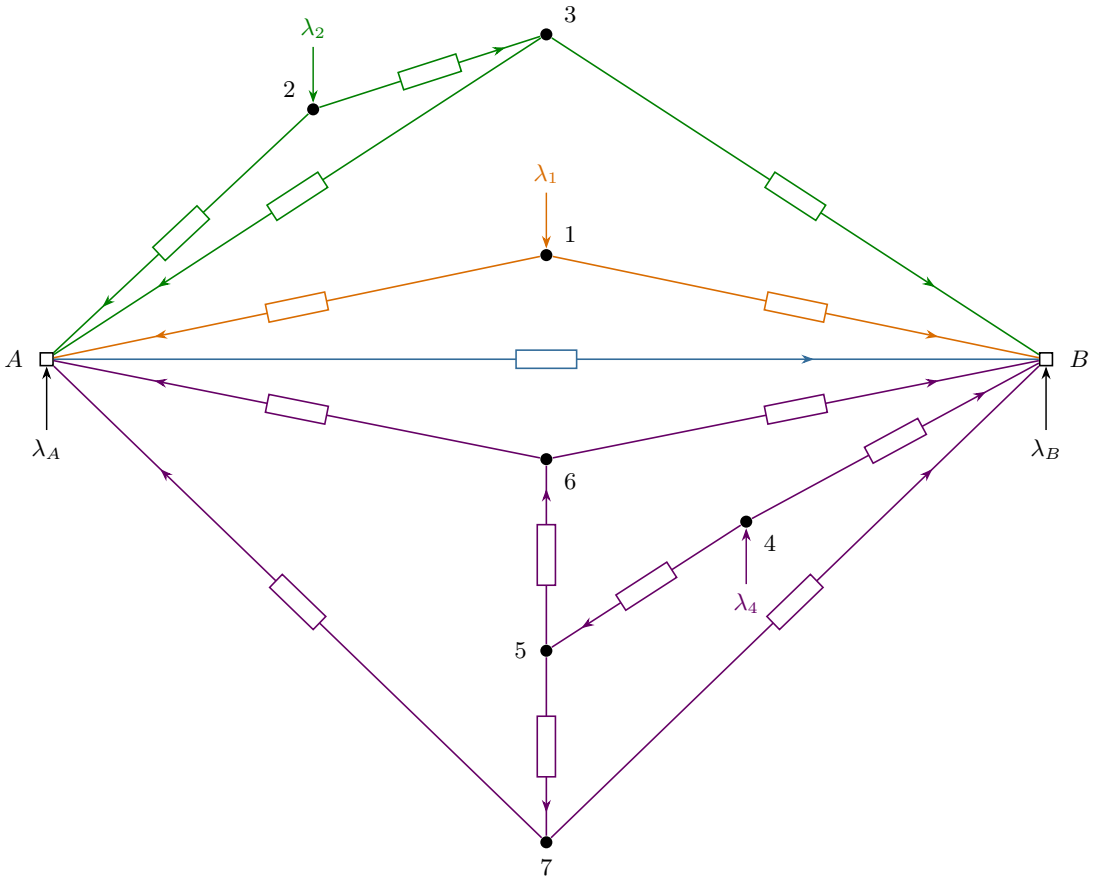


Figure 7.4: A seven-loop product graph with three trees, each in its own colour, between the insertions A and B; the marked edge AB is blue. Each tree has one more spoke than vertices, and its source enters at its only vertex of degree two: 1, 2 and 4. Leaves of degree three are joined to both insertions, and the branching vertex 5 has no spoke.

Proof. A cubic node has three terminals. In T_j the $|T_j|-1$ internal resistors wire $2(|T_j|-1)$ terminals and the $|T_j|+1$ spokes wire $|T_j|+1$ more, so the numbers of resistors at its nodes add up to

$$\sum_{v\in T_j} \deg_\Gamma v = 2(|T_j|-1)+(|T_j|+1) = 3|T_j|-1. \tag{7.6}$$

Every node has two or three resistors by Proposition 7.2, so exactly one node has two, and

$$\underbrace{3|T_j|}_{\text{terminals}} - \underbrace{2(|T_j|-1)}_{\text{internal resistor ends}} - \underbrace{(|T_j|+1)}_{\text{spokes}} = 1 \tag{7.7}$$

terminal stays free, at that node: each tree carries a single source, injected at its node with two resistors. □

When T_j instead contains one superpotential vertex with $\deg_\Gamma \bar W$ contracted fields out of $\deg \bar W$ monomial factors and all other vertices are cubic, (7.6) modifies to

$$\#\{\text{degree-two cubic vertices}\} = \underbrace{3(|T_j|-1)}_{\text{three per cubic vertex}} - \underbrace{(3|T_j|-1-\deg_\Gamma \bar W)}_{\substack{\text{their degrees,}\\ \text{by (7.6)}}} = \deg_\Gamma \bar W - 2,$$

$$\begin{aligned} \#\{\text{fields surviving at } \bar W\} &= \deg \bar W - \deg_\Gamma \bar W, \\ \#\{\text{fields surviving in } T_j\} &= \underbrace{3(|T_j|-1)+\deg \bar W}_{\text{fields}} - \underbrace{(3|T_j|-1)}_{\substack{\text{contracted ends,}\\ \text{(7.6)}}} = \deg \bar W - 2. \end{aligned} \tag{7.8}$$

At low loop orders ($n \leq 2$), all marked product forests belong to the fan and spider kernels, whereas from $n = 3$ onward general product forests also require star–triangle reductions (Section 7.3). As shown in Section 4, the calculation of Q_n in non-renormalizable theories involves only the spider kernel, which we evaluate next.

7.2 Spiders

A superpotential vertex of degree greater than three can join several arms (Proposition 4.13).

Definition 7.4 (Spiders)**.** A *spider* consists of paths meeting only at a vertex A, together with a second vertex B joined to every vertex of these paths, including A. In the supercharge Q_n graphs, A carries $\bar{W}$ and $B = \mathcal{O}$. The ith arm contains ℓ_i cubic vertices, labelled $v_{i,1}, \ldots, v_{i,\ell_i}$ from its outer end towards A; its length is ℓ_i and the loop number is $n = \sum_i \ell_i$.

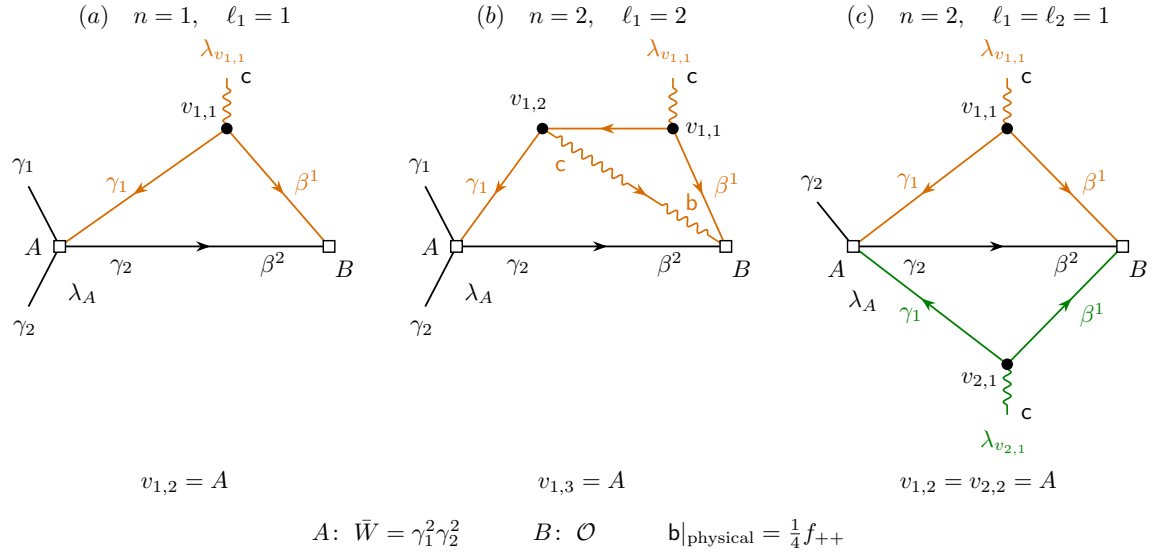


Figure 7.5: One- and two-loop spiders for $\bar{W} = \gamma_1^2\gamma_2^2$: (a) $\ell_1 = 1$; (b) $\ell_1 = 2$; (c) $\ell_1 = \ell_2 = 1$. The superpotential vertex is A and the operator insertion is B; each arm, with its spokes, has its own colour. Arm indices increase towards $A = v_{i,\ell_i+1}$. Each $v_{i,1}$ carries an output c; the output matter at A supplies λ_A.

On arm i, $v_{i,1}$ carries the source, v_{i,ℓ_i} is adjacent to A, and $v_{i,\ell_i+1} = A$. The remaining arm vertices are fully contracted:

$$\begin{aligned} \Gamma_1 &= \big((v_{i,j}, v_{i,j+1}), (v_{i,j}, B)\big)_{i=1,\ldots,a;\ j=1,\ldots,\ell_i} \cup \{(A, B)\}, \\ &\ell_1 + \cdots + \ell_a = n, \qquad \lambda_{v_{i,j},+\dot{a}} = \delta_{j1}\lambda_{v_{i,1},+\dot{a}}. \end{aligned} \tag{7.9}$$

All computations of Q_n involve only the spider kernel.

The computation of the spider kernel is similar to repeating the computation of the fan kernel on each path.

Each arm is reduced as a fan is, from its outer end (Figure 7.6).

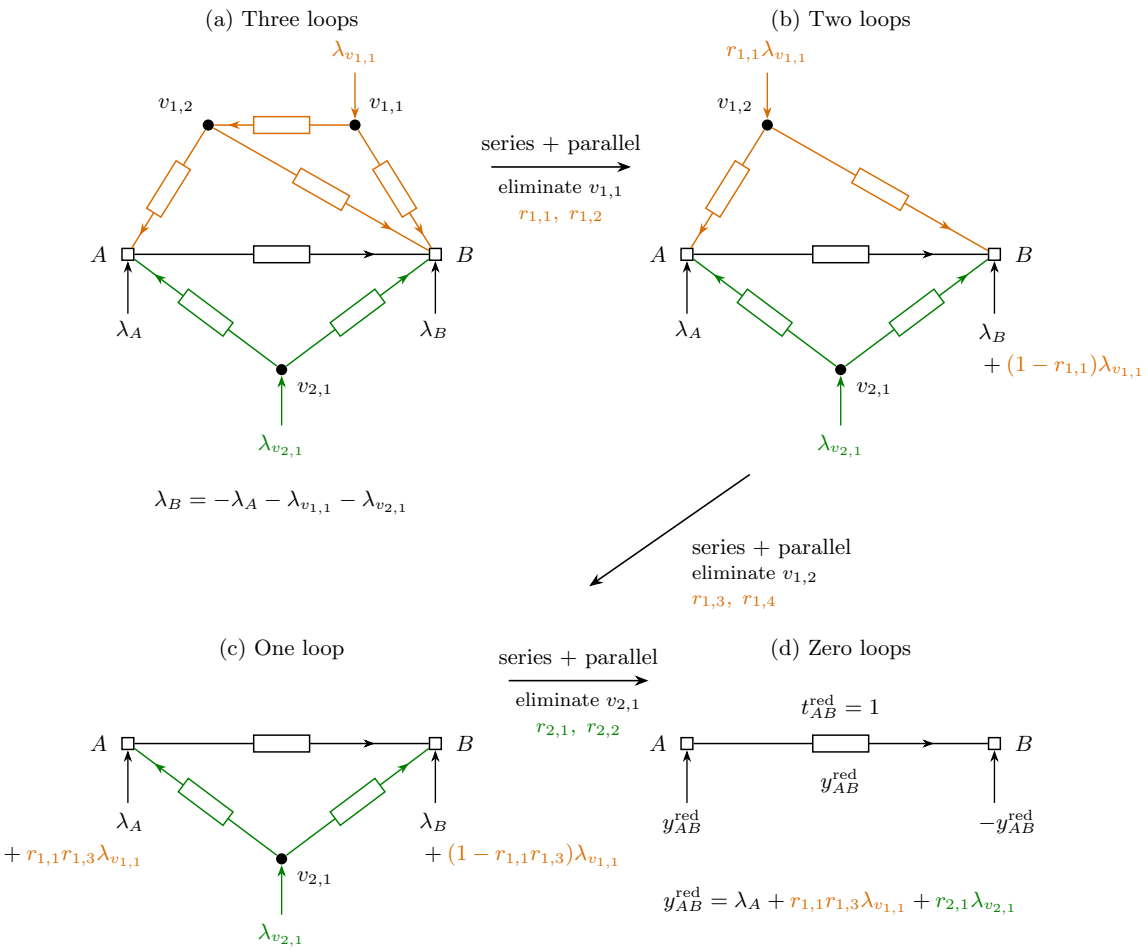


Figure 7.6: Reduction of the three-loop spider with arms $\ell_1 = 2$ (orange) and $\ell_2 = 1$ (green). Each step eliminates an arm vertex by a series step, which passes the fraction $r_{i,2j-1}$ of its source along the arm and the rest to B, and a parallel merge with the next spoke or with AB; each step lowers the loop number by one. The last panel is the unit resistor carrying (7.11).

Each merge lowers the loop number by one. When every arm is reduced, its branch and the original edge AB merge into one resistor, whose Schwinger time is fixed to one. We evaluate the iterated kernel operators in the joint coordinates

$$0 < r_{i,j} < 1 \quad (j < 2\ell_i), \qquad r_{i,2\ell_i} > 0, \qquad \sum_i r_{i,2\ell_i} < 1. \tag{7.10}$$

Here $r_{i,2j-1}$ is the fraction of the source at $v_{i,j}$, after the earlier eliminations, that the series step passes to $v_{i,j+1}$, the rest going to B;

for $j < \ell_i$, $r_{i,2j}$ is the fraction of the merged spoke current at $v_{i,j+1}$ carried by the new branch;

and $r_{i,2\ell_i}$ is the fraction of the final current carried by the branch of arm i, the original edge AB carrying $1 - \sum_i r_{i,2\ell_i}$.

The final resistor carries the reduced current y_{AB}^{red}, which by conservation at A is the source collected at A: λ_A and, from each arm, the fraction of its source passed along the whole arm,

$$y_{AB,+\dot{a}}^{\mathrm{red}} = \lambda_{A,+\dot{a}} + \sum_i \left(\prod_{k=1}^{\ell_i} r_{i,2k-1} \right) \lambda_{v_{i,1},+\dot{a}}. \tag{7.11}$$

Reversing the reductions splits y_{AB}^{red} back into the original currents (7.18)–(7.19).

Deleting the arms one at a time gives the kernel:

$$
\begin{aligned}
&\mathcal{I}_\Gamma(\lambda; z) \\
&= \left[\prod_{j=1}^{\ell_1, \longrightarrow} \left\{ \begin{array}{l} \displaystyle\int_0^1 dr_1 \int_0^1 dr_2 \, e^{-\lambda_{v_{1,j},+\dot{a}}[r_1 z_{v_{1,j} v_{1,j+1}} + (1-r_1) z_{v_{1,j} B}]^{+\dot{a}}} \\ \qquad \times \quad \underbrace{r_2 \lambda_{v_{1,j},+}{}^{\dot{b}} \dfrac{\partial}{\partial z^{\mathrm{red},+\dot{b}}_{v_{1,j+1} B}}}_{\text{current-area Jacobian insertion}} \end{array} \right\} \right] \\
&\quad \times \underbrace{\mathcal{I}_{\Gamma \setminus \{v_{1,1}, \ldots, v_{1,\ell_1}\}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}})}_{(n-\ell_1)\text{-loop kernel}} \\
&= \left[\prod_{i=1}^{2, \longrightarrow} \prod_{j=1}^{\ell_i, \longrightarrow} \left\{ \begin{array}{l} \displaystyle\int_0^1 dr_1 \int_0^1 dr_2 \, e^{-\lambda_{v_{i,j},+\dot{a}}[r_1 z_{v_{i,j} v_{i,j+1}} + (1-r_1) z_{v_{i,j} B}]^{+\dot{a}}} \\ \qquad \times r_2 \lambda_{v_{i,j},+}{}^{\dot{b}} \dfrac{\partial}{\partial z^{\mathrm{red},+\dot{b}}_{v_{i,j+1} B}} \end{array} \right\} \right] \\
&\quad \times \underbrace{\mathcal{I}_{\Gamma \setminus \{v_{1,1}, \ldots, v_{1,\ell_1}, v_{2,1}, \ldots, v_{2,\ell_2}\}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}})}_{(n-\ell_1-\ell_2)\text{-loop kernel}} \\
&= \cdots \\
&= \int_{(7.10)} \prod_{i=1}^{a} \prod_{j=1}^{2\ell_i} dr_{i,j} \, \underbrace{\left(1 - \sum_i r_{i,2\ell_i}\right)}_{\text{final parallel merge}} \\
&\quad \times \prod_{i=1}^{a} \underbrace{r_{i,2\ell_i}^{\ell_i - 1} \prod_{j=1}^{\ell_i - 1} \left[r_{i,2j-1}^{\ell_i - j} \, r_{i,2j}^{j-1} \, (1 - r_{i,2j}) \right] \left[\lambda_{v_{i,1},+\dot{a}} y^{\mathrm{red}}_{AB,+}{}^{\dot{a}} \right]^{\ell_i}}_{\text{diagonal blocks of arm } i} \; e^{-\sum_e y_{e,+\dot{a}}(r) z_e^{+\dot{a}}}.
\end{aligned}
\tag{7.12}
$$

In the first two lines, the ordered products are compositions of the displayed integral operators, with the first vertex of each arm outermost. Each factor uses the current graph and its own local r_1, r_2 of Theorem 5.1, with $v = v_{i,j}$, $u = v_{i,j+1}$ and $w = B$; each derivative acts on the entire expression to its right, including the later exponential factors and displacement substitutions,

$$
\begin{aligned}
\lambda^{\mathrm{red}}_u &= \lambda_u + r_1 \lambda_v, \qquad \lambda^{\mathrm{red}}_B = \lambda_B + (1 - r_1) \lambda_v, \\
z^{\mathrm{red}}_{uB} &= r_2 z_{uB} + (1 - r_2)(z_{vB} - z_{vu}), \qquad z^{\mathrm{red}}_e = z_e \quad (e \neq uB).
\end{aligned}
$$

Deleting an arm includes these source transfers and parallel merges; the second line is omitted if there is only one arm. Let us verify how the iterated operators produce the last line of (7.12) in the joint coordinates (7.10).

Evaluating the iterated arm and book operators

Eliminate the interior vertices $v_{i,1}, \ldots, v_{i,\ell_i - 1}$, arm by arm, followed by the terminal vertices $v_{1,\ell_1}, \ldots, v_{a-1,\ell_{a-1}}$, leaving the final triangle (v_{a,ℓ_a}, A, B) whose one-loop kernel is (5.3). Along arm i, the internal arm sources vanish as in (7.9), so with local parameters $(r_1, r_2) = (r_{i,2j-1}, 1 - r_{i,2j})$ for $j < \ell_i$,

$$
\lambda^{\mathrm{red}}_{v_{i,j+1},+\dot{a}} = r_{i,2j-1} \lambda^{\mathrm{red}}_{v_{i,j},+\dot{a}} \quad (j < \ell_i), \qquad \lambda^{\mathrm{red}}_{v_{i,j},+\dot{a}} = \left(\prod_{k=1}^{j-1} r_{i,2k-1} \right) \lambda_{v_{i,1},+\dot{a}} \quad (1 \leq j \leq \ell_i). \tag{7.13}
$$

Because every later source on the same arm is proportional to $\lambda_{v_{i,1}}$, an earlier directional derivative $\lambda_{v_{i,1},+}{}^{\dot a}\partial/\partial z^{\mathrm{red},+\dot a}_{v_{i,k}B}$ annihilates its source phase ($\lambda_{v_{i,1},+\dot a}\lambda^{\mathrm{red}}_{v_{i,k},+}{}^{\dot a}=0$) and passes through the affine spoke-displacement update $z^{\mathrm{red}}_{v_{i,k+1}B}=(1-r_{i,2k})z_{v_{i,k+1}B}+r_{i,2k}(z^{\mathrm{red}}_{v_{i,k}B}-z_{v_{i,k}v_{i,k+1}})$ as

$$\lambda_{v_{i,1},+}{}^{\dot a}\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{v_{i,k}B}}=r_{i,2k}\lambda_{v_{i,1},+}{}^{\dot a}\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{v_{i,k+1}B}}\qquad(k<\ell_i).$$

Repeating for $j=1,\dots,\ell_i-1$ on every arm leaves the a-page book on A,B with page vertices $v_{1,\ell_1},\dots,v_{a,\ell_a}$ carrying sources $\lambda_i=\lambda^{\mathrm{red}}_{v_{i,\ell_i}}$. Removing pages $1,\dots,a-1$ onto (a,A,B), the later page phases are independent of z^{red}_{AB}, while each parallel merge multiplies $\partial/\partial z^{\mathrm{red},+\dot a}_{AB}$ by $r_2|_{v=k}$:

$$\begin{aligned}\mathcal{I}_{\mathrm{book}}(\lambda;z)=&\int_{(0,1)^{2a-2}}\prod_{i=1}^{a-1}\left[d(r_1|_{v=i})\,d(r_2|_{v=i})\right]e^{-\sum_{i=1}^{a-1}\lambda_{i,+\dot a}[(r_1|_{v=i})z_{iA}+(1-r_1|_{v=i})z_{iB}]^{+\dot a}}\\&\times\prod_{i=1}^{a-1}\left[\left(\prod_{k=i}^{a-1}r_2|_{v=k}\right)\lambda_{i,+}{}^{\dot a_i}\frac{\partial}{\partial z^{\mathrm{red},+\dot a_i}_{AB}}\right]\mathcal{I}_{\triangle}(\lambda^{\mathrm{red}};z^{\mathrm{red}}).\end{aligned}\tag{7.14}$$

On the remaining triangle (a,A,B) with coordinates (r_1,r_2) of (5.3), put $r_1|_{v=a}=r_1/(1-r_2)$ and $r_2|_{v=a}=1-r_2$. Passing to the simplex current shares $r_{i,2\ell_i}$ via

$$r_1|_{v=i}=r_{i,2\ell_i-1},\qquad r_2|_{v=i}=\frac{1-\sum_{k=i}^{a}r_{k,2\ell_k}}{1-\sum_{k=i+1}^{a}r_{k,2\ell_k}},\tag{7.15}$$

the telescoping product $\prod_{i=1}^{a}(r_2|_{v=i})^i\,|d(r_2|_{v=i})|=(1-\sum_{i=1}^{a}r_{i,2\ell_i})\prod_{i=1}^{a}dr_{i,2\ell_i}$ gives the book kernel

$$\mathcal{I}_{\mathrm{book}}(\lambda;z)=\int_{\substack{0<r_{i,2\ell_i-1}<1\\ r_{i,2\ell_i}>0,\ \sum_i r_{i,2\ell_i}<1}}\prod_{i=1}^{a}(dr_{i,2\ell_i-1}\,dr_{i,2\ell_i})\left(1-\sum_{i=1}^{a}r_{i,2\ell_i}\right)\prod_{i=1}^{a}(\lambda_{i,+\dot a_i}y^{\mathrm{red}}_{AB,+}{}^{\dot a_i})e^{-\sum_e y_{e,+\dot b}z_e^{+\dot b}}.\tag{7.16}$$

Finally, at the book stage z^{red}_{AB} contains the terminal spoke of arm i with coefficient $r_{i,2\ell_i}$, so the interior vertex $v_{i,j}$ ($j<\ell_i$) contributes $(1-r_{i,2j})(\prod_{k=1}^{j-1}r_{i,2k-1})(\prod_{k=j+1}^{\ell_i-1}r_{i,2k})r_{i,2\ell_i}(\lambda_{v_{i,1},+\dot a}y^{\mathrm{red}}_{AB,+}{}^{\dot a})$, while the terminal page v_{i,ℓ_i} contributes $(\prod_{k=1}^{\ell_i-1}r_{i,2k-1})(\lambda_{v_{i,1},+\dot a}y^{\mathrm{red}}_{AB,+}{}^{\dot a})$. Multiplying these ℓ_i factors along arm i,

$$\left[\prod_{j=1}^{\ell_i-1}(1-r_{i,2j})\left(\prod_{k=1}^{j-1}r_{i,2k-1}\right)\left(\prod_{k=j+1}^{\ell_i-1}r_{i,2k}\right)r_{i,2\ell_i}\right]\left(\prod_{k=1}^{\ell_i-1}r_{i,2k-1}\right)=r_{i,2\ell_i}^{\ell_i-1}\prod_{j=1}^{\ell_i-1}\left[r_{i,2j-1}^{\ell_i-j}r_{i,2j}^{j-1}(1-r_{i,2j})\right],\tag{7.17}$$

yields the last equality of (7.12).

Reconstructing the edge currents

Reversing the current update (5.11) from each terminal page v_{i,ℓ_i} outwards via $y^{\mathrm{red}}_{v_{i,\ell_i}B}=(1-r_{i,2\ell_i-1})(\prod_{k=1}^{\ell_i-1}r_{i,2k-1})\lambda_{v_{i,1}}+r_{i,2\ell_i}y^{\mathrm{red}}_{AB}$ and $y^{\mathrm{red}}_{v_{i,j}B}=(1-r_{i,2j-1})\lambda^{\mathrm{red}}_{v_{i,j}}+r_{i,2j}y^{\mathrm{red}}_{v_{i,j+1}B}$ ($j<\ell_i$) gives

$$y^{\mathrm{red}}_{v_{i,j}B,+\dot a}=\sum_{r=j}^{\ell_i}\left(\prod_{k=j}^{r-1}r_{i,2k}\right)(1-r_{i,2r-1})\left(\prod_{k=1}^{r-1}r_{i,2k-1}\right)\lambda_{v_{i,1},+\dot a}+\left(\prod_{k=j}^{\ell_i-1}r_{i,2k}\right)r_{i,2\ell_i}\,y^{\mathrm{red}}_{AB,+\dot a},\tag{7.18}$$

and hence the original spoke, central, and arm currents:

$$\begin{aligned}&y_{v_{i,j}v_{i,j+1},+\dot a}=\lambda^{\mathrm{red}}_{v_{i,j},+\dot a}-y^{\mathrm{red}}_{v_{i,j}B,+\dot a},&&y_{v_{i,1}B,+\dot a}=y^{\mathrm{red}}_{v_{i,1}B,+\dot a},\\&y_{v_{i,j}B,+\dot a}=(1-r_{i,2j-2})y^{\mathrm{red}}_{v_{i,j}B,+\dot a}\quad(j\ge2),&&y_{AB,+\dot a}=\left(1-\sum_i r_{i,2\ell_i}\right)y^{\mathrm{red}}_{AB,+\dot a},\end{aligned}\tag{7.19}$$

$$
\begin{aligned}
y_{v_{i,j}v_{i,j+1},+\dot a} = &\sum_{s=j}^{\ell_i-1}\left(\prod_{k=1}^{s} r_{i,2k-1}\right)\left(\prod_{k=j}^{s-1} r_{i,2k}\right)(1-r_{i,2s})\,\lambda_{v_{i,1},+\dot a}\\
&+\left(\prod_{k=1}^{\ell_i} r_{i,2k-1}\right)\left(\prod_{k=j}^{\ell_i-1} r_{i,2k}\right)\lambda_{v_{i,1},+\dot a}-\left(\prod_{k=j}^{\ell_i} r_{i,2k}\right)y^{\mathrm{red}}_{AB,+\dot a},
\end{aligned}
\tag{7.20}
$$

with loop-displacement split

$$
\sum_e y_{e,+\dot a}z_e^{+\dot a}=\lambda_{A,+\dot a}z_{AB}^{+\dot a}+\sum_i \lambda_{v_{i,1},+\dot a}z_{v_{i,1}B}^{+\dot a}+\sum_{i=1}^{a}\sum_{j=1}^{\ell_i} y_{v_{i,j}v_{i,j+1},+\dot a}(z_{v_{i,j}v_{i,j+1}}+z_{v_{i,j+1}B}-z_{v_{i,j}B})^{+\dot a}\quad (v_{i,\ell_i+1}=A).
\tag{7.21}
$$

The cubic arm vertices carry no derivatives, so $z_{v_{i,j}v_{i,j+1}}=0$ for $j<\ell_i$; the spokes, the last arm edges and AB keep general displacements if we want to compute results with holomorphic derivatives.

The exponential then translates the outputs at $v_{i,1}$ and A:

$$
-\sum_e y_{e,+\dot a}(r)z_e^{+\dot a}=\lambda_{A,+\dot a}x_A^{+\dot a}(r,z)+\sum_{i=1}^{a}\lambda_{v_{i,1},+\dot a}x_{v_{i,1}}^{+\dot a}(r,z),
\tag{7.22}
$$

with translated positions

$$
\begin{aligned}
x_A^{+\dot a}(r,z)=&-\left(1-\sum_i r_{i,2\ell_i}\right)z_{AB}^{+\dot a}\\
&-\sum_{i=1}^{a} r_{i,2\ell_i}\left[\sum_{j=1}^{\ell_i}(1-r_{i,2j-2})\left(\prod_{k=j}^{\ell_i-1} r_{i,2k}\right)z_{v_{i,j}B}^{+\dot a}-z_{v_{i,\ell_i}A}^{+\dot a}\right],\\
x_{v_{i,1}}^{+\dot a}(r,z)=&-\sum_{s=1}^{\ell_i}(1-r_{i,2s-1})\left(\prod_{k=1}^{s-1} r_{i,2k-1}\right)\sum_{j=1}^{s}(1-r_{i,2j-2})\left(\prod_{k=j}^{s-1} r_{i,2k}\right)z_{v_{i,j}B}^{+\dot a}\\
&+\left(\prod_{k=1}^{\ell_i} r_{i,2k-1}\right)(x_A^{+\dot a}(r,z)-z_{v_{i,\ell_i}A}^{+\dot a}),\qquad r_{i,0}=0.
\end{aligned}
\tag{7.23}
$$

For nonnegative integers $\kappa_{i,j}$ and $\bar\kappa_{i,2j}$, the weighted monomial moments on (7.10) are

$$
\begin{aligned}
&\int_{(7.10)}\prod_{i,j}dr_{i,j}\left(1-\sum_i r_{i,2\ell_i}\right)\prod_{i=1}^{a}\left[r_{i,2\ell_i}^{\ell_i-1+\kappa_{i,2\ell_i}}\prod_{j=1}^{\ell_i} r_{i,2j-1}^{\ell_i-j+\kappa_{i,2j-1}}\prod_{j=1}^{\ell_i-1} r_{i,2j}^{j-1+\kappa_{i,2j}}(1-r_{i,2j})^{1+\bar\kappa_{i,2j}}\right]\\
&=\frac{\prod_i(\ell_i-1+\kappa_{i,2\ell_i})!}{(n+1+\sum_i\kappa_{i,2\ell_i})!}\prod_{i=1}^{a}\left[\prod_{j=1}^{\ell_i}\frac{1}{\ell_i-j+1+\kappa_{i,2j-1}}\prod_{j=1}^{\ell_i-1}\frac{(j-1+\kappa_{i,2j})!\,(1+\bar\kappa_{i,2j})!}{(j+1+\kappa_{i,2j}+\bar\kappa_{i,2j})!}\right].
\end{aligned}
\tag{7.24}
$$

We organize the final result: Sum over nonnegative integers m_{iA}, m_{ik} $(k\neq i)$ with $m_{iA}+\sum_{k\neq i}m_{ik}=\ell_i$, p_{ijs} $(1\le j\le s\le \ell_i)$, and q_{ijA}, q_{ijk} $(1\le j\le \ell_i,\ 1\le k\le a)$, and write $q_{ij}=q_{ijA}+\sum_k q_{ijk}$ and $M_i=\sum_{k\neq i}m_{ki}+\sum_{k=1}^{a}\sum_{j=1}^{\ell_k}q_{kji}$.

The multinomial exponents m_{iA} and m_{ik} choose the λ_A and $\lambda_{v_{k,1}}$ terms in $[\lambda_{v_{i,1}}y_{AB}^{\mathrm{red}}]^{\ell_i}$. In the displacement exponential, p_{ijs} counts the sth $\lambda_{v_{i,1}}$ term of (7.20), including $s=\ell_i$, while q_{ijA} and q_{ijk} count the λ_A and $(\prod_{h=1}^{\ell_k}r_{k,2h-1})\lambda_{v_{k,1}}$ terms of y_{AB}^{red} in its last term.

The full Taylor expansion is

$$\begin{aligned}
\mathcal{I}_\Gamma(\lambda;z) = e^{-\lambda_{A,+\dot a} z_{AB}^{+\dot a} - \sum_i \lambda_{v_{i,1},+\dot a} z_{v_{i,1}B}^{+\dot a}} \sum_{m,p,q} \prod_{i=1}^{a} \Bigg\{ & \\
\ell_i! \, \frac{(\lambda_{v_{i,1},+\dot b} \lambda_{A,+}{}^{\dot b})^{m_{iA}}}{m_{iA}!} \prod_{k\neq i} \frac{(\lambda_{v_{i,1},+\dot c} \lambda_{v_{k,1},+}{}^{\dot c})^{m_{ik}}}{m_{ik}!} & \\
\times \prod_{j=1}^{\ell_i} \Bigg[\prod_{s=j}^{\ell_i} \frac{\left[-\lambda_{v_{i,1},+\dot d} (z_{v_{i,j}v_{i,j+1}} + z_{v_{i,j+1}B} - z_{v_{i,j}B})^{+\dot d}\right]^{p_{ijs}}}{p_{ijs}!} & \\
\times \frac{\left[\lambda_{A,+\dot e} (z_{v_{i,j}v_{i,j+1}} + z_{v_{i,j+1}B} - z_{v_{i,j}B})^{+\dot e}\right]^{q_{ijA}}}{q_{ijA}!} & \\
\times \prod_{k=1}^{a} \frac{\left[\lambda_{v_{k,1},+\dot f} (z_{v_{i,j}v_{i,j+1}} + z_{v_{i,j+1}B} - z_{v_{i,j}B})^{+\dot f}\right]^{q_{ijk}}}{q_{ijk}!} \Bigg] \Bigg\} & \\
\times \frac{\prod_i \left(\ell_i - 1 + \sum_j q_{ij}\right)!}{\left(n + 1 + \sum_{i,j} q_{ij}\right)!} \prod_{i=1}^{a} \prod_{t=1}^{\ell_i} \frac{1}{\ell_i - t + 1 + M_i + \sum_{s=t}^{\ell_i} \sum_{j=1}^{s} p_{ijs}} & \\
\times \prod_{i=1}^{a} \prod_{t=1}^{\ell_i - 1} \frac{\left(t - 1 + \sum_{j=1}^{t} \left[q_{ij} + \sum_{s=t+1}^{\ell_i} p_{ijs}\right]\right)! \, \left(1 + \sum_{j=1}^{t} p_{ijt}\right)!}{\left(t + 1 + \sum_{j=1}^{t} \left[q_{ij} + \sum_{s=t}^{\ell_i} p_{ijs}\right]\right)!}. &
\end{aligned} \tag{7.25}$$

For the physical displacements, $z_{v_{i,j}v_{i,j+1}} = 0$ for $j < \ell_i$. Taking one arm gives the Taylor coefficients of (5.26), which is exactly the fan kernel result. (A fan is a special type of spider.)

7.3 Kernels of general product forests

As noted in Section 7.1, spiders do not exhaust the graphs of the quantum product: other graphs (general forests) occur from three loops on and are the majority from four loops on, where the trees also begin to branch (Table 7.1).[21]

Table 7.1: Census of the marked graphs through seven loops, up to isomorphism fixing A, B. The fan and spider columns exclude the book.

n	graphs	books	fans	spiders	type I′,	not type I′,	branching	$A \leftrightarrow B$
				(≥ 3 arms)	not spiders	paths	trees	symmetric
1	1	1	0	0	0	0	0	1
2	3	1	2	0	0	0	0	1
3	8	1	4	0	0	3	0	2
4	23	1	6	2	1	11	2	3
5	65	1	6	6	3	38	11	5
6	194	1	8	12	8	112	53	8
7	576	1	8	20	16	313	218	14

Recall that a graph of *type I′* is built from one edge by adding vertices joined to both endpoints of an existing edge [13, Definition 4.12].

[21] From three loops on, some of these graphs are Laman graphs not of type I′ [13, Definition 4.12]: in the three-loop graph with edges $12, 13, 2A, 2B, 3A, 3B, AB$, the only degree-two vertex, 1, has neighbours that are not joined. These reductions therefore reach beyond the recursion of [13, Theorem 4.13].

Proposition 7.5 (Product graphs of type I′)**.** *Let* Γ *contain* AB*, interaction trees* T_j*, and at most one spoke per vertex to each insertion, with* $|E_j| = |T_j| + 1$ *spokes. Then* Γ *is of type I′ if and only if the endpoints of every internal edge share an insertion neighbour. For cubic vertices, each tree is then an arm with a common insertion neighbour, its source at one endpoint, and both insertion spokes at the other; for* $n \geq 3$*,* Γ *is a spider if and only if all arms of length at least two share the same insertion neighbour.*

Proof. Every edge of a type I′ graph lies in a triangle, and a triangle containing an internal edge must meet an insertion since the internal graph is a forest. Conversely, the common-neighbour condition gives each interaction vertex a spoke; if every leaf had two spokes, a tree with $L \geq 2$ leaves would have $|E_j| \geq 2L + (|T_j| - L) \geq |T_j| + 2$, a contradiction, so some leaf has one spoke and is removed by a series–parallel step. Induction proves the first assertion. For cubic vertices, tree degree three has zero spokes by (7.5), so every tree is a path along which the common insertion propagates; sharing that insertion across all arms of length ≥ 2 is precisely the spider condition. □

Star–triangle kernel operators

To reduce trees outside type I′, we supplement the series–parallel operator (Theorem 5.1) with two star–triangle kernel operators.

Theorem 7.6 (Star–triangle, the first case: two edges present)**.** *In a Laman circuit, let* v *have degree three, neighbours* u, w, x*, and source* λ_v*. Assume* ux, wx *are present and* uw *is absent, and let* $r_1, r_2 \in (0,1)$ *be the fractions of the merged* ux, wx *currents carried by the old* ux, wx *edges. The reduction removes one vertex, two edges, and one loop, preserving the Laman condition. For* $\lambda_v = 0$*,*

$$\mathcal{I}_\Gamma(\lambda; z) = \int_0^1 dr_1 \int_0^1 dr_2 \underbrace{\left(-r_1 r_2 \frac{\partial}{\partial z_{ux}^{\mathrm{red},+\dot{a}}} \frac{\partial}{\partial z_{wx,\dot{a}}^{\mathrm{red},+}} \right)}_{\text{local current Jacobian}} \mathcal{I}_{\Gamma^{\mathrm{red}}}(\lambda; z^{\mathrm{red}}), \tag{7.26}$$

with reduced displacements

$$z_{uw}^{\mathrm{red}} = z_{vw} - z_{vu}, \quad z_{ux}^{\mathrm{red}} = r_1 z_{ux} + (1 - r_1)(z_{vx} - z_{vu}), \quad z_{wx}^{\mathrm{red}} = r_2 z_{wx} + (1 - r_2)(z_{vx} - z_{vw}),$$

and $z_e^{\mathrm{red}} = z_e$ *for* $e \neq uw, ux, wx$*.*

Proof for $\lambda_v = 0$*.* The star–triangle replacement and the two parallel merges on ux, wx give $t_{uw}^{\mathrm{red}} = t_{vu} + t_{vw} + t_{vu}t_{vw}/t_{vx}$, $r_1 = t_{ux}^{\mathrm{red}}/t_{ux}$, and $r_2 = t_{wx}^{\mathrm{red}}/t_{wx}$, with resistance-Jacobian orientation sign -1, and current map

$$\begin{aligned} y_{vu} &= -y_{uw}^{\mathrm{red}} - (1 - r_1) y_{ux}^{\mathrm{red}}, & y_{vw} &= y_{uw}^{\mathrm{red}} - (1 - r_2) y_{wx}^{\mathrm{red}}, \\ y_{vx} &= (1 - r_1) y_{ux}^{\mathrm{red}} + (1 - r_2) y_{wx}^{\mathrm{red}}, & y_{ux} &= r_1 y_{ux}^{\mathrm{red}}, \quad y_{wx} = r_2 y_{wx}^{\mathrm{red}}. \end{aligned} \tag{7.27}$$

Choose a reduced spanning tree containing ux, wx. Conservation at fixed λ makes $d(y_{ux,+\dot{a}}^{\mathrm{red}} + y_{uw,+\dot{a}}^{\mathrm{red}})$ and $d(y_{wx,+\dot{a}}^{\mathrm{red}} - y_{uw,+\dot{a}}^{\mathrm{red}})$ linear combinations of the other retained chord-current differentials; modulo those differentials, $dy_{ux,+\dot{a}} \equiv y_{ux,+\dot{a}}^{\mathrm{red}} dr_1 - r_1 dy_{uw,+\dot{a}}^{\mathrm{red}}$ and $dy_{wx,+\dot{a}} \equiv y_{wx,+\dot{a}}^{\mathrm{red}} dr_2 + r_2 dy_{uw,+\dot{a}}^{\mathrm{red}}$, so

$$(-dy_{ux,++} \wedge dy_{ux,+-}) \wedge (-dy_{wx,++} \wedge dy_{wx,+-}) \equiv -r_1 r_2 (y_{ux,+\dot{a}}^{\mathrm{red}} y_{wx,+}^{\mathrm{red}\ \dot{a}}) dr_1 \wedge dr_2 \wedge (-dy_{uw,++}^{\mathrm{red}} \wedge dy_{uw,+-}^{\mathrm{red}}).$$

Including the resistance-change orientation sign and replacing the paired currents by displacement derivatives yields (7.26). □

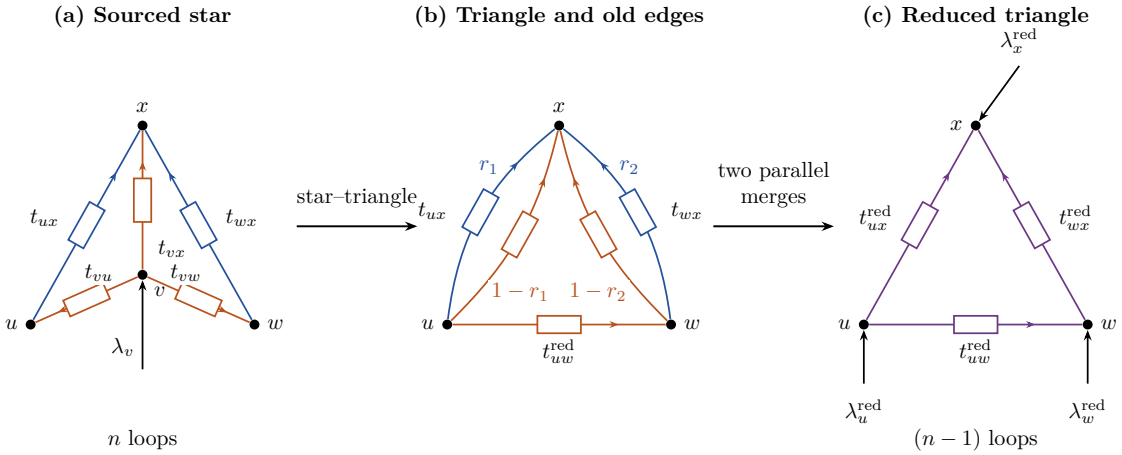


Figure 7.7: Sourced star–triangle reduction. The source at v is transferred to u, w, x; the blue and orange labels are fractions of the final merged currents.

When v carries a nonzero source λ_v (Figure 7.7), the unmerged star–triangle transformation distributes λ_v to u, w, x in proportion to the opposite triangle resistances:

$$\Sigma = t^{\triangle}_{uw} + t^{\triangle}_{ux} + t^{\triangle}_{wx}, \qquad \lambda^{\mathrm{red}}_u = \lambda_u + \frac{t^{\triangle}_{wx}}{\Sigma}\lambda_v, \quad \lambda^{\mathrm{red}}_w = \lambda_w + \frac{t^{\triangle}_{ux}}{\Sigma}\lambda_v, \quad \lambda^{\mathrm{red}}_x = \lambda_x + \frac{t^{\triangle}_{uw}}{\Sigma}\lambda_v, \tag{7.28}$$

where after the parallel merges $t^{\triangle}_{uw} = t^{\mathrm{red}}_{uw}$, $t^{\triangle}_{ux} = t^{\mathrm{red}}_{ux}/(1-r_1)$, and $t^{\triangle}_{wx} = t^{\mathrm{red}}_{wx}/(1-r_2)$, so that

$$\Sigma = t^{\mathrm{red}}_{uw} + \frac{t^{\mathrm{red}}_{ux}}{1-r_1} + \frac{t^{\mathrm{red}}_{wx}}{1-r_2}. \tag{7.29}$$

Because λ^{red} now depends on (r_1, r_2) and the reduced resistances, the sourced kernel retains the resistance-dependent source transfer inside the projective integral over $\mathbb{RP}^{2n-2}_{>0}$:

$$\mathcal{I}_\Gamma(\lambda; z) = -\int_0^1 dr_1 \int_0^1 dr_2 \int_{\mathbb{RP}^{2n-2}_{>0}} \exp\left[-\frac{\lambda_{v,+\dot a}}{\Sigma}\left(\frac{t^{\mathrm{red}}_{wx} z_{vu}}{1-r_2} + \frac{t^{\mathrm{red}}_{ux} z_{vw}}{1-r_1} + t^{\mathrm{red}}_{uw} z_{vx}\right)^{+\dot a} - \sum_{e\in\Gamma_1^{\mathrm{red}}} y^{\mathrm{red}}_{e,+\dot a} z_e^{\mathrm{red},+\dot a}\right]$$
$$\times\, \iota_{\partial/\partial r_2}\iota_{\partial/\partial r_1}\Big[\big(-d(r_1 y^{\mathrm{red}}_{ux,++}) \wedge d(r_1 y^{\mathrm{red}}_{ux,+-})\big) \wedge \big(-d(r_2 y^{\mathrm{red}}_{wx,++}) \wedge d(r_2 y^{\mathrm{red}}_{wx,+-})\big) \wedge \bigwedge_{\substack{e\in\Gamma_1^{\mathrm{red}}\setminus T^{\mathrm{red}} \\ e\neq uw}} (-dy^{\mathrm{red}}_{e,++} \wedge \tag{7.30}$$

Theorem 7.7 (Star–triangle, the second case: two star–triangle reductions at a time)**.** *In a Laman circuit, let v_1, v_2 be source-free degree-three vertices joined to u, A, B, with AB present and uA, uB absent (Figure 7.8). Let $r_1, r_2 \in (0,1)$ be the conductance fractions contributed by the v_1 star to uA, uB (with $1-r_1, 1-r_2$ from v_2), and let $r_3, r_4, 1-r_3-r_4$ be the conductance fractions on AB contributed by v_1, v_2 and the old edge AB. The reduction removes two vertices, four edges and two loops, preserving the Laman condition:*

$$\begin{aligned}\mathcal{I}_\Gamma(\lambda; z) = \int_0^1 dr_1 \int_0^1 dr_2 \int_{\substack{r_3>0,\ r_4>0 \\ r_3+r_4<1}} dr_3\, dr_4\, (r_1-r_2)(1-r_3-r_4) \\ \times \frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{uA}} \frac{\partial}{\partial z^{\mathrm{red},+}_{AB,\dot a}} \frac{\partial}{\partial z^{\mathrm{red},+\dot c}_{uB}} \frac{\partial}{\partial z^{\mathrm{red},+}_{AB,\dot c}} \mathcal{I}_{\Gamma^{\mathrm{red}}}(\lambda; z^{\mathrm{red}}),\end{aligned} \tag{7.31}$$

with reduced displacements $z^{\mathrm{red}}_{uA} = r_1(z_{v_1A} - z_{v_1u}) + (1-r_1)(z_{v_2A} - z_{v_2u})$, $z^{\mathrm{red}}_{uB} = r_2(z_{v_1B} - z_{v_1u}) + (1-r_2)(z_{v_2B} - z_{v_2u})$, $z^{\mathrm{red}}_{AB} = (1-r_3-r_4)z_{AB} + r_3(z_{v_1B} - z_{v_1A}) + r_4(z_{v_2B} - z_{v_2A})$, and $z^{\mathrm{red}}_e = z_e$ otherwise.

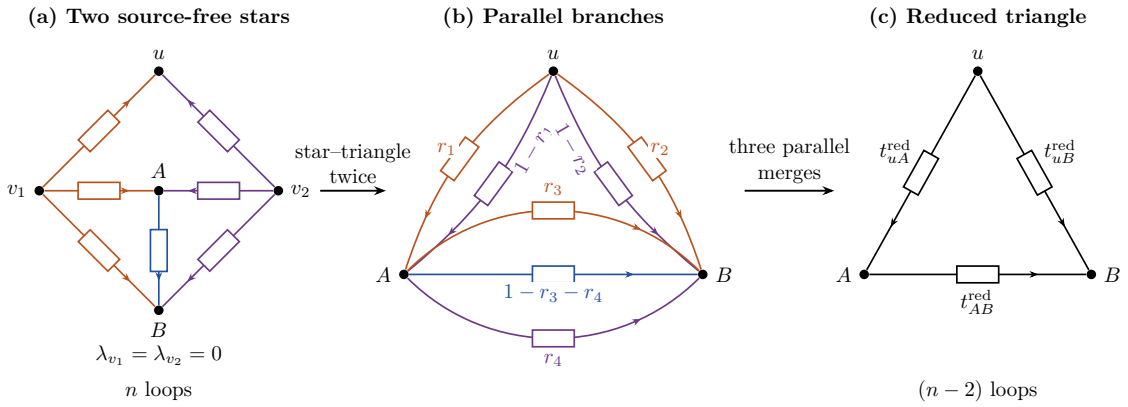


Figure 7.8: Two source-free stars reduce to a triangle by two star–triangle moves and three parallel merges. Orange and purple branches come from v_1 and v_2; blue is the old AB edge. The labels in the middle panel are fractions of the final merged currents.

Proof. The two star–triangle changes and three parallel merges preserve orientation, with current map

$$\begin{aligned} y_{v_1u} &= -r_1 y^{\mathrm{red}}_{uA} - r_2 y^{\mathrm{red}}_{uB}, \quad & y_{v_2u} &= -(1-r_1) y^{\mathrm{red}}_{uA} - (1-r_2) y^{\mathrm{red}}_{uB}, \\ y_{v_1A} &= r_1 y^{\mathrm{red}}_{uA} - r_3 y^{\mathrm{red}}_{AB}, & y_{v_2A} &= (1-r_1) y^{\mathrm{red}}_{uA} - r_4 y^{\mathrm{red}}_{AB}, \\ y_{v_1B} &= r_2 y^{\mathrm{red}}_{uB} + r_3 y^{\mathrm{red}}_{AB}, & y_{v_2B} &= (1-r_2) y^{\mathrm{red}}_{uB} + r_4 y^{\mathrm{red}}_{AB}, \qquad y_{AB} = (1-r_3-r_4) y^{\mathrm{red}}_{AB}. \end{aligned} \tag{7.32}$$

Choose a reduced spanning tree containing uA, uB. Modulo the differentials of the reduced chords other than AB, conservation at fixed λ yields

$$\begin{aligned} &dy_{AB,+\dot{+}} \wedge dy_{AB,+\dot{-}} \wedge dy_{v_1B,+\dot{+}} \wedge dy_{v_1B,+\dot{-}} \wedge dy_{v_2A,+\dot{+}} \wedge dy_{v_2A,+\dot{-}} \\ &\equiv (r_1 - r_2)(1 - r_3 - r_4)\big(y^{\mathrm{red}}_{uA,+\dot{a}} y^{\mathrm{red}}_{AB,+}{}^{\dot{a}}\big)\big(y^{\mathrm{red}}_{uB,+\dot{b}} y^{\mathrm{red}}_{AB,+}{}^{\dot{b}}\big)\, dr_1 \wedge dr_2 \wedge dr_3 \wedge dr_4 \wedge dy^{\mathrm{red}}_{AB,+\dot{+}} \wedge dy^{\mathrm{red}}_{AB,+\dot{-}}, \end{aligned}$$

from cancelling $(1-r_3-r_4)\big[(1-r_3-r_4) - (1-r_1-r_4) + (r_3-r_2)\big] = (r_1-r_2)(1-r_3-r_4)$. The three loop-area minus signs reduce to the single minus sign on the retained loop, and the four displacement derivatives insert the current product. □

Recursive reduction of product forests

The reduction of each tree in the forest can now be carried out independently from its leaves to its root.

Definition 7.8 (Rooted interaction trees)**.** Root a cubic interaction tree at its unique vertex of full graph degree two. The parent of a non-root vertex v is its neighbour on the path to the root; the other tree neighbours are its children, and vertices with the same parent are siblings. A spoke at a vertex u is an edge uA or uB.

The subtree of a vertex consists of the vertices whose path to the root passes through it.

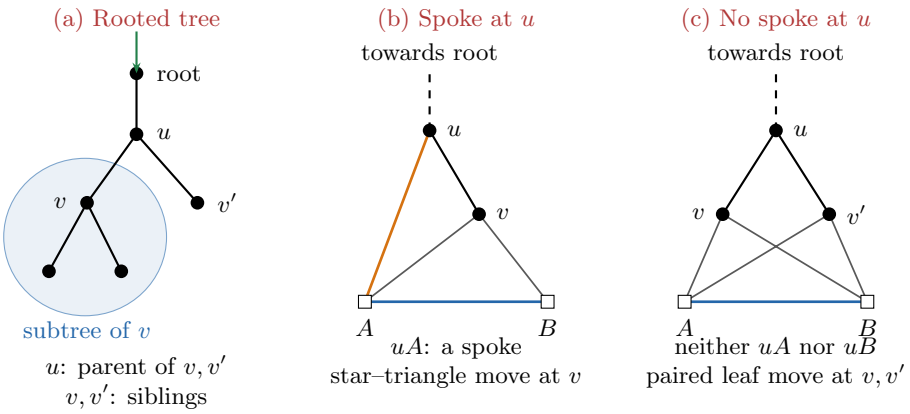


Figure 7.9: Rooted-tree terminology and the two reduction neighborhoods. (a) The shaded subtree contains v and its descendants; u is the parent of v, v', which are siblings. (b) The edge uA is a spoke; after the descendants of v are removed, a star–triangle move removes v. (c) The parent u has no spoke; after both descendant subtrees are removed, v, v' are eliminated together. Dashed edges continue towards the root; the blue edge is AB.

Proposition 7.9 (Kernels of cubic product forests)**.** *Let Γ be a nonzero marked graph of the quantum product with cubic interaction vertices (Proposition 7.3), and root each tree at its vertex of degree two, its only source.*

(i) *If Γ is of type I′, series–parallel moves (Theorem 5.1) along each arm reduce it to the marked resistor, which then carries*

$$y^{\mathrm{red}}_{AB,+\dot{a}} = \lambda_{A,+\dot{a}} + \sum_{\text{arms at } B} \left(\prod_{k=1}^{\ell_i} r_{i,2k-1} \right) \lambda_{v_{i,1},+\dot{a}} + \sum_{\text{arms at } A} \left(1 - \prod_{k=1}^{\ell_i} r_{i,2k-1} \right) \lambda_{v_{i,1},+\dot{a}}. \tag{7.33}$$

summed over the arms wired to B and to A. The reduction is similar to that in Figure 7.6, the reduction of spiders.

(ii) *Every Γ is reduced to the marked resistor by removing each vertex after all the vertices of its subtree. A source-free vertex is removed by a star–triangle move (Theorem 7.6) when its tree neighbor towards the root has a spoke, and together with its sibling (Theorem 7.7) when that neighbor has none; the root is removed last, by a series–parallel move.*

(iii) *The kernel of Γ is an integral over $2n$ coordinates, ranging over intervals and simplices, of a polynomial density times the exponential of polynomial currents (the displacements), and each Taylor coefficient in the displacements is a finite sum of the Dirichlet simplex moments*

$$\int_{\substack{r_j>0 \\ \sum_{j=1}^{\dim\Delta} r_j<1}} \prod_{j=1}^{\dim\Delta} (r_j^{k_j}\, dr_j) = \frac{\prod_{j=1}^{\dim\Delta} k_j!}{\left(\dim\Delta + \sum_{j=1}^{\dim\Delta} k_j\right)!}. \tag{7.34}$$

(The essential point is that no "spanning tree denominators" remain and the calculation is polynomial.)

Proof. (i) By Proposition 7.5, every tree is an arm $v_{i,1} - \cdots - v_{i,\ell_i}$ whose nodes are all wired to one insertion, with the source at $v_{i,1}$ and a resistor from v_{i,ℓ_i} to each insertion. On an arm wired to B, the two resistors of $v_{i,1}$ end at $v_{i,2}$ and B, which are joined, so Theorem 5.1 removes $v_{i,1}$ and passes the fraction $r_{i,1}$ of its source to $v_{i,2}$, which then has two resistors in turn. The source that reaches the last node of the arm is $\prod_k r_{i,2k-1}\lambda_{v_{i,1}}$, and the last move passes it to A across the marked resistor, as for the spider (7.11). On an arm wired to A the same moves run with A and B exchanged, and the fraction of the source that reaches A is $1 - \prod_k r_{i,2k-1}$. Conservation at A gives (7.33).

(ii) A source-free vertex v whose subtree is gone has three resistors, to its tree neighbour u towards the root, to A and to B: its own spokes and the sides left by the moves at its children, merged in parallel. If u has a spoke, to $X \in \{A, B\}$, two sides of the triangle that replaces v, namely uX and AB, merge with resistors already there, and Theorem 7.6 removes v. See Figure 7.10(a). If u has none, then by (7.5) u has a second child v', and once the subtree of v' is gone, Theorem 7.7 removes v and v' together. (The two star–triangle transformations at v and v' are performed together.) In both cases u is then wired to A and to B. The root, once its subtree is gone, has two resistors, to A and to B, which the marked resistor joins, and Theorem 5.1 removes it across AB.

(iii) Each move integrates the kernel of the smaller circuit over its ratios with a polynomial (not rational) factor. The currents of the larger circuit are linear in those of the smaller one with polynomial coefficients, and the series–parallel moves pass sources on with the weights r_1 and $1 - r_1$. A move removes one loop with two coordinates, or two loops with four, so the coordinates number $2n$, and the edge currents are recovered in reverse reduction order by (5.11), (7.27), and (7.32). By induction on the operators (5.10), (7.26), and (7.31) (which remove $1, 1, 2$ loops using $1, 2, 4$ displacement derivatives respectively), the displacement-degree-m piece has coordinate degree $\deg_{r_j} \le m + 2n - 1$ in every r_j, so expanding the exponential expresses each Taylor coefficient as a finite sum of at most $(m + 2n)^{2n}$ moments (7.34). □

Example 7.10 (Figure 7.10(a))**.** For $\lambda_1 = \lambda_3 = 0$ and $\lambda_B = -\lambda_A - \lambda_2$, apply (7.26) at 3 and 1, then (5.10) at 2. The successive ordered edge lists are

$$\begin{aligned}
\Gamma_1 &= (12, 13, 1A, 2B, 3A, 3B, AB),\\
(\Gamma^{\mathrm{red}})_1 &= (1B, 12, 1A, 2B, AB),\\
(\Gamma^{\mathrm{red\,red}})_1 &= (2A, 2B, AB),\\
(\Gamma^{\mathrm{red\,red\,red}})_1 &= (AB).
\end{aligned}$$

Use r_1, r_2 for the first star–triangle move, r_3, r_4 for the second, and r_5, r_6 for the series–parallel move, all in $(0, 1)$. Their displacement substitutions and source transfer are

$$\begin{aligned}
z_{1B}^{\mathrm{red}} &= z_{13} + z_{3B}, & z_{1A}^{\mathrm{red}} &= r_1 z_{1A} + (1 - r_1)(z_{13} + z_{3A}),\\
z_{AB}^{\mathrm{red}} &= r_2 z_{AB} + (1 - r_2)(z_{3B} - z_{3A}), & z_{12}^{\mathrm{red}} &= z_{12}, \quad z_{2B}^{\mathrm{red}} = z_{2B},\\
z_{2A}^{\mathrm{red\,red}} &= z_{1A}^{\mathrm{red}} - z_{12}, & z_{2B}^{\mathrm{red\,red}} &= r_3 z_{2B} + (1 - r_3)(z_{1B}^{\mathrm{red}} - z_{12}),\\
z_{AB}^{\mathrm{red\,red}} &= r_4 z_{AB}^{\mathrm{red}} + (1 - r_4)(z_{1B}^{\mathrm{red}} - z_{1A}^{\mathrm{red}}),\\
z_{AB}^{\mathrm{red\,red\,red}} &= r_6 z_{AB}^{\mathrm{red\,red}} + (1 - r_6)(z_{2B}^{\mathrm{red\,red}} - z_{2A}^{\mathrm{red\,red}}),\\
\lambda_A^{\mathrm{red\,red\,red}} &= \lambda_A + r_5 \lambda_2, & \lambda_B^{\mathrm{red\,red\,red}} &= \lambda_B + (1 - r_5)\lambda_2.
\end{aligned}$$

Derivatives act before these substitutions. With $z_{BA}^{\mathrm{red}} = -z_{AB}^{\mathrm{red}}$, the chain rule gives

$$\begin{aligned}
-\frac{\partial}{\partial z_{1A}^{\mathrm{red},+\dot a}} \frac{\partial}{\partial z_{BA,\dot a}^{\mathrm{red},+}} &= -\underbrace{\left(\frac{\partial}{\partial z_{2A}^{\mathrm{red\,red},+\dot a}} - (1 - r_4)\frac{\partial}{\partial z_{AB}^{\mathrm{red\,red},+\dot a}}\right)}_{\text{displacement Jacobian}} \underbrace{\left(-r_4 \frac{\partial}{\partial z_{AB,\dot a}^{\mathrm{red\,red},+}}\right)}_{\text{displacement Jacobian}}\\
&= r_4 \frac{\partial}{\partial z_{2A}^{\mathrm{red\,red},+\dot a}} \frac{\partial}{\partial z_{AB,\dot a}^{\mathrm{red\,red},+}}.
\end{aligned}$$

$$\mathcal{I}_\Gamma(\lambda;z) = -\int_{(0,1)^2} dr_1\,dr_2\,r_1r_2\,\frac{\partial}{\partial z_{1A}^{\mathrm{red},+\dot a}}\frac{\partial}{\partial z_{BA,\dot a}^{\mathrm{red},+}}\underbrace{\mathcal{I}_{\Gamma^{\mathrm{red}}}(\lambda;z^{\mathrm{red}})}_{\text{two loops}}$$

$$= -\int_{(0,1)^4}\prod_{j=1}^{4} dr_j\,r_1r_2r_3r_4^2 \times \frac{\partial}{\partial z_{2A}^{\mathrm{red\,red},+\dot a}}\frac{\partial}{\partial z_{AB,\dot a}^{\mathrm{red\,red},+}}\frac{\partial}{\partial z_{2B}^{\mathrm{red\,red},+\dot c}}\frac{\partial}{\partial z_{AB,\dot c}^{\mathrm{red\,red},+}} \times \underbrace{\mathcal{I}_{\Gamma^{\mathrm{red\,red}}}(\lambda;z^{\mathrm{red\,red}})}_{\text{one loop}}$$

$$= -(\lambda_{2,+\dot a}\lambda_{A,+}{}^{\dot a})^3\int_{(0,1)^6}\prod_{j=1}^{6} dr_j\,r_1r_2r_3r_4^2r_5(1-r_5)r_6^3 \times e^{-\lambda_{2,+\dot b}\left[r_5 z_{2A}^{\mathrm{red\,red}}+(1-r_5)z_{2B}^{\mathrm{red\,red}}\right]^{+\dot b}}\underbrace{e^{-(\lambda_A+r_5\lambda_2)_{+\dot c}z_{AB}^{\mathrm{red\,red\,red},+\dot c}}}_{\text{kernel of the final resistor}}.$$

$$\mathcal{I}_\Gamma(\lambda;0) = -(\lambda_{2,+\dot a}\lambda_{A,+}{}^{\dot a})^3\left(\frac12\right)^3\frac13\frac16\frac14$$
$$= -\frac{1}{576}(\lambda_{2,+\dot a}\lambda_{A,+}{}^{\dot a})^3.$$

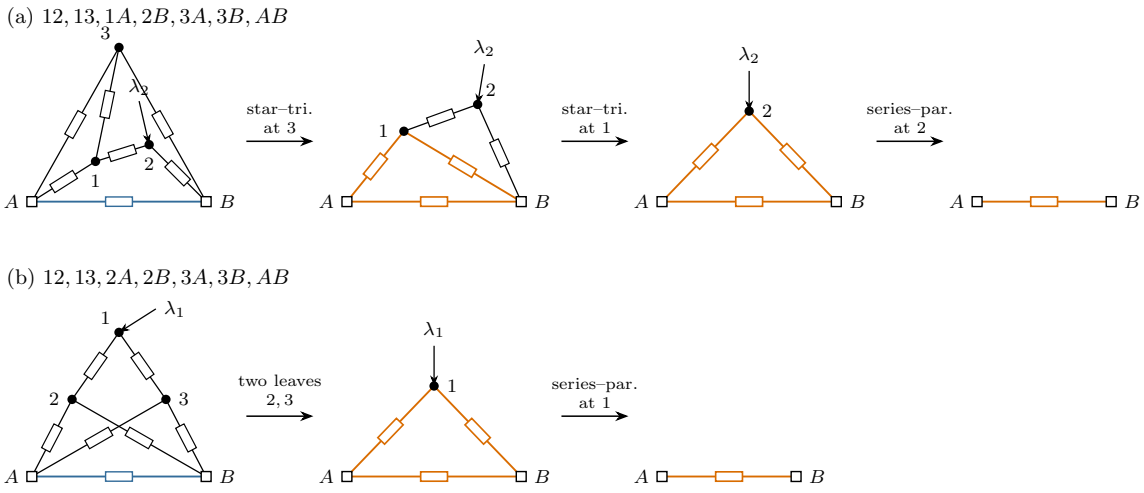


Figure 7.10: The three-loop graphs not of type I′, up to the exchange of A and B, reduced as in Proposition 7.9(ii); orange: the resistors created or merged by the preceding move, blue: the marked resistor. (a) Star–triangle moves (Theorem 7.6) at 3 and at 1, then a series–parallel move (Theorem 5.1) at the root 2. (b) The leaves $2,3$ hang from the root 1, which has no spoke, and are removed together (Theorem 7.7); the root is removed last.

Proposition 7.11 (Trees with a superpotential vertex)**.** *Let a tree T_j contain one superpotential vertex, with $\deg_\Gamma \bar W$ resistors and $\deg \bar W$ fields, its other vertices being cubic. Then T_j has $\deg_\Gamma \bar W - 2$ cubic vertices of degree two, each carrying a source, and the superpotential vertex carries a source when $\deg \bar W > \deg_\Gamma \bar W$. Series–parallel moves pass every source on with the weights r_1 and $1-r_1$, so graphs of type I′ keep polynomial currents for any placement of the sources. A star–triangle move at a vertex with a source distributes it with the rational weights (7.28), and the currents become rational functions of the coordinates.*

Proof. The counts are (7.8): the terminals of the cubic nodes and of the superpotential node, minus the ends of the resistors, leave $\deg \bar W - 2$ free terminals, $\deg \bar W - \deg_\Gamma \bar W$ of them at the superpotential node and one at each cubic node with two resistors. Theorem 5.1 holds for any

source at the removed node, with the polynomial weights r_1 and $1-r_1$. At a star with source λ_v, the source divides among the three outer nodes in proportion to the opposite sides of the triangle, (7.28), and these ratios are rational in the coordinates. □

Example 7.12 (Sourced star–triangle moves)**.** In the graph of Figure 7.10(b), let 2 be a superpotential vertex with $\deg_\Gamma \bar{W} = 3 < \deg \bar{W}$, so that 1, 2, A and B carry sources while $\lambda_3 = 0$. The neighbours 2, 3 of the degree-two vertex 1 are not joined, and the star–triangle move at the only source-free vertex, 3, merges only the side AB. A series step at 1 and this move each leave the complete circuit on four nodes, all of them with sources, so every elimination order needs a star–triangle move at a vertex with a source (Figure 7.11).

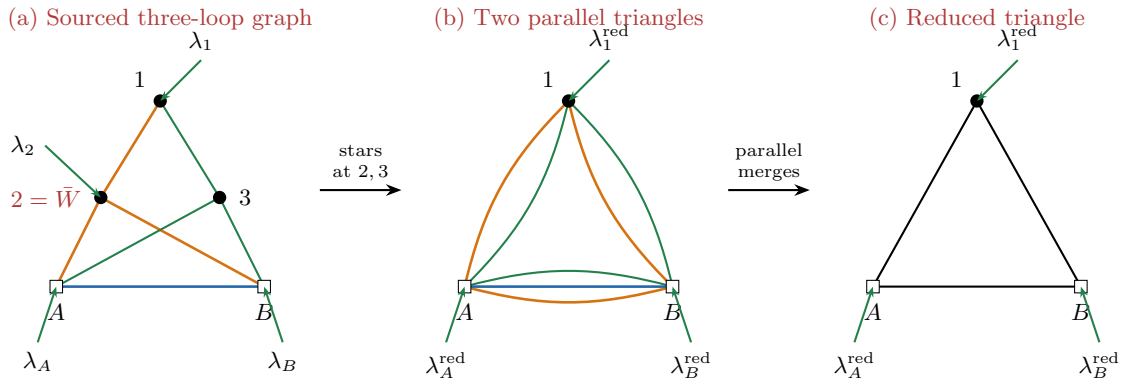


Figure 7.11: The three-loop sourced example, with 2 carrying $\bar{W}$ and $\lambda_2 \neq 0$. Removing the stars at 2 (orange) and 3 (green, $\lambda_3 = 0$) gives two triangles in parallel with the original marked edge AB (blue). Parallel merging leaves the triangle on $1, A, B$ with sources $\lambda_1^{\mathrm{red}}, \lambda_A^{\mathrm{red}}, \lambda_B^{\mathrm{red}}$.

Orienting and ordering the edges as $(21, 2A, 2B, 31, 3A, 3B, AB)$ and applying the two star–triangle changes of Theorem 7.7 while retaining the source transfer (7.28) at 2, the merged triangle has $t_{1A}^{\mathrm{red}} = 1 - r_5 - r_6$, $t_{1B}^{\mathrm{red}} = r_5$, $t_{AB}^{\mathrm{red}} = r_6$, and $\Sigma = (1 - r_5 - r_6)/r_1 + r_5/r_2 + r_6/r_3$. Unlike the cubic-forest integrands of Proposition 7.9(iii), the local loop-current Jacobian here retains a non-cancelling rational denominator in Σ. Nevertheless, combining the quadratic edge-contraction identity of the graph kernels [9, Section 3.2], [5, Theorem 7] (obtained by subdividing $AB \mapsto Av, vB$, where four contractions reduce to fans (5.19)) with the source-free boundary value $\mathcal{I}_\Gamma(\lambda; 0)|_{\lambda_A = \lambda_3 = 0} = -(\lambda_{1,+\dot{a}} \lambda_{2,+}{}^{\dot{a}})^3/576$ evaluates the zero-displacement integral in closed form:

$$\mathcal{I}_\Gamma(\lambda;0)|_{\lambda_3=0} = \frac{\lambda_{1,+\dot{a}}\lambda_{2,+}{}^{\dot{a}}}{576}\Big[5(\lambda_{1,+\dot{b}}\lambda_{A,+}{}^{\dot{b}})^2 + 12(\lambda_{1,+\dot{b}}\lambda_{A,+}{}^{\dot{b}})(\lambda_{2,+\dot{c}}\lambda_{A,+}{}^{\dot{c}}) - (\lambda_{1,+\dot{b}}\lambda_{2,+}{}^{\dot{b}})(\lambda_{1,+\dot{c}}\lambda_{A,+}{}^{\dot{c}}) + 3(\lambda_{2,+\dot{b}}\lambda_{A,+}{}^{\dot{b}})^2 + 3(\lambda_{1,+\dot{b}}\lambda_{2,+}{}^{\dot{b}})(\lambda_{2,+\dot{c}}\lambda_{A,+}{}^{\dot{c}}) - (\lambda_{1,+\dot{b}}\lambda_{2,+}{}^{\dot{b}})^2\Big]. \tag{7.35}$$

Table 7.2 counts these graphs through seven loops.

Table 7.2: Product graphs whose trees have at most one superpotential node, all other nodes cubic, up to isomorphism fixing A, B. Columns 3 and 6: superpotential node with at least four resistors; last column: those of column 6 reduced without a sourced star–triangle step.

n	graphs	$\bar{W}$ node with ≥ 4 resistors	type I′	spiders	$\bar{W}$ node with ≥ 4 resistors, not type I′	reduced without a sourced star–triangle
3	11	3	8	8	0	0
4	46	23	23	19	10	4

n	graphs	$\bar{W}$ node with	type I′	spiders	$\bar{W}$ node with ≥ 4	reduced without a
5	200	135	59	37	92	29
6	874	680	152	71	557	144
7	3765	3189	370	125	2864	610

8 Quantum supercharges and $\mathcal{N} = 4$ Yang–Mills theory

Combining the graph kernels with the BV pairing gives the quantum supercharge. For a general interaction, let W_Γ denote the ordered fibre contractions, including their Koszul signs. Restoring our scalar normalization (3.55) gives

$$Q_n\mathcal{O} = \frac{(-1)^n}{(n+1)!\,(8\pi^2)^n} \sum_{\Gamma \text{ labelled, rooted, Laman}} \Big[\mathcal{I}_\Gamma(\lambda; z) W_\Gamma[\mathcal{I}^{\otimes(n+1)}, \mathcal{O}]\Big]\Big|_{\text{jets, } z_{v,i}=0}. \tag{8.1}$$

At $n = 0$ the single-edge graph gives $Q_0\mathcal{O} = \{\mathcal{I}, \mathcal{O}\}_0$, which is the tree-level (classical) SUSY transformation.

The n-loop Q_n increases the number of derivatives by $2n$ (and by more if $\mathcal{I}$ contains derivatives):

Under $\lambda_{v,+\dot{a}} \to c\,\lambda_{v,+\dot{a}}$ and $z_e^{+\dot{a}} \to z_e^{+\dot{a}}/c$ at fixed $\dot{a}$, the displacement exponential in (3.64) is invariant and the current form, with n factors $dy_{e,+\dot{a}}$, scales as c^n. Since λ_v differentiates output fields and each z_e-derivative takes a derivative from an input letter, Q_n raises the numbers of $\partial_{+\dot{+}}$ and $\partial_{+\dot{-}}$ derivatives by n each, $2n$ in total. (A similar analysis holds in all dimensions.)

An operator with $d_{\dot{+}}$ and $d_{\dot{-}}$ holomorphic derivatives can be Q_n-exact (and exact under the successive differential d_n of Section 9) only if $n \leq \min(d_{\dot{+}}, d_{\dot{-}})$, so finitely many orders decide whether it becomes quantum exact.

In this section we present the main result: the loop corrections to the supercharge in $\mathcal{N} = 4$ SYM.

In $\mathcal{N} = 4$ SYM, all the holomorphic fields can be organized into one superfield C [4, Equation (3.52)].

The functional derivatives we defined at displaced points are normalized by

$$\frac{\delta C^B(y, \theta')}{\delta C^A(y', \theta)} = \delta_A{}^B\, \delta^{(4)}(y - y') \prod_{I=1}^{3} (\theta_I - \theta'_I), \tag{8.2}$$

8.1 The all-order anomalous SUSY transformation

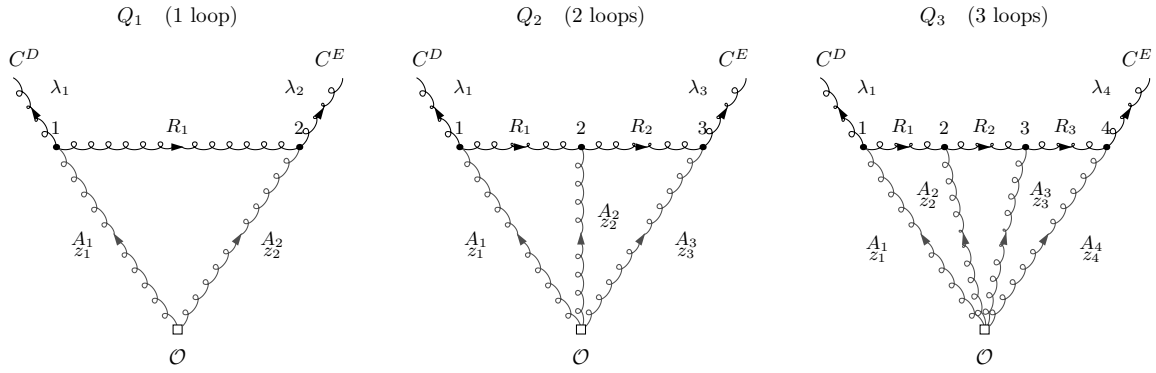


Figure 8.1: One-, two- and three-loop supercharge fans in $\mathcal{N}=4$ Yang–Mills theory. The spoke labelled A_j, z_j contracts an input $C^{A_j}(z_j,\theta)$ at $\mathcal{O}$. The path contracts the colour indices $R_1,\dots,R_n$ and leaves the endpoint fields C^D, C^E, carrying λ_1, λ_{n+1}.

Theorem 8.1. *For the superfield action* (A.3) *and an operator at x, the n-loop supercharge for $n \geq 1$ is*

$$\begin{aligned} Q_n = {} & \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n} \int_{(5.13)} dr_1 \cdots dr_{2n} \left[\prod_{j=1}^{n-1} r_{2j-1}^{n-j} r_{2j} (1-r_{2j})^{j-1} \right] r_{2n}^{n-1} \int d^3\theta \int \prod_{j=1}^{n+1} d^4 z_j \\ & \times C_{A_1 D R_1} \left(\prod_{j=2}^{n} C_{A_j R_{j-1} R_j} \right) C_{A_{n+1} R_n E} \\ & \times (\partial_{+\dot{a}_1} \cdots \partial_{+\dot{a}_n} C^D)(x + x_1(r,z), \theta) (\partial_+{}^{\dot{a}_1} \cdots \partial_+{}^{\dot{a}_n} C^E)(x + x_{n+1}(r,z), \theta) \\ & \times \frac{\delta}{\delta C^{A_1}(x+z_1,\theta)} \cdots \frac{\delta}{\delta C^{A_{n+1}}(x+z_{n+1},\theta)}, \end{aligned} \tag{8.3}$$

where the output fields are differentiated before they are evaluated at the translated positions.

$$\begin{aligned} x_1^{+\dot{a}}(r,z) = {} & \sum_{j=1}^{n} (1-r_{2j-1}) \left(\prod_{k=1}^{j-1} r_{2k-1} \right) \\ & \times \left[\left(\prod_{k=1}^{j-1} (1-r_{2k}) \right) z_1^{+\dot{a}} + \sum_{i=2}^{j} r_{2i-2} \left(\prod_{k=i}^{j-1} (1-r_{2k}) \right) z_i^{+\dot{a}} \right] \\ & + \left(\prod_{k=1}^{n} r_{2k-1} \right) z_{n+1}^{+\dot{a}}, \\ x_{n+1}^{+\dot{a}}(r,z) = {} & r_{2n} \left[\left(\prod_{k=1}^{n-1} (1-r_{2k}) \right) z_1^{+\dot{a}} + \sum_{i=2}^{n} r_{2i-2} \left(\prod_{k=i}^{n-1} (1-r_{2k}) \right) z_i^{+\dot{a}} \right] \\ & + (1-r_{2n}) z_{n+1}^{+\dot{a}}. \end{aligned}$$

The (fan) kernel and the complete Taylor expansion in the z_j are derived and listed in Section 5.

Proof.

The cubic term of (A.3) gives the vertex iC_{ABC}, which includes the 3! assignments of its three legs, and the rooted fan has symmetry factor 1/2 for the reversal of its interaction path.

The functional derivatives in (8.3) attach the spokes to the input letters; only the two endpoint fields remain uncontracted, so $\lambda_2 = \cdots = \lambda_n = 0$.

In the endpoint kernel (5.22), the substitution (3.32) gives, with $\partial_{1,+\dot a}$ and $\partial_{n+1,+\dot a}$ acting on the first and second output fields, respectively,

$$
\begin{aligned}
(\lambda_{1,+\dot a}\lambda_{n+1,+}{}^{\dot a})^n &\longmapsto 4^{-n}\partial_{1,+\dot a_1}\cdots\partial_{1,+\dot a_n}\partial_{n+1,+}{}^{\dot a_1}\cdots\partial_{n+1,+}{}^{\dot a_n},\\
e^{\lambda_{1,+\dot a}x_1^{+\dot a}(r,z)+\lambda_{n+1,+\dot a}x_{n+1}^{+\dot a}(r,z)} &\longmapsto e^{-\frac12 x_1^{+\dot a}(r,z)\partial_{1,+\dot a}-\frac12 x_{n+1}^{+\dot a}(r,z)\partial_{n+1,+\dot a}}.
\end{aligned}
$$

$$
\begin{aligned}
&e^{-\frac12 x_1^{+\dot a}(r,z)\partial_{1,+\dot a}-\frac12 x_{n+1}^{+\dot a}(r,z)\partial_{n+1,+\dot a}}C^D(x,\theta)C^E(x,\theta)\\
&\quad= C^D(x+x_1(r,z),\theta)C^E(x+x_{n+1}(r,z),\theta), \qquad \frac{\partial}{\partial x^{+\dot a}} = -\frac12\partial_{+\dot a}.
\end{aligned}
$$

$$
\underbrace{\frac12}_{\text{path reversal}}\ \underbrace{i^{n+1}}_{\text{cubic vertices}}\ \underbrace{(-8\pi^2)^{-n}}_{\text{scalar-kernel normalization}}\ \underbrace{4^{-n}}_{\text{output derivatives}} = \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n}.
$$

The path contracts the colour indices in the order $C_{A_1DR_1}(\prod_{j=2}^n C_{A_jR_{j-1}R_j})C_{A_{n+1}R_nE}$; inserting these factors and the density (5.22) gives (8.3). □

In practice, the colour factor can be simplified further with the trace relations of $SU(N)$ and of the other gauge groups [14].

For the zero-jet sector:

$$
\begin{aligned}
Q_n\big|_{\text{input jets}=0} = \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n (n!)^3(n+1)}\int d^3\theta\, C_{A_1DR_1}\left(\prod_{j=2}^n C_{A_jR_{j-1}R_j}\right)C_{A_{n+1}R_nE}\\
\times(\partial_{+\dot a_1}\cdots\partial_{+\dot a_n}C^D)(\partial_+{}^{\dot a_1}\cdots\partial_+{}^{\dot a_n}C^E)\prod_{j=1}^{n+1}\frac{\delta}{\delta C^{A_j}(\theta)}.
\end{aligned}
\tag{8.4}
$$

8.2 One- and two-loop examples in $\mathcal{N}=4$ Yang–Mills theory

The general-n operator (8.3) can look abstract. We also display the two-loop result, a problem that has recently attracted attention [3, Section 3.2.3].

$$
\begin{aligned}
Q_2 = &-\frac{i}{2(32\pi^2)^2}\int_0^1 dr_1\int_0^1 dr_2\int_0^1 dr_4\int_0^{1-r_4} dr_3\, r_1r_2r_4\int d^3\theta\int\prod_{j=1}^3 d^4z_j\\
&\times C_{A_1DR}C_{A_2RS}C_{A_3SE}\,(\partial_{+\dot a}\partial_{+\dot b}C^D)(x+x_1(r,z),\theta)(\partial_+{}^{\dot a}\partial_+{}^{\dot b}C^E)(x+x_3(r,z),\theta)\\
&\times\frac{\delta}{\delta C^{A_1}(x+z_1,\theta)}\frac{\delta}{\delta C^{A_2}(x+z_2,\theta)}\frac{\delta}{\delta C^{A_3}(x+z_3,\theta)},
\end{aligned}
\tag{8.5}
$$

where

$$
\begin{aligned}
x_1^{+\dot a}(r,z) &= (1-r_1)z_1^{+\dot a}+r_1(1-r_3)\big[(1-r_2)z_1^{+\dot a}+r_2z_2^{+\dot a}\big]+r_1r_3z_3^{+\dot a},\\
x_3^{+\dot a}(r,z) &= r_4\big[(1-r_2)z_1^{+\dot a}+r_2z_2^{+\dot a}\big]+(1-r_4)z_3^{+\dot a}.
\end{aligned}
$$

This result extends previous results and agrees with them.

For example, the first two components below agree with [4, Equations (3.66) and (3.67)] up to normalization:[22]

[22] On the physical Lagrangian subspace, with $\delta^{AB}(\gamma^I)_B \leftrightarrow \bar\phi_I{}^A$, $(\beta_I)^A \leftrightarrow -\frac{i}{\sqrt2}\psi_+^{IA}$, $\delta^{AB}\mathsf{b}_B \leftrightarrow \frac14 f_{++}^A$, $\partial_{+\dot a}\mathsf{c}^A \leftrightarrow 2\bar\lambda_{\dot a}^A$ and $\partial_{+\dot a}\partial_{+\dot b}\mathsf{c}^A \leftrightarrow 2\nabla_{+(\dot a}\bar\lambda_{\dot b)}^A$, equation (8.6) becomes $Q_1(\bar\phi_1{}^A\psi_+^{1B}) = \frac{i\sqrt2\,g^2}{16\pi^2}C^A{}_{RD}C^{BR}{}_E\,\bar\lambda_{\dot a}^D\bar\lambda^{E\dot a}$, the component form of the Konishi anomaly [7].

In component form:

$$Q_1(\delta^{AA'}(\gamma^1)_{A'}(\beta_1)^B) = \frac{g^2}{64\pi^2} C_{ARD}C_{BRE}(\partial_{+\dot a}\mathsf{c}^D)(\partial_+{}^{\dot a}\mathsf{c}^E), \tag{8.6}$$

$$Q_1((\beta_1)^A(\beta_2)^B) = \frac{ig}{64\pi^2}(C_{ARD}C_{BRE} + C_{BRD}C_{ARE})(\partial_{+\dot a}\mathsf{c}^D)\,\delta^{EF}\partial_+{}^{\dot a}(\gamma^3)_F, \tag{8.7}$$

$$Q_2((\beta_1)^{A_1}(\beta_2)^{A_2}(\beta_3)^{A_3}) = -\frac{ig^2}{48(32\pi^2)^2}\sum_{\sigma\in S_3} C_{A_{\sigma(1)}DR}C_{A_{\sigma(2)}RS}C_{A_{\sigma(3)}SE}(\partial_{+\dot a}\partial_{+\dot b}\mathsf{c}^D)(\partial_+{}^{\dot a}\partial_+{}^{\dot b}\mathsf{c}^E). \tag{8.8}$$

9 Multi-graviton completions and obstruction equations

Quantum multiplication constructs perturbatively closed multi-graviton operators in $\mathcal{N}=4$ Yang–Mills theory. We prove their closure at arbitrary loop order (Theorem 9.1) and, for illustration, evaluate an explicit product (Section 9.2).[23]

We then review the framework that uses the operators Q_n to test which classical classes are lifted: the successive transfers of Section 9.3 compute the perturbative obstructions of classical representatives and decide which of their closed completions are quantum coboundaries. The algorithm and the explicit results are in the supplementary material [14].

Since every graviton class has a closed completion, no differential maps one graviton class to another (Section 9.3); a lift pairs a fortuitous class with a graviton class or with another fortuitous class. This makes the quantum-correction test much lighter than a brute-force sweep. Quantum closure of a classical representative $\mathcal{O}_0$ means that there exist finite corrections $\mathcal{O}_1, \mathcal{O}_2, \dots$ with $\mathbf{Q}(\mathcal{O}_0 + \hbar\mathcal{O}_1 + \hbar^2\mathcal{O}_2 + \cdots) = 0$ at every order in $\hbar$; we remind the reader that $\mathcal{O}_0$ itself is in general not annihilated by $\mathbf{Q}$ (Section 9.3). Nonperturbative corrections can also modify the cohomology.

9.1 Perturbative closure of multi-gravitons

The single-graviton representatives of Chang and Yin [1, Section 4] are components of $\mathrm{Tr}(dC)^p$, where

$$dC = dx^{+\dot a}\frac{\partial C}{\partial x^{+\dot a}} + d\theta_I\frac{\partial_L C}{\partial\theta_I}. \tag{9.1}$$

Theorem 9.1. *Let $G_1, \dots, G_m$ be single-graviton representatives. Every iterated quantum product $\mathcal{O}$ of these operators, with any fixed bracketing, satisfies*

$$\mathbf{Q}\mathcal{O} = 0, \qquad \mathcal{O}|_{\hbar=0} = G_1\cdots G_m. \tag{9.2}$$

For each fixed component, its perturbative completion has finitely many terms.

Proof. The classical differential acts on a single graviton by

$$Q_0(dC) = [C, dC], \qquad Q_0\,\mathrm{Tr}(dC)^p = \mathrm{Tr}[C, (dC)^p] = 0. \tag{9.3}$$

Let us first consider a single graviton G.

[23] The theorem proves that multi-graviton classes stay closed; however, a closed multi-graviton can still become exact, e.g., if there is a nonclosed fortuitous class with the allowed charges. The most direct one-loop example is the lifting of a black-hole class by the Konishi correction: in the $SO(7)$ theory, the one-loop supercharge pairs a multi-graviton with a fortuitous class [2, Section 3], and by the theorem the multi-graviton is the exact member of the pair. Section 2.3.2 and [14, supplement, Section S3] give the one-loop lifts in $SO(2N+1)$ and their relation to S-duality. There are many more one-loop lifts in $SO(2N+1)$ than in $SU(N)$.

At order $n \geq 1$, the auxiliary θ part of the Q_n kernel reads:

$$\prod_{j=1}^{n} \prod_{I=1}^{3} (\theta_I^{(j)} - \theta_I^{(j+1)}). \tag{9.4}$$

A nonzero coincident value requires at least $3n$ odd derivatives. Each of the $n+1$ contracted factors of dC supplies at most one, leaving degree $3n - (n+1) = 2n - 1 > 0$ in the relative odd coordinates. The kernel therefore vanishes when those coordinates coincide, so all single gravitons are $\mathbf{Q}$-closed at every perturbative order.

$$Q_0 G = 0, \qquad Q_n G = 0 \quad (n \geq 1). \tag{9.5}$$

Applying (6.13) at each binary vertex of a quantum product gives

$$\mathbf{Q}(A \star B) = (\mathbf{Q}A) \star B + (-1)^{|A|} A \star (\mathbf{Q}B) = 0 \qquad (\mathbf{Q}A = \mathbf{Q}B = 0). \tag{9.6}$$

Induction proves the first assertion. The leading term of each binary product is ordinary multiplication, giving the stated classical limit [1, Section 5].

Finiteness follows from the finiteness corollary in Section 6.3. □

For example, for two single gravitons G_1 and G_2, the explicit quantum product shows that $G_1 G_2$ admits a quantum completion (the quantum completion is always nonunique, which is exactly the renormalization freedom), namely $G_1 \star G_2$; equivalently, $G_1 G_2$ is not lifted by a failure of $\mathbf{Q}$-closure.

The graviton part of the cohomology. At finite rank the classical cohomology consists of a graviton part, spanned by multi-graviton classes, and the remaining fortuitous classes [19]. Theorem 9.1 supplies every class of the graviton part with an explicit closed completion: write its classical representative as a polynomial in single-graviton representatives and replace each product by a quantum product. By Theorem 6.4, the resulting class is independent of the bracketing and of the order of the factors. (With more auxiliary pairs one can also define quantum products of three or more factors compatible with $\mathbf{Q}$; they differ from the iterated products $(G_1 \star G_2) \star G_3$ and $G_1 \star (G_2 \star G_3)$ by $\mathbf{Q}$-exact terms.) The same construction applies to a graviton-dressed fortuitous class $G\,\mathcal{O}_0$: once the fortuitous core $\mathcal{O}_0$ has a closed completion $\mathcal{O}[\mathcal{O}_0]$, the product $G \star \mathcal{O}[\mathcal{O}_0]$ is closed.

9.2 An explicit example

Take the two single-graviton representatives

$$A = \delta^{XY} (\gamma^1)_X (\gamma^1)_Y, \qquad B = (\gamma^2)_X (\beta_1)^X, \qquad \mathcal{O}_0 = AB. \tag{9.7}$$

Equation (A.2), with left differentiation, identifies both operators as components of $\mathrm{Tr}(dC)^2$:

$$\begin{aligned} \frac{\partial_L C^A}{\partial \theta_2}\bigg|_{\theta_1=\theta_2=0} &= \delta^{AX} (\gamma^2)_X - \theta_3 (\beta_1)^A, \\ \delta_{AB} \frac{\partial_L C^A}{\partial \theta_2} \frac{\partial_L C^B}{\partial \theta_2}\bigg|_{\theta_1=\theta_2=0} &= \delta^{XY} (\gamma^2)_X (\gamma^2)_Y - 2\theta_3 (\gamma^2)_X (\beta_1)^X. \end{aligned} \tag{9.8}$$

$$\begin{aligned} A &= \delta_{AB} \frac{\partial_L C^A}{\partial \theta_1} \frac{\partial_L C^B}{\partial \theta_1}\bigg|_{\theta=0}, \\ B &= -\frac{1}{2} \frac{\partial_L}{\partial \theta_3} \left(\delta_{AB} \frac{\partial_L C^A}{\partial \theta_2} \frac{\partial_L C^B}{\partial \theta_2} \right)\bigg|_{\theta=0}. \end{aligned} \tag{9.9}$$

Gauge invariance gives $Q_0 A = 0$. For B, the superpotential term gives

$$Q_0 B = -iC^{XYZ}(\gamma^2)_X(\gamma^2)_Y(\gamma^3)_Z = 0. \tag{9.10}$$

Both A and B are automatically Q_1-closed by Theorem 9.1.

Using (8.3), $\mathcal{O}_0$ itself is not Q_1-closed:

$$Q_1\mathcal{O}_0 = \frac{g^2}{32\pi^2} C^X{}_{RD} C^{YR}{}_E (\gamma^1)_X(\gamma^2)_Y(\partial_{+\dot a}\mathsf{c}^D)(\partial_+{}^{\dot a}\mathsf{c}^E). \tag{9.11}$$

Without the quantum product, whether $\mathcal{O}_0$ is lifted needs to be examined: one has to implement an algorithm to test whether the right-hand side of (9.11) is Q_0-exact. If the answer is yes, RHS $= Q_0\mathcal{O}'$, then $\mathcal{O}_0 - \hbar\mathcal{O}'$ is closed through one loop and the class survives at one loop (it is not **Q**-exact since it has no derivatives); if the answer is no, $\mathcal{O}_0$ is lifted together with another member of the Q_0-cohomology.

However, with the quantum product, we can explicitly answer the question: $\mathcal{O}_0$ is not lifted.

Taking $\mathcal{O}_1 = [\hbar](A \star B)$, we obtain

$$\begin{aligned}
\mathcal{O}_1 &= \{\mathcal{I}, A, B\}^{\gamma,0} \\
&= \frac{i}{8\pi^2} 2(ig)(-1)\frac{1}{2}\frac{1}{4} C_D{}^{XY}(\partial_{+\dot a}\mathsf{c}^D)(\partial_+{}^{\dot a}(\gamma^1)_X)(\gamma^2)_Y \\
&= \frac{g}{32\pi^2} C_D{}^{XY}(\partial_{+\dot a}\mathsf{c}^D)(\partial_+{}^{\dot a}(\gamma^1)_X)(\gamma^2)_Y.
\end{aligned} \tag{9.12}$$

Let us verify that $Q_0\mathcal{O}_1 + Q_1\mathcal{O}_0 = 0$.

Expanding $Q_0\mathcal{O}_1$ gives

$$\begin{aligned}
Q_0\mathcal{O}_1 = \frac{g^2}{32\pi^2} C_D{}^{XY}\Big[& -C^D{}_{EF}(\partial_{+\dot a}\mathsf{c}^E)\mathsf{c}^F(\partial_+{}^{\dot a}(\gamma^1)_X)(\gamma^2)_Y \\
& + C_{XE}{}^F(\partial_{+\dot a}\mathsf{c}^D)(\partial_+{}^{\dot a}\mathsf{c}^E)(\gamma^1)_F(\gamma^2)_Y \\
& + C_{XE}{}^F(\partial_{+\dot a}\mathsf{c}^D)\mathsf{c}^E(\partial_+{}^{\dot a}(\gamma^1)_F)(\gamma^2)_Y \\
& + C_{YE}{}^F(\partial_{+\dot a}\mathsf{c}^D)(\partial_+{}^{\dot a}(\gamma^1)_X)\mathsf{c}^E(\gamma^2)_F\Big].
\end{aligned} \tag{9.13}$$

The first, third and fourth lines cancel by invariance of C_{DXY} (the Jacobi identity):

$$C_{RX}{}^T C_{TYZ} + C_{RY}{}^T C_{XTZ} + C_{RZ}{}^T C_{XYT} = 0. \tag{9.14}$$

The remaining term is

$$\begin{aligned}
Q_0\mathcal{O}_1 &= \frac{g^2}{32\pi^2} C_D{}^{XY} C_{XE}{}^F (\gamma^1)_F(\gamma^2)_Y(\partial_{+\dot a}\mathsf{c}^D)(\partial_+{}^{\dot a}\mathsf{c}^E) \\
&= -\frac{g^2}{32\pi^2} C^X{}_{RD} C^{YR}{}_E (\gamma^1)_X(\gamma^2)_Y(\partial_{+\dot a}\mathsf{c}^D)(\partial_+{}^{\dot a}\mathsf{c}^E) \\
&= -Q_1\mathcal{O}_0.
\end{aligned} \tag{9.15}$$

The one-loop term of the quantum product therefore cancels the variation of the ordinary product of the two gravitons. The component degrees $\deg_\theta \mathcal{O}_0 = 5$ and $\deg_\theta \mathcal{O}_1 = 2$ make the quantum product terminate after its one-loop term:

$$\begin{gathered}
\mathcal{O} = A \star B = \mathcal{O}_0 + \hbar\mathcal{O}_1, \qquad Q_{n\geq 2}\mathcal{O}_0 = Q_{n\geq 1}\mathcal{O}_1 = 0, \\
\mathbf{Q}\mathcal{O} = \hbar(Q_1\mathcal{O}_0 + Q_0\mathcal{O}_1) = 0.
\end{gathered} \tag{9.16}$$

The displayed one-loop completion is the full perturbative product.

For other, more complicated classes, the two-loop term $Q_2\mathcal{O}_0$ might not vanish automatically, but the quantum product provides the completion at every order.

9.3 Homotopy transfer and successive obstructions

Having derived the full Q_2 and Q_n results, we now review the foundations of homotopy transfer.

$$\mathbf{Q}(\mathcal{O}_0+\hbar\mathcal{O}_1+\hbar^2\mathcal{O}_2+\cdots)=0 \quad\Longleftrightarrow\quad \sum_{\substack{j=0\\ \text{loop orders}}}^{s} Q_j\mathcal{O}_{s-j}=0 \qquad (s\geq 0). \tag{9.17}$$

Successive contractions. Let E_0 be the complex of holomorphic local operators and set $d_0=Q_0$. The pages are iterated cohomologies,

$$\begin{gathered} E_1=H(Q_0), \qquad E_2=H\big(H(Q_0),d_1\big), \qquad E_3=H\big(H(H(Q_0),d_1),d_2\big),\\ \ldots, \qquad E_{r+1}=H(E_r,d_r), \end{gathered} \tag{9.18}$$

The first differential is $d_1[\mathcal{O}_0]_1=[Q_1\mathcal{O}_0]_1$, and $Q_1^2=-(Q_0Q_2+Q_2Q_0)$ gives $d_1^2=0$ on $H(Q_0)$. For $r\geq 2$, the page obstruction includes the corrections to the representative:

$$d_r[\mathcal{O}_0]_r=\left[\sum_{\substack{j=1\\ \text{loop orders}}}^{r} Q_j\mathcal{O}_{r-j}\right]_r, \qquad \sum_{\substack{j=0\\ \text{loop orders}}}^{s} Q_j\mathcal{O}_{s-j}=0 \quad (0\leq s<r). \tag{9.19}$$

Successive transfer constructs these maps from $\mathbf{Q}$; the calculation below gives $d_r^2=0$ on E_r. Choose an inclusion $i_r:E_{r+1}\to E_r$ of cohomology representatives, a projection $\pi_r:E_r\to E_{r+1}$, and a homotopy $h_r:E_r\to E_r$ of degree -1, satisfying

$$\begin{gathered} \pi_r i_r=1, \qquad d_r i_r=\pi_r d_r=0, \qquad d_r h_r+h_r d_r=1-i_r\pi_r,\\ h_r i_r=\pi_r h_r=h_r^2=0. \end{gathered} \tag{9.20}$$

For a d_r-closed representative $a\in E_r$, the contraction separates its chosen cohomology representative from its exact part:

$$d_r a=0 \quad\Longrightarrow\quad a=i_r\pi_r a+\underbrace{d_r h_r a}_{\text{exact part}}, \qquad Q_0\mathcal{O}_0=0 \quad\Longrightarrow\quad \mathcal{O}_0=i_0\pi_0\mathcal{O}_0+Q_0h_0\mathcal{O}_0. \tag{9.21}$$

Write $\mathbf{Q}^{(0)}=\mathbf{Q}$. The superscript (r) counts transfers, while Q_n is the original n-loop coefficient; $[\hbar^j]$ extracts the coefficient of $\hbar^j$.

$$\begin{aligned} \mathbf{Q}^{(0)}&=Q_0+\hbar Q_1+\hbar^2Q_2+\cdots,\\ \mathbf{Q}^{(r)}&=d_r+\hbar\big([\hbar]\mathbf{Q}^{(r)}\big)+\hbar^2\big([\hbar^2]\mathbf{Q}^{(r)}\big)+\cdots, \qquad d_r=[\hbar^0]\mathbf{Q}^{(r)},\\ Q_n&:E_0^k\to E_0^{k+1}, \qquad \mathbf{Q}^{(r)}:E_r^k[[\hbar]]\to E_r^{k+1}[[\hbar]], \qquad d_r:E_r^k\to E_r^{k+1}. \end{aligned} \tag{9.22}$$

Here k is the cohomological degree and $E_r[[\hbar]]$ consists of formal series with coefficients in E_r; the coefficient $[\hbar^0]$ is the part independent of $\hbar$.

The transferred differential and corrected maps are

$$\begin{aligned} i_r(\hbar)&=\big[1+h_r(\mathbf{Q}^{(r)}-d_r)\big]^{-1}i_r,\\ \pi_r(\hbar)&=\pi_r\big[1+(\mathbf{Q}^{(r)}-d_r)h_r\big]^{-1},\\ h_r(\hbar)&=\big[1+h_r(\mathbf{Q}^{(r)}-d_r)\big]^{-1}h_r,\\ \mathbf{Q}^{(r+1)}&=\frac{1}{\hbar}\pi_r(\mathbf{Q}^{(r)}-d_r)i_r(\hbar), \qquad d_{r+1}=\mathbf{Q}^{(r+1)}\Big|_{\hbar=0}. \end{aligned} \tag{9.23}$$

$$
\begin{aligned}
E^k_{r+1} &= \frac{\ker(d_r : E^k_r \to E^{k+1}_r)}{\operatorname{im}(d_r : E^{k-1}_r \to E^k_r)},\\
E_1 \xrightarrow{i_0} E_0 \xrightarrow{Q_1} E_0 \xrightarrow{\pi_0} E_1, &\qquad d_1 = \pi_0 Q_1 i_0,\\
i_r(\hbar) &= i_r - \hbar h_r([\hbar]\mathbf{Q}^{(r)}) i_r + \cdots .
\end{aligned}
\tag{9.24}
$$

$$
\begin{aligned}
\mathbf{Q}^{(r)} i_r(\hbar) &= \hbar i_r(\hbar)\mathbf{Q}^{(r+1)},\\
\pi_r(\hbar)\mathbf{Q}^{(r)} &= \hbar \mathbf{Q}^{(r+1)}\pi_r(\hbar),\\
\mathbf{Q}^{(r)} h_r(\hbar) + h_r(\hbar)\mathbf{Q}^{(r)} &= 1 - i_r(\hbar)\pi_r(\hbar),\\
\pi_r(\hbar) i_r(\hbar) &= 1, \qquad\qquad (\mathbf{Q}^{(r+1)})^2 = 0.
\end{aligned}
\tag{9.25}
$$

Starting from $\mathbf{Q}^2 = 0$, these identities prove nilpotence successively:

$$
\begin{aligned}
0 &= (\mathbf{Q}^{(r)})^2 i_r(\hbar) = \hbar^2 i_r(\hbar)(\mathbf{Q}^{(r+1)})^2,\\
0 &= \hbar^2 \pi_r(\hbar) i_r(\hbar)(\mathbf{Q}^{(r+1)})^2 = \hbar^2 (\mathbf{Q}^{(r+1)})^2,\\
(\mathbf{Q}^{(r+1)})^2 &= 0, \qquad d^2_{r+1} = [\hbar^0](\mathbf{Q}^{(r+1)})^2 = 0,\\
&\operatorname{im} d_{r+1} \subseteq \ker d_{r+1}.
\end{aligned}
\tag{9.26}
$$

The third line follows by equating coefficients of formal series.

The first four obstruction maps make the successive transfers explicit.

The first obstruction maps. Writing $[\hbar^k]$ for the coefficient of $\hbar^k$, the first transfer has coefficients

$$
\begin{aligned}
[\hbar^0]\mathbf{Q}^{(1)} &= \pi_0 Q_1 i_0,\\
[\hbar]\mathbf{Q}^{(1)} &= \pi_0 (Q_2 - Q_1 h_0 Q_1) i_0,\\
[\hbar^2]\mathbf{Q}^{(1)} &= \pi_0 (Q_3 - Q_2 h_0 Q_1 - Q_1 h_0 Q_2 + Q_1 h_0 Q_1 h_0 Q_1) i_0.
\end{aligned}
\tag{9.27}
$$

The maps π_0, i_0, h_0 form the original contraction of the full physical operator complex onto its classical Q_0-cohomology.

In particular,

$$
\begin{aligned}
d_{r+1} &= \pi_r([\hbar]\mathbf{Q}^{(r)}) i_r,\\
d_1 &= \pi_0([\hbar]\mathbf{Q}^{(0)}) i_0,\\
d_2 &= \pi_1([\hbar]\mathbf{Q}^{(1)}) i_1,\\
d_3 &= \pi_2([\hbar]\mathbf{Q}^{(2)}) i_2,\\
d_4 &= \pi_3([\hbar]\mathbf{Q}^{(3)}) i_3.
\end{aligned}
\tag{9.28}
$$

$$
\begin{aligned}
d_1 &= \pi_0 Q_1 i_0,\\
d_2 &= \pi_1 \pi_0 (Q_2 - Q_1 h_0 Q_1) i_0 i_1,\\
d_3 &= \pi_2 \pi_1 \Big([\hbar^2]\mathbf{Q}^{(1)} - ([\hbar]\mathbf{Q}^{(1)}) h_1([\hbar]\mathbf{Q}^{(1)}) \Big) i_1 i_2,\\
d_4 &= \pi_3 \pi_2 \Big([\hbar^2]\mathbf{Q}^{(2)} - ([\hbar]\mathbf{Q}^{(2)}) h_2([\hbar]\mathbf{Q}^{(2)}) \Big) i_2 i_3.
\end{aligned}
\tag{9.29}
$$

The coefficients on each page are obtained from the same expansion:

$$
\begin{aligned}
[\hbar]\mathbf{Q}^{(r+1)} &= \pi_r \Big([\hbar^2]\mathbf{Q}^{(r)} - ([\hbar]\mathbf{Q}^{(r)}) h_r([\hbar]\mathbf{Q}^{(r)}) \Big) i_r,\\
[\hbar^2]\mathbf{Q}^{(r+1)} &= \pi_r \Big([\hbar^3]\mathbf{Q}^{(r)} - ([\hbar^2]\mathbf{Q}^{(r)}) h_r([\hbar]\mathbf{Q}^{(r)})\\
&\qquad - ([\hbar]\mathbf{Q}^{(r)}) h_r([\hbar^2]\mathbf{Q}^{(r)}) + ([\hbar]\mathbf{Q}^{(r)}) h_r([\hbar]\mathbf{Q}^{(r)}) h_r([\hbar]\mathbf{Q}^{(r)}) \Big) i_r.
\end{aligned}
\tag{9.30}
$$

$$
\begin{aligned}
[\hbar^3]\mathbf{Q}^{(1)} = \pi_0\Big(&Q_4 - Q_3h_0Q_1 - Q_2h_0Q_2 - Q_1h_0Q_3 \\
&+ Q_2h_0Q_1h_0Q_1 + Q_1h_0Q_2h_0Q_1 + Q_1h_0Q_1h_0Q_2 \\
&- Q_1h_0Q_1h_0Q_1h_0Q_1\Big)i_0.
\end{aligned}
\tag{9.31}
$$

The second term in d_2 is the contribution of the first correction to the representative. To see this directly, take $\mathcal{O}_0 = i_0 i_1 a$ with $a \in E_2$, and choose $\mathcal{O}_1 = -h_0 Q_1 \mathcal{O}_0$. Then

$$
\begin{aligned}
Q_0\mathcal{O}_1 &= -(1 - i_0\pi_0)Q_1\mathcal{O}_0 = -Q_1\mathcal{O}_0, \\
Q_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1) &= -Q_1^2\mathcal{O}_0 - Q_1Q_0\mathcal{O}_1 = 0, \\
\pi_1\pi_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1) &= d_2 a.
\end{aligned}
\tag{9.32}
$$

Equivalently, with $[\cdot]_r$ denoting a class in E_r,

$$
d_2[\mathcal{O}_0]_2 = [Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1]_2. \tag{9.33}
$$

Changing $\mathcal{O}_1$ by a classical cocycle changes the projected obstruction by a d_1-boundary, so it does not affect the d_2 obstruction equation.

Corrected representatives. For a class $a \in E_3$, the condition $d_2 i_2 a = 0$ gives the following choice of corrections through two loops:

$$
\begin{aligned}
\mathcal{O}_0 &= i_0 i_1 i_2 a, \\
\mathcal{O}_1 &= -h_0 Q_1 \mathcal{O}_0 - i_0 h_1 \pi_0 (Q_2 - Q_1 h_0 Q_1)\mathcal{O}_0, \\
\mathcal{O}_2 &= -h_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1).
\end{aligned}
\tag{9.34}
$$

$$
\begin{gathered}
\{\mathcal{O}_1 : Q_0\mathcal{O}_1 = -Q_1\mathcal{O}_0\} = -h_0Q_1\mathcal{O}_0 + \ker Q_0, \\
Q_0\mathcal{O}_2 = -(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1), \qquad \pi_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1) = 0, \\
Q_0\Big[-i_0h_1\pi_0(Q_2 - Q_1h_0Q_1)\mathcal{O}_0\Big] = 0.
\end{gathered}
\tag{9.35}
$$

The second-order equation selects a Q_0-closed addition to the first-order solution:

$$
\begin{aligned}
\pi_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1) &= (1 - d_1h_1)\pi_0(Q_2 - Q_1h_0Q_1)\mathcal{O}_0 \\
&= i_1\pi_1\pi_0(Q_2 - Q_1h_0Q_1)\mathcal{O}_0 \\
&= i_1 d_2 i_2 a = 0, \qquad a \in E_3.
\end{aligned}
\tag{9.36}
$$

The choice of $\mathcal{O}_2$ above solves the second-order equation; a further Q_0-closed term gives another such choice.

For a homogeneous component in $\mathcal{N} = 4$ theory, the grading of Appendix A gives

$$
\begin{aligned}
&\deg_\theta \mathcal{O}_s = \deg_\theta \mathcal{O}_0 - 3s, \qquad \mathcal{O}_s = 0 \quad \text{if } 3s > \deg_\theta \mathcal{O}_0, \\
&\deg_\theta(d_r[\mathcal{O}_0]_r) = \deg_\theta \mathcal{O}_0 - 3r \quad \text{whenever } d_r[\mathcal{O}_0]_r \neq 0.
\end{aligned}
\tag{9.37}
$$

On every page the same construction gives

$$
\mathbf{Q}\, i_0(\hbar)\cdots i_{r-1}(\hbar) = \hbar^r i_0(\hbar)\cdots i_{r-1}(\hbar)\mathbf{Q}^{(r)}. \tag{9.38}
$$

For $a \in E_{r+1}$, choose $\mathcal{O}_s$ as the coefficient of $\hbar^s$ in $i_0(\hbar)\cdots i_{r-1}(\hbar) i_r a$. Then

$$
\sum_{\substack{k=0 \\ \text{loop orders}}}^{s} Q_k\mathcal{O}_{s-k} = 0 \quad (0 \le s \le r), \qquad d_{r+1}[\mathcal{O}_0]_{r+1} = \left[\sum_{\substack{k=1 \\ \text{loop orders}}}^{r+1} Q_k\mathcal{O}_{r+1-k}\right]_{r+1}. \tag{9.39}
$$

For $a \in E_4$, expand $i_0(\hbar)i_1(\hbar)i_2(\hbar)i_3 a$ to first order:

$$\begin{aligned}
[\hbar] i_r(\hbar) &= -h_r([\hbar]\mathbf{Q}^{(r)}) i_r, \\
\mathcal{O}_0 &= i_0 i_1 i_2 i_3 a, \\
\mathcal{O}_1 &= -h_0 Q_1 \mathcal{O}_0 - i_0 h_1([\hbar]\mathbf{Q}^{(1)}) i_1 i_2 i_3 a \\
&\quad - i_0 i_1 h_2([\hbar]\mathbf{Q}^{(2)}) i_2 i_3 a.
\end{aligned} \tag{9.40}$$

$$\begin{aligned}
Q_0 i_0 i_1 h_2 = 0, \qquad d_1 \pi_0 i_0 i_1 h_2 = d_1 i_1 h_2 = 0, \\
\pi_2([\hbar]\mathbf{Q}^{(2)}) i_2 i_3 a = d_3 i_3 a = 0, \\
d_2\Big[-h_2([\hbar]\mathbf{Q}^{(2)}) i_2 i_3 a\Big] = -([\hbar]\mathbf{Q}^{(2)}) i_2 i_3 a.
\end{aligned} \tag{9.41}$$

The last term in $\mathcal{O}_1$ preserves the preceding closure equations and supplies the next compatibility condition.

Products and classical representatives. Suppose the operators $A = \sum_{s<r} \hbar^s A_s$ and $B = \sum_{s<r} \hbar^s B_s$ already satisfy the closure equations through order $r-1$. Write $\mathcal{O}_s = [\hbar^s](A \star B)$. The leading term is $\mathcal{O}_0 = A_0 B_0$. Taking coefficients in (6.13) gives

$$\begin{aligned}
&\sum_{\substack{k=0 \\ \text{loop orders}}}^{s} Q_k \mathcal{O}_{s-k} = 0 \qquad (0 \le s < r), \\
Q_0 \mathcal{O}_r + &\sum_{\substack{k=1 \\ \text{loop orders}}}^{r} Q_k \mathcal{O}_{r-k} = \Big(\sum_{\substack{k=1 \\ \text{loop orders}}}^{r} Q_k A_{r-k} \Big) B_0 \\
&\qquad + (-1)^{|A_0|} A_0 \Big(\sum_{\substack{k=1 \\ \text{loop orders}}}^{r} Q_k B_{r-k} \Big).
\end{aligned} \tag{9.42}$$

Passing to E_r gives

$$\begin{aligned}
[A_0]_r [B_0]_r &= [A_0 B_0]_r, \\
d_r([A_0]_r [B_0]_r) &= (d_r[A_0]_r)[B_0]_r + (-1)^{|A_0|}[A_0]_r (d_r[B_0]_r).
\end{aligned} \tag{9.43}$$

Starting from multiplication on $H(Q_0)$, this identity gives an associative graded-commutative product on each successive page.

$$E_{r+1}^k = \frac{\ker d_r^k}{\operatorname{im} d_r^{k-1}}, \qquad d_r^k : E_r^k \to E_r^{k+1}. \tag{9.44}$$

$$\begin{aligned}
d_r[A_0]_r = d_r[B_0]_r = 0 \quad &\Longrightarrow \quad d_r[A_0 B_0]_r = 0, \\
[A_0]_r = d_r a, \quad d_r[B_0]_r = 0 \quad &\Longrightarrow \quad [A_0 B_0]_r = d_r(a[B_0]_r), \\
[A_0]_{r+1}[B_0]_{r+1} &= [A_0 B_0]_{r+1}, \\
[A_0]_{r+1}[B_0]_{r+1} = 0 \quad &\Longleftrightarrow \quad [A_0 B_0]_r \in \operatorname{im} d_r.
\end{aligned} \tag{9.45}$$

Higher differentials on the graviton part. Applying the projection identity in (9.25) on successive pages gives

$$\pi_{r-1}(\hbar) \cdots \pi_0(\hbar)\, \mathbf{Q} = \hbar^r\, \mathbf{Q}^{(r)}\, \pi_{r-1}(\hbar) \cdots \pi_0(\hbar). \tag{9.46}$$

If $\mathbf{Q}\mathcal{O} = 0$, dividing (9.46) applied to $\mathcal{O}$ by $\hbar^r$ gives $\mathbf{Q}^{(r)}\pi_{r-1}(\hbar)\cdots\pi_0(\hbar)\mathcal{O} = 0$. Its constant term is $d_r\pi_{r-1}\cdots\pi_0\mathcal{O}_0$, and by induction on r the projection $\pi_{r-1}\cdots\pi_0\mathcal{O}_0$ is the class $[\mathcal{O}_0]_r$. Hence, a class with a closed completion satisfies

$$d_r[\mathcal{O}_0]_r = 0 \qquad (r \geq 1). \tag{9.47}$$

By Theorem 9.1, every d_r with $r \geq 1$ vanishes on the graviton part. For $d_r^k : E_r^k \to E_r^{k+1}$, the dimensions satisfy

$$\begin{aligned} \operatorname{rank} d_r^k &= \dim E_r^k - \dim\ker d_r^k, \\ \dim E_{r+1}^k &= \dim E_r^k - \operatorname{rank} d_r^k - \operatorname{rank} d_r^{k-1}. \end{aligned} \tag{9.48}$$

$$\begin{aligned} \operatorname{rank} d_r^k = \dim \left\{ [\mathcal{O}_0]_1 \in E_1^k : \begin{array}{c} \exists\, \mathcal{O}_1, \dots, \mathcal{O}_{r-1}, \\ \sum\limits_{\substack{j=0 \\ \text{loop orders}}}^{s} Q_j \mathcal{O}_{s-j} = 0 \quad (0 \leq s < r) \end{array} \right\} \\ - \dim \left\{ [\mathcal{O}_0]_1 \in E_1^k : \begin{array}{c} \exists\, \mathcal{O}_1, \dots, \mathcal{O}_{r}, \\ \sum\limits_{\substack{j=0 \\ \text{loop orders}}}^{s} Q_j \mathcal{O}_{s-j} = 0 \quad (0 \leq s \leq r) \end{array} \right\}. \end{aligned} \tag{9.49}$$

$$d_r[\mathcal{O}_0]_r = 0 \quad (\mathcal{O}_0 \text{ a multi-graviton}), \qquad [\mathcal{O}_0]_{r+1} = 0 \iff [\mathcal{O}_0]_r \in \operatorname{im} d_r^{k-1}. \tag{9.50}$$

The extension spaces are nested and contain the graviton part, so for any finite number R of pages,

$$\sum_{\substack{r=1 \\ \text{pages}}}^{R} \operatorname{rank} d_r^k \leq \dim E_1^k - \dim \operatorname{span} \left\{ [\mathcal{O}_0]_1 \in E_1^k : \mathcal{O}_0 \text{ is a multi-graviton} \right\}. \tag{9.51}$$

Fortuitous classes are sparse among the classical classes known so far [19, 20], so the successive differentials are extremely sparse on the known cohomology.

For completed operators satisfying

$$\mathbf{Q}X = \hbar^r \mathcal{O}, \qquad \mathbf{Q}\mathcal{O} = 0, \qquad \mathcal{O} = \mathbf{Q}(\hbar^{-r}X), \tag{9.52}$$

the last equality holds after inverting $\hbar$. Over formal power series, the first equality records a class annihilated by $\hbar^r$. The successive pages record these d_r-exact classes at their respective orders in $\hbar$.

$$\mathbf{Q}X = \hbar^r\mathcal{O} \implies d_r[X_0]_r = [\mathcal{O}_0]_r, \qquad [\mathcal{O}_0]_{r+1} = 0, \qquad X_0 = [\hbar^0]X. \tag{9.53}$$

Proposition 9.2 (Validity of successive homotopy transfer)**.**

Let $\mathbf{Q}^{(r)}$ be a formal differential on $E_r[[\hbar]]$, with $(\mathbf{Q}^{(r)})^2 = 0$ and constant coefficient d_r, and choose a contraction as in (9.20).

The maps defined by (9.23) are well defined coefficientwise and satisfy all identities in (9.25); consequently the successive transfers and the representative chain (9.38) hold.

For the homogeneous $\mathcal{N} = 4$ contractions of (A.13), both inverse series terminate on every fixed homogeneous polynomial input.

Proof.

The starting contraction can be chosen degree by degree: select representatives of $\ker d_r / \operatorname{im} d_r$ and a complement of $\ker d_r$ in E_r.

Take i_r to be the representative inclusion, π_r the projection onto these representatives followed by their cohomology classes, and h_r the inverse of d_r from its image to the chosen

complement, extended by zero on the representatives and the complement.

On the representatives, boundaries and complement, $d_r h_r + h_r d_r$ acts respectively as $0, 1, 1$, while $1 - i_r \pi_r$ has the same action; the remaining identities in (9.20) follow from these definitions.

Since $\mathbf{Q}^{(r)} - d_r$ has zero constant coefficient, the inverses are the following formal series, with k counting powers of the displayed operator:

$$\begin{aligned} \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} &= \sum_{\substack{k \geq 0 \\ \text{operator powers}}} (-1)^k \big[h_r(\mathbf{Q}^{(r)} - d_r)\big]^k, \\ \big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big]^{-1} &= \sum_{\substack{k \geq 0 \\ \text{operator powers}}} (-1)^k \big[(\mathbf{Q}^{(r)} - d_r)h_r\big]^k. \end{aligned} \tag{9.54}$$

At coefficient order $\hbar^n$, terms with $k > n$ vanish, so each coefficient is a finite sum; multiplying the series by its denominator cancels adjacent powers and gives the identity.

$$\begin{aligned} i_r(\hbar) &= i_r - h_r(\hbar)(\mathbf{Q}^{(r)} - d_r)i_r, \\ \pi_r(\hbar) &= \pi_r - \pi_r(\mathbf{Q}^{(r)} - d_r)h_r(\hbar), \\ h_r(\hbar) &= \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} h_r \\ &= h_r \big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big]^{-1}. \end{aligned} \tag{9.55}$$

The last equality follows term by term from associativity of composition; using $h_r^2 = h_r i_r = \pi_r h_r = 0$ gives

$$\begin{aligned} h_r\big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} &= h_r, \\ h_r(\hbar)^2 = \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} h_r^2 &= 0, \\ h_r(\hbar) i_r = 0, \qquad \pi_r h_r(\hbar) &= 0. \end{aligned} \tag{9.56}$$

$$\begin{aligned} \pi_r(\hbar) i_r(\hbar) = \pi_r i_r &- \pi_r h_r(\hbar)(\mathbf{Q}^{(r)} - d_r) i_r \\ &- \pi_r(\mathbf{Q}^{(r)} - d_r) h_r(\hbar) i_r \\ &+ \pi_r(\mathbf{Q}^{(r)} - d_r) h_r(\hbar)^2 (\mathbf{Q}^{(r)} - d_r) i_r = 1, \\ h_r(\hbar) i_r(\hbar) = 0, \qquad \pi_r(\hbar) h_r(\hbar) &= 0. \end{aligned} \tag{9.57}$$

From $(\mathbf{Q}^{(r)})^2 = d_r^2 = 0$,

$$\begin{aligned} &(\mathbf{Q}^{(r)} - d_r)\mathbf{Q}^{(r)} + \mathbf{Q}^{(r)}(\mathbf{Q}^{(r)} - d_r) \\ &= -d_r \mathbf{Q}^{(r)} - \mathbf{Q}^{(r)} d_r = (\mathbf{Q}^{(r)} - d_r)^2. \end{aligned} \tag{9.58}$$

Multiplication by the two inverse denominators gives the explicit homotopy calculation:

$$\begin{aligned} &\big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big](\mathbf{Q}^{(r)} h_r(\hbar) + h_r(\hbar)\mathbf{Q}^{(r)}) \\ &\qquad \times \big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big] \\ &= \mathbf{Q}^{(r)} h_r + h_r \mathbf{Q}^{(r)} \\ &\quad + h_r\big[(\mathbf{Q}^{(r)} - d_r)\mathbf{Q}^{(r)} + \mathbf{Q}^{(r)}(\mathbf{Q}^{(r)} - d_r)\big] h_r \\ &= 1 - i_r \pi_r + (\mathbf{Q}^{(r)} - d_r)h_r + h_r(\mathbf{Q}^{(r)} - d_r) \\ &\quad + h_r(\mathbf{Q}^{(r)} - d_r)^2 h_r \\ &= \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]\big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big] - i_r \pi_r. \end{aligned} \tag{9.59}$$

Applying the inverses already established yields

$$\mathbf{Q}^{(r)}h_r(\hbar) + h_r(\hbar)\mathbf{Q}^{(r)} = 1 - i_r(\hbar)\pi_r(\hbar). \tag{9.60}$$

$$\begin{aligned}\mathbf{Q}^{(r)}i_r(\hbar)\pi_r(\hbar) &= \mathbf{Q}^{(r)} - \mathbf{Q}^{(r)}h_r(\hbar)\mathbf{Q}^{(r)}\\ &= i_r(\hbar)\pi_r(\hbar)\mathbf{Q}^{(r)},\\ h_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar) &= \big[1 - i_r(\hbar)\pi_r(\hbar) - \mathbf{Q}^{(r)}h_r(\hbar)\big]i_r(\hbar) = 0.\end{aligned} \tag{9.61}$$

The differential induced on $E_{r+1}[[\hbar]]$ is therefore

$$\begin{aligned}\pi_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar) &= \pi_r\mathbf{Q}^{(r)}i_r(\hbar)\\ &\quad - \pi_r(\mathbf{Q}^{(r)} - d_r)h_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar)\\ &= \pi_r(\mathbf{Q}^{(r)} - d_r)i_r(\hbar) = \hbar\mathbf{Q}^{(r+1)}.\end{aligned} \tag{9.62}$$

The retraction and invariance identities now give both chain maps:

$$\begin{aligned}\mathbf{Q}^{(r)}i_r(\hbar) &= i_r(\hbar)\pi_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar)\\ &= \hbar i_r(\hbar)\mathbf{Q}^{(r+1)},\\ \pi_r(\hbar)\mathbf{Q}^{(r)} &= \pi_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar)\pi_r(\hbar)\\ &= \hbar\mathbf{Q}^{(r+1)}\pi_r(\hbar).\end{aligned} \tag{9.63}$$

$$\begin{aligned}0 = \pi_r(\hbar)(\mathbf{Q}^{(r)})^2 i_r(\hbar) &= \hbar^2(\mathbf{Q}^{(r+1)})^2,\\ (\mathbf{Q}^{(r+1)})^2 = 0, \qquad d_{r+1}^2 &= [\hbar^0](\mathbf{Q}^{(r+1)})^2 = 0,\\ d_{r+1} &= \pi_r\big([\hbar]\mathbf{Q}^{(r)}\big)i_r.\end{aligned} \tag{9.64}$$

Multiplication by $\hbar^2$ shifts formal-series coefficients injectively, giving the second line; iterating the proved inclusion identity gives (9.38).

For a homogeneous polynomial $a \in E_r$, $\deg_\theta a$ is the component degree (A.7); the homogeneous $\mathcal{N} = 4$ contractions (A.13) give, for $m \geq 1$ and nonzero outputs,

$$\begin{aligned}\deg_\theta\Big(h_r\big([\hbar^m]\mathbf{Q}^{(r)}\big)a\Big) &= \deg_\theta a - 3m,\\ \deg_\theta\Big(\big([\hbar^m]\mathbf{Q}^{(r)}\big)h_r a\Big) &= \deg_\theta a - 3m.\end{aligned} \tag{9.65}$$

Nonnegative component degree therefore implies

$$\begin{aligned}\big[h_r(\mathbf{Q}^{(r)} - d_r)\big]^k a = \big[(\mathbf{Q}^{(r)} - d_r)h_r\big]^k a &= 0 \qquad (3k > \deg_\theta a),\\ \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} a &= \sum_{\substack{0\le k\le \lfloor \deg_\theta a/3\rfloor\\ \text{operator powers}}} (-1)^k\big[h_r(\mathbf{Q}^{(r)} - d_r)\big]^k a,\\ \big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big]^{-1} a &= \sum_{\substack{0\le k\le \lfloor \deg_\theta a/3\rfloor\\ \text{operator powers}}} (-1)^k\big[(\mathbf{Q}^{(r)} - d_r)h_r\big]^k a.\end{aligned} \tag{9.66}$$

Each surviving coefficient of order $\hbar^n$ lowers the component degree by $3n$, so both expressions are polynomials in $\hbar$ of degree at most $\lfloor \deg_\theta a/3\rfloor$. □

10 Discussion

We close with three directions.

The physical theory. All our results are obtained in the holomorphically twisted theory. It remains to understand the same structures directly in the physical theory: the supersymmetry transformation of composite operators and, more generally, the full BV transformation that includes every symmetry together with its ghosts and auxiliary fields. This requires complete control of the counterterms and of the ultraviolet divergences of composite operators, and would give a complete account of quantum anomalies in this generalized sense and of the exact structure of local operators in supersymmetric gauge theories. It is considerably harder than the twisted computation suggests.

Lifting beyond one loop. The most immediate question is whether $\mathcal{N}=4$ Yang–Mills theory has a two-loop lift at all. In the ranges that we have scanned, the $\mathcal{N}=4$ theories have a nonzero d_1 in nineteen sectors, one for $SU(2)$ [20], sixteen for $SO(7)$ through $E+J_L=11$, the first of them found in [2], and one each for $SO(9)$ and $SO(11)$, and no nonzero d_r with $r\geq 2$ [14, supplement, Section S3.3]. By contrast, the complete census of the $SU(3)$ Leigh–Strassler theory through $E+J_L=11$ has 53 pairs of sectors with $d_1\neq 0$ and 12 with $d_2\neq 0$ [14, supplement, Section S6.2]. The scans should be pushed to higher levels and ranks, and the algorithm improved. Most of our ranks are computed modulo a large prime; the results marked [$\mathbb{Q}$] are certified over the rational numbers, and the others remain to be certified in the same way. Nonperturbative lifts of perturbative cohomology classes by instantons [3, 4], and their interplay with the higher brackets and the quantum product, are also worth studying.

Other dimensions and the gravity dual. The circuit reduction is not tied to four dimensions: the graph integrals and their current equations exist on $\mathbb{C}^d$ and in holomorphic–topological theories [5, 54], and the iterated network reductions should evaluate the corresponding Feynman graphs. It would also be interesting to identify the counterparts of the loop corrections, of the quantum product and of lifting in the gravity dual.

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A Conventions, the $\mathcal{N}=4$ superfield and charge sectors

A.1 The $\mathcal{N}=4$ Yang–Mills superfield

The $\mathcal{N}=4$ SYM theory has the superpotential

$$\bar{W}(\gamma,0)=\frac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F. \tag{A.1}$$

For the twist by Q^4_-, the three adjoint chiral multiplets can be further combined with the gauge fields into one superfield $C^A(\theta)$ [1, 4].

The odd coordinates obey $\theta_I\bar{\partial}=-\bar{\partial}\theta_I$, with $\int d^3\theta\,\theta_1\theta_2\theta_3=1$ and $\varepsilon^{123}=\varepsilon_{123}=1$.

$$C^A(\theta)=ig\,\mathsf{c}^A+\theta_I\delta^{AB}(\gamma^I)_B-\frac{1}{2}\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A+\frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B, \tag{A.2}$$

The physical meaning of each θ_I is $Q^I_+C^A=\partial_LC^A/\partial\theta_I$.

With $(t_A^{\mathrm{adj}})^B{}_C = iC^B{}_{AC}$, the action and differential are

$$\begin{aligned} S_{\mathrm{tw}} &= \int_{\mathbb{C}^2} \Omega \wedge \left[\int d^3\theta \left(\frac{1}{2}\delta_{AB} C^A \bar{\partial} C^B + \frac{i}{3!} C_{ABC} C^A C^B C^C \right) \right]_{(0,2)}, \\ Q_0 C^A &= \frac{i}{2} C^A{}_{BC} C^B C^C. \end{aligned} \tag{A.3}$$

The left fibre derivative and odd integration reproduce the component pairing,

$$\begin{aligned} \frac{\delta}{\delta C^A(\theta)} &= \frac{\theta_1\theta_2\theta_3}{ig} \frac{\partial_L}{\partial \mathsf{c}^A} - \frac{1}{2}\varepsilon^{IJK}\theta_J\theta_K \frac{\partial}{\partial(\gamma^I)_A} - \theta_I \frac{\partial_L}{\partial(\beta_I)^A} - ig\frac{\partial}{\partial \mathsf{b}_A}, \\ \frac{\delta C^B(\theta^{(2)})}{\delta C^A(\theta^{(1)})} &= \delta_A{}^B \prod_{I=1}^{3} (\theta_I^{(1)} - \theta_I^{(2)}). \end{aligned} \tag{A.4}$$

$$\Delta_{vw} = -\delta^{DE} \int d^3\theta \, \frac{\delta}{\delta C_v^D(\theta)} \frac{\delta}{\delta C_w^E(\theta)}. \tag{A.5}$$

A.2 Charge sectors and the grading of the loop corrections

For $\mathcal{N} = 4$ Yang–Mills theory, we organize the cochains by their letter weights and derivative counts.

Counting the vertices and contractions gives the grading of each loop correction, which determines its target sector and bounds the possible obstruction orders.

For the $\mathcal{N} = 4$ letters, θ_I counts the Ith odd-coordinate weight in C, and $n_\pm$ counts $\partial_{+\pm}$ derivatives.

For a monomial with Y letters, let e_I be the Ith unit vector and write

$$\begin{aligned} &\lambda = (\theta_1, \theta_2, \theta_3; n_+, n_-), \qquad \Theta = \theta_1 + \theta_2 + \theta_3, \qquad \mathsf{d} = (1,1,1;-1,-1), \\ &\theta(\gamma^I) = e_I, \qquad \theta(\beta_I) = (1,1,1) - e_I, \qquad \theta(\mathsf{b}) = (1,1,1), \qquad \theta(\mathsf{c}) = 0, \end{aligned} \tag{A.6}$$

$C^Y(\lambda)$, also written $(\lambda)_Y$, is the cochain sector with weight λ and Y letters. Its component degree is $\Theta = \deg_\theta$.

$$\deg_\theta \mathsf{c} = 0, \quad \deg_\theta \gamma = 1, \quad \deg_\theta \beta = 2, \quad \deg_\theta \mathsf{b} = 3, \qquad \deg_\theta(\partial_{+\dot{a}} X) = \deg_\theta X, \tag{A.7}$$

At order n, a product graph has n cubic vertices and $2n$ physical edges, each pairing component degrees that sum to three.

$$\deg_\theta \mathcal{O}_n = \deg_\theta \mathcal{O}_0 + 3n - 6n = \deg_\theta \mathcal{O}_0 - 3n, \qquad \mathcal{O}_n = 0 \quad \text{if } 3n > \deg_\theta \mathcal{O}_0. \tag{A.8}$$

The degree losses add under iterated products.

For a nonzero $Q_n X$, the $n+1$ cubic vertices and $2n+1$ edges give

$$\deg_\theta(Q_n X) = \deg_\theta X + 3(n+1) - 3(2n+1) = \deg_\theta X - 3n \tag{A.9}$$

The physical charges are the dimension E, the three $SU(4)_R$ Cartan charges R_I, and the spins J_L, J_R.

$$\begin{aligned} E &= \tfrac{1}{2}(\Theta + Y) + n_+ + n_-, \qquad R_I = \theta_I - \tfrac{1}{2}(\Theta - Y), \\ J_L &= \tfrac{1}{2}(\Theta - Y + n_+ + n_-), \qquad J_R = \tfrac{1}{2}(n_+ - n_-), \\ E &= R_1 + R_2 + R_3 + 2J_L, \qquad 2(E + J_L) = 2\Theta + 3(n_+ + n_-). \end{aligned} \tag{A.10}$$

Table A.1: Letters of $\mathcal{N}=4$ Yang–Mills theory and their charges.

letter	physical field	θ	(n_+,n_-)	E	(R_1,R_2,R_3)	J_L	J_R
γ^I	scalar $\bar{\phi}_I$	e_I	$(0,0)$	1	e_I	0	0
β_I	fermion ψ^I_+	$(1,1,1)-e_I$	$(0,0)$	$\frac{3}{2}$	$(\frac{1}{2},\frac{1}{2},\frac{1}{2})-e_I$	$\frac{1}{2}$	0
b	field strength f_{++}	$(1,1,1)$	$(0,0)$	2	$(0,0,0)$	1	0
$\partial_{+\pm}\mathsf{c}$	gaugino $\bar{\lambda}_\pm$	0	$(1,0),(0,1)$	$\frac{3}{2}$	$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$	0	$\pm\frac{1}{2}$
$\partial_{+\pm}$	covariant derivative $\nabla_{+\pm}$	0	$(1,0),(0,1)$	1	$(0,0,0)$	$\frac{1}{2}$	$\pm\frac{1}{2}$

The level is $E+J_L$. Equation (A.10) makes its set of sectors finite.

Example Threshold charges

The $SU(2)$ threshold class $\mathcal{O}_0$ [20, 21] has two γ, five β and zero derivatives.

$$\lambda=(4,4,4;0,0),\quad Y=7,$$
$$E=\frac{19}{2},\qquad R_I=\frac{3}{2},\qquad J_L=\frac{5}{2},\qquad E+J_L=12.$$

The current form (3.64) has degree n in each dotted component, giving

$$\mathcal{I}_\Gamma(\lambda_{v,+\dot{a}}\mapsto c\lambda_{v,+\dot{a}};\ z_e^{+\dot{a}}\mapsto c^{-1}z_e^{+\dot{a}})=c^n\,\mathcal{I}_\Gamma(\lambda;z),\qquad c>0. \tag{A.11}$$

The λ powers supply output derivatives and the z powers remove input derivatives. Hence Q_n adds n derivatives in each direction [3, Equations (3.44)–(3.45)].

Q_0 preserves both totals. Choose homogeneous contractions (i_r,π_r,h_r) in (9.23).

Proposition A.1.

For $n\geq 0$ and $r\geq 1$,

$$Q_n: C^Y(\lambda)\to C^{Y+1-n}(\lambda-n\mathsf{d}),\qquad d_r: E_r^Y(\lambda)\to E_r^{Y+1-r}(\lambda-r\mathsf{d}). \tag{A.12}$$

Both maps have $(\Delta E;\Delta R_I;\Delta J_L,\Delta J_R)=(\frac{1}{2};\frac{1}{2},\frac{1}{2},\frac{1}{2};-\frac{1}{2},0)$.

Proof

$$\Delta Y=3(n+1)-2(2n+1)=1-n,\qquad \theta_I(Q_nX)-\theta_I(X)=(n+1)-(2n+1)=-n.$$

The vertex and edge counts give the letter and flavour shifts. Equation (A.11) supplies n derivatives in each direction.

The homogeneous contractions and transferred coefficients obey

$$\begin{aligned} h_r &: E_r^Y(\lambda)\longrightarrow E_r^{Y+r-1}(\lambda+r\mathsf{d}),\\ [\hbar^m]\mathbf{Q}^{(r)} &: E_r^Y(\lambda)\longrightarrow E_r^{Y+1-r-m}(\lambda-(r+m)\mathsf{d}). \end{aligned} \tag{A.13}$$

For $[\hbar^m]\mathbf{Q}^{(r+1)}$, let ℓ count perturbation factors. Their orders sum to $m+1$, and the term contains $\ell-1$ copies of h_r.

$$\begin{aligned} \Delta Y &= \ell(1-r)-(m+1)+(\ell-1)(r-1)=-r-m,\\ \Delta\lambda &= -(\ell r+m+1)\mathsf{d}+(\ell-1)r\mathsf{d}=-(r+m+1)\mathsf{d}. \end{aligned} \tag{A.14}$$

This proves the induction from $\mathbf{Q}^{(0)}=\mathbf{Q}$. Set $m=0$ for d_{r+1} and insert the shifts into

(A.10).

$$\Delta\Theta = -3n, \qquad \Delta(n_+ + n_-) = 2n, \qquad (\Delta E; \Delta R_I; \Delta J_L, \Delta J_R) = \left(\frac{1}{2}; \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, 0\right).$$

□

The sectors $(\lambda + k\mathsf{d})_{Y+k}$, $k \in \mathbb{Z}$, have the same physical charges. The cochain grading distinguishes the loop orders.

Corollary A.2.

An outgoing d_r requires $r \leq \min_I \theta_I$, and an incoming d_r requires $r \leq \min(n_+, n_-)$.

A class with zero input derivatives has no incoming d_r for $r \geq 1$.

Proof

$$\theta_I - r \geq 0 \iff r \leq \min_I \theta_I, \qquad n_\pm - r \geq 0 \iff r \leq \min(n_+, n_-).$$

□

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